exploring cell types with the allen institute

12 November 2026

allen institute/

allen institute

agenda/morning

- 9:00-9:30 | check-in / breakfast
- **9:30-9:40 | welcome, logistics, and overview of allen institute**
- 9:40-10:05 | cell typing at allen institute: taxonomies, data sets, and use cases
- 10:05-10:30 | human and mammalian brain atlas (HMBA)
- 10:30-10:40 | break
- 10:40-11:05 | localizing cell types with spatial transcriptomics
- 11:05-11:30 | cell-type targeted enhancer development
- 11:30-11:55 | Seattle Alzheimer's Disease Brain Cell Atlas in the brain health accelerator
- 11:55-12:05 | Overview of brain knowledge platform (BKP) tools & resources
- 12:05-12:10 | Preview of afternoon
- 12:10-1:10 | lunch

github.com/AllenInstitute/SPOCTT/tree/main/2026_Additional_Presentations

agenda/afternoon – breakout rooms

Session 1: connecting multimodal data to brain cell types

Room 1255-1265

1:10-1:50 | defining and using common coordinate frameworks (reference atlases, mesoscale)

1:50-2:00 | patch-seq overview and tools for analysis

2:00-2:20 | Troubleshooting Annotation with the BKP

2:20-2:30 | break

2:30-2:40 | ABC Atlas demo & external use case examples

2:40-3:40 | user exploration of allen institute tools

3:40-4:00 | feedback survey

4:00 – 5:00 | additional time for user exploration

Session 2: analysis of your own sequencing data using allen institute resources.

Room 1165 - 1185

1:10-1:35 | Using a reference cell type taxonomy to understand Alzheimer's disease

1:35-2:00 | cross-species alignment and comparative analyses

2:00-2:20 | other interactive tools for -omics data exploration (CellxGene, Cytosplore, etc.)

2:20-2:30 | break

2:30-2:40 | ABC Atlas demo & external use case examples

2:40-3:40 | user exploration of allen institute tools

3:40-4:00 | feedback survey

4:00 – 5:00 | additional time for user exploration

agenda/afternoon **– Baltimore**

1:10-1:45 Integrating with multimodal data

1:45-2:30 Analyzing own data with allen tools

2:30-2:40 Break

2:40-3:00 ABC Atlas demo & external use case examples

3:00-3:30 User exploration of allen institute tools

3:30-3:45 Tool feedback and Discussion

3:45-4:00 Workshop Survey / closing of formal program

4:00-5:00 Additional time for user exploration

workshop introduction/

- welcome
- logistics
- allen institute overview

alleninstitute/

about us/staff at North Bethesda

allen institute

Aaron Garcia
Human & mammalian brain atlas

Ashwin Bhandiwad
Common coordinate framework

Kyle Travaglini
Seattle Alzheimer's Project

Lauren Alfiler & Rachel Hostetler
General help for anything

Meghan Turner
Spatial transcriptomics

Morgan Wirthlin
Cross-species epigenomics

Will Laird
Viral tool development

Thank you to our NIH hosts!

Christie Espinoza
Olivier Berton
Rajtarun Madangopal
Katherine Savell
Falk Lohoff
Xylena Reed

about us/staff at Baltimore location

allen institute

Jeremy Miller
taxonomy& MapMyCells

Irika Sinha
data coordination

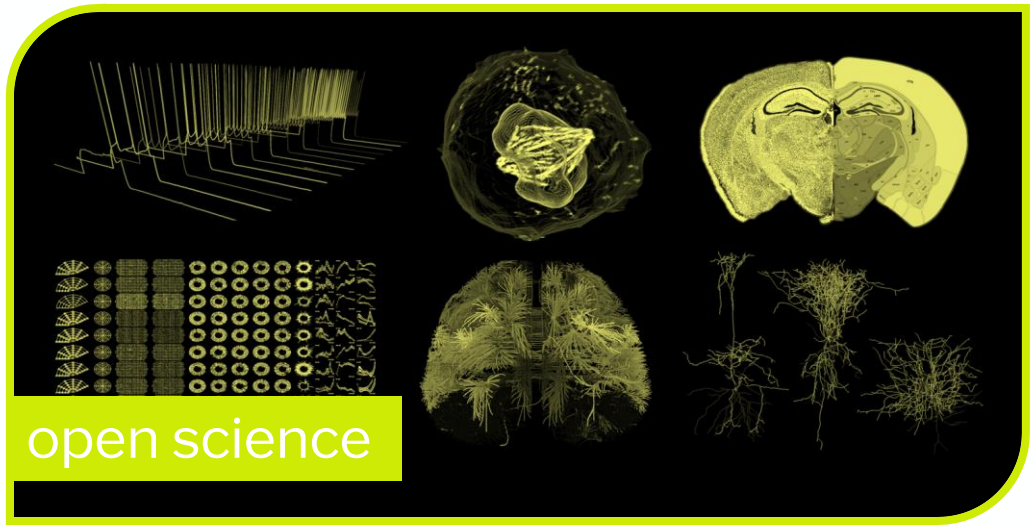
Elysha Fiabane
ABC Atlas

allen institute/

- founding date: September 16, 2003
- founders: Paul G. Allen; Jody Allen
- employees: ~900
- non-profit institution
  - no university affiliation (or degrees)
  - no traditional labs with single PI
- foundational science

understanding life
advancing health

allen institute/ our core principles



allen institute/ funding & structure

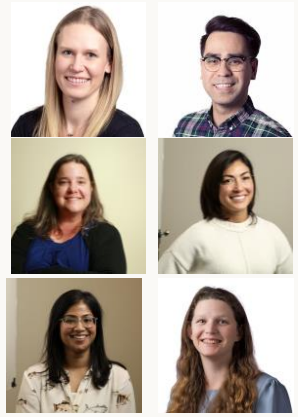
- funding types
  - majority private funding, particularly ”Fund for Science & Technology” – part of Paul G. Allen’s estates
  - some public funding (primarily NIH Grants)
    - no K series (training grants)
    - mainly U & R series

We are privileged in our funding support so we want to ensure all our resources can be used by the community

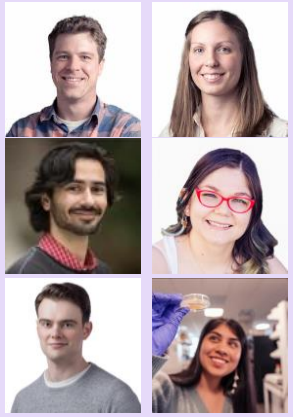
- our structure is very different from standard academic structure (ie 1 PI with lab of 3-10, all focus on project related to PI’s interest or field)
- allen institute structure
  - multiple teams who all work on same big project
  - projects are determined by multiple scientific leaders, along with input from our scientific advisory council

team science/ spatial transcriptomics example

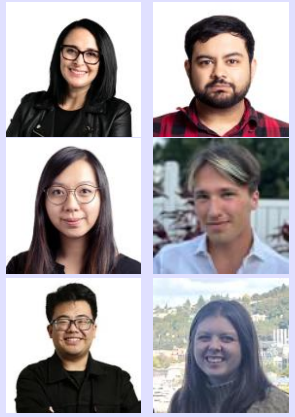
animal care teams



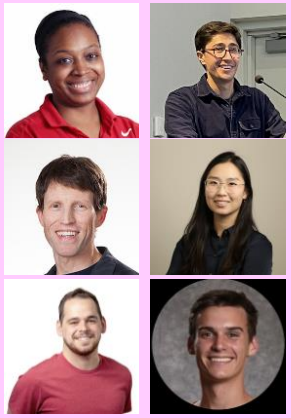
tissue processing



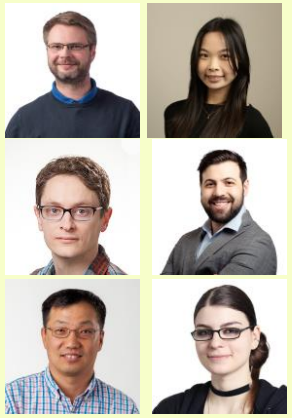
histology



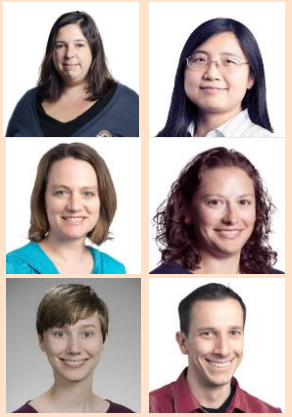
imaging



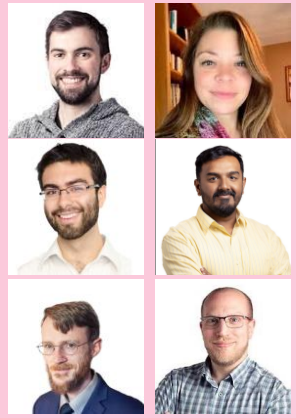
spatial bioinformatics



informatics & analysis



data & Technology



additional teams

project management

scientific support

research compliance

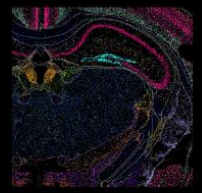
finance

shipping & receiving

+ MANY more folks & teams not pictured

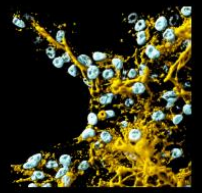
allen institute/ scientific accelerators

explore our accelerators/



brain science/

Mapping every cell, connection, and circuit in the brain—openly shared with the world.

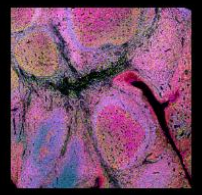


cell science/

Decoding how cells become tissues, then programming that knowledge into powerful new research tools.

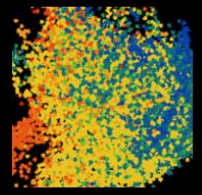
neural dynamics/

Revealing the brain's hidden algorithms that transform neural activity into real-world behavior.



immunology/

Creating the deepest open reference for the healthy human immune system ever built.

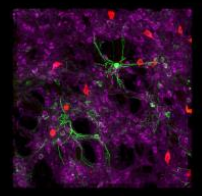


synthetic biology/

Engineering cells to record their own histories, transforming how we understand disease over time.

ai biodesign/

Advancing evolution to explore biology's full design space and advance human health



brain health/

Accelerating human brain research and new therapeutic strategies for neurological conditions

alleninstitute.org/our-research

big science/

brain science

human brain atlas

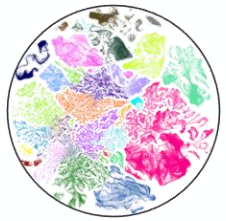
cell science

Paul G. Allen Frontiers Group

immunology

neural dynamics

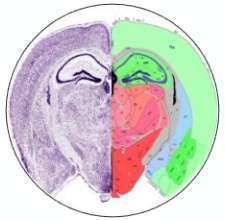
allen brain cell atlas



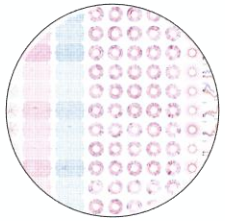
new

brain health

2003 2026

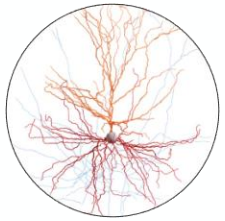


mouse brain atlas

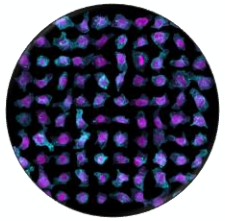


MindScope

connectivity atlas



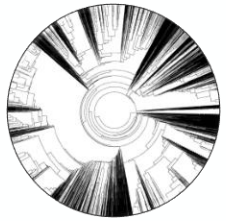
cell types database & brain observatory



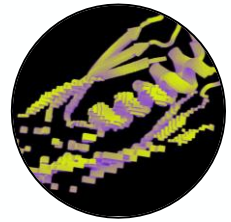
allen cell explorer

Alzheimer's disease brain cell atlas

human immune system explorer



Seattle Hub for Synthetic Biology



AI biodesign

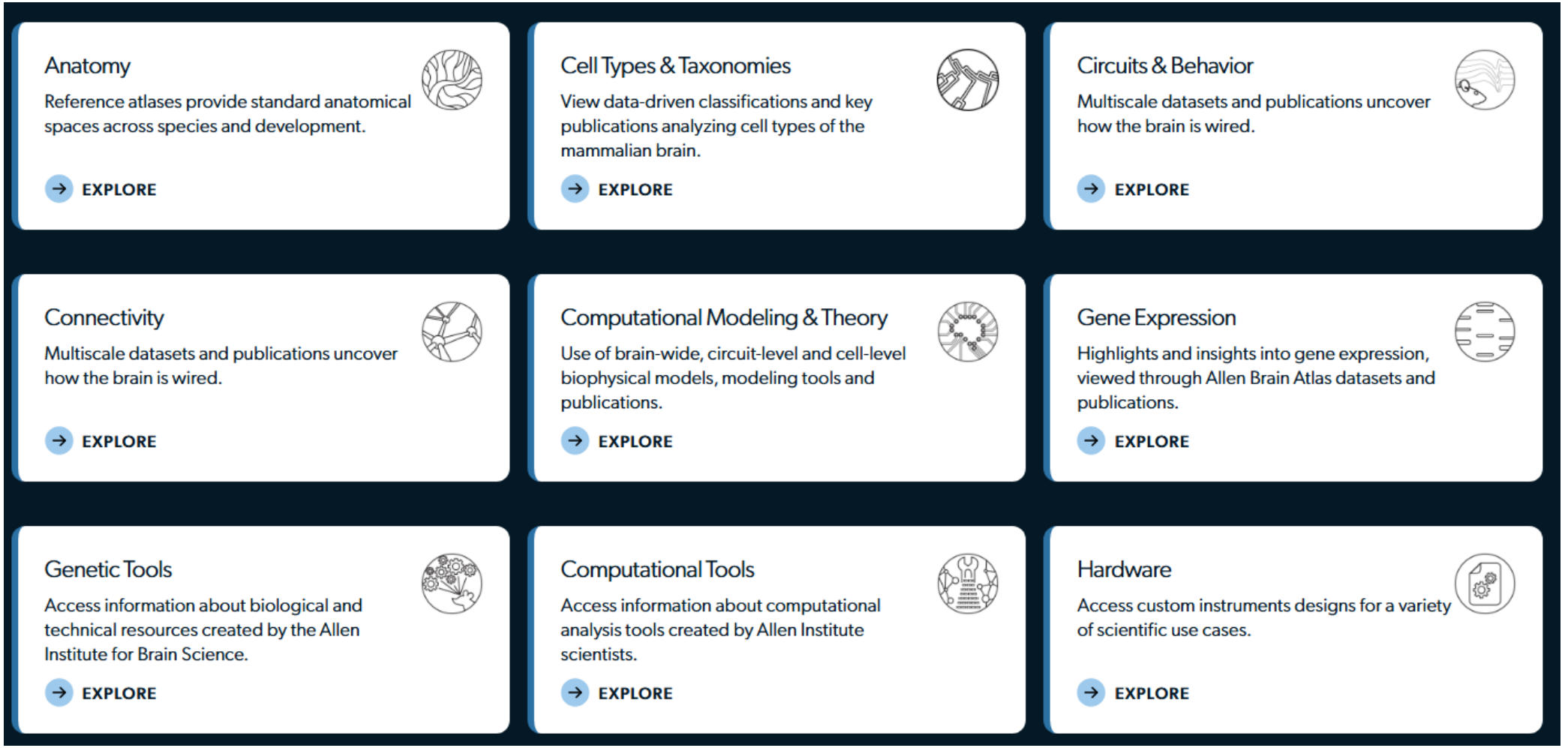
new

brain science is a major focus

open science/

- The goal of open science is to make scientific research data, tools, and knowledge available and accessible to the public
- This is more than just publication!
  - Data is available and fully public
  - Published protocols (find us on protocols.io)
  - Develop open-access tools for data use and access
  - Provide resources & trainings to help use the data & tools
  - Following FAIR (Findable, Accessible, Interoperable, Reproducible) Principles

all FREE on Allen Brain Map/ brain-map.org

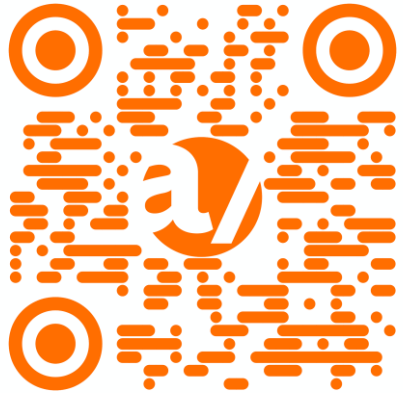


careers/

- we hire straight out of graduate school – no post-doc required!
  - this is typically a Scientist I
  - average “Scientist I” salary is about $80k-$100k USD
  - some non-research post-PhD roles (e.g., PM)
  - each position will be clear if it supports Visa or not - it is position dependent
- hiring is year-round, although sometimes more open spots during the start of the new year
- we often look for specific skill sets to fill open positions; the skill set depends on the position.

NOTE: We don’t have any undergraduate internships available for international students.

As of now, we also do not have any PhD-level internships.



all careers are posted online
alleninstitute.org/careers

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cell typing at the Allen Institute: taxonomies, data sets, and use cases/

Rachel Hostetler, Scientist I

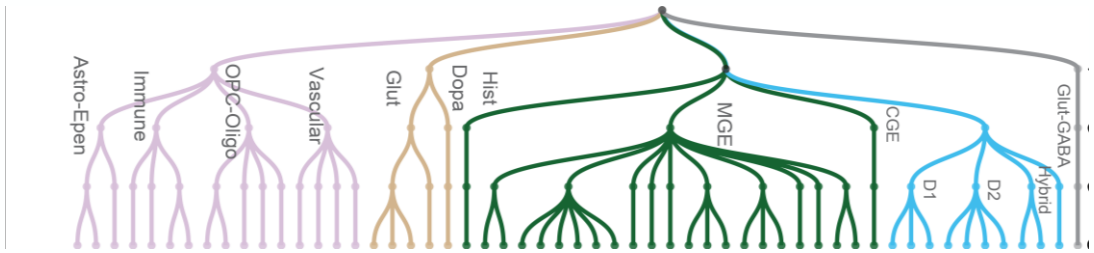
allen institute/

allen institute

key methods for cell type atlasing/

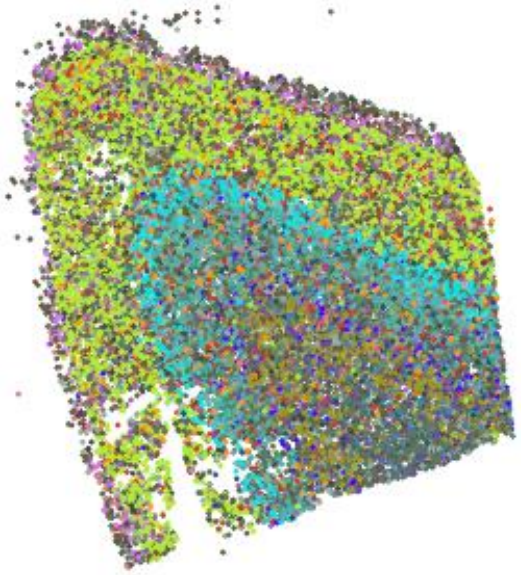
Single cell -omics

- All genes per experiment
- Up to millions of cells per study
- Dissection-level resolution
- Used for defining cell types



Spatial transcriptomics

- <1000 genes per data set
- Cellular resolution
- A few regions (human) or brain-wide (mouse)
- Can assign types to all cells



key methods for cell type atlasing/

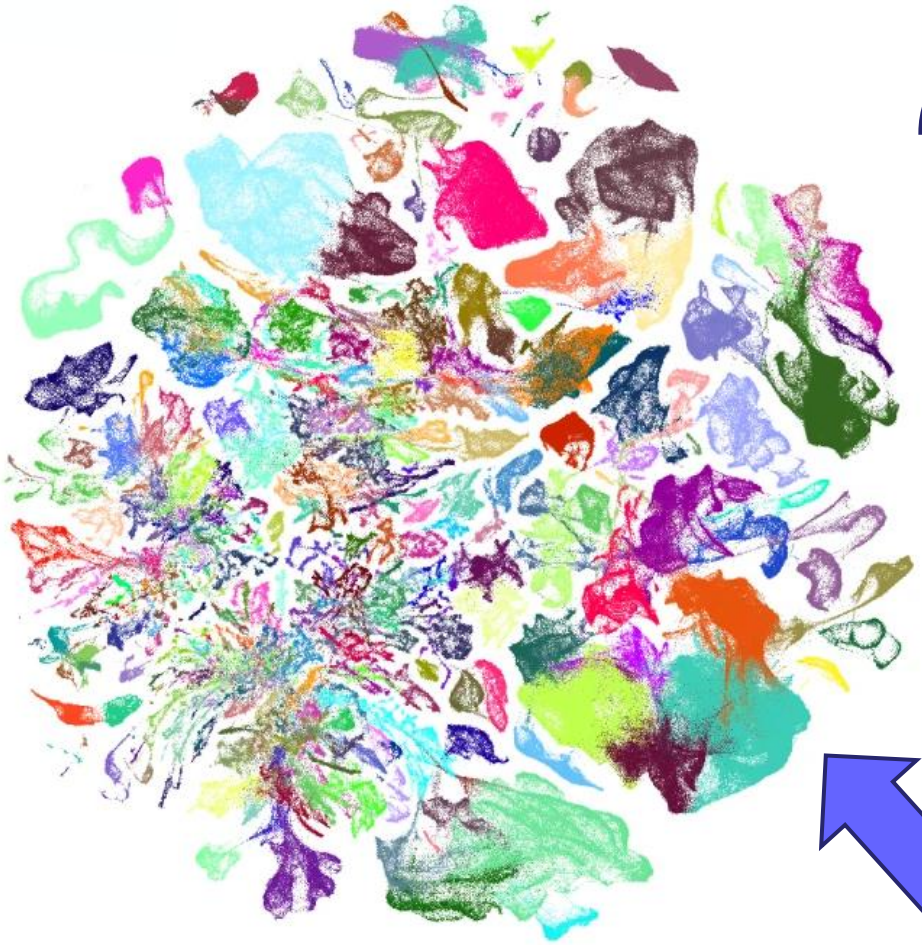
Single cell -omics

Spatial transcriptomics

Adapted from Shalek and Regev (2016)

defining and annotating cell type taxonomies/

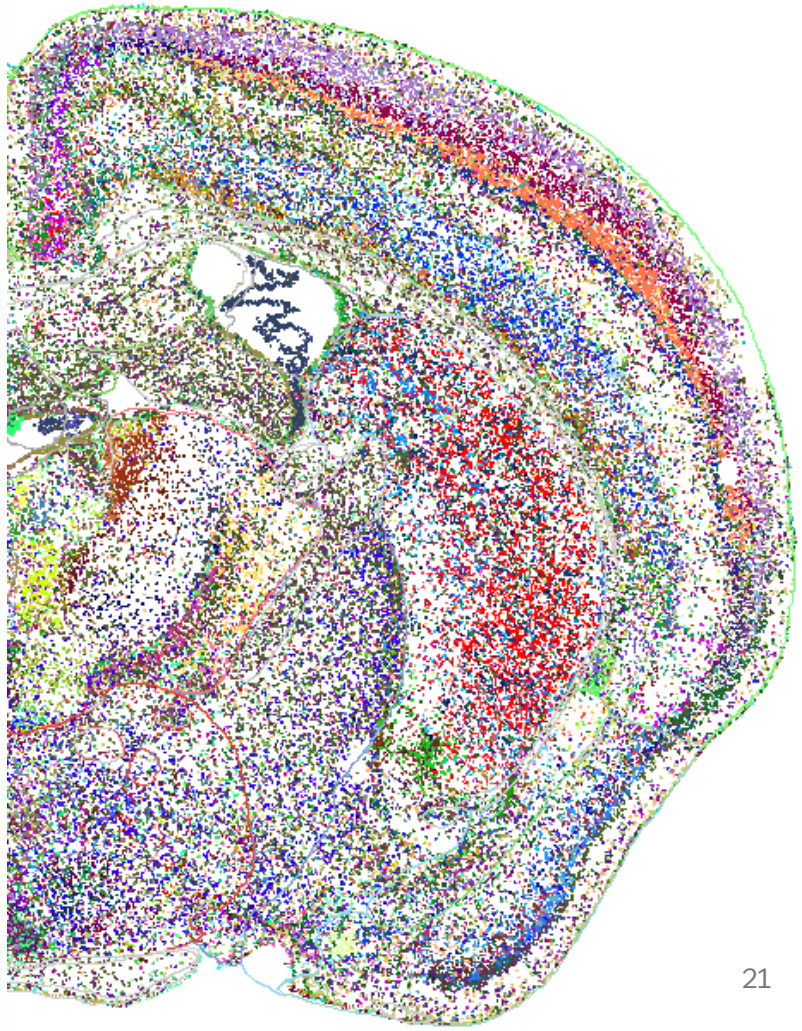
Single cell -omics:
What are the cell types



Iterate on analysis for quality control and cell type naming.

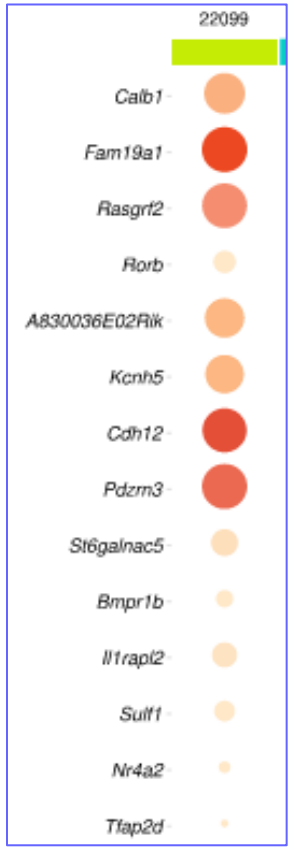
Other modalities add context

Spatial transcriptomics:
Where are the cell types

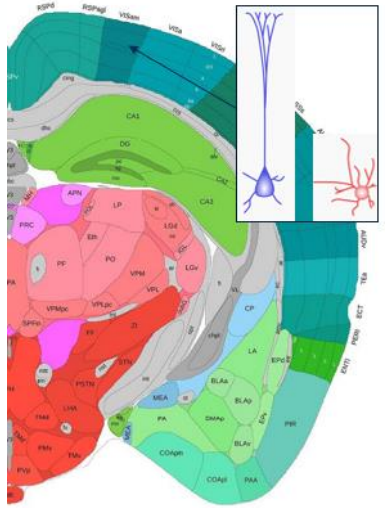


data from other modalities can be anchored to gene expression and cell location/

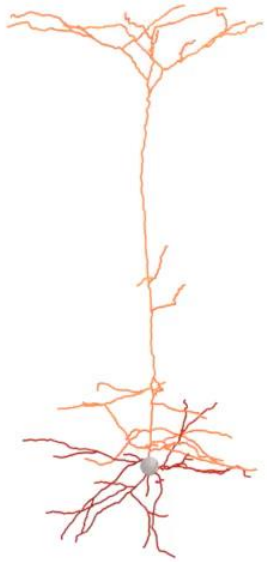
Gene expression
(transcriptomics)



Cell location
(anatomy)



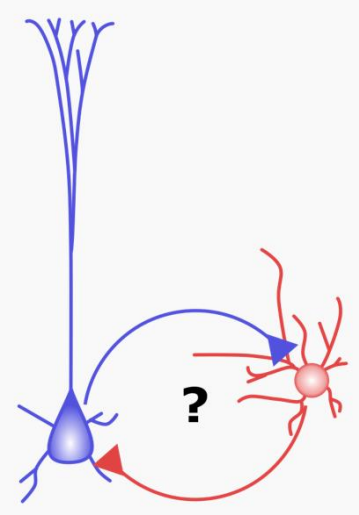
Cell shape
(morphology)



Firing properties
(electrophysiology)

Chromatic accessibility
(epigenetics / targeting)

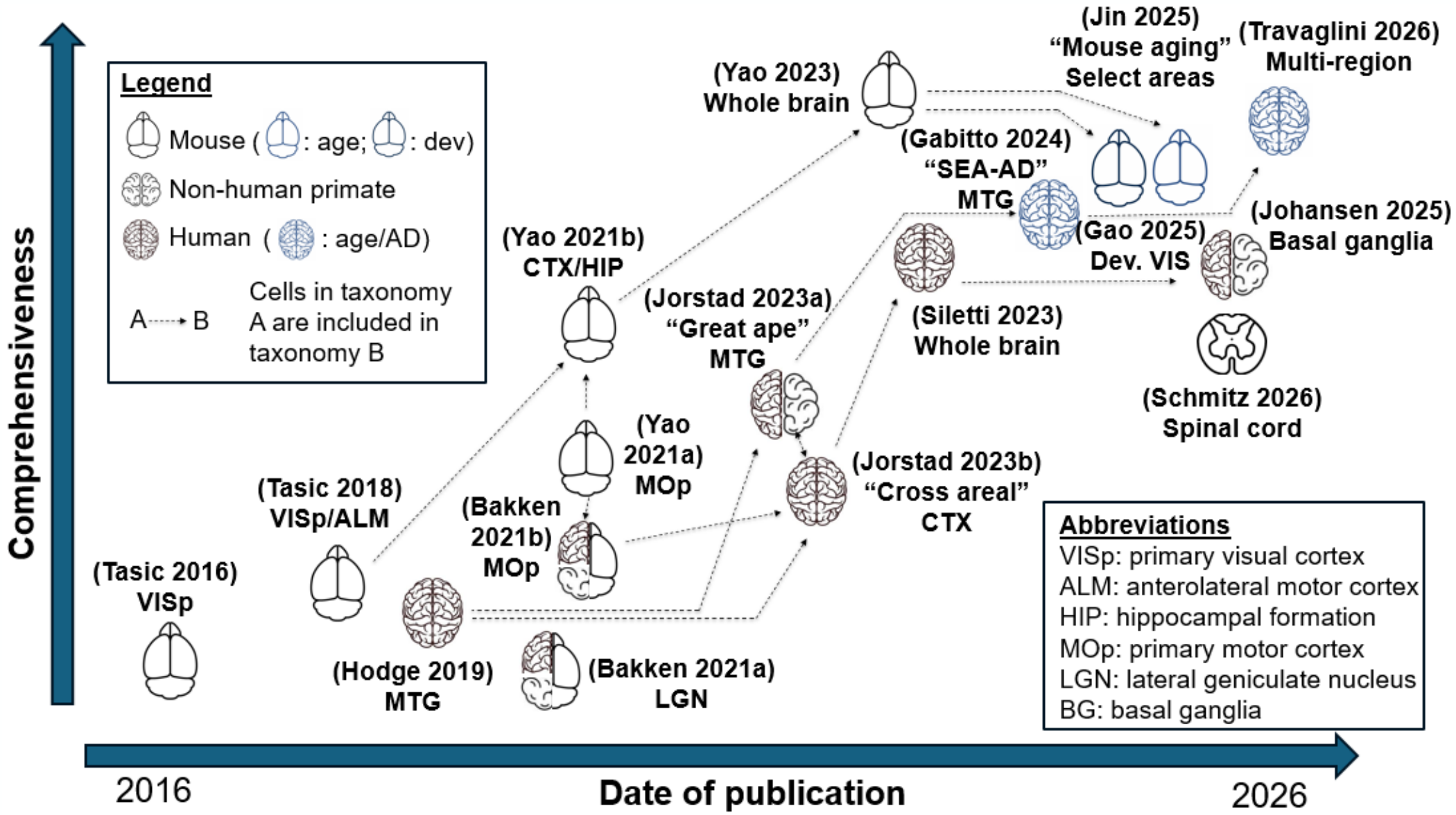
Cell communication
(connectivity)



Cell function
(in response to behavior)

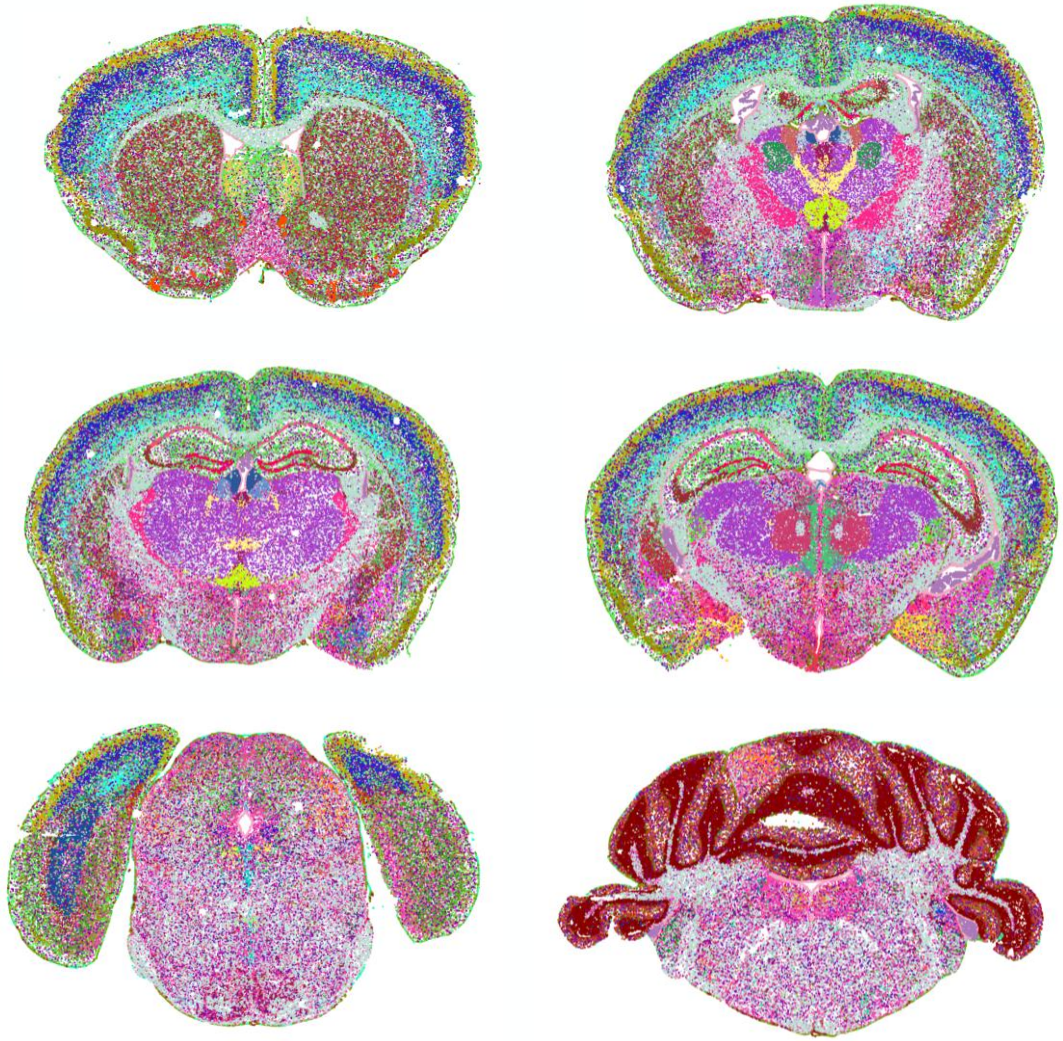
You'll hear about some of these data modalities later in the day!

more than a decade of cell typing efforts at the Allen Institute span brain regions and species/

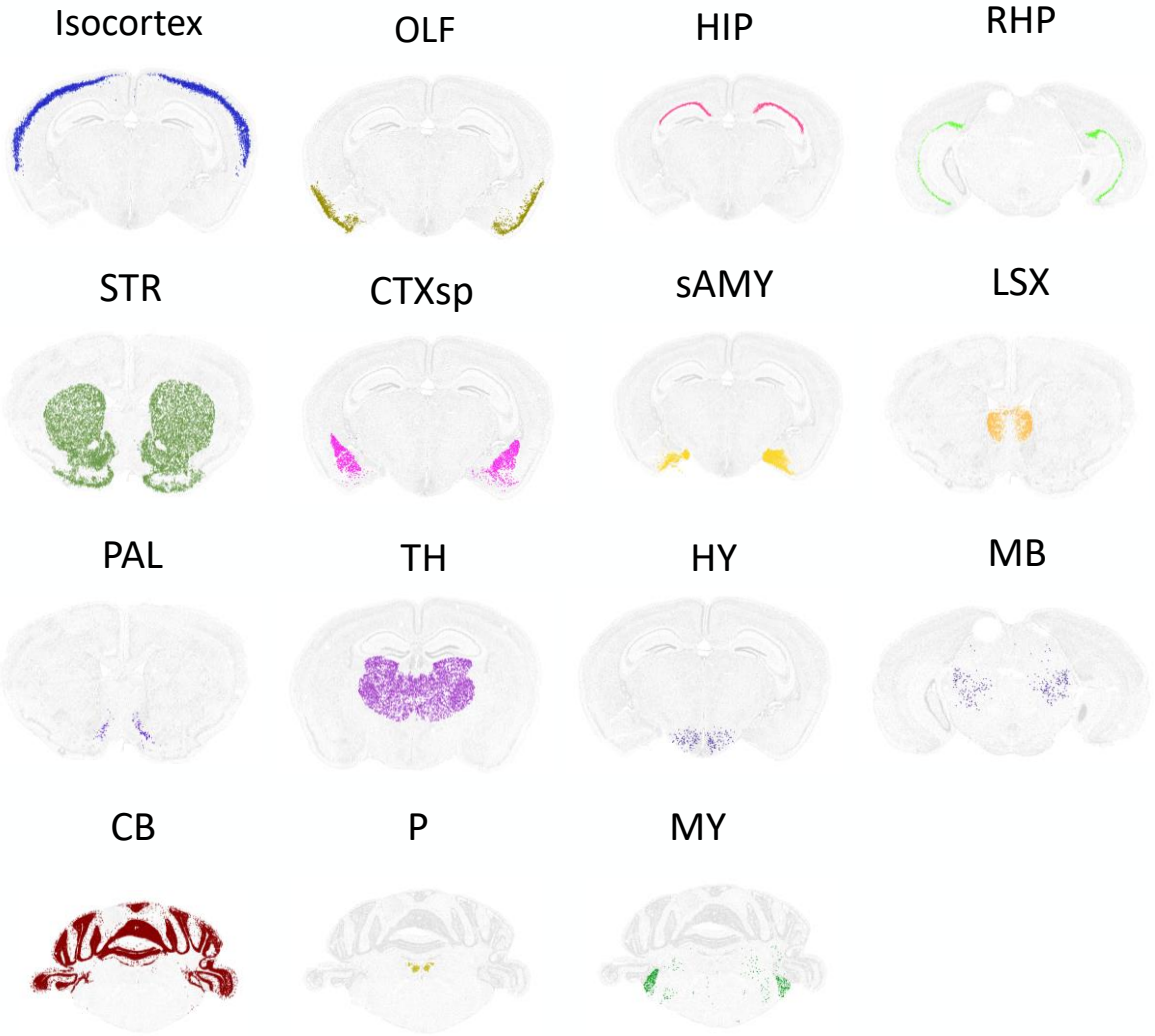


whole mouse brain (cell type) atlas/ created by combining single cell and spatial transcriptomics brain-wide/

(Cells color-coded by cell type)

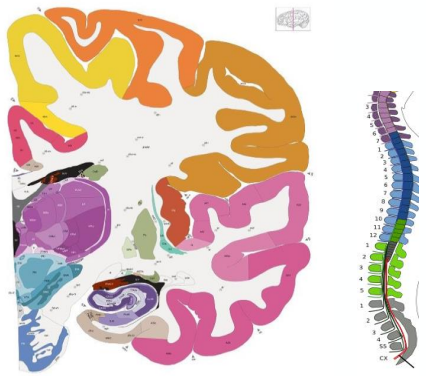


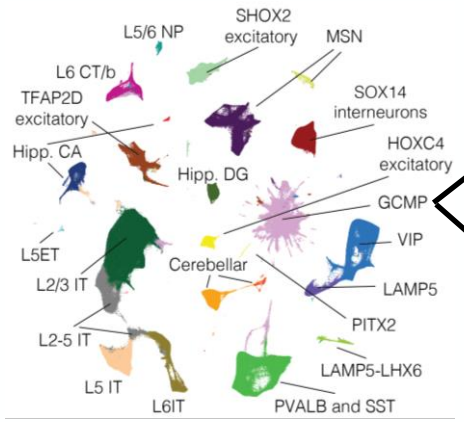
(Cells color-coded by brain region)

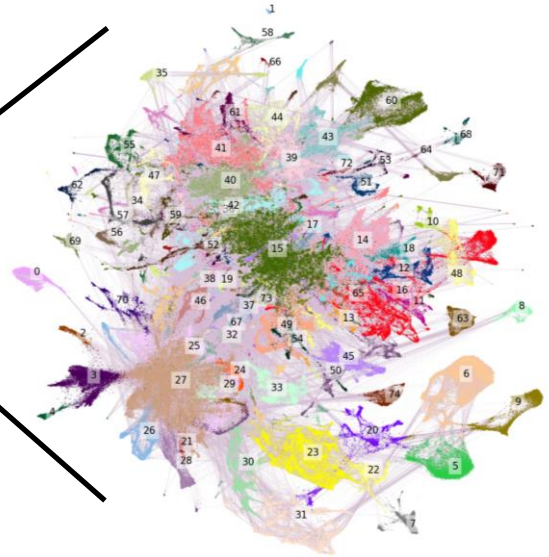


Yao et al., 2023, Nature

efforts to define cell types across the whole human brain are ongoing/



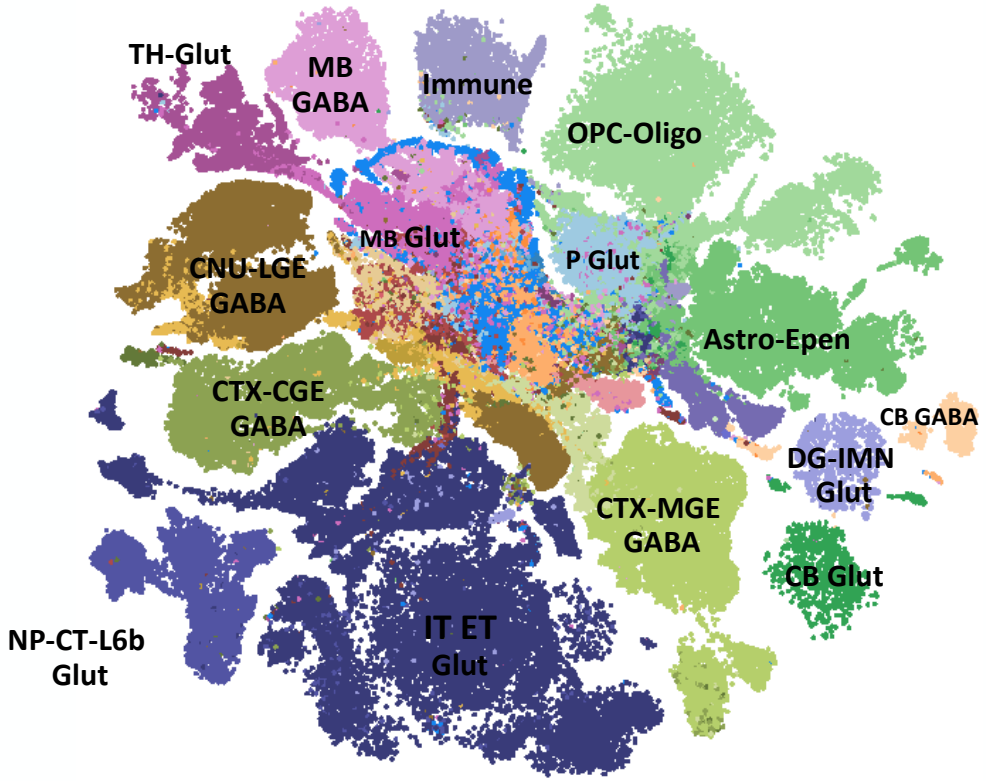




Published:

- ~3,000 cell types in the whole brain based on ~3,000,000 cells
- Minimal spatial transcriptomics data

Siletti et al., 2023, Science



In process:

- Add more single nucleus RNA-seq data (now >5,000 types)
- Spatial transcriptomics

(expected late 2027)

studies of healthy adult brain cells act as anchor for defining cell types in development, aging, and disease/

Create cell type atlases in **healthy** adult mouse, human, and non-human primate brain

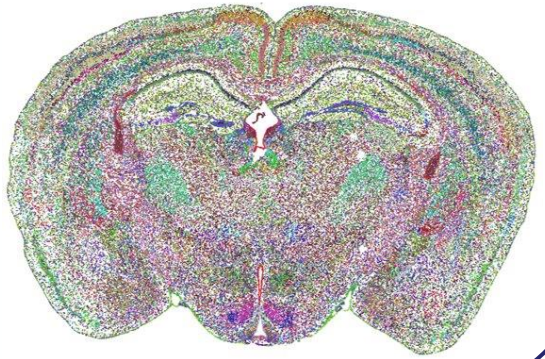
BICCN

BICAN

The BRAIN Initiative®

HMBA

Example ➔ whole mouse spatial atlas (MERFISH)



Extend such atlases to the aged human brain to understand early molecular and cellular changes in Alzheimer's disease.

SEA-AD

The Brain Health Accelerator program will expand these efforts to both understand and treat multiple brain disorders.

datasets available for exploration and mapping in the brain knowledge platform/

mouse

- **Whole Mouse Brain** (scRNAseq and MERFISH data)
  - Yao et al., 2023, Zhang et al., 2023 – Nature
- **Consensus Whole Mouse Brain** (scRNAseq)
  - Yao et al., 2023, Langlieb et al., 2023 – Nature
- **Aging Mouse** (scRNAseq)
  - Jin et al., 2025 – Nature.
- **Developing Mouse Visual Cortex** (scRNAseq)
  - Gao et al., 2025 – Nature

human & primate

- **Whole Human Brain** (snRNAseq)
  - Siletti et al., 2023 – Science
- **Human Parkinson's Disease** (ASAP Human Postmortem BrainSeq) (snRNAseq)
  - (No publication—collaboration with MJFF/ASAP)
- **Cross-species HMBA Basal Ganglia** (snRNAseq and MERFISH data)
  - Johansen & Fu et al, 2026; Hewitt & Turner et al, 2026 - bioRxiv
- **Human Alzheimer's Disease (SEA-AD)** – Multiple regions (snRNAseq) and MTG (MERFISH data)
  - Gabitto & Travaglini et al., 2024 – Nature Neuroscience
  - Travaglini, Gabitto, and Ding et al., 2026 – bioRxiv
  - Kana et al 2026 – bioRxiv (Caudate)

You'll hear a lot more about HOW to work with these data sets later in the day!

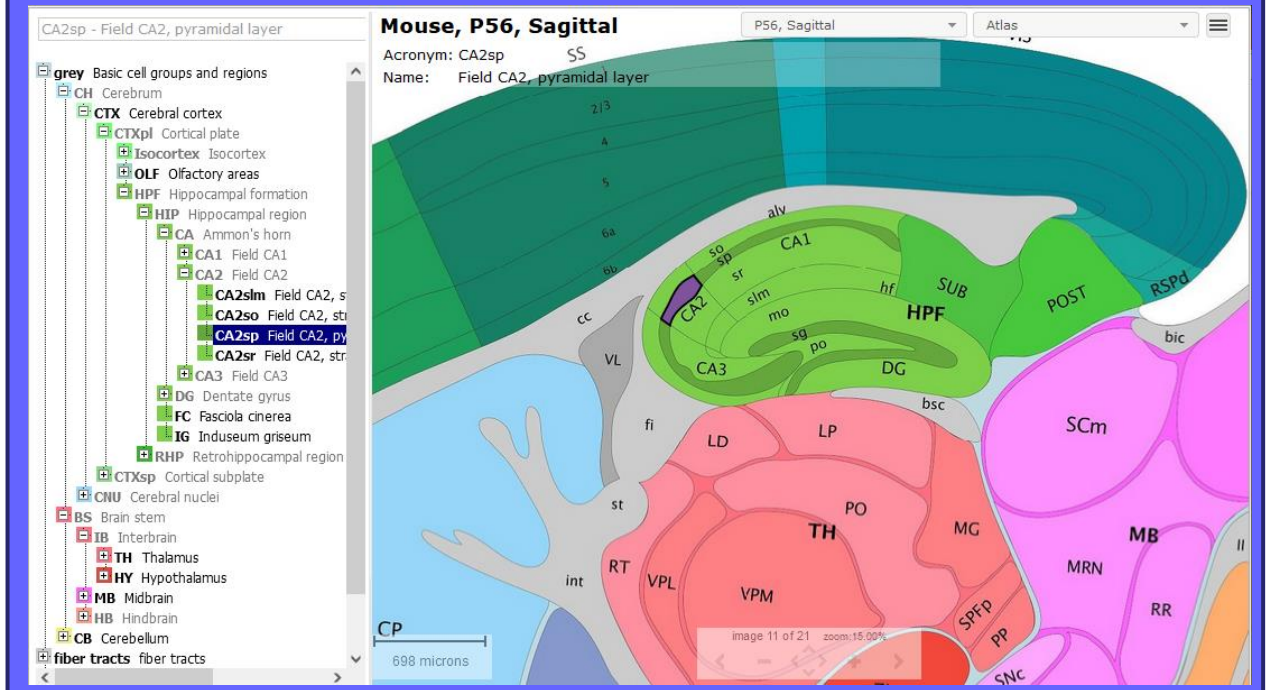
allen brain map use cases

alleninstitute/

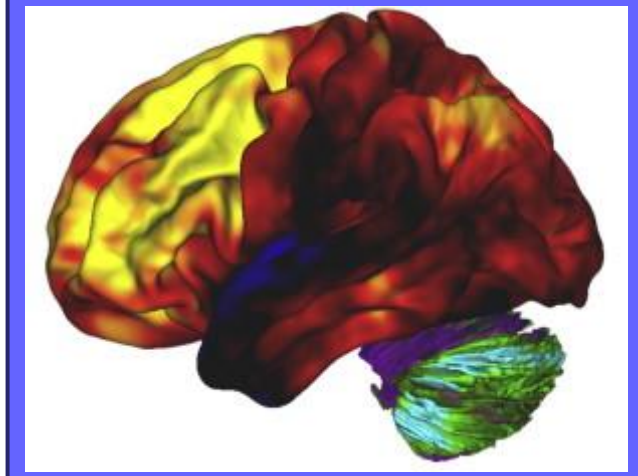
use cases/localization

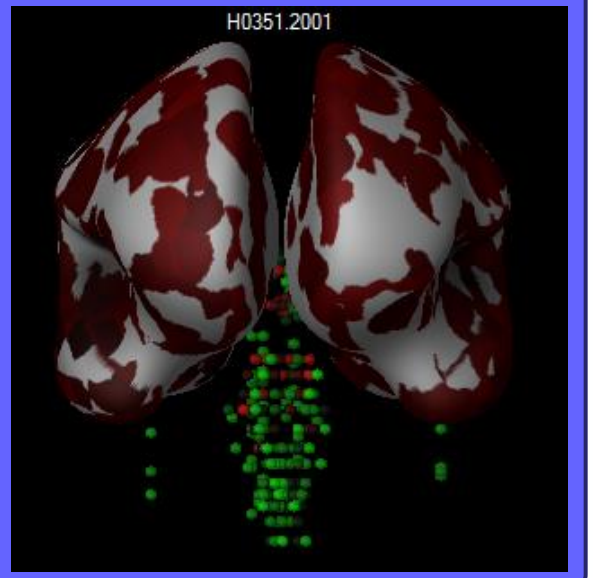
- What is this brain region called and where in the brain is it located?
- How does it connect to other brain regions in mouse? In human?

Allen Brain Reference Atlas (mouse/human/NHP)

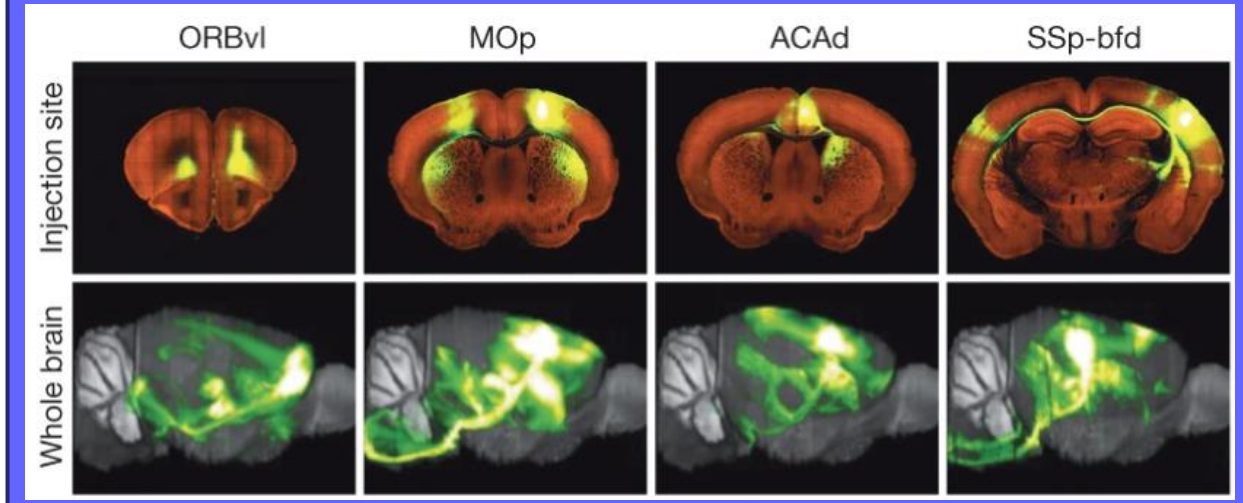


(Indirectly by linking structural MRI and gene expression from Allen Human Brain Atlas with external fMRI)





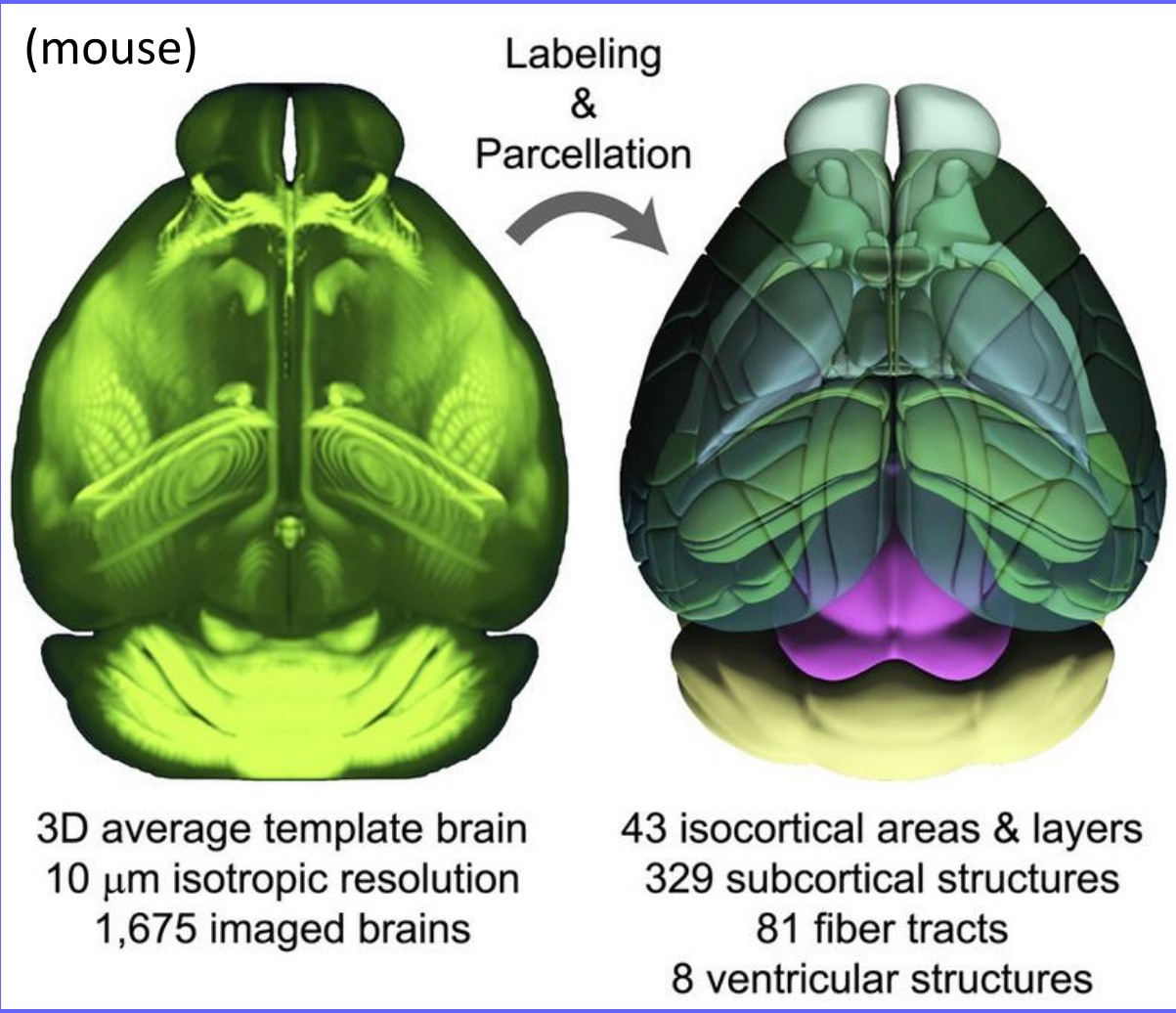
Allen Connectivity Atlas (mouse)



use cases/localization

- What is this brain region called and where in the brain is it located?
- How does it connect to other brain regions in mouse? In human?
- How can we align brain regions (and any other data modalities) to the same set of coordinate?

Allen Common Coordinate Frameworks (CCF)

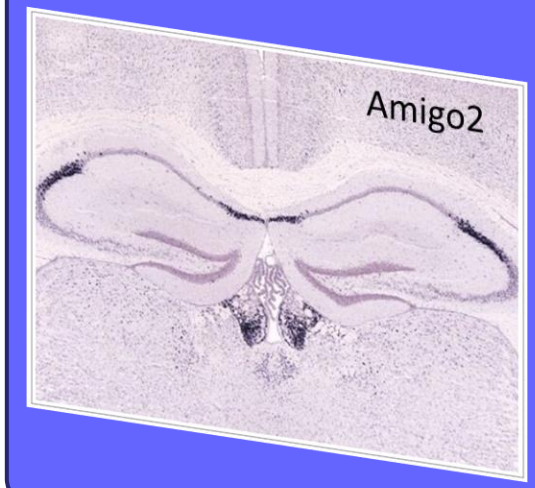


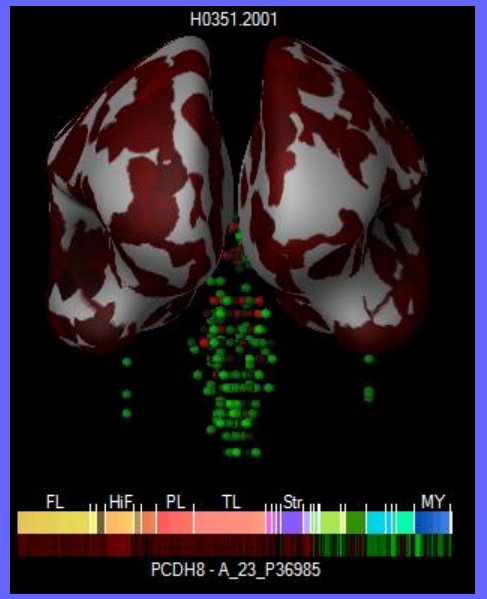
(CCFs in human and NHP based on MRI data exist or are in progress via HMBA.)

use cases/gene expression

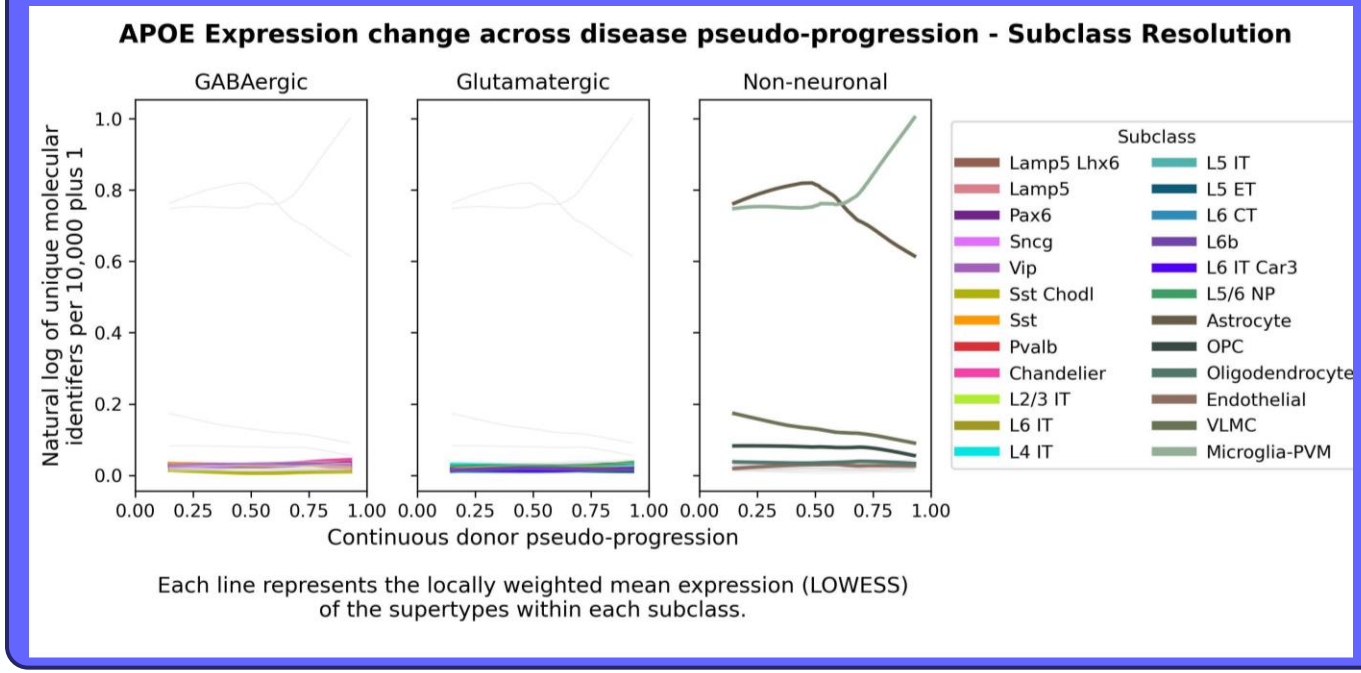
- Where in the brain is my gene expressed?
- And in which cell types?
- How does this change in AD?

Historic atlases

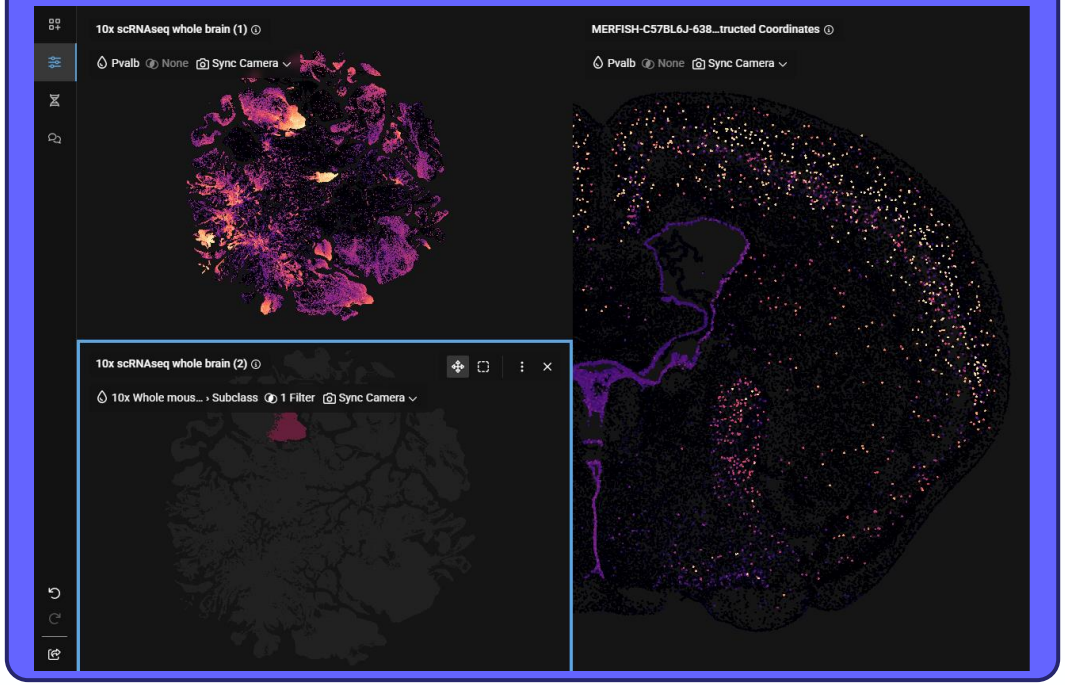




SEA-AD gene trajectory viewer



Allen Brain Cell Atlas



use cases/gene expression

- Where in the brain is my gene expressed?
- And in which cell types?
- How does this change in AD?
- Which other genes are patterned similarly?
- And how does this change across species?

External tools like CELLxGENE (or code)

APPLICATION
Collections | Datasets | Gene Expression | Cell Guide | Differential Expression NEW
CENSUS
API | Models
Explore Your Data | Help & Documentation

Filters

- Assay
- Cell Type
- Consortia
- Allen Institute for Brain Scie... ×
- Development Stage
- Disease
- Organism
- Publication
- Publication Date
- Self-Reported Ethnicity
- Sex
- Suspension Type
- Tissue

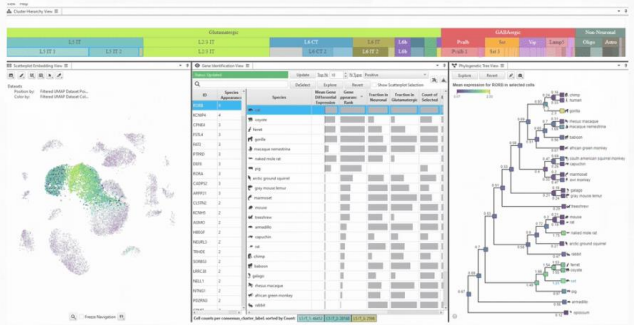
Collections 13 of 387	Publication	Tissue	Disease	Organism
SEA-AD: Seattle Alzheimer's Disease Brain Cell Atlas	Gabitto et al. (2024) Nat Neurosci	dorsolateral prefrontal cortex middle temporal gyrus	dementia normal	Homo sapiens
A molecularly defined and spatially resolved cell atlas of the whole mouse brain	Zhang et al. (2023) Nature	13 tissues	normal	Mus musculus
A single-cell transcriptomic atlas spanning the adult rhesus macaque brain	Chiou et al. (2023) Sci. Adv.	27 tissues	normal	Macaca mulatta
Transcriptomic cytoarchitecture reveals principles of human neocortex organization	Jorstad et al. (2023) Science	9 tissues	normal	Homo sapiens
Comparative transcriptomics reveals human-specific cortical features	Jorstad et al. (2023) Science	middle temporal gyrus	normal	3 organisms

External tools like Cytosplore viewer

Cytosplore Viewer | Team | Download | Projects | Cytosplore

Cross-Species Analysis in Single-Cell Transcriptomics Studies

EvoViewer and Simian projects provide comparative and evolutionary visualizations across species.

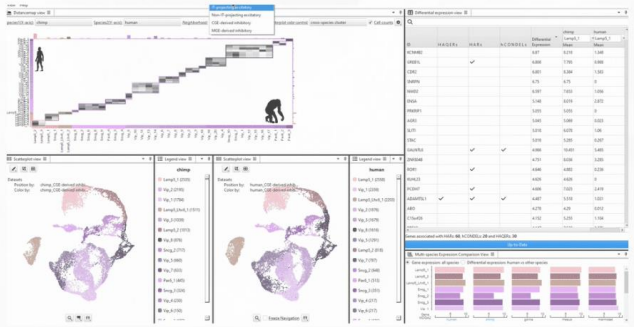


EvoViewer

Project File: Other/Evo Viewer in the Cytosplore Viewer
Project Contact: Soumyadeep

Cytosplore EvoViewer enables interactive exploration of single-cell transcriptomic datasets across species. It links marker gene expression with comparative evolutionary information, allowing users to detect subtle similarities and differences in gene regulation and cellular composition. Users can analyze cross-species relationships, compare evolutionary traits, and investigate the impact of marker genes on biological processes.

Website | Publication: IEEE Pacific Visualization



Simian Viewer

Project File: Other/Simian Viewer in the Cytosplore Viewer
Project Contact: Soumyadeep

Cytosplore Simian Viewer provides interactive visualization of multi-species single-cell transcriptomic datasets. It links cell type annotations and gene expression across human, macaque, and marmoset, enabling comparative analysis of cellular and molecular features. Users can explore species-specific and conserved gene expression patterns, analyze cell type similarities and differences, and derive evolutionary insights.

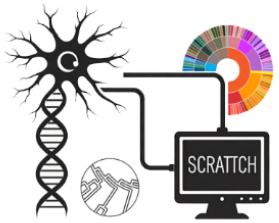
Website | Publication: Science | Publication: Eurographics VCBM

use cases/cell types

- How do I localize cells in the brain?
- How do I define my own cell types (in cochlea?)?
- How do I assign Allen Institute cell type labels to my own cells?

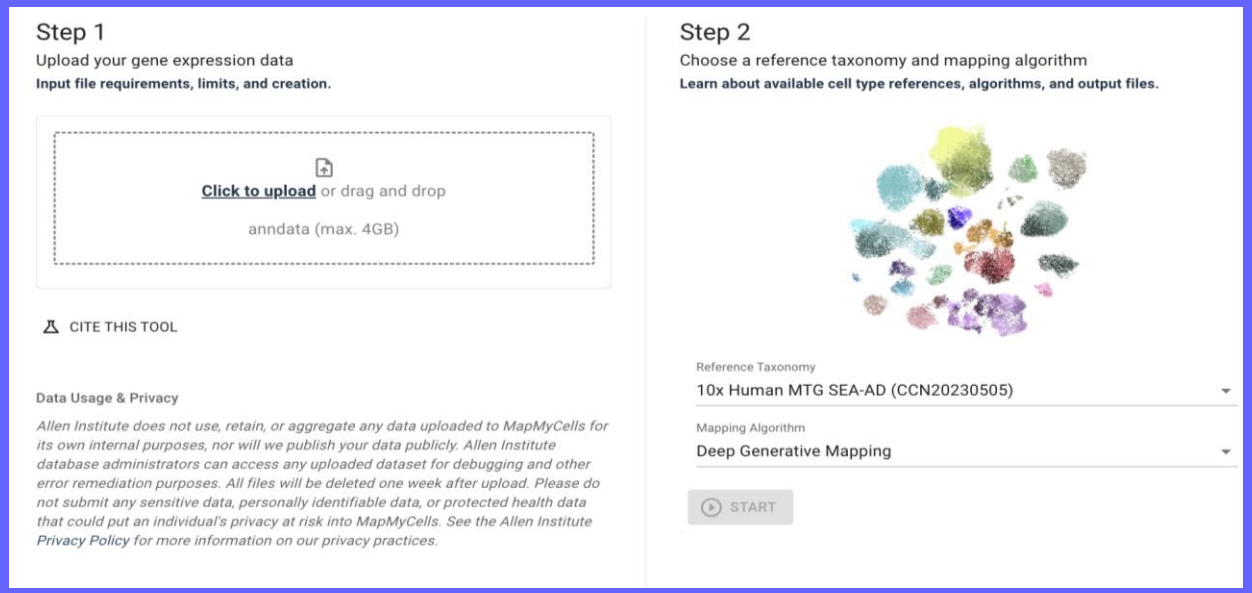
scrattch

Single-**c**ell **R**NA-seq **a**nalysis for **t**ranscriptomic **t**ype **ch**aracterization (or "scrattch") is a suite of R and python scripts from the Allen Institute for Brain Science. The core R libraries are linked in the umbrella package called `scrattch`, which is modeled after the `tidyverse` package. You can use `scrattch` to automatically install or update some of the underlying packages and can run the remaining packages in docker environments. This page describes all core and adjacent scrattch content.

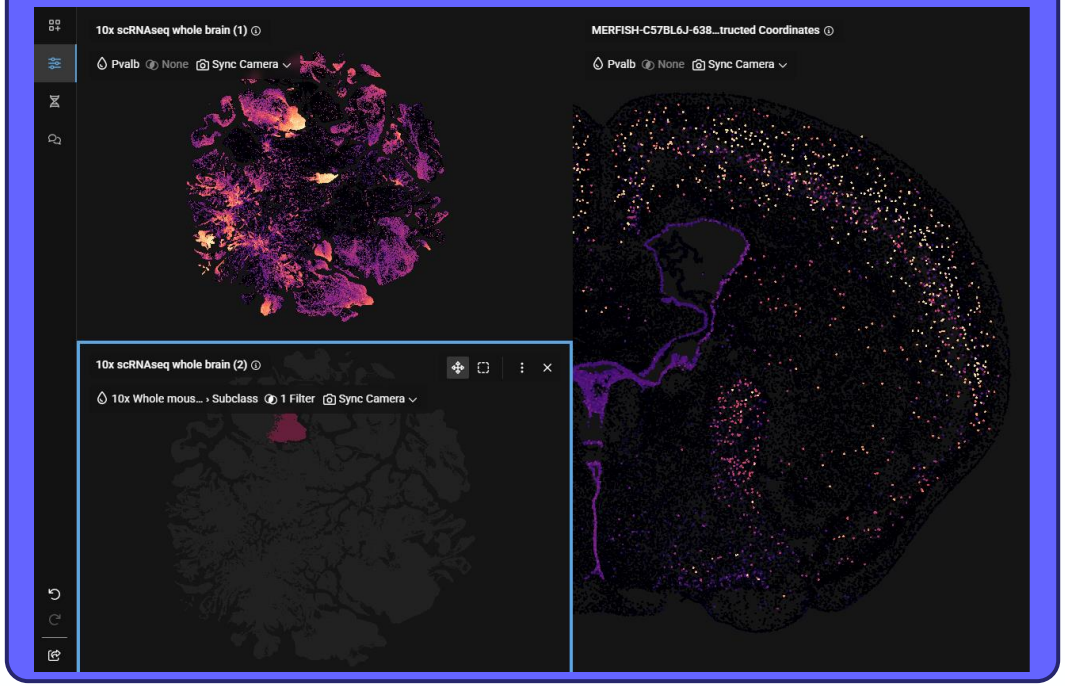


- **Clustering**
- **Taxonomy building**
- **Visualization**
- **Mapping**

MapMyCells



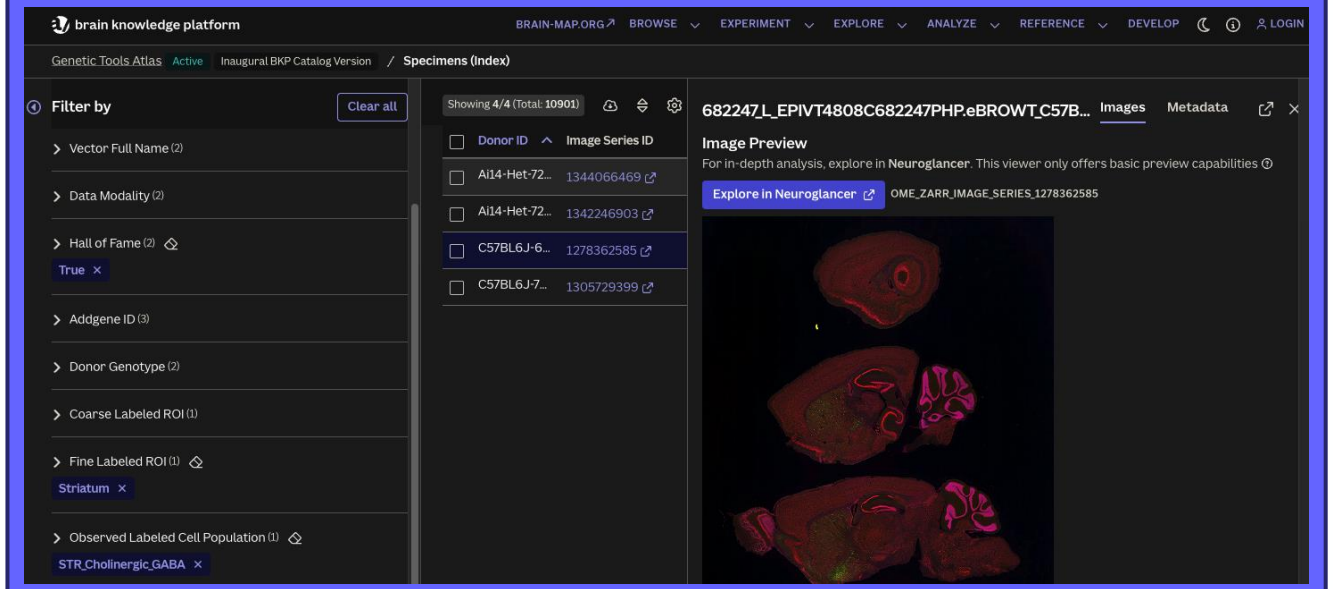
Allen Brain Cell Atlas



use cases/cell types

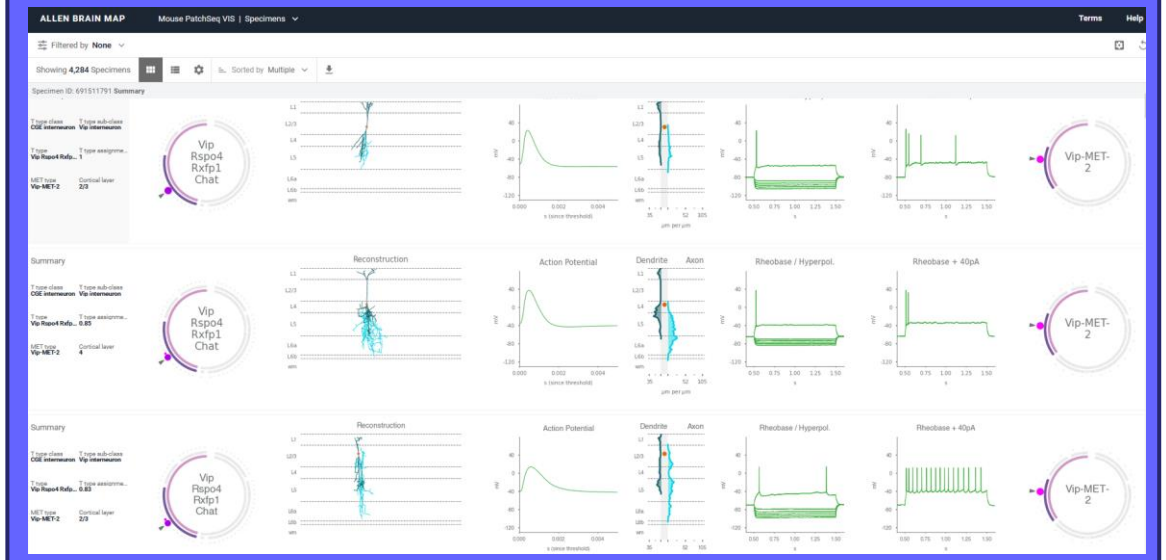
- How do I localize cells in the brain?
- How do I define my own cell types (in cochlea?)?
- How do I assign Allen Institute cell type labels to my own cells?
- Which genetic tools target specific cell types?
- What are their morphoelectric features?

Genetic tools atlas

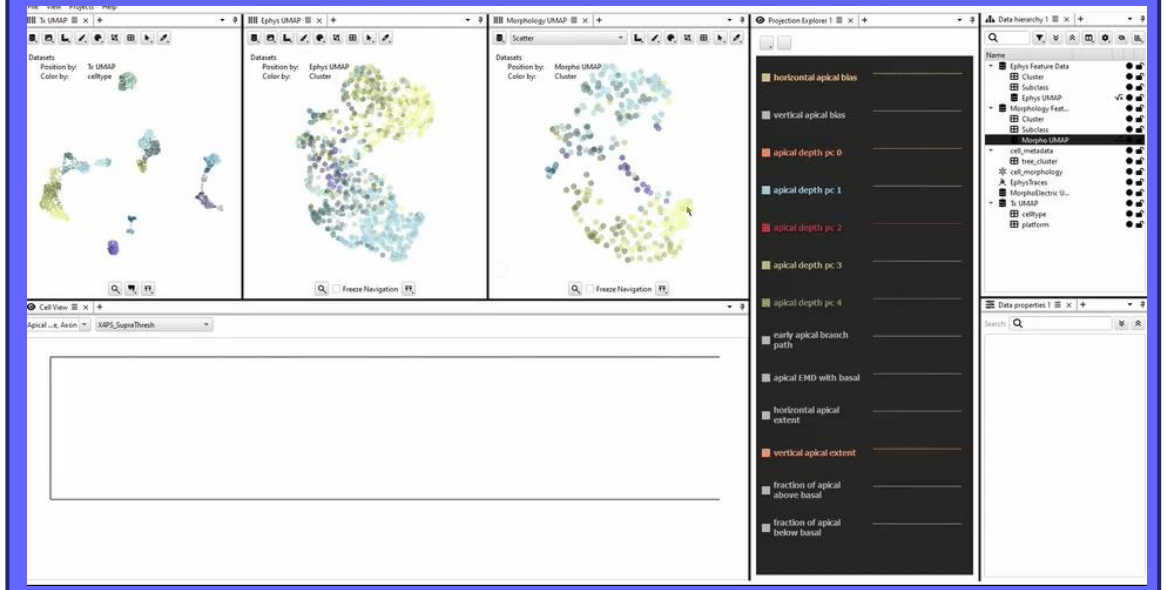


Cell Type Knowledge Explorer (not shown)

Patch-seq Viewer



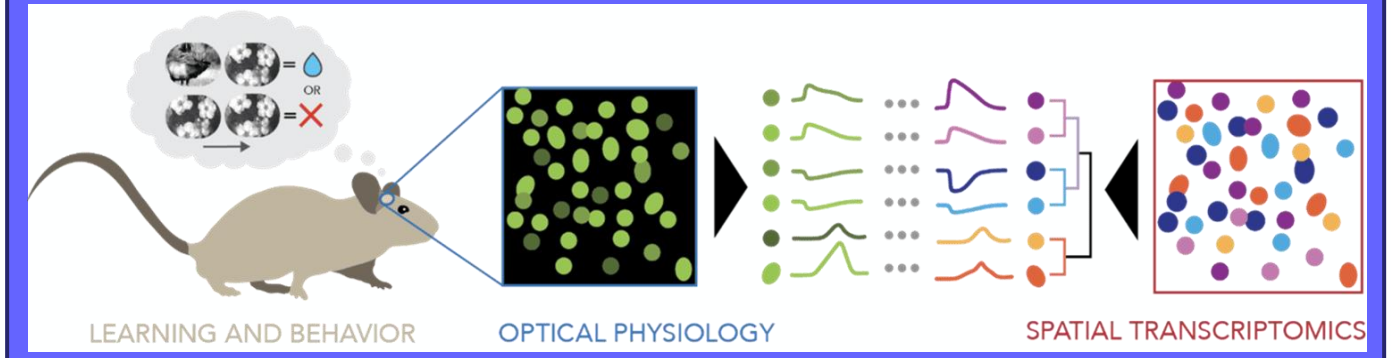
Cytosplore Viewer



use cases/cell types

- What are good cell type markers?
- How do different cell types respond to different visual behaviors?

allen institute/neural dynamics

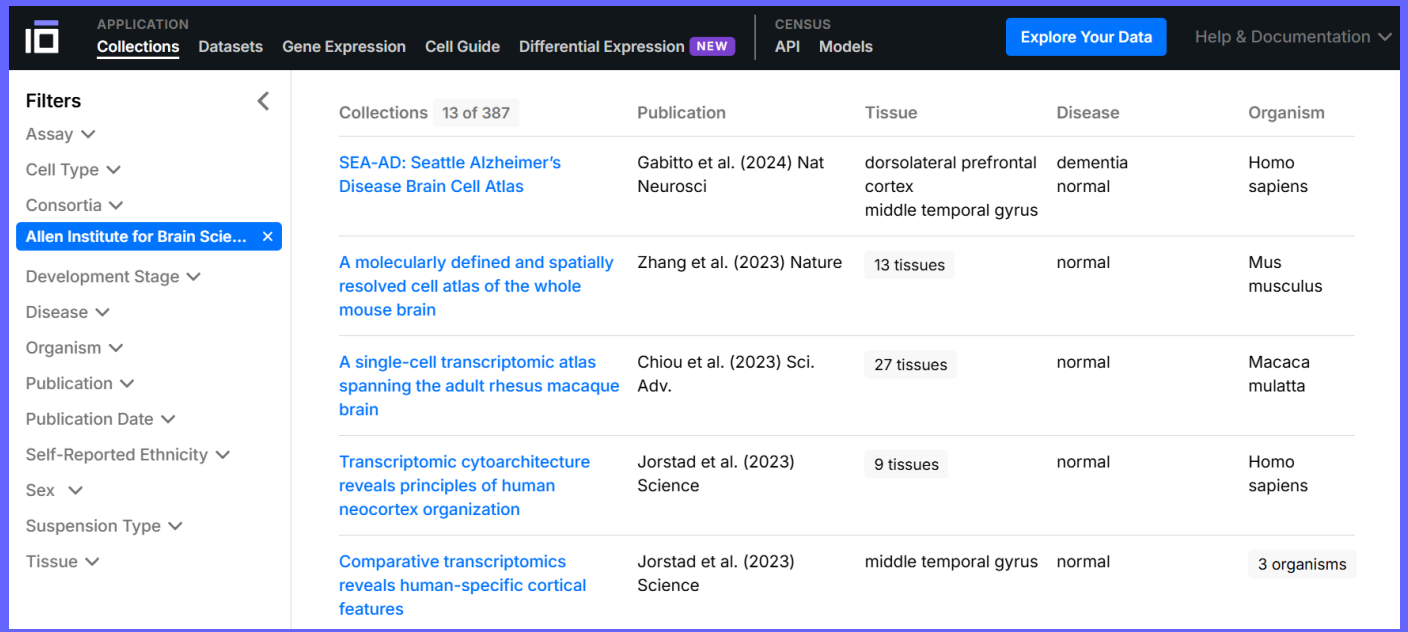


Mouse does something

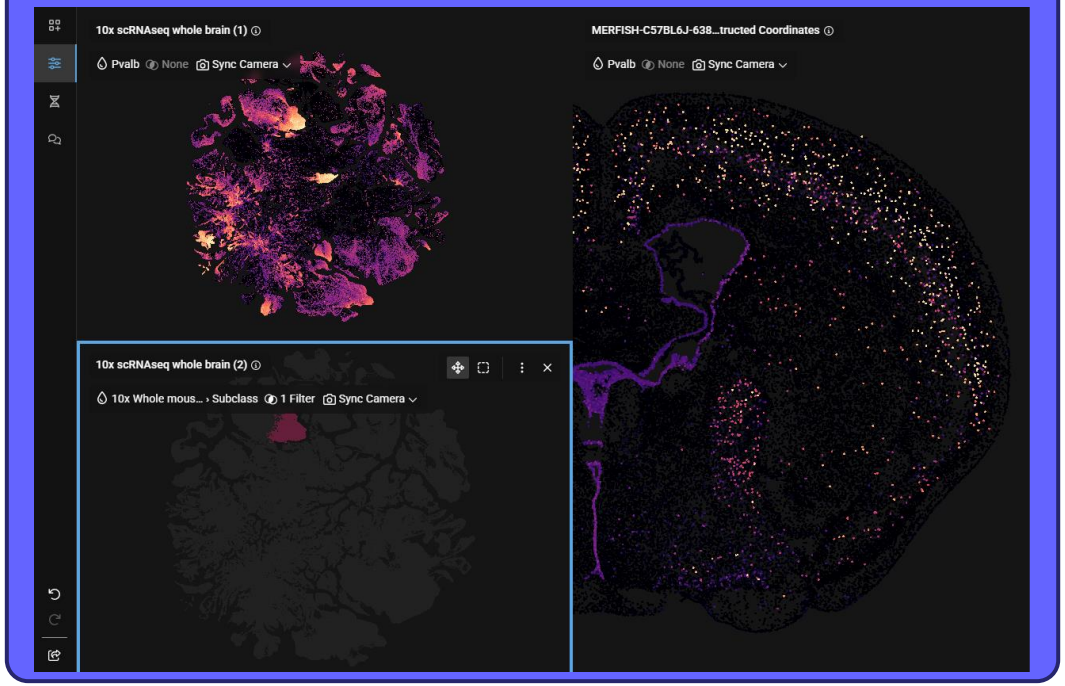
Measure cells while mouse does something

After the experiments, see what kind of cells were measured

External tools like CELLxGENE (or code)



Allen Brain Cell Atlas

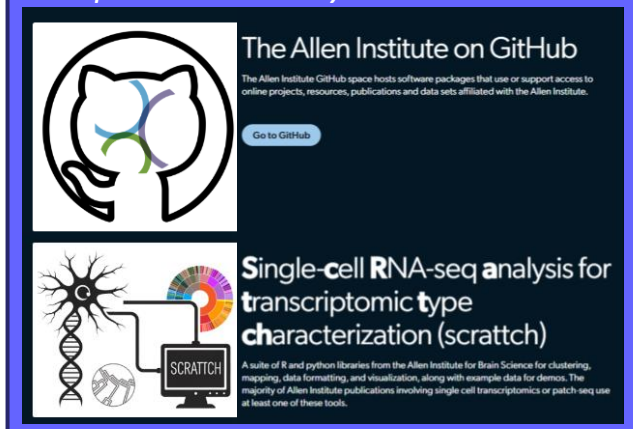


use cases/data access

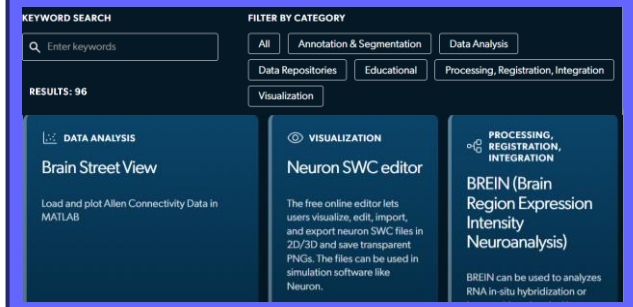
- How do I access the data myself?
- What resources exist for doing analyses off-line using python or R?

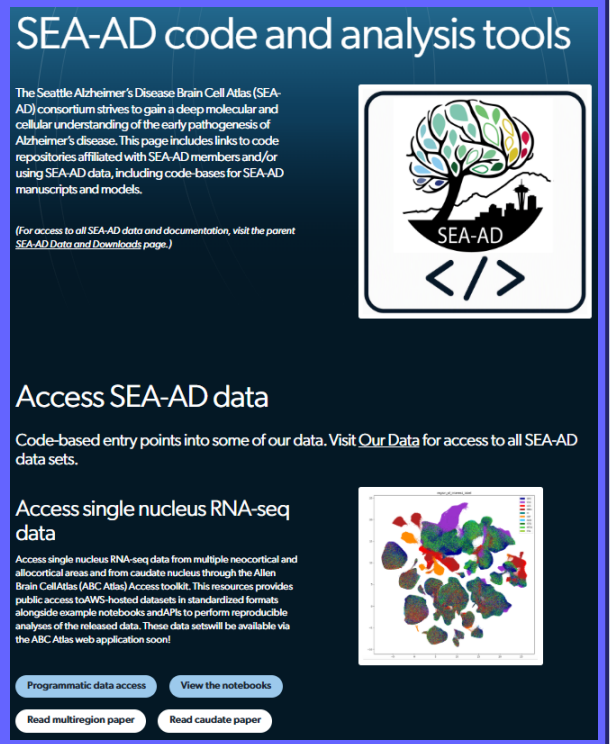
Code landing pages

Computational Analysis Toolbox



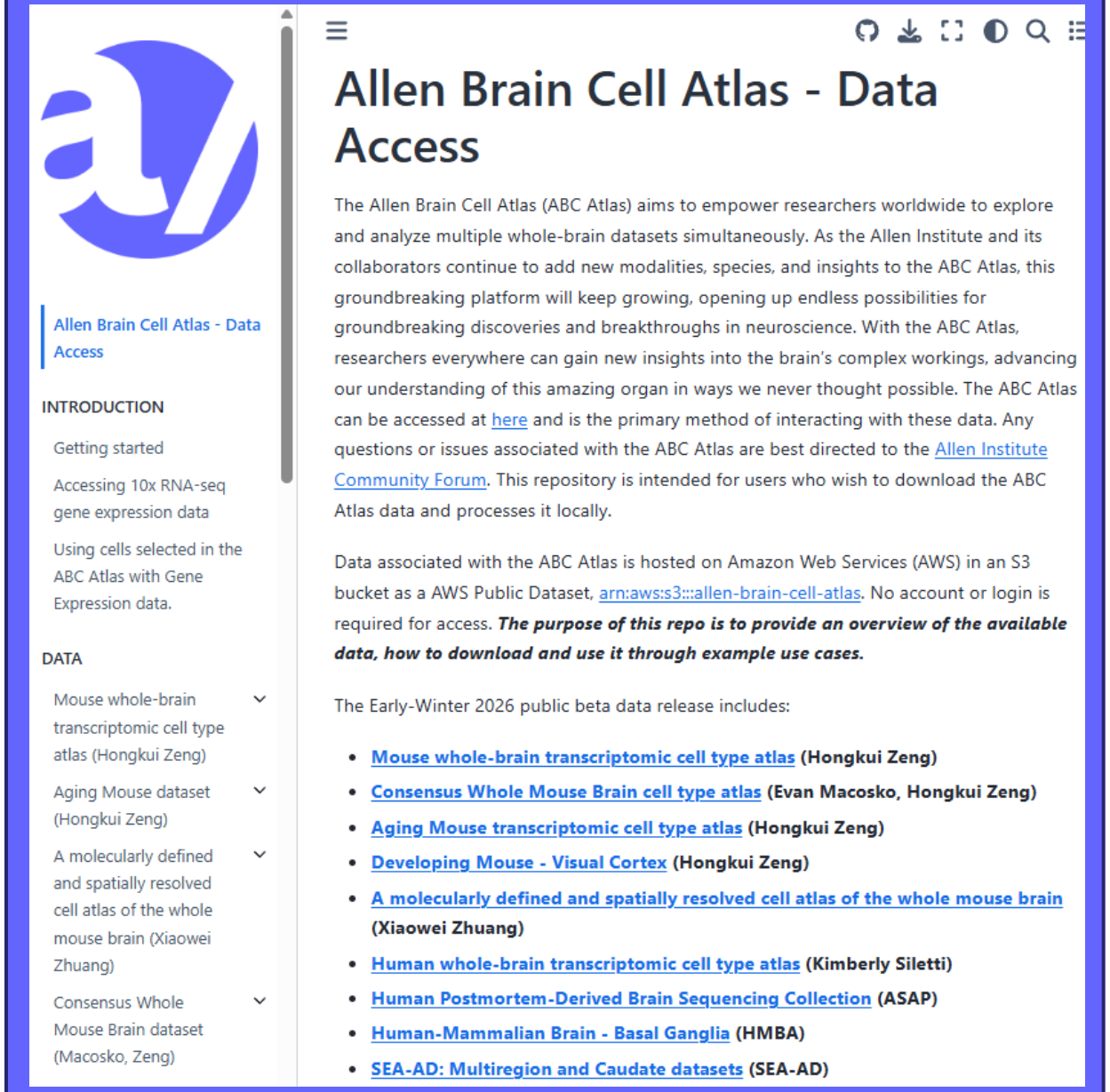
Community Tools





ABC Atlas Access

(and resource-specific download pages not shown)



questions?

allen institute/

agenda/morning

- 9:00-9:30 | check-in / breakfast
- 9:30-9:40 | welcome, logistics, and overview of allen institute
- 9:40-10:05 | cell typing at allen institute: taxonomies, data sets, and use cases
- **10:05-10:30 | human and mammalian brain atlas (HMBA)**
- 10:30-10:40 | break
- 10:40-11:05 | localizing cell types with spatial transcriptomics
- 11:05-11:30 | cell-type targeted enhancer development
- 11:30-11:55 | Seattle Alzheimer's Disease Brain Cell Atlas in the brain health accelerator
- 11:55-12:05 | Overview of brain knowledge platform (BKP) tools & resources
- 12:05-12:10 | Preview of afternoon
- 12:10-1:10 | lunch

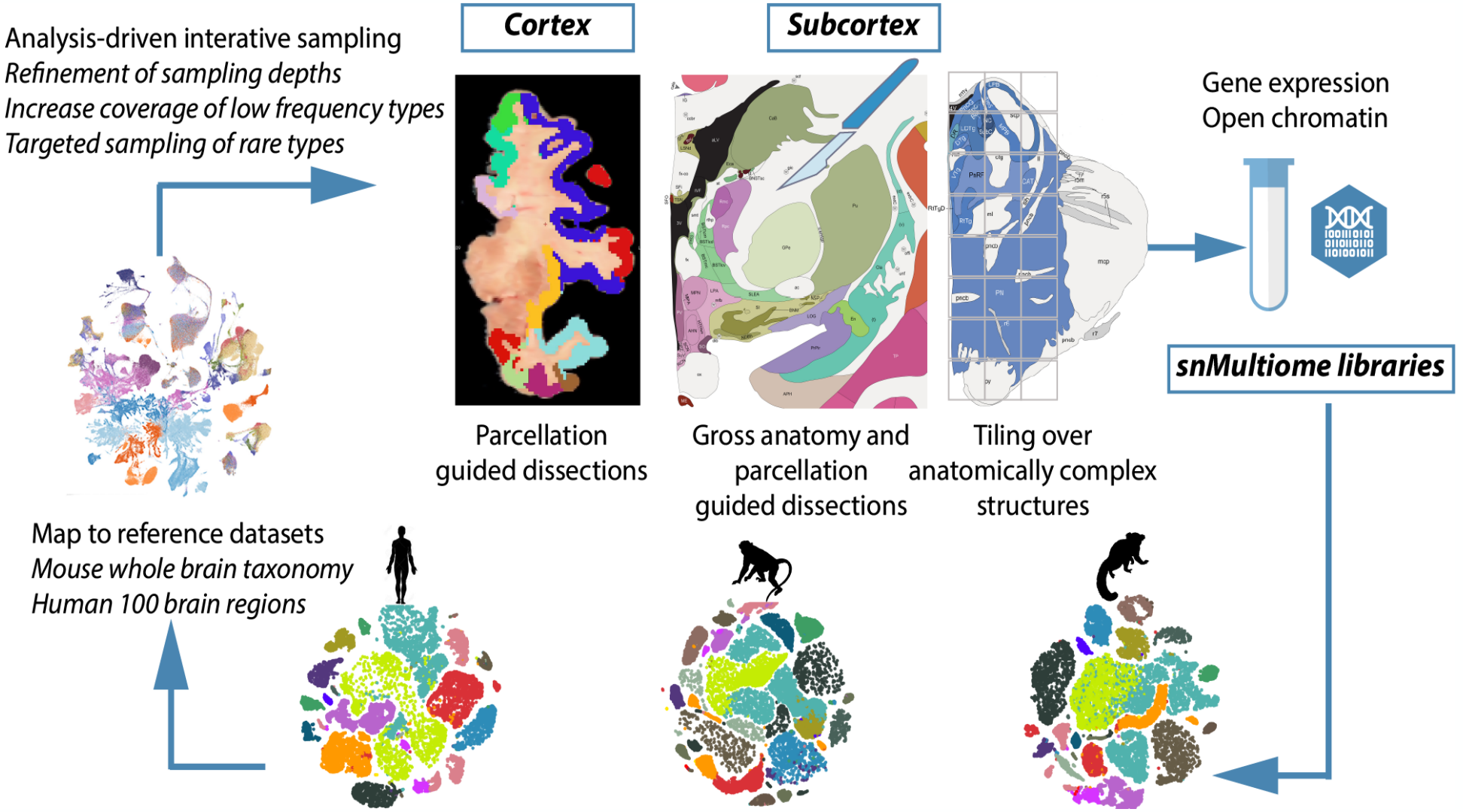
The human and mammalian brain atlas (HMBA)/

Aaron Garcia, Scientist II

allen institute/

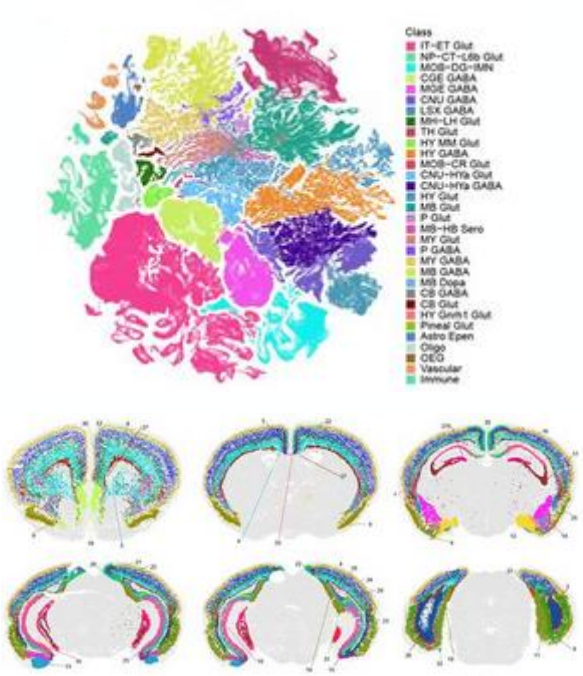
allen institute

HMBA overview/ building on previous datasets to create comprehensive, multiomic atlases of primate brains



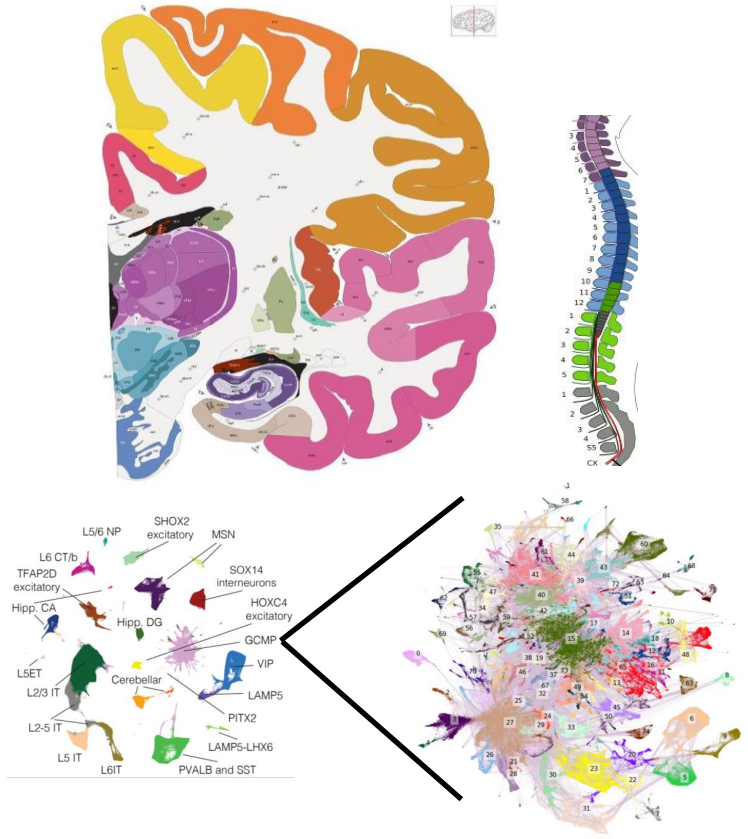
HMBA overview/ building on previous datasets to create comprehensive, multiomic atlases of primate brains

Mouse whole brain atlas



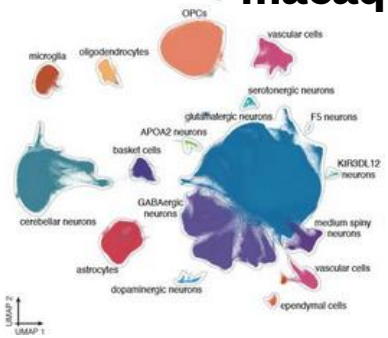
~7M cells profiled with scRNA-seq, spatially resolved dataset of ~4.3M cells with MERFISH, 5200 clusters

Human brain survey



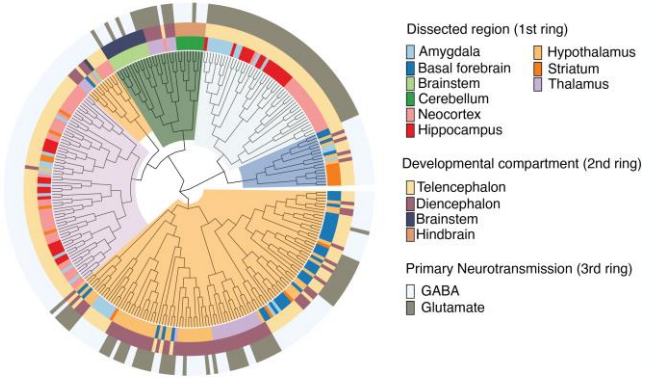
100 dissections across forebrain, midbrain, hindbrain, ~3M nuclei, >3000 clusters

Single cell multiomic atlas of rhesus macaque brain



30 brain regions, transcriptomic and epigenomic profiling, 4.2M cells

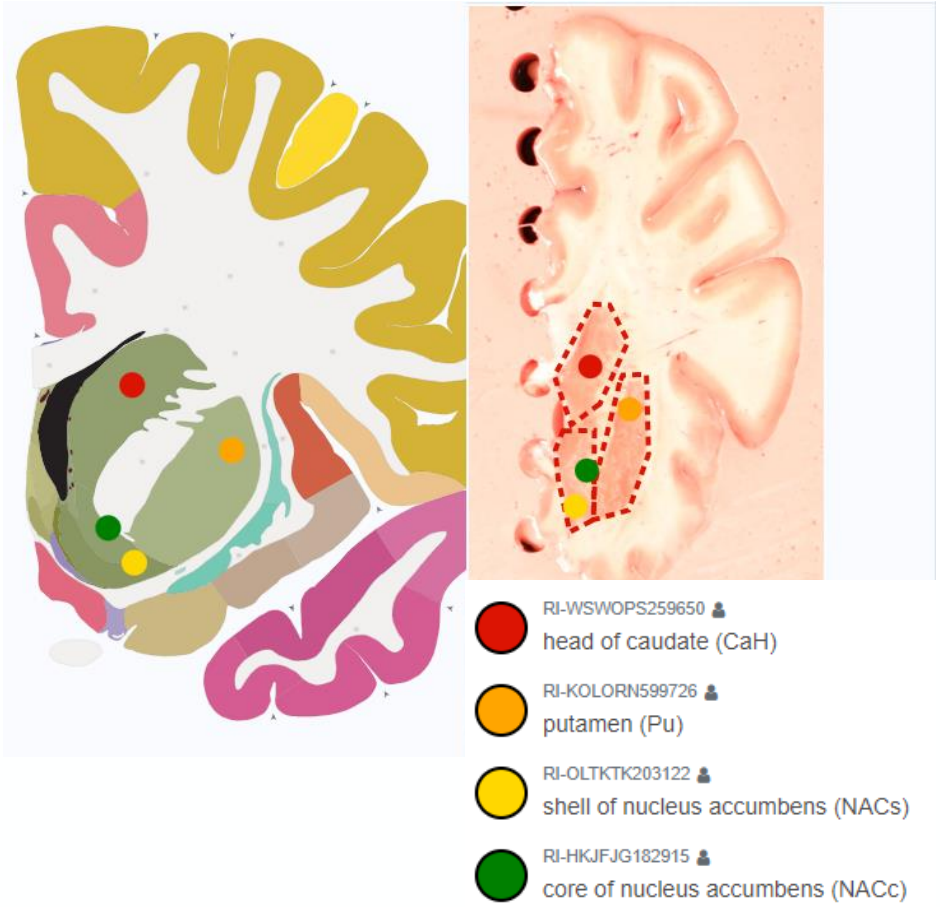
Marmoset brain cell census



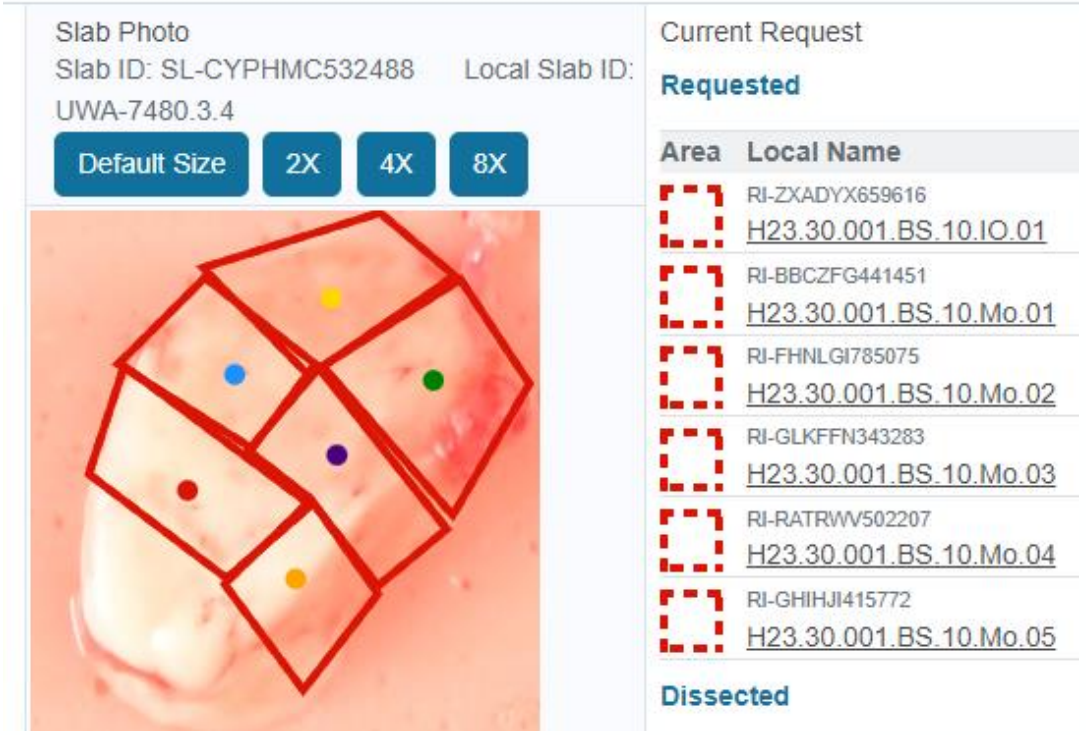
2.4M nuclei, 8 cortical and 8 subcortical areas, smFISH quantification of striatal & cortical interneurons

HMBA dissections/atlas-guided- & tiling-based-sampling for comprehensive coverage of subcortical brain regions

Atlas-guided sampling of subcortical regions



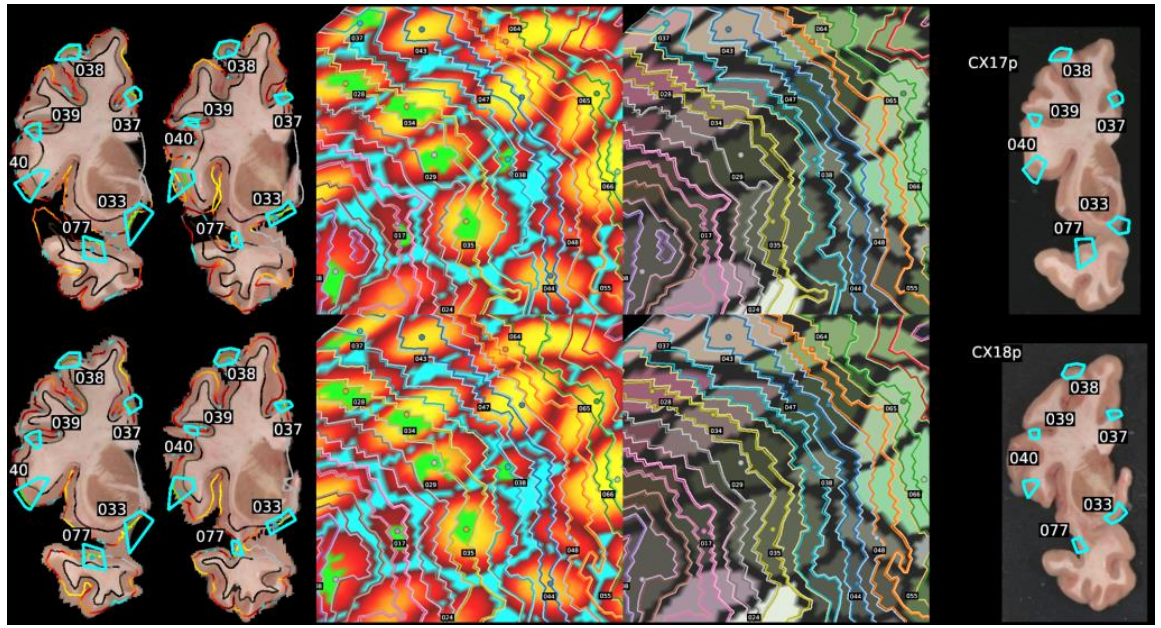
Tiled sampling of brainstem tissue



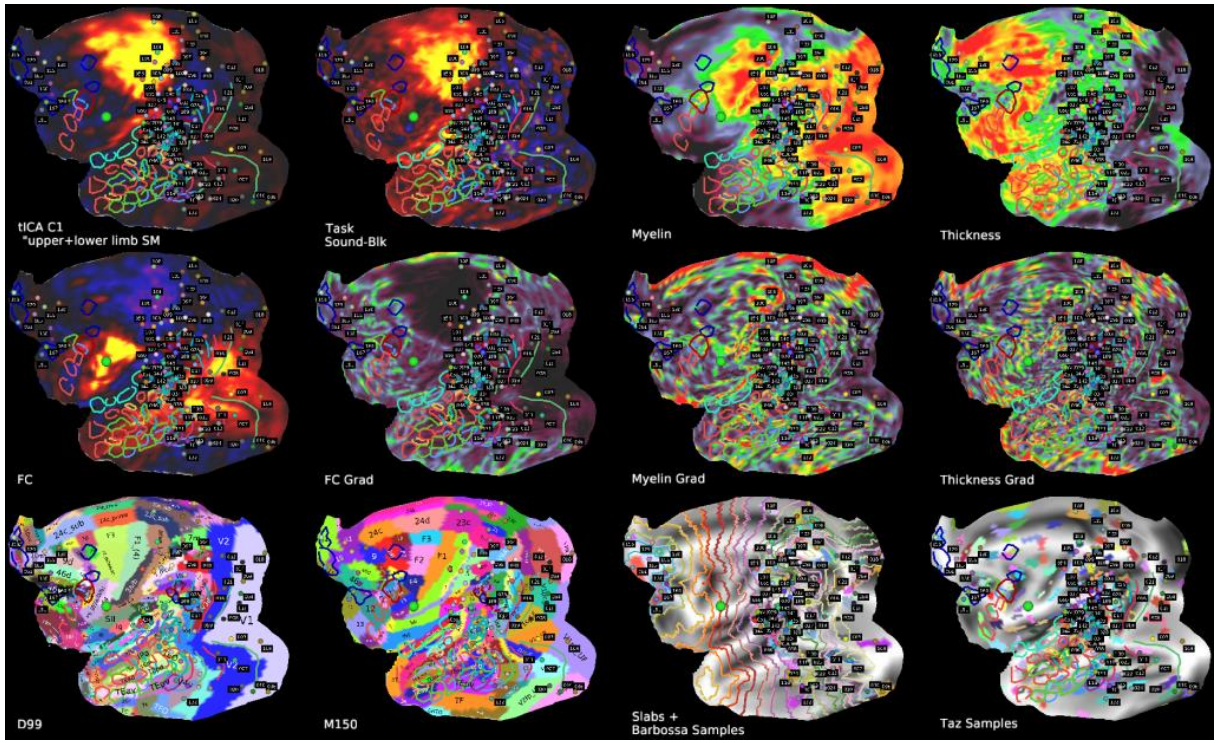
UTHealth Houston | Specimen Portal | Seq Library Portal

HMBA dissections/ MRI-guided cortical sampling

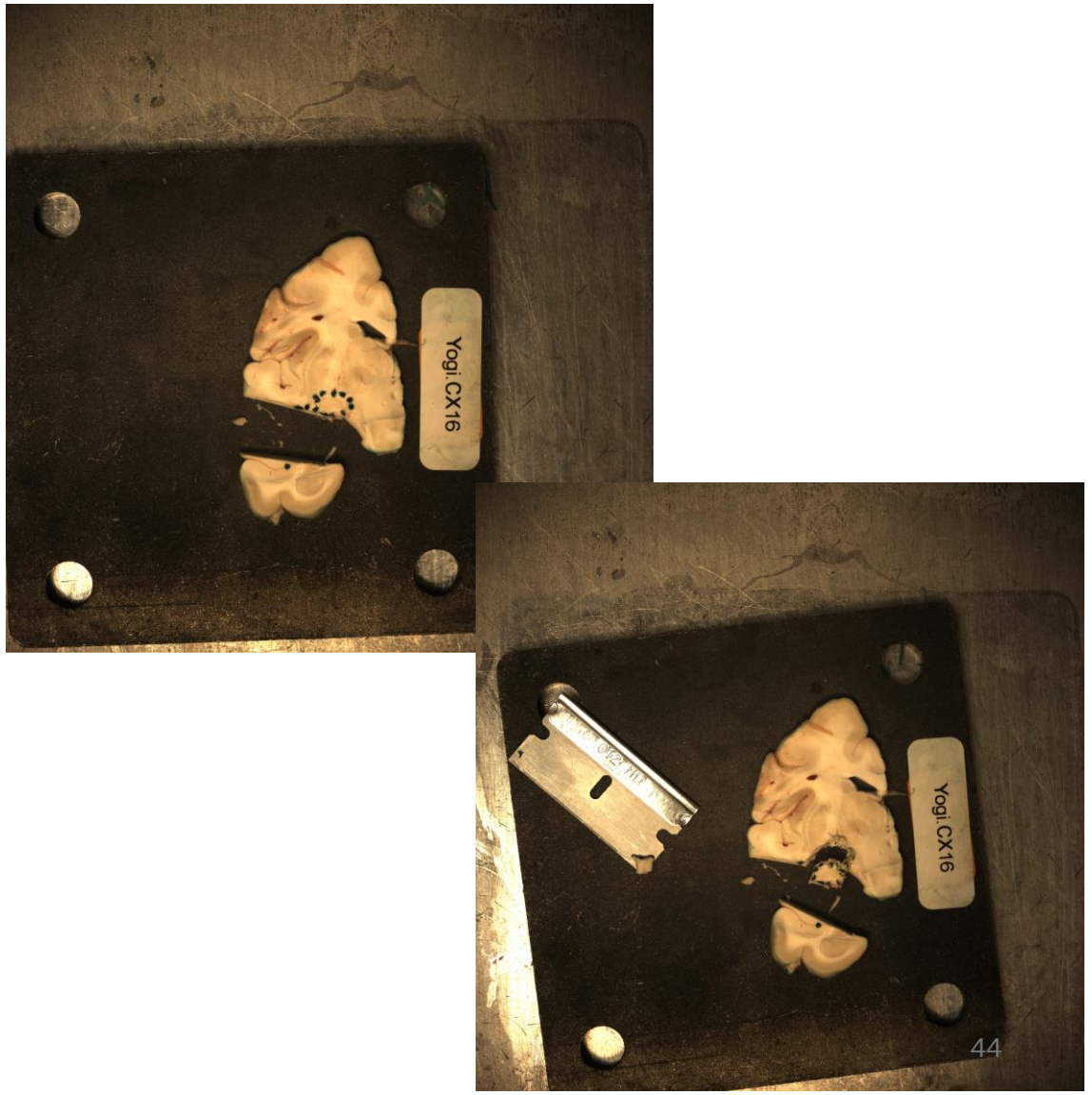
Probability-driven ROIs from hyperfine MRI mapped to volumetric atlases for human cortical sampling



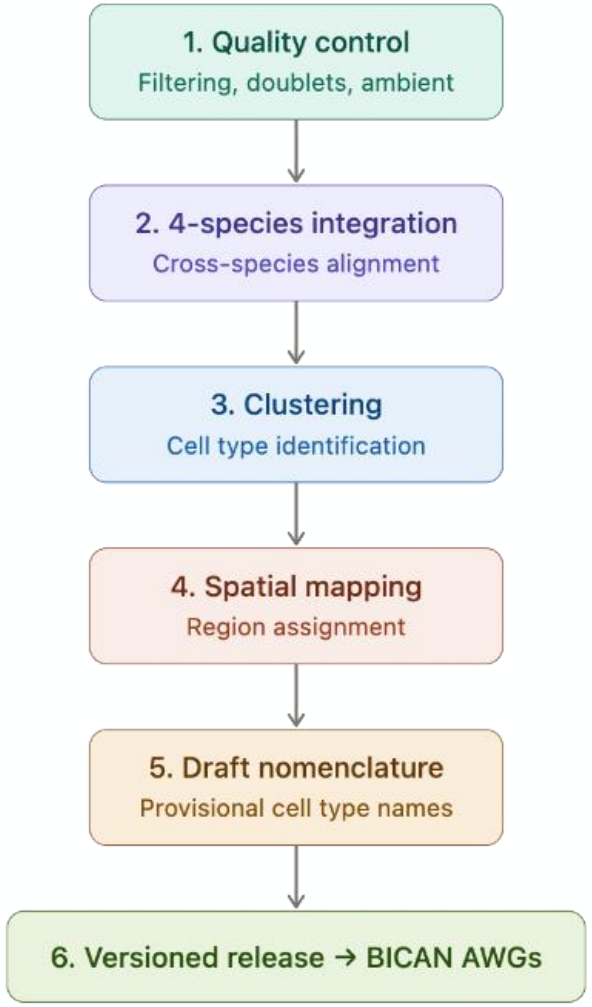
fMRI-guided sampling of 160 macaque cortical regions



HMBA dissections/ Tissues dissected on a chilled plate with comprehensive photo-documentation



HMBA data workflow/ versioned releases supplemented by new data and regionally-focused community annotation groups



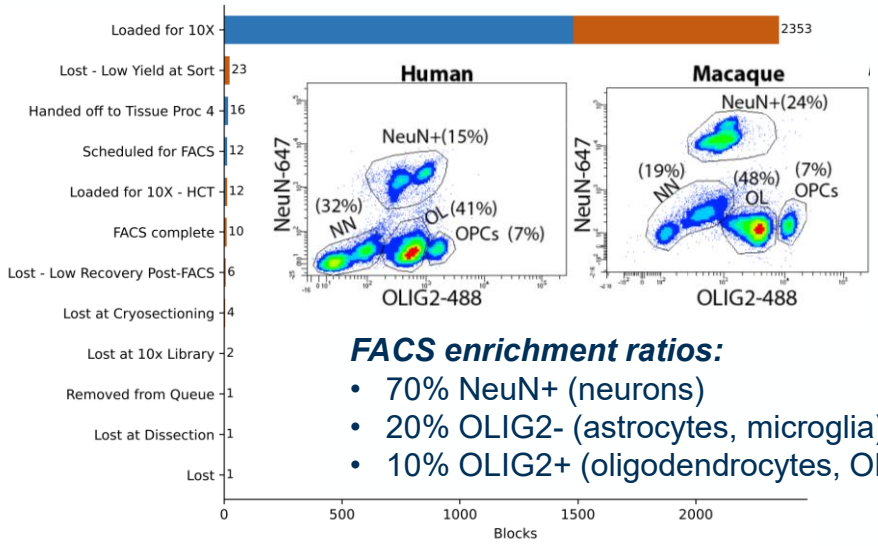
- HMBA v1: subcortex, sparse cortex (2026)
  - Whole cerebrum spatial tx from Marmoset (n=2) and Macaque (n=1-2)
  - Rough draft (v0.5) for refinement by regional annotation groups

Neocortex	Rebecca Hodge
Medial temporal lobe	Aaron Garcia, Yuanyuan Fu
Basal ganglia	Nelson Johansen, Yuanyuan Fu
Thalamus	Brian Long, Meghan Turner
Hypothalamus	Fenna Krienen, Stephanie Seeman
Midbrain, Medulla, Pons	Zizhen Yao, Cindy van Velthoven, Hongkui Zeng
Cerebellum	Rebecca Hodge
Spinal cord	Matthew Schmitz, Nelson Johansen
Non-neuronal WB	Yuanyuan Fu

- HMBA v2: +macaque cortex, more subcortex (2027)
- HMBA v3: +human cortex, +mouse brainstem 5x profiling, +/- human/NHP brainstem spatial tx (2028, pending data completion)

HMBA data generation progress/

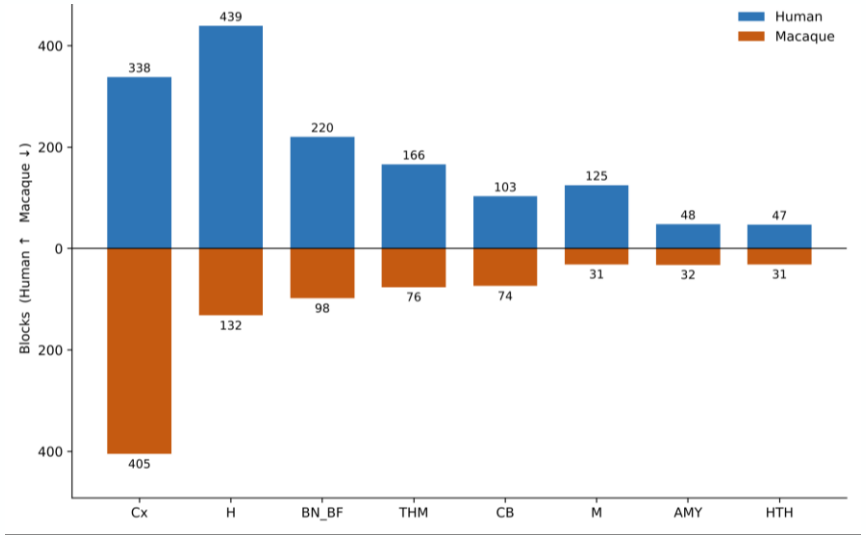
Sample Block Status



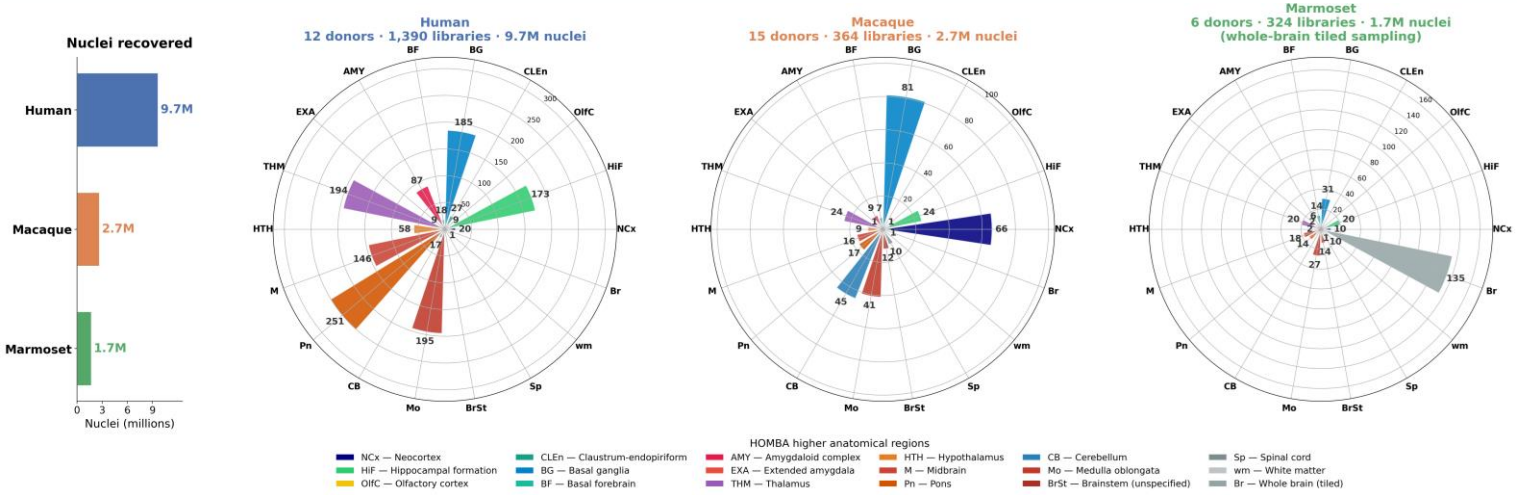
FACS enrichment ratios:

- 70% NeuN+ (neurons)
- 20% OLIG2- (astrocytes, microglia)
- 10% OLIG2+ (oligodendrocytes, OPCs)

Regional Study Set Distributions

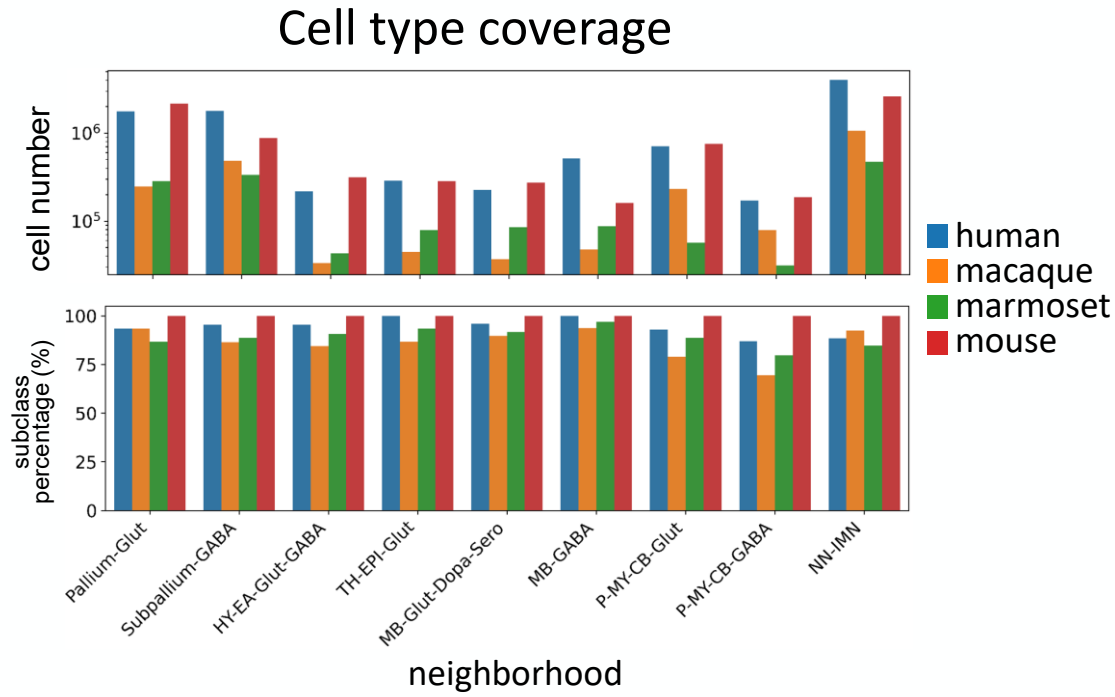


Sequenced Data Overview – HMBA v0.5 (July 2026)



Marmoset samples generated by lab of Fenna Krienen (Princeton)

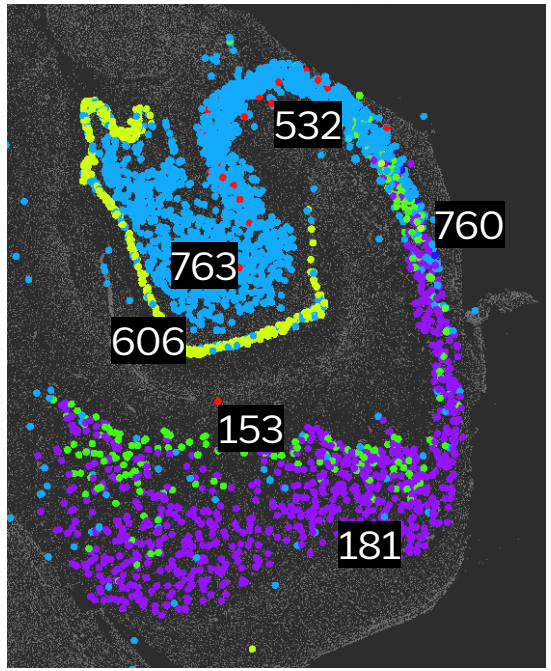
HMBA atlas v1/ integrated cross-species subcortical data with sparse cortical coverage



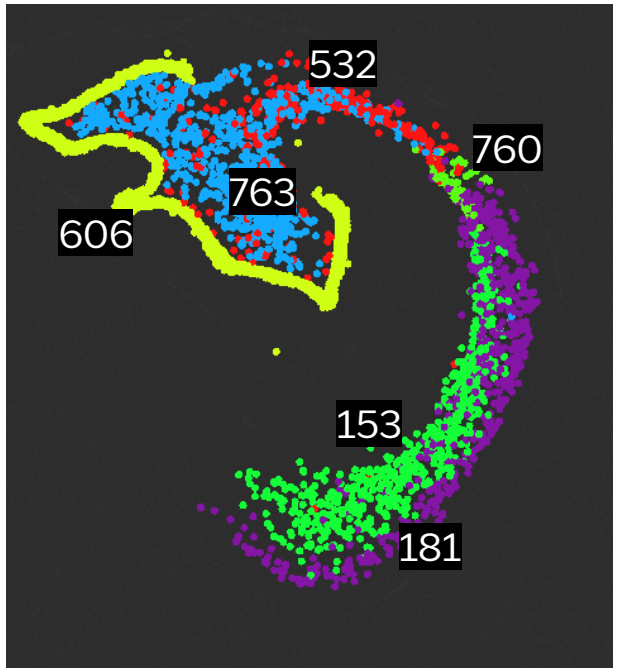
HMBA beyond RNAseq/ other modalities

Spatial Transcriptomics:
91.5 million cells from **250 sections** sampling **6 brains** across **3 species**

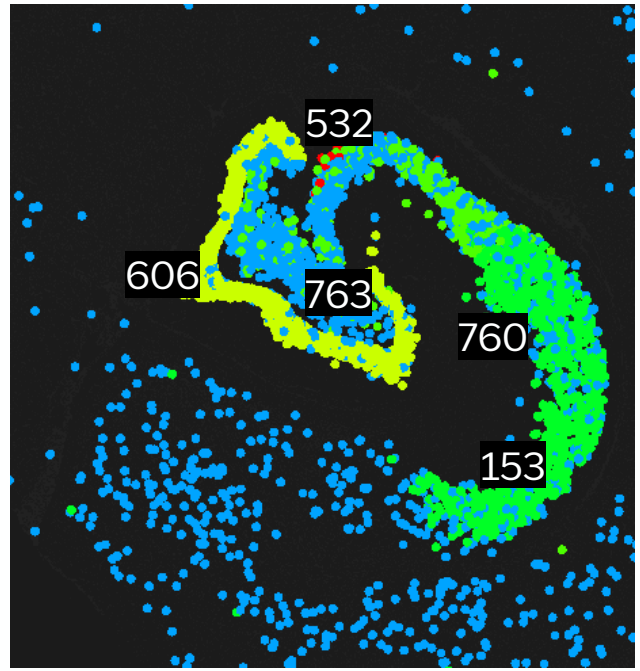
Human



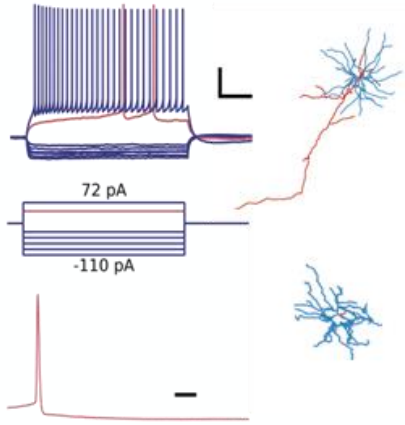
Macaque

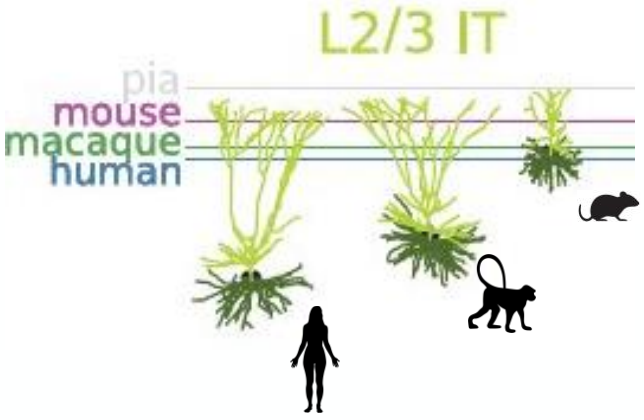


Marmoset



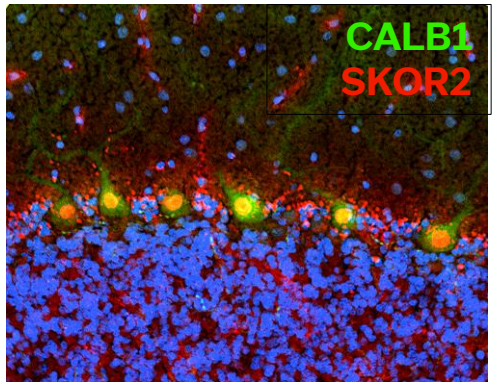
Patch-Seq:
~6000 cells recorded from V1, Hippocampus, Basal Ganglia, and Temporal Gyrus



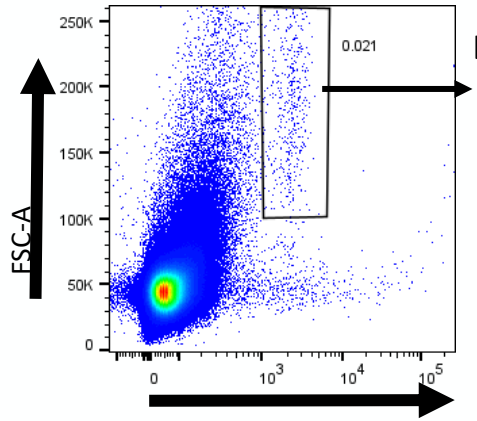


HMBA beyond atlassing/ other efforts

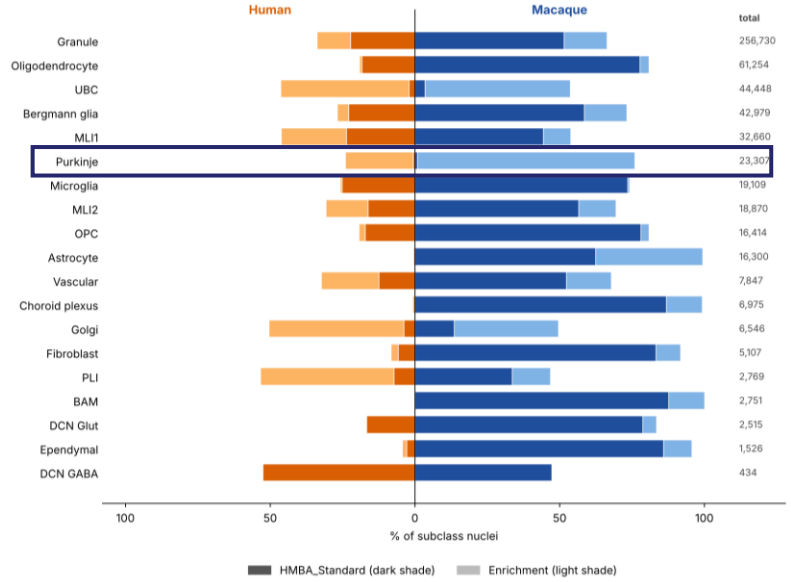
- **Integration with external datasets**
  - Seattle Alzheimer's Disease (SEA-AD)
  - Slide-tag: whole transcriptome spatial transcriptomics
- **Enrichment for rare populations in RNA-seq datasets**
  - Purkinje cells (shown)
  - Microglia
  - Vascular cells
  - Rare brainstem populations
- **Evolutionary and Population Variation**



FACS Sorting for SKOR2+ nuclei



Large, SKOR2+ cell somas



Want to learn more? Visit PSTR232.17/DD7 at SFN

Community annotation efforts/ interested in joining?

- We want to engage the community to help with annotating cell types
- Regional annotation groups focused on subsets of the brain
- Provide feedback on nomenclature, biological properties, robustness of cell types
- Data will be made available in accessible tools (snRNA-seq+Spatial Transcriptomics)
- Talk to me or email if you are interested in participating **aaron.garcia@alleninstitute.org**

allen institute/Brain Health

Ed Lein

Rebecca Hodge

Trygve Bakken

Brian Kalmbach

Nelson Johansen

Song-Lin Ding

Jonathan Ting

Jeremy Miller

Susan Sunkin

Lauren Kruse

allen institute/Brain Science

Hongkui Zeng

Kimberly Smith

Zizhen Yao

Jennie Close

Brian Long

Tim Jarsky

Stephanie Seeman

Brian Lee

Staci Sorensen

Shoaib Mufti

Mike Hawrylycz

Tyler Mollenkopf

Jimena Meladze

UCSD

C. Dirk Keene

UW

Tom Grabowski

Christine MacDonald

Tim Brown

Kimiko Domoto-Reilly

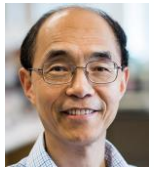
MGH

Bruce Fischl

Juan Eugenio Iglesias

MIT

Nancy Kanwisher


Guoping Feng

RU

Winrich Freiwald

WashU

David Van Essen

Matt Glasser

INSERM

Henry Kennedy

RIKEN

Takuya Hayashi

Princeton

Fenna Krienen

ASU

Noah Snyder-Mackler

UPenn

Michael Platt

Stanford

Andreas Tolias

BCM

Xiaolong Jiang

Leiden

Boudewijn Lelieveldt

Sanger

David Osumi-Sutherland

Special Mention:

Shu (Daisy) Dan

Maddy Hewitt

Meghan Turner

Yuanyuan Fu

Rachel Dalley

Xiao-Ping Lui

Burke Rosen

Faraz Yazdani/ Dingzhi Hu

+ all of the amazing teams at each institution!

NIH award UM1MH130981-01

questions?

allen institute/

alleninstitute/
thank you/

agenda/morning

- 9:00-9:30 | check-in / breakfast
- 9:30-9:40 | welcome, logistics, and overview of allen institute
- 9:40-10:05 | cell typing at allen institute: taxonomies, data sets, and use cases
- 10:05-10:30 | human and mammalian brain atlas (HMBA)
- **10:30-10:40 | break**
- 10:40-11:05 | localizing cell types with spatial transcriptomics
- 11:05-11:30 | cell-type targeted enhancer development
- 11:30-11:55 | Seattle Alzheimer's Disease Brain Cell Atlas in the brain health accelerator
- 11:55-12:05 | Overview of brain knowledge platform (BKP) tools & resources
- 12:05-12:10 | Preview of afternoon
- 12:10-1:10 | lunch

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from taxonomy to atlas/ localizing cell types with spatial transcriptomics

Meghan Turner, Scientist II

allen institute/

allen institute

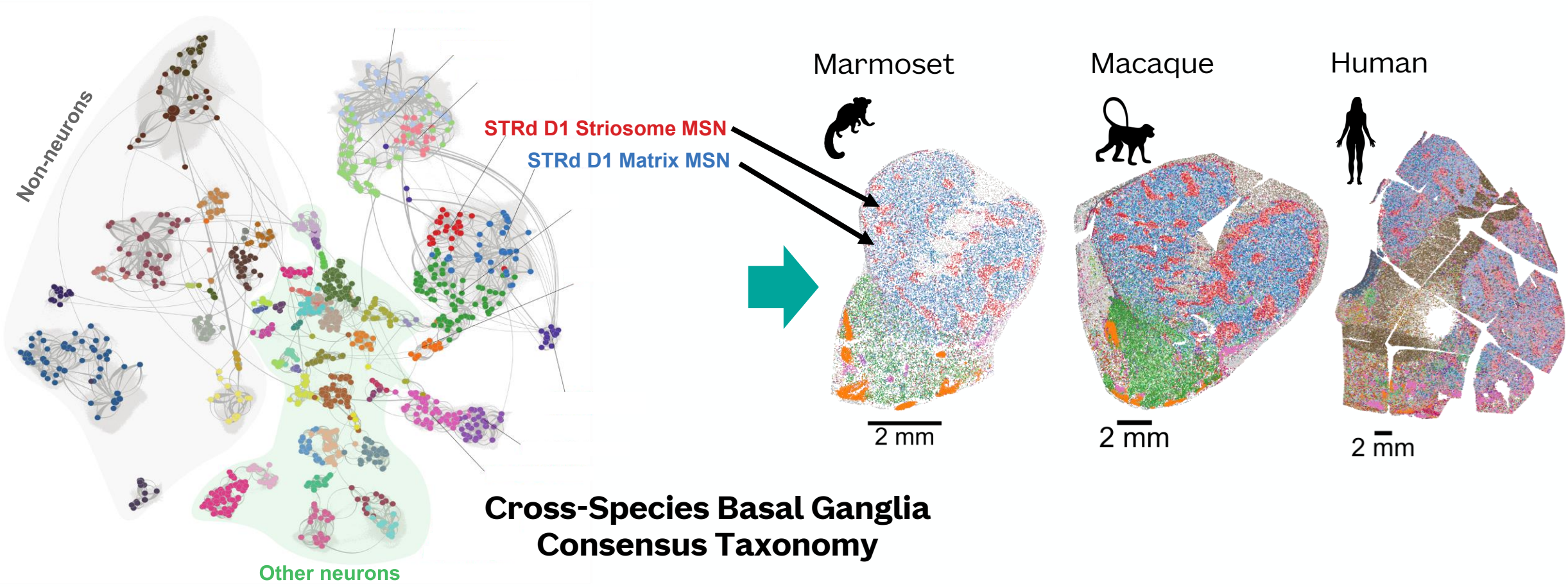
why spatial transcriptomics/ putting together the puzzle of cell types in the brain

Puzzle pieces sorted by color and pattern … now in their proper spatial context!

Slide courtesy of Maddy Hewitt; image source: https://www.puzzlewarehouse.com/blog/shaped-puzzle-strategy-for-beginners/t

why spatial transcriptomics/ placing single cell transcriptomics in anatomical context

Individual cells from single-nuclei RNA sequencing (snRNA-seq) grouped by gene expression similarity … now in their proper anatomical context!

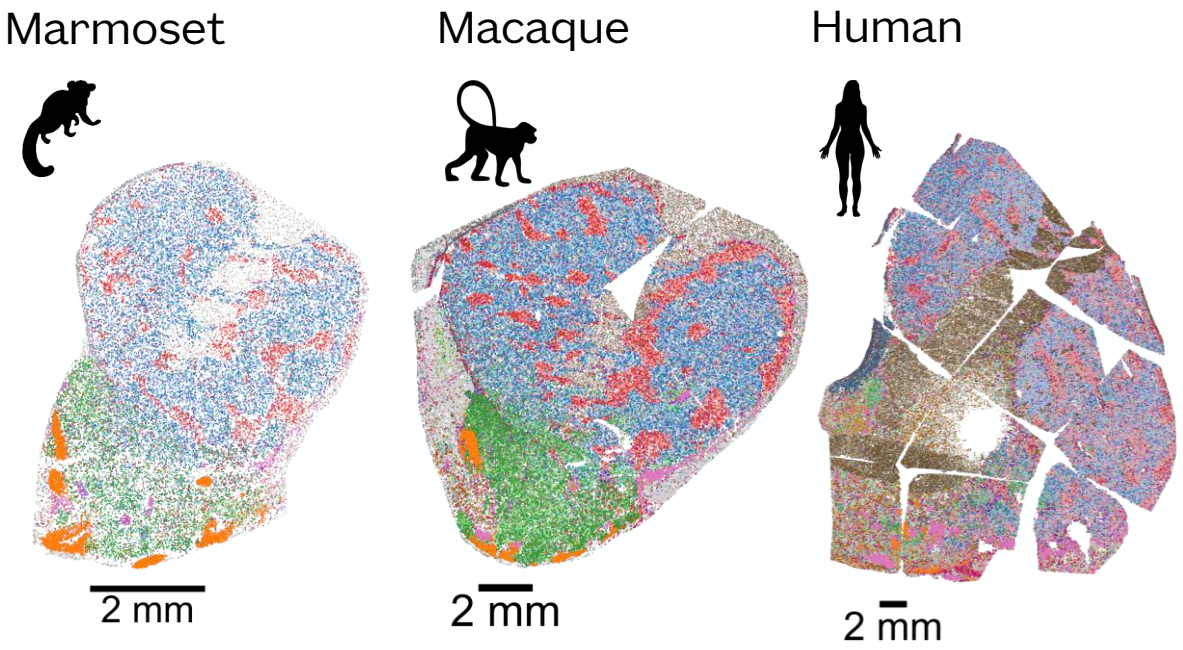


Cross-Species Basal Ganglia Consensus Taxonomy

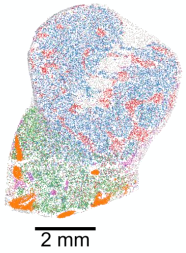
Johansen, Fu, ..., Lein, Bakken. 2026. DOI: 10.1016/j.cell.2026.10.012

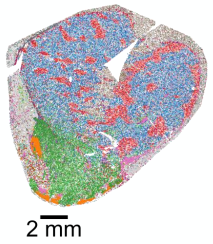
Hewitt, Turner, ..., Close, Long, Seeman. 2026. DOI: 10.1016/j.cell.2026.10.023

why spatial transcriptomics/ link transcriptomic cell types to brain structure and other data modalities



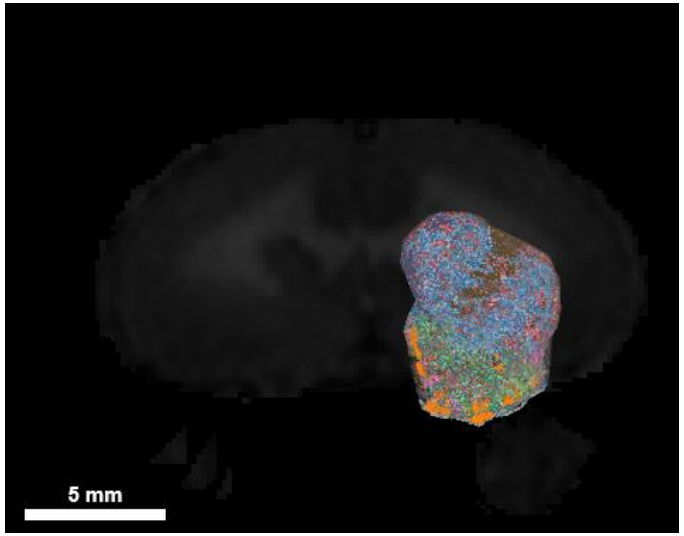
why spatial transcriptomics/ link transcriptomic cell types to brain structure and other data modalities



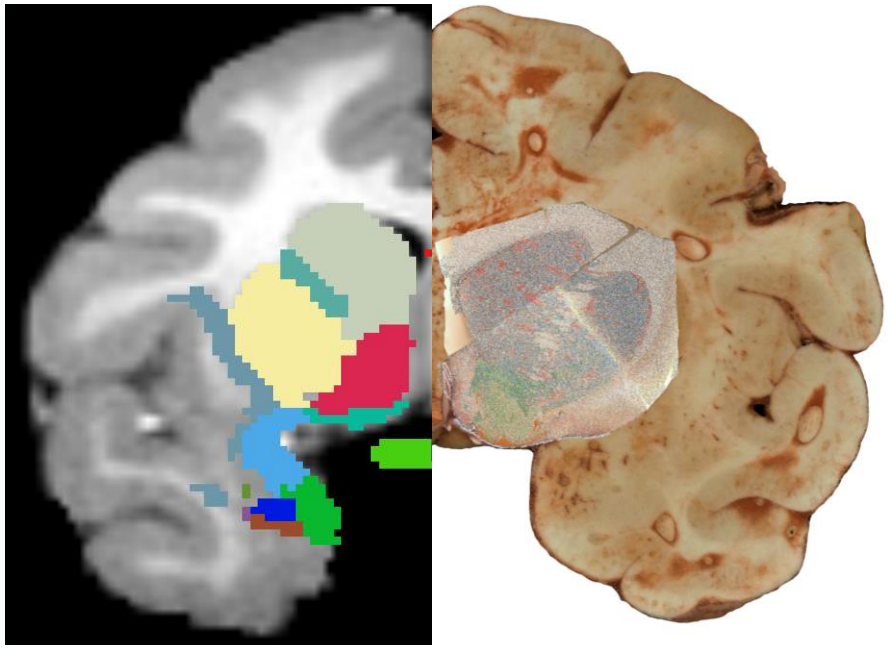


why spatial transcriptomics/ link transcriptomic cell types to brain structure and other data modalities

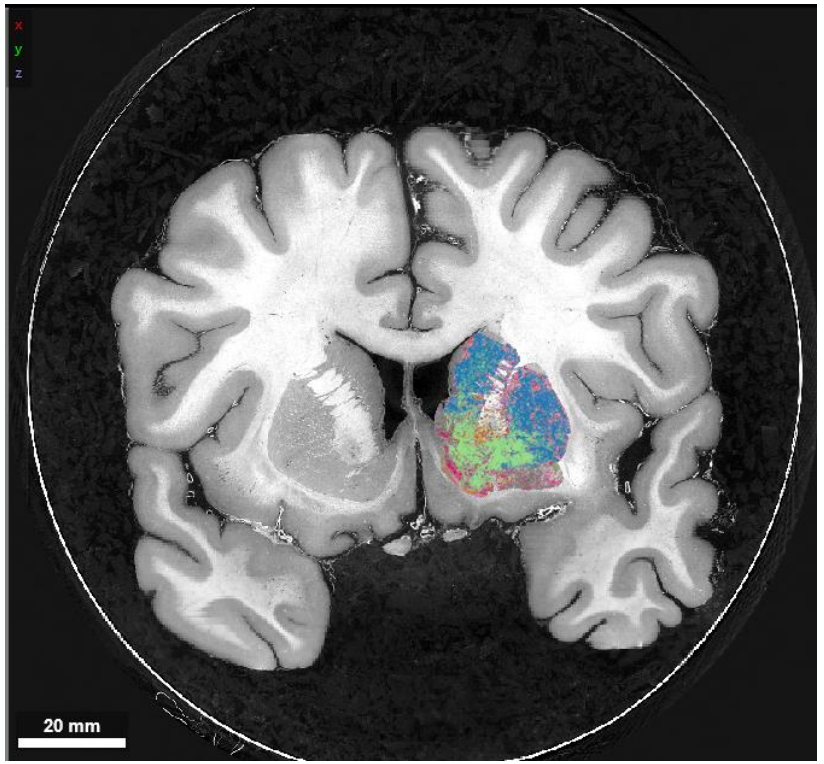
Marmoset



Macaque



Human

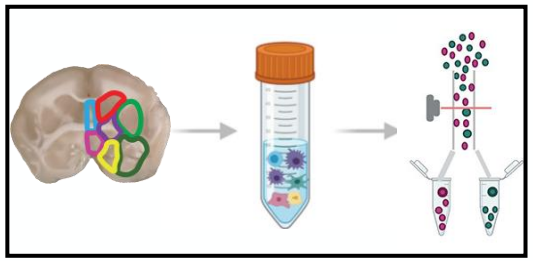


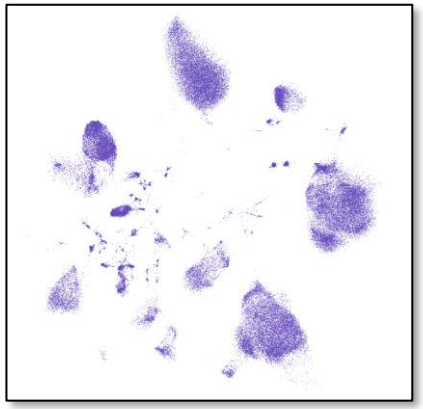
Spatial transcriptomics is the critical link from cell types to:

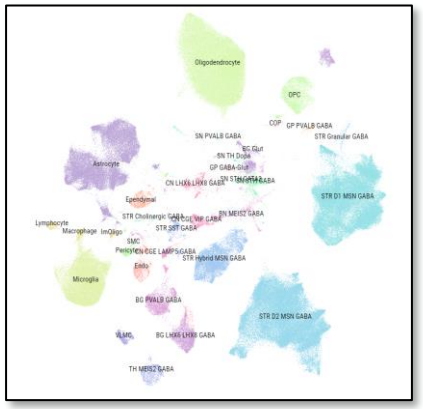
- Functional MRI (fMRI) measurements = infer brain activity during behavior or tasks
- Classic neuroanatomical definitions & literature
- Targets for Deep Brain Stimulation (DBS) in human patients

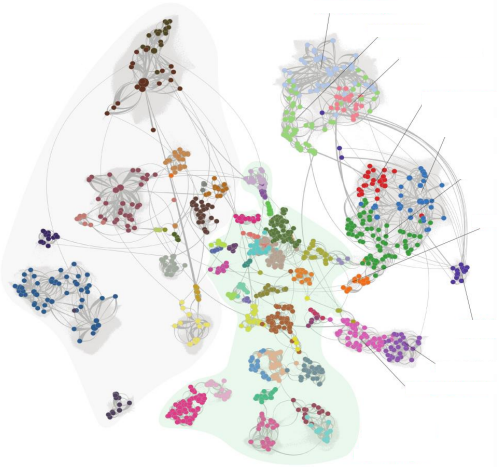
making a reference cell type taxonomy/ RNA-seq data

Collect single-cell/nucleus RNA-seq data → QC → Computationally cluster RNA-seq data → Initial labels added to RNA-seq clusters → Draft RNA-seq reference taxonomy







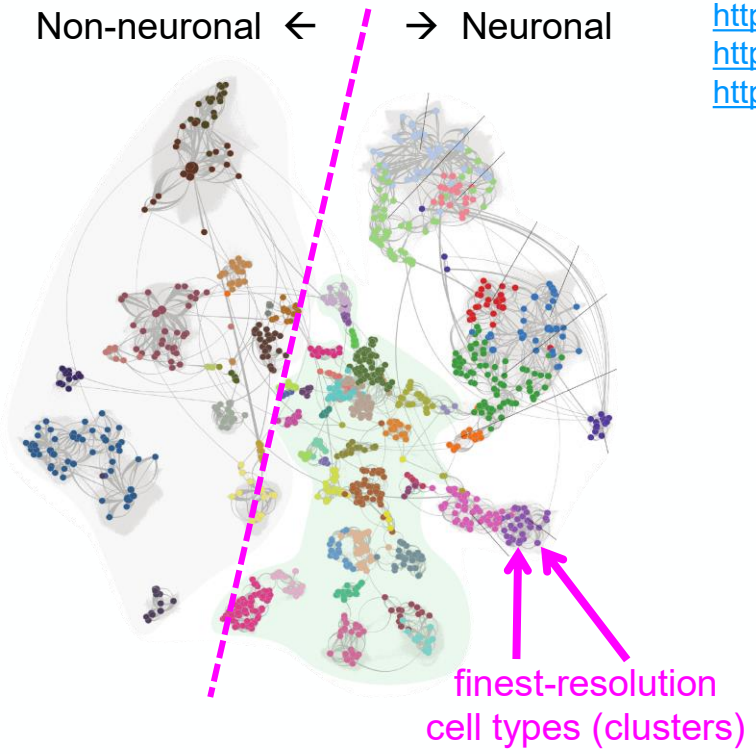


All steps where spatial transcriptomics interfaces with RNA-seq data and helps shape the final reference taxonomy

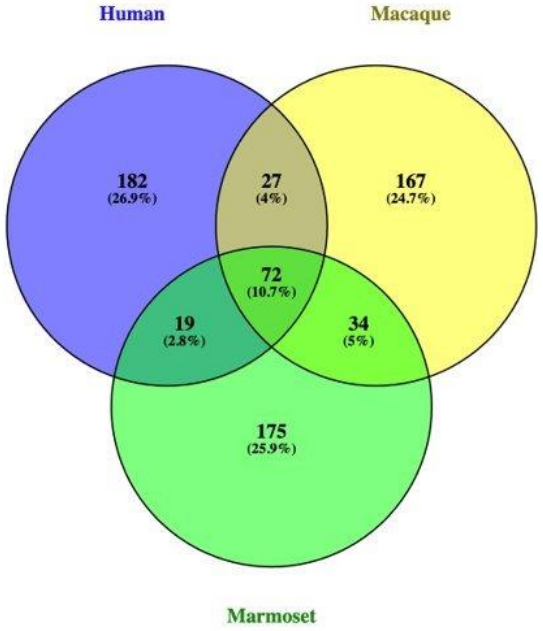
spatial transcriptomics data collection/ gene panel design

1) Leverage existing RNA-seq reference taxonomies to select preliminary gene panel (e.g. 300 genes)

a) Seed algorithms with well-known marker genes.
b) Computationally select genes to differentiate **cluster** level using tools like geneBasis, mfishtools, & NS-Forest.
c) Check if key genes were kept (e.g. cross-species markers).

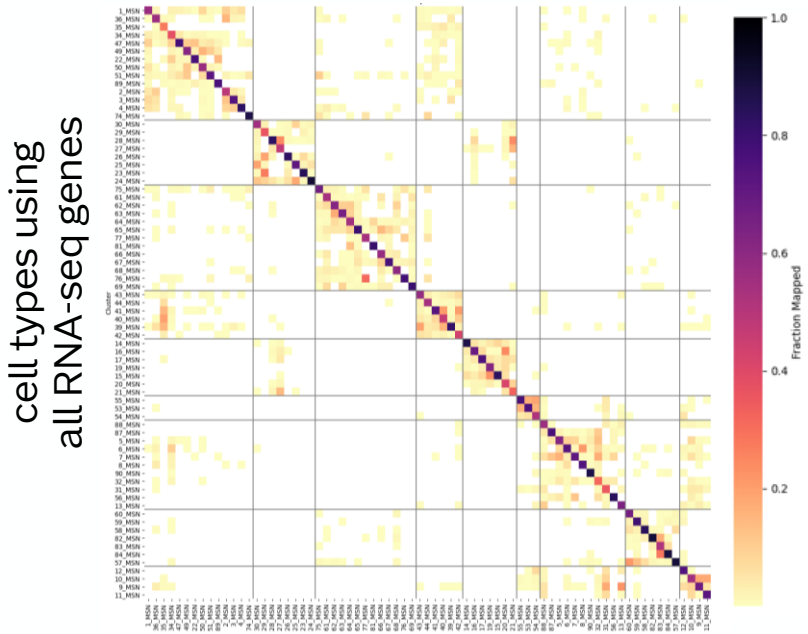


https://alleninstitute.github.io/mfishtools/
https://github.com/MarioniLab/geneBasisR
https://github.com/JCVenterInstitute/NSForest

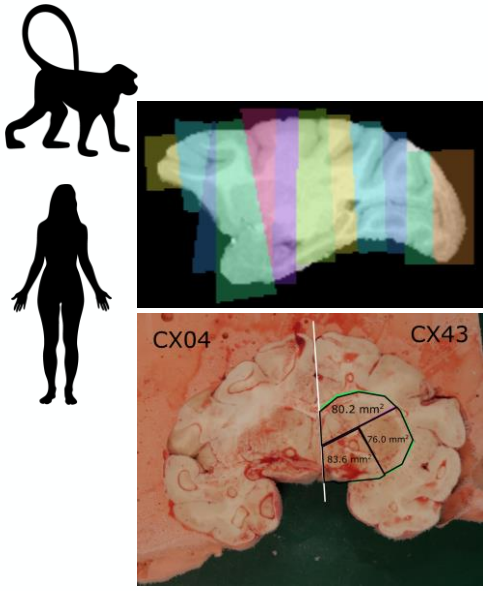


2) Assess cell type recovery using self-mapping of RNA-seq data

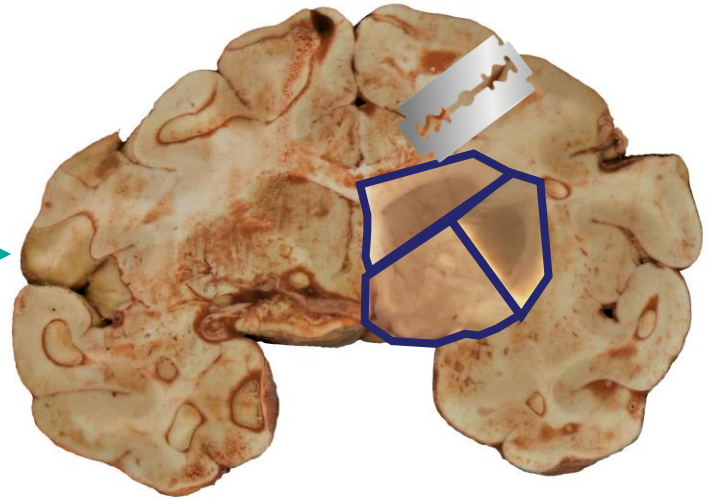
a) Subset reference RNA-seq dataset to the proposed 300-gene spatial panel.
b) Quantify how many cells map to same cluster.
c) If necessary, return to (1) and iterate.



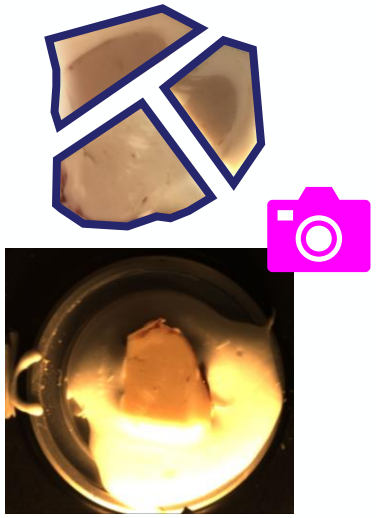
spatial transcriptomics data collection/ sample preparation



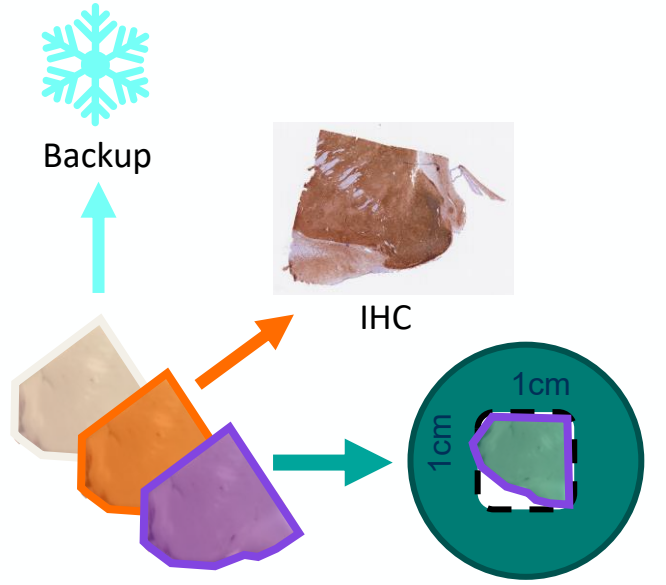
Brain divided into coronal slabs & fresh frozen



Slabs cut into blocks containing ROIs

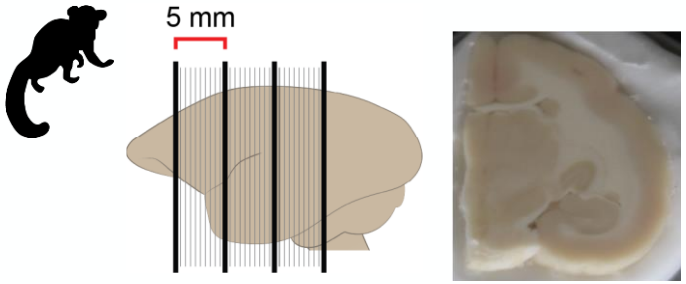


Blocks embedded & photodocumented



Blocks are sectioned

MERSCOPE coverslip



Brain divided into coronal slabs & fresh frozen

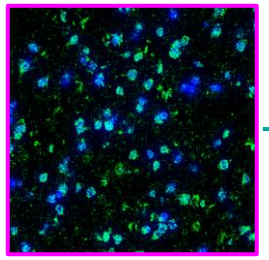
No blocking necessary – intact hemisphere sectioned

Xenium slide

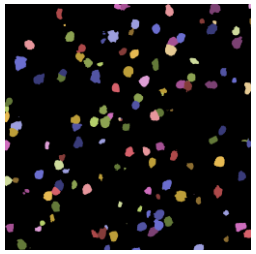
spatial transcriptomics data collection/ data processing

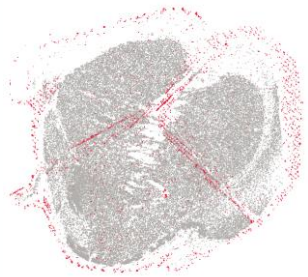
Cell Segmentation & Quality Control

Cell stains



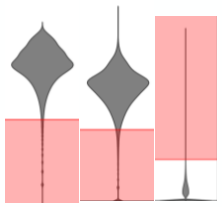
Segmented cells



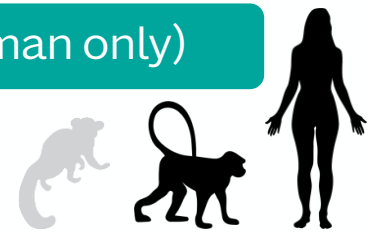


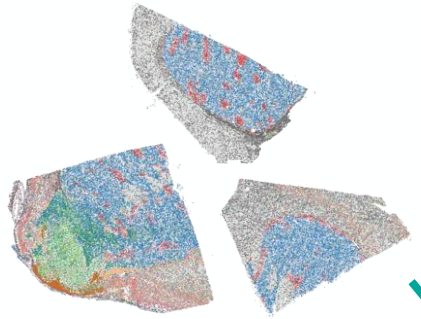
- DAPI
- All-transcripts
- 18S RNA*

Per-cell Quality control

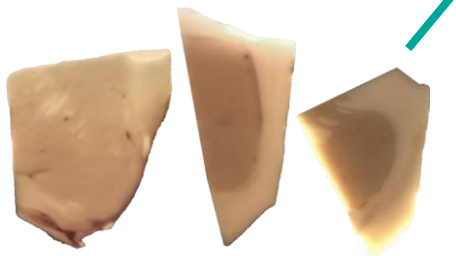


Block Stitching & Registration (macaque & human only)

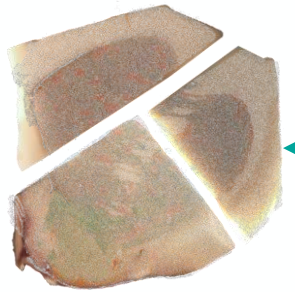




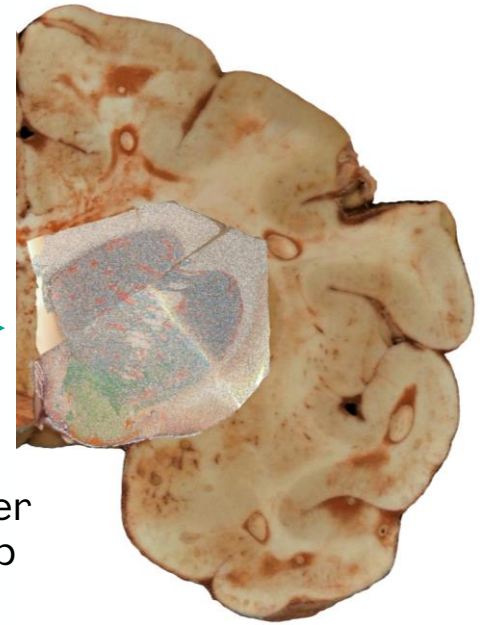
Segmented cells with preliminary cell type labels



Blockface images



Align to each other and to frozen slab image



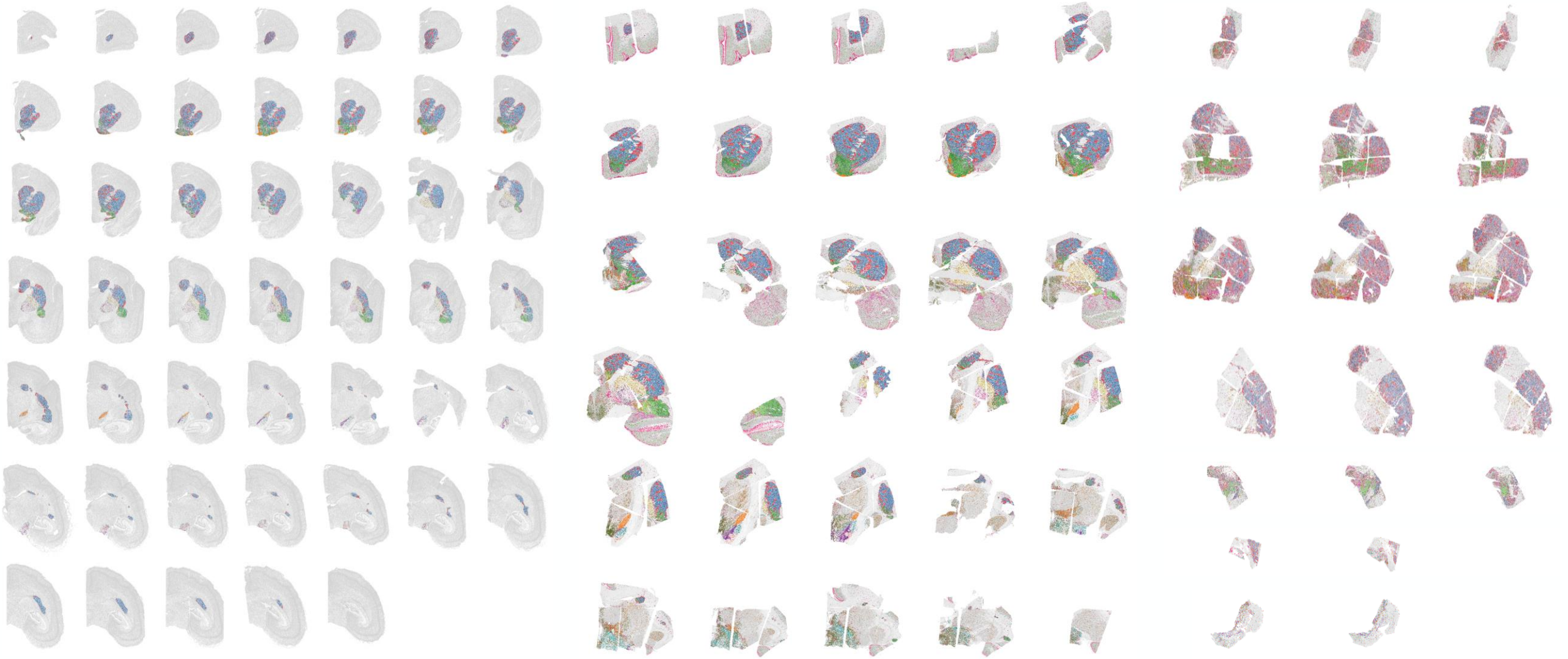
Hewitt, Turner, ..., Close, Long, Seeman. 2026. DOI: 10.1016/j.cell.2026.10.023

spatial transcriptomics data collection/ completed dataset of human and nonhuman primate basal ganglia

Marmoset – 80 cm^2 of imaged area

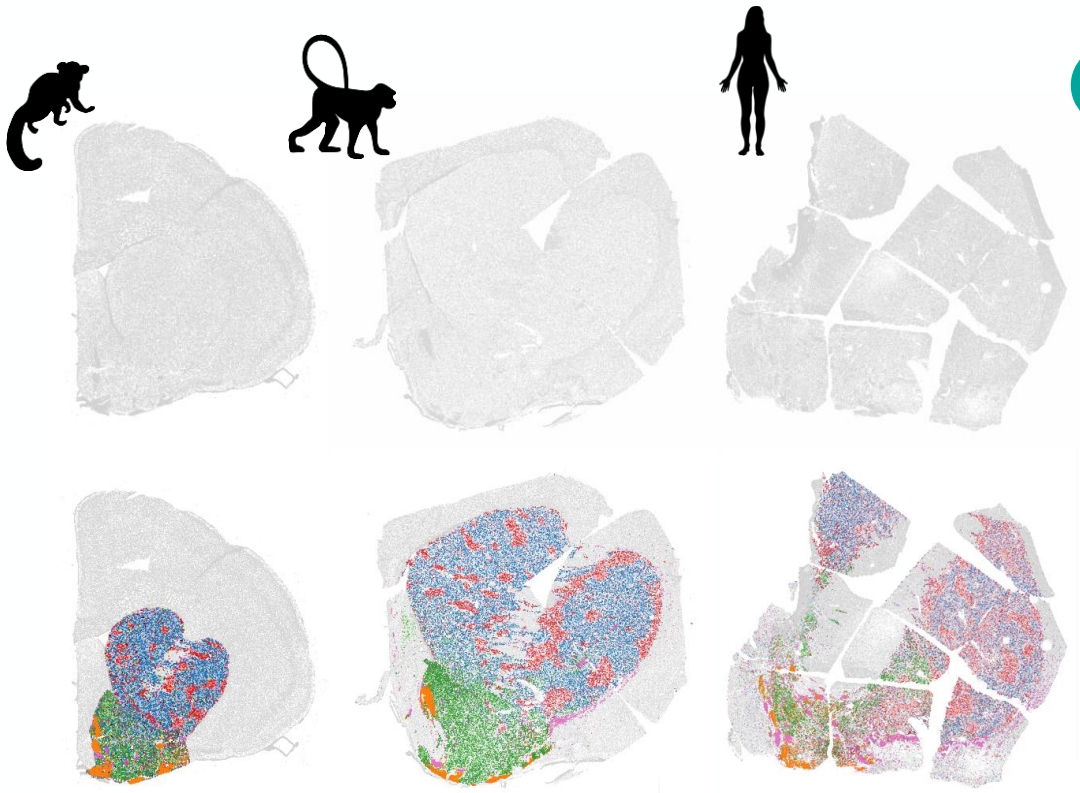
Macaque – 100 cm^2 of imaged area

Human – 100 cm^2 of imaged area

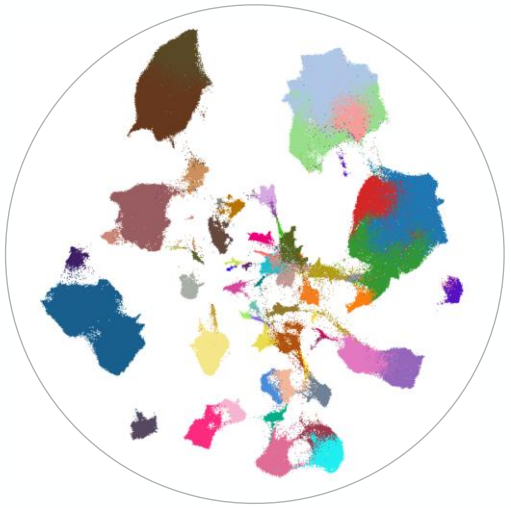


Hewitt, Turner, ..., Close, Long, Seeman. 2026. DOI: 10.1016/j.cell.2026.10.023

from taxonomy to atlas/ leveraging the spatial transcriptomics data to annotate basal ganglia cell types



1 Map spatial cells to species-specific RNA-seq clusters ★



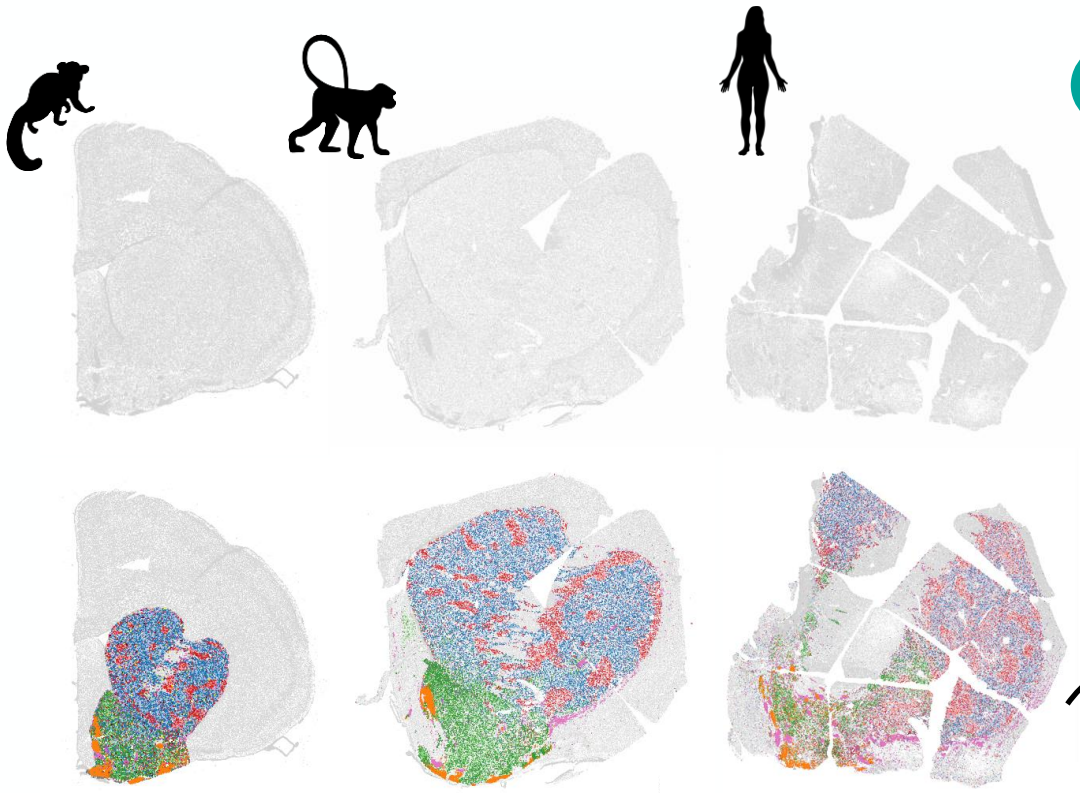
Examples of label transfer (aka "mapping") methods for spatial transcriptomics data:

★ **MapMyCells** – correlation to cluster gene expression means
- fast; can run on just new cells
- https://brain-map.org/bkp/analyze/mapmycells

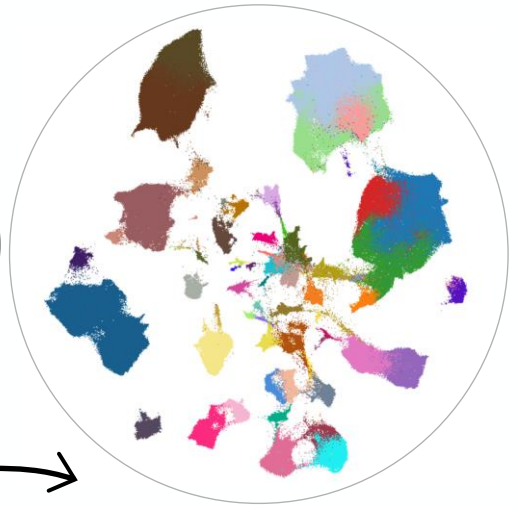
- **scVI** – integration of RNA-seq & spatial cells into common latent space + KNN or RandomForest classifier for label transfer
  - time-intensive; usually re-run on full dataset as new data comes in

Johansen, Fu, ..., Lein, Bakken. 2026. DOI: 10.1016/j.cell.2026.10.012
Hewitt, Turner, ..., Close, Long, Seeman. 2026. DOI: 10.1016/j.cell.2026.10.023

from taxonomy to atlas/ leveraging the spatial transcriptomics data to annotate basal ganglia cell types



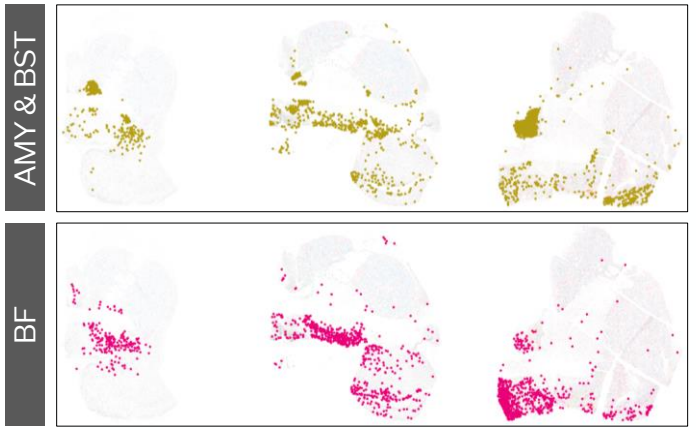
1 Map spatial cells to species-specific RNA-seq clusters



2 Update Group annotations to reflect anatomical localization (e.g. split or merge Groups, adjust cluster membership)

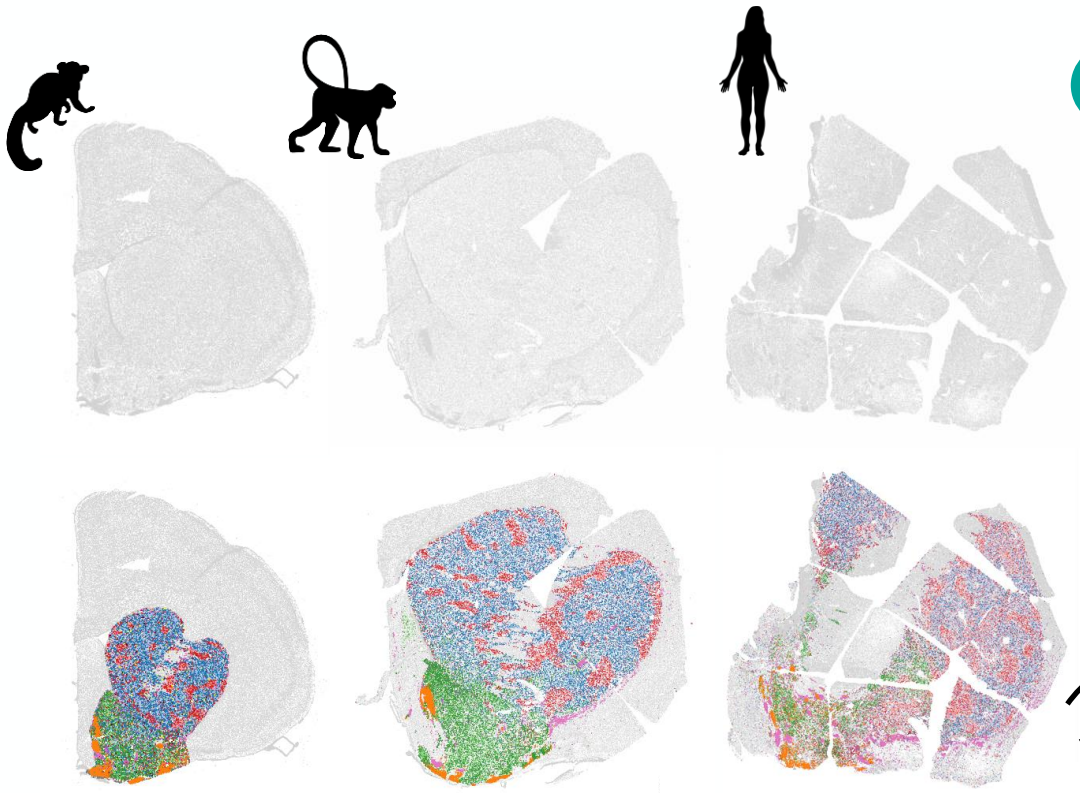
Regional Specialization → Split Groups

Non-BG cell types → Adjacent Groups

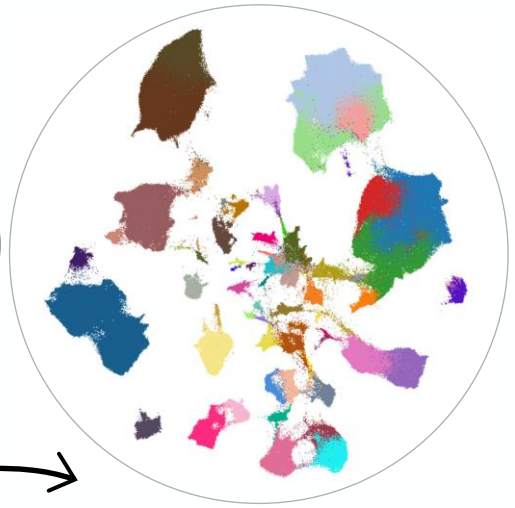


Johansen, Fu, ..., Lein, Bakken. 2026. DOI: 10.1016/j.cell.2026.10.012
Hewitt, Turner, ..., Close, Long, Seeman. 2026. DOI: 10.1016/j.cell.2026.10.023

from taxonomy to atlas/ leveraging the spatial transcriptomics data to annotate basal ganglia cell types



1 Map spatial cells to species-specific RNA-seq clusters



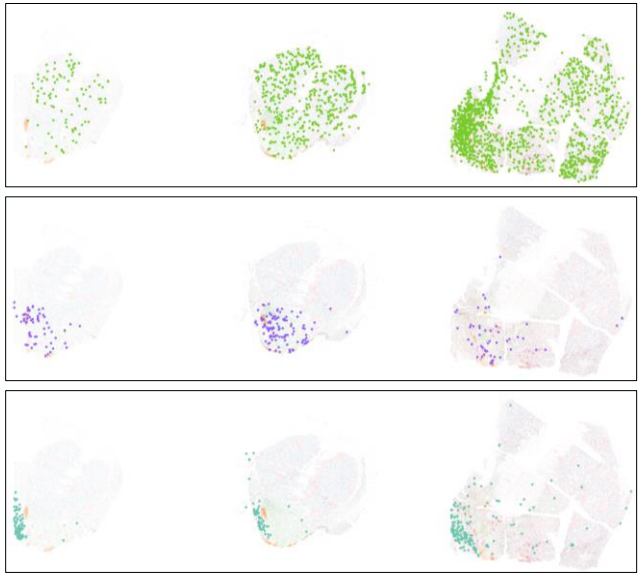
2 Update Group annotations to reflect anatomical localization (e.g. split or merge Groups, adjust cluster membership)

3 Update Group labels on spatial cells

∞ Iterate 2 & 3 to generate final BG Consensus Taxonomy

Regional Specialization → Split Groups

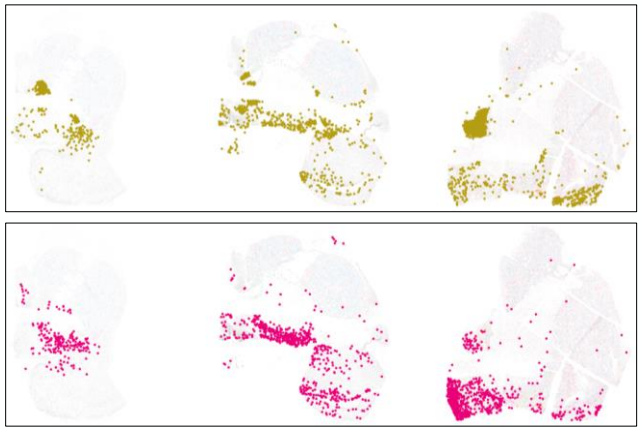
Cholinergic Interneuron Groups



Non-BG cell types → Adjacent Groups

AMY & BST

BF



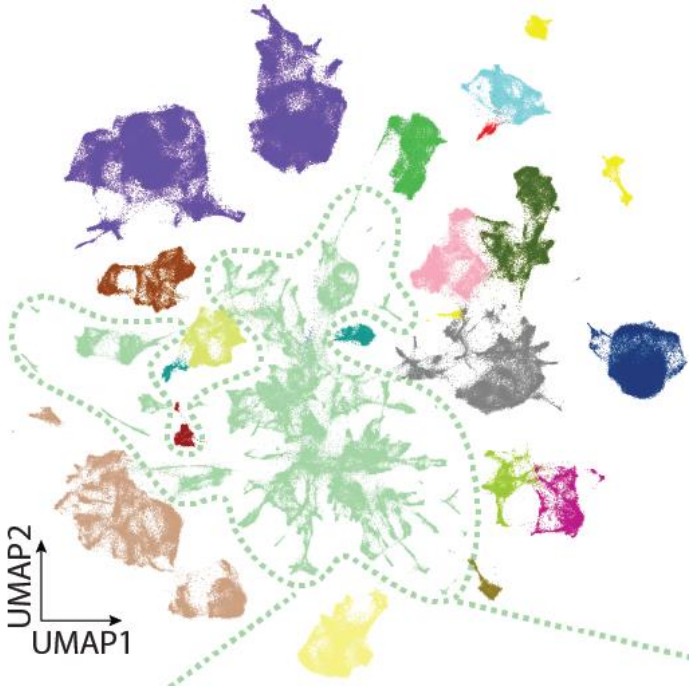
Johansen, Fu, ..., Lein, Bakken. 2026. DOI: 10.1016/j.cell.2026.10.012
Hewitt, Turner, ..., Close, Long, Seeman. 2026. DOI: 10.1016/j.cell.2026.10.023

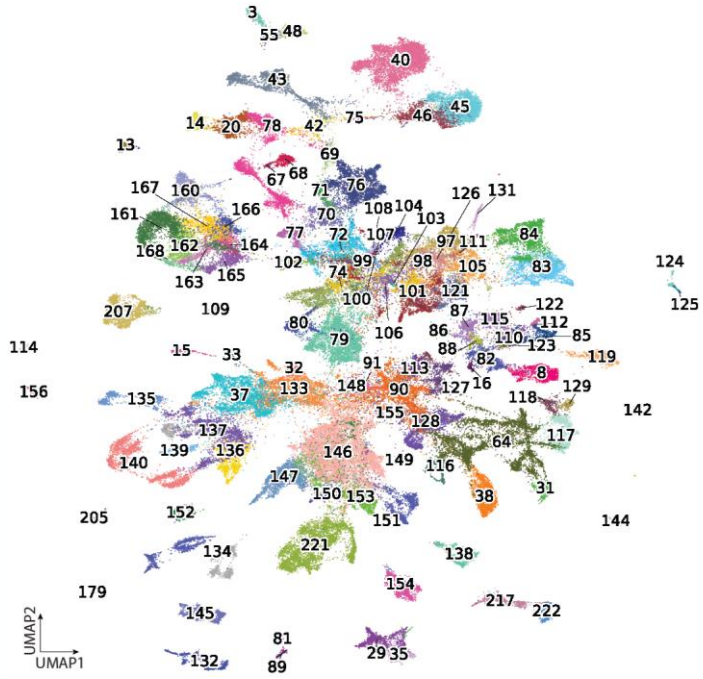
from taxonomy to atlas/ leveraging marmoset spatial transcriptomics data to improve human cell type annotations

Coarse annotation in current whole-brain human reference, especially highly diverse “splatter”

Mapping to marmoset spatial data provided detailed anatomical localization

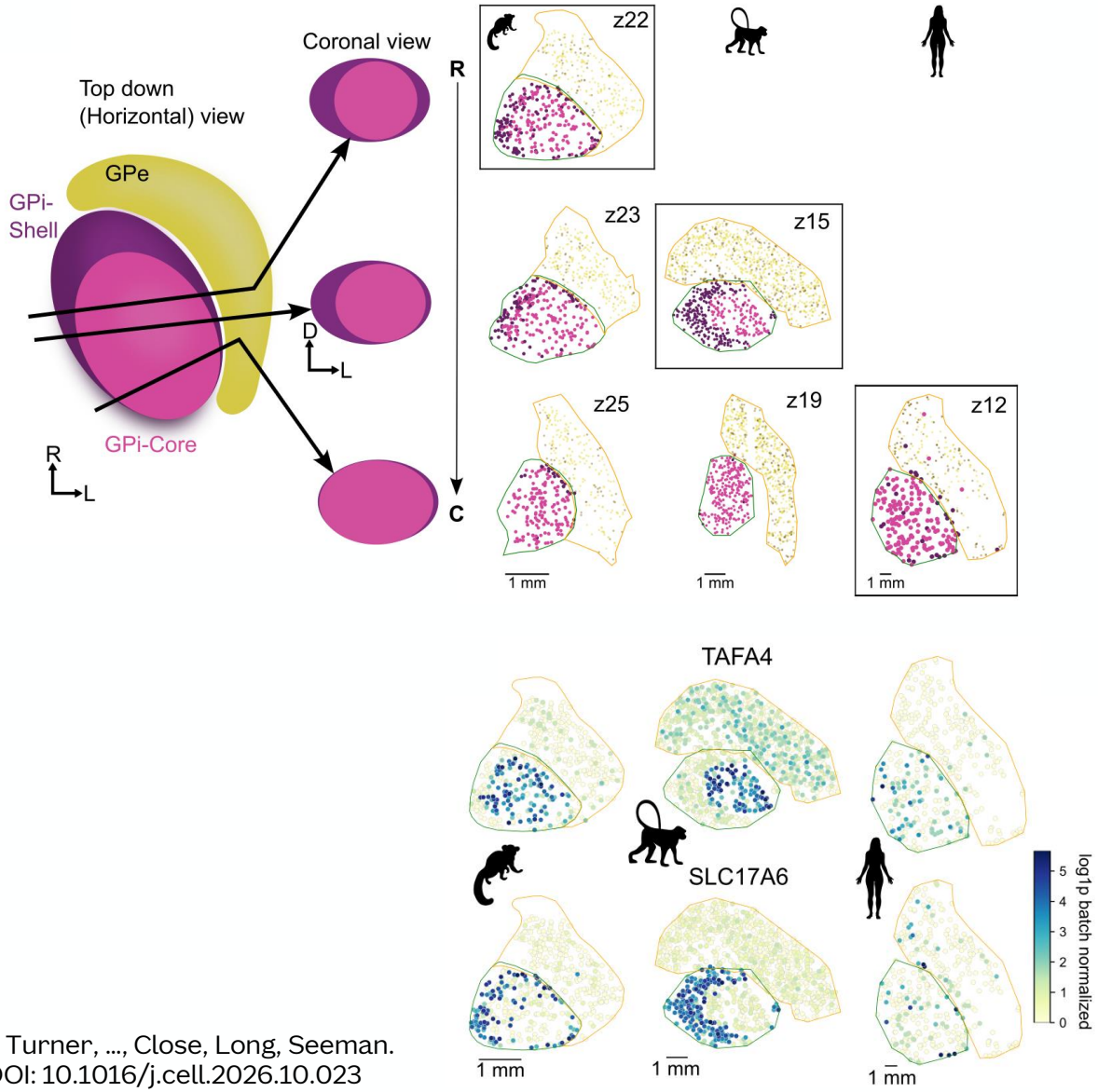
Resulted in 117 anatomically annotated cell type Groups spanning 11 anatomical regions





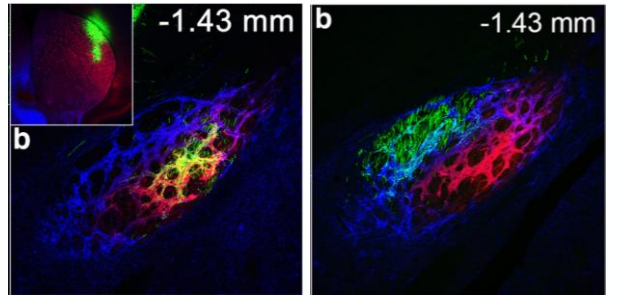
Dan, Turner, DeBerardine, ..., Long, Krienan. 2026. DOI:10.1016/j.cell.2026.10.007

from atlas to scientific findings/ novel core-shell organization of GPi cell types in primates



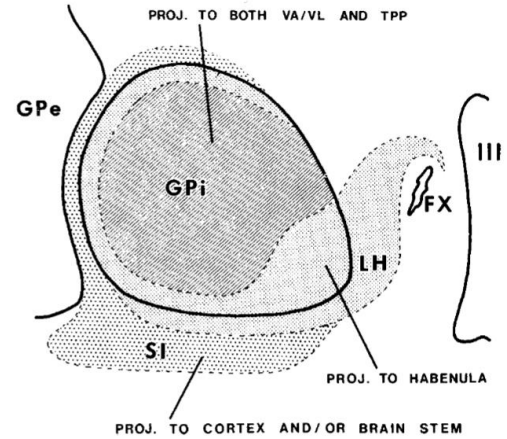
Hewitt, Turner, ..., Close, Long, Seeman. 2026. DOI: 10.1016/j.cell.2026.10.023

GPi Core-Shell cell type organization mirrors known topographical projections



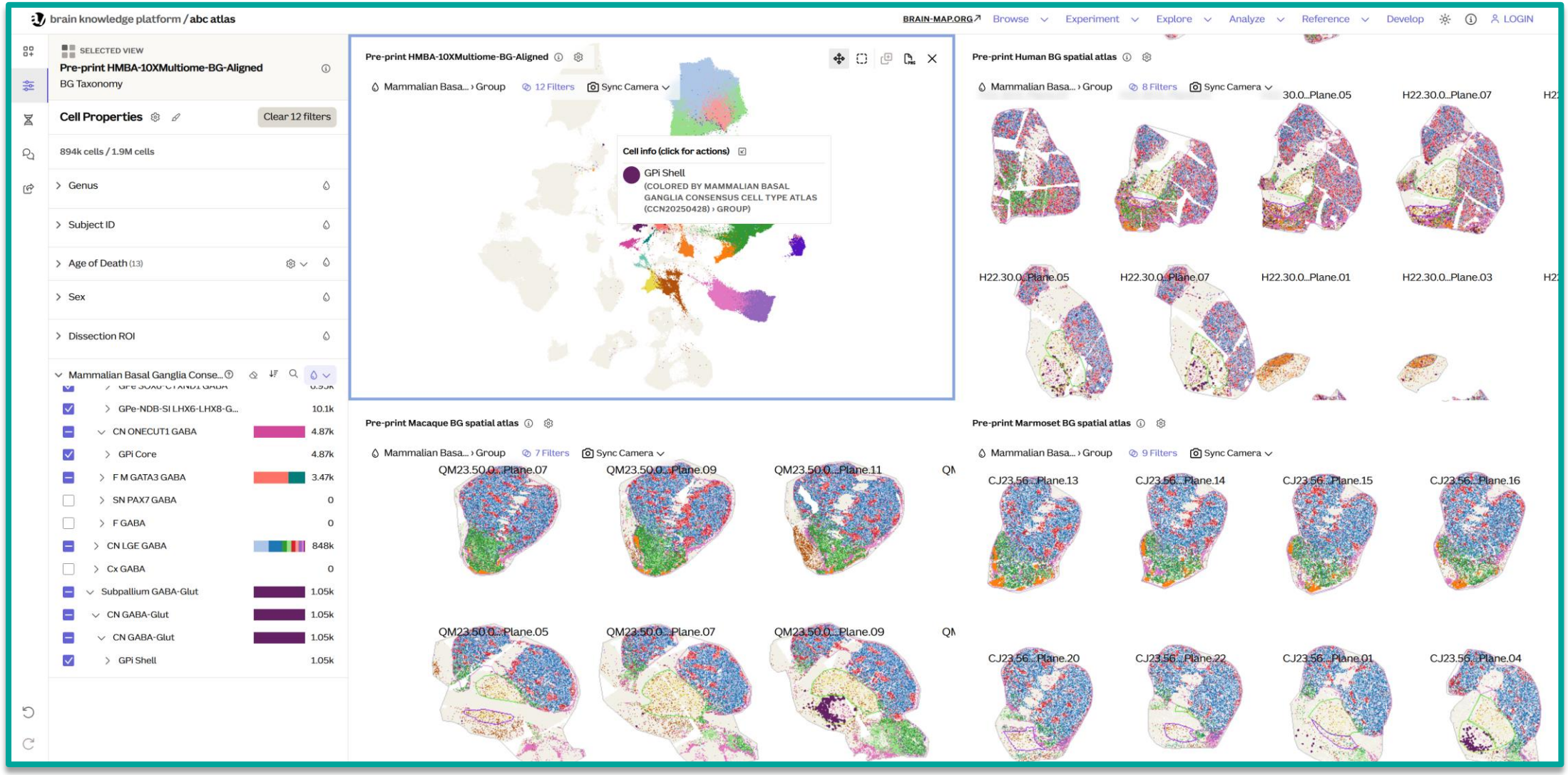
shell | core | topographical projections

Miyamoto & Fukuda, 2022.



(Parent and De Bellefeuille, 1982)

from atlas to open resource / ABC Atlas



https://knowledge.brain-map.org/abcatlas

from basal ganglia to whole brain/ regional community annotation working groups

We're seeking community input on cell type annotation of HMBA's whole-brain single-cell and spatial transcriptomics data, including:

- Neuroanatomical boundaries
- Topographical connectivity patterns
- Specialized cell type(s) with known circuit function but unannotated molecular characteristics
- Spatial cell type patterns that mirror projection or connectivity maps

Regional Working Groups
Basal Ganglia, Basal Forebrain
Cerebellum
Hypothalamus
Medial Temporal Lobe
Midbrain, Medulla, Pons
Neocortex
Non-neuronal Whole-Brain
Spinal Cord
Thalamus

Email Aaron Garcia or me (meghan.turner@alleninstitute.org) if you're interested in joining a working group!

acknowledgements/

HMBA Spatial Transcriptomics
Maddy Hewitt
Meghan Turner
Jennie Close
Brian Long
Stephanie Seeman

Spatial Transcriptomics
Jack Waters
Delissa McMillen
Michael Kunst
Naomi Martin
Christine Kim
Paul Olsen
Shea Ransford
Lindsey Ching
Jacob Quon

Program Management
Susan Sunkin
Lauren Kruse
Chelsea Pagan

Histology
Jeanelle Ariza Torres
Alice Pom
Augustin Ruiz
Angela Ayala
Stuard Barta
Alana Oyama
Alvin Huang
Maddy Bixby
Kimberly Smith

HMBA Multi-omics
Nelson Johansen
Yuanyuan Fu
Mathew Schmitz
Inkar Karpen
Trygve Bakken
Aaron D. Garcia
Rebecca Hodge

Leadership
Hongkui Zeng
Ed S. Lein

Tissue Procurement/ Processing (Marmoset)
(Guoping Feng Lab, MIT)
Kirsten Levandowski
Qiangge Zhang

Marmoset Multi-omics
(Krienen Lab, Princeton Univ.)
Fenna Krienen
Shu (Daisy) Dan
Mike DeBerardine
Lakme Caceres
Jess Schembri
Victor Nieto-Caballero
Christopher Cardenas
Ning Wang

Data & Technology
Mike Huang
Scott Daniel
Chris Morrison
Ashwin Bhandiwad
Raymond Sanchez
Elysha Fiabane
Tyler Mollenkopf
... and many more!

Tissue Procurement/ Processing
Nick Dee
Tamara Casper
Eric Thorland
Atlas Jordan
Christopher Nyweide
Una Noski
Rachel McCue
Julie Nyhus

RNAseq
Kimberly Smith
Amy Torkelson
Darren Bertagnoli
Thanh Cardenas
Rebecca Ferrer
Junitta Guzman
Katelyn James
Chistine Rimorin
Michael Tieu

Microscopy Core
Jack Waters
Sam Hastings
Mike Taormina
Zoe Juneau
Tulip O'Neill
Nadezhda Dotson
Kiet Ngo

FACS
Rebecca Hodge
Nadiya Shapovalova
Danielle Jones
Nick Peña

Bioinformatics Core
Anish Chakka
Jeff Goldy
Rushil Chakrabarty
Dana Rocha
Beagan Nguy
Kimberly Smith

Funding
National Institute of Mental Health
Award #UM1MH130981-01

questions?

allen institute/

alleninstitute/
thank you/

agenda/morning

- 9:00-9:30 | check-in / breakfast
- 9:30-9:40 | welcome, logistics, and overview of allen institute
- 9:40-10:05 | cell typing at allen institute: taxonomies, data sets, and use cases
- 10:05-10:30 | human and mammalian brain atlas (HMBA)
- 10:30-10:40 | break
- 10:40-11:05 | localizing cell types with spatial transcriptomics
- **11:05-11:30 | cell-type targeted enhancer development**
- 11:30-11:55 | Seattle Alzheimer's Disease Brain Cell Atlas in the brain health accelerator
- 11:55-12:05 | Overview of brain knowledge platform (BKP) tools & resources
- 12:05-12:10 | Preview of afternoon
- 12:10-1:10 | lunch

enhancer design for cell type-specific targeting/ examples from the basal ganglia and beyond

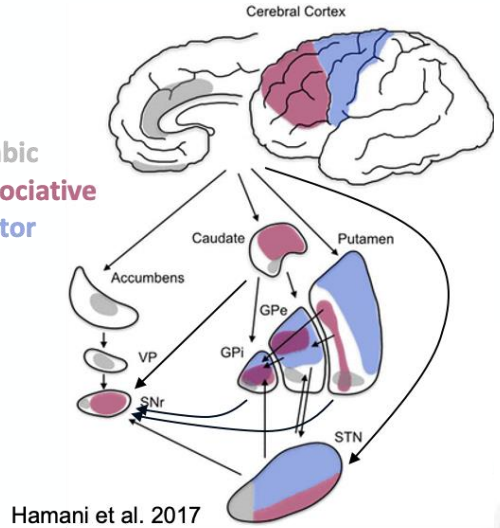
Morgan Wirthlin, Scientist II

Will Laird, Research Associate II

allen institute/

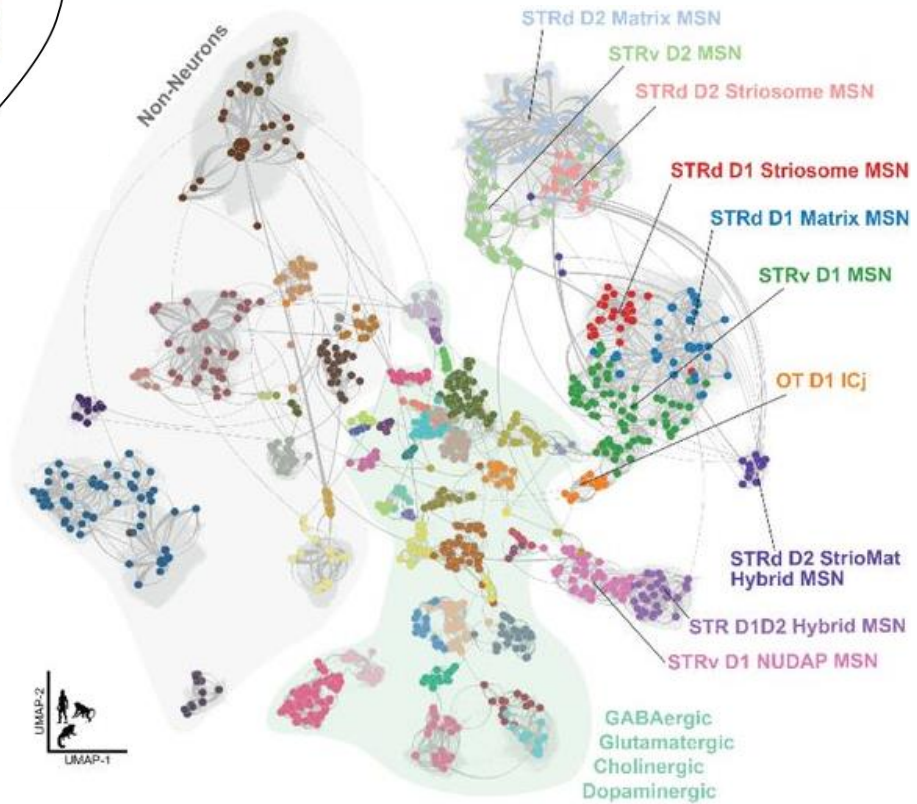
allen institute

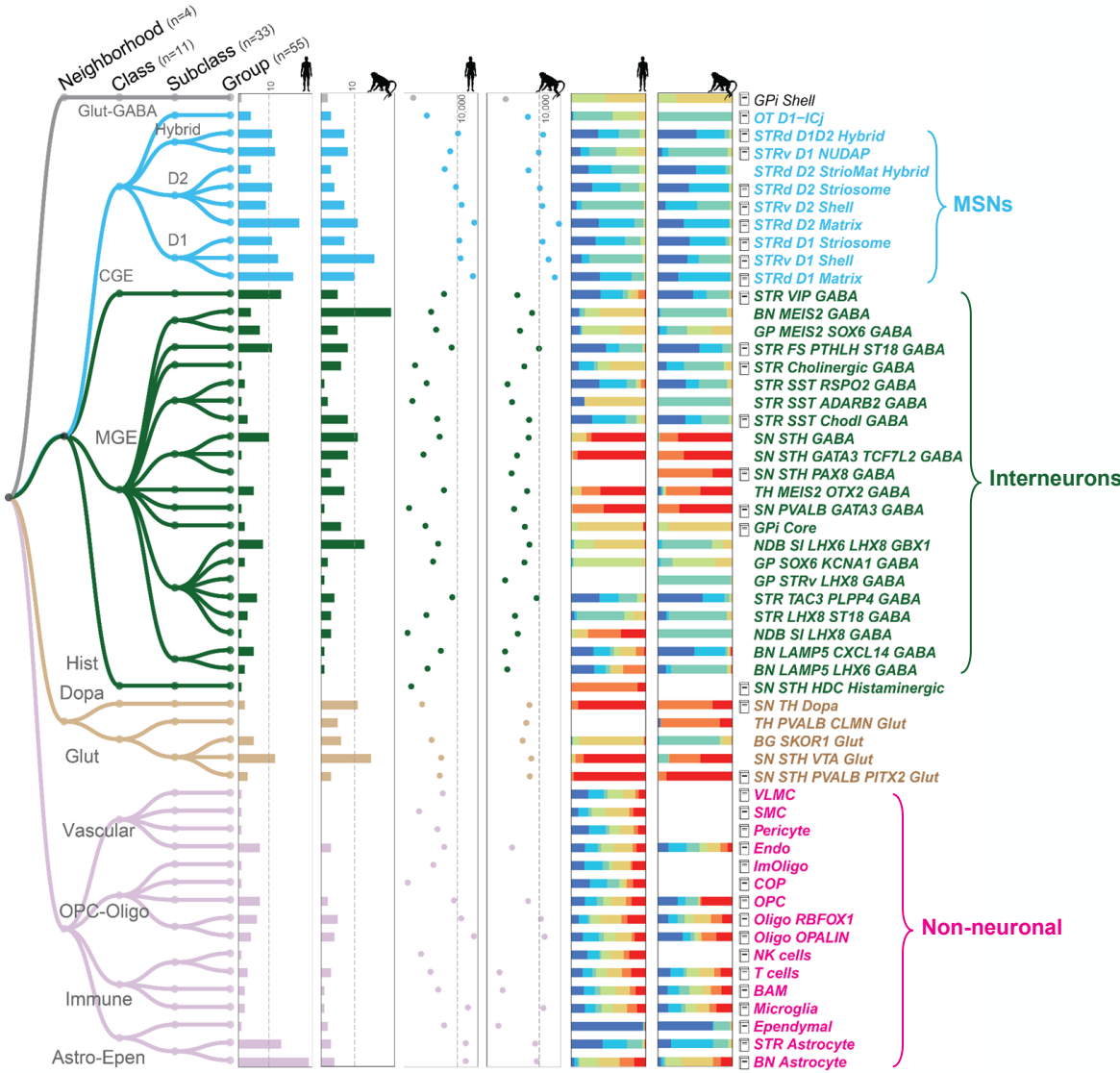
human and NHP consensus basal ganglia taxonomy/



Hamani et al. 2017

Cross-species integration



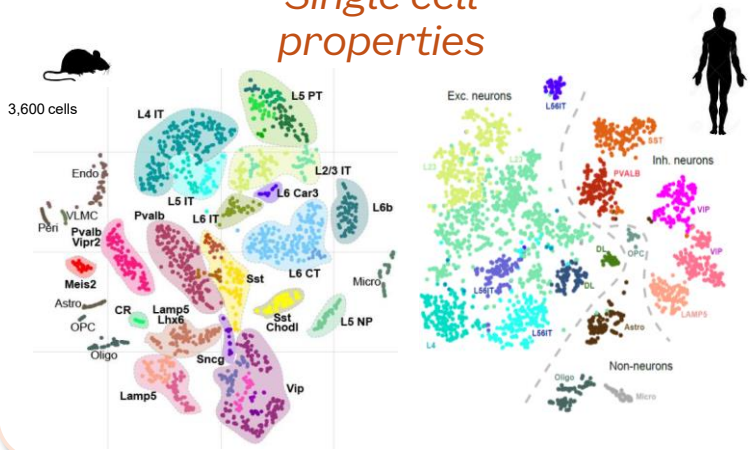


Johansen, Fu, et al. Cell (in press)

cell types taxonomies –> genetic tools/

Define cell types

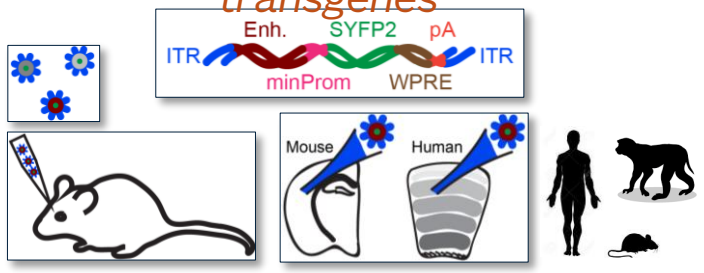
Single cell properties



Enable experimental access

> Monitoring
> Perturbing
> Treating

Recombinant enhancer AAVs and transgenes



Publicly available:

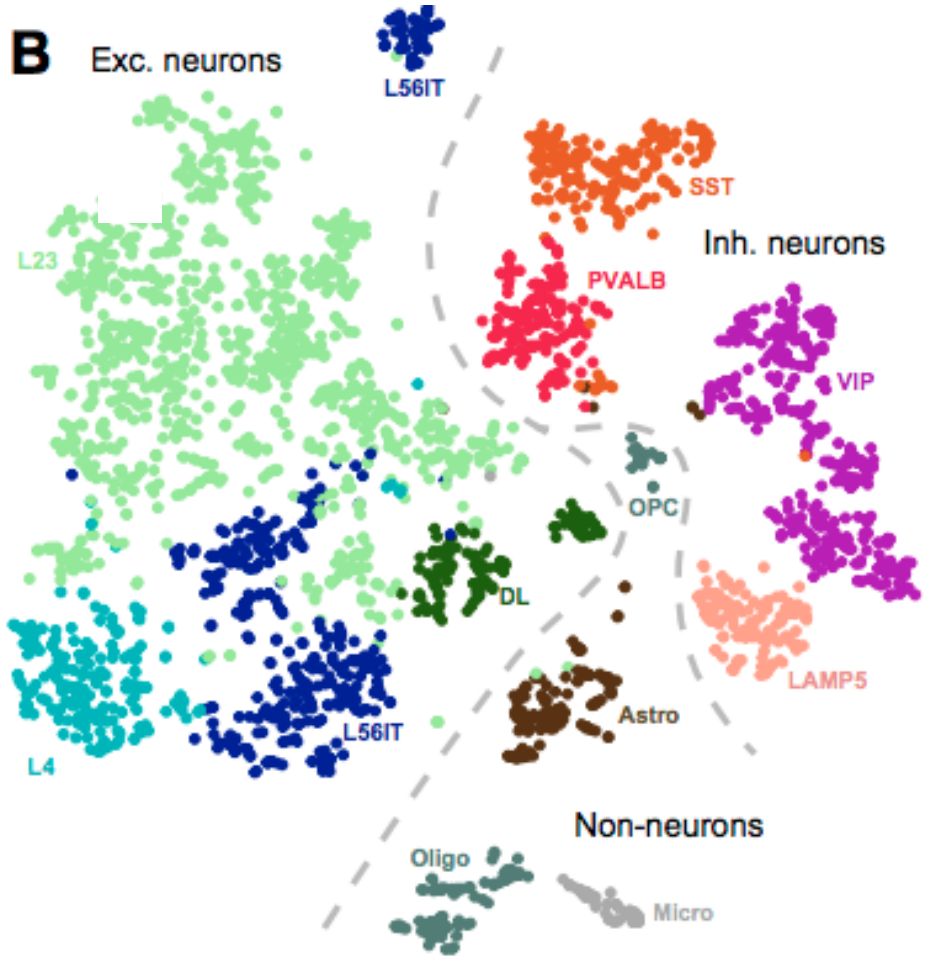
- **Datasets**
- **Taxonomies** (cell-type definition)
- **Reagents**
- **Recipes** for targeting cell classes/subclasses/types in the mouse (and primate) brain

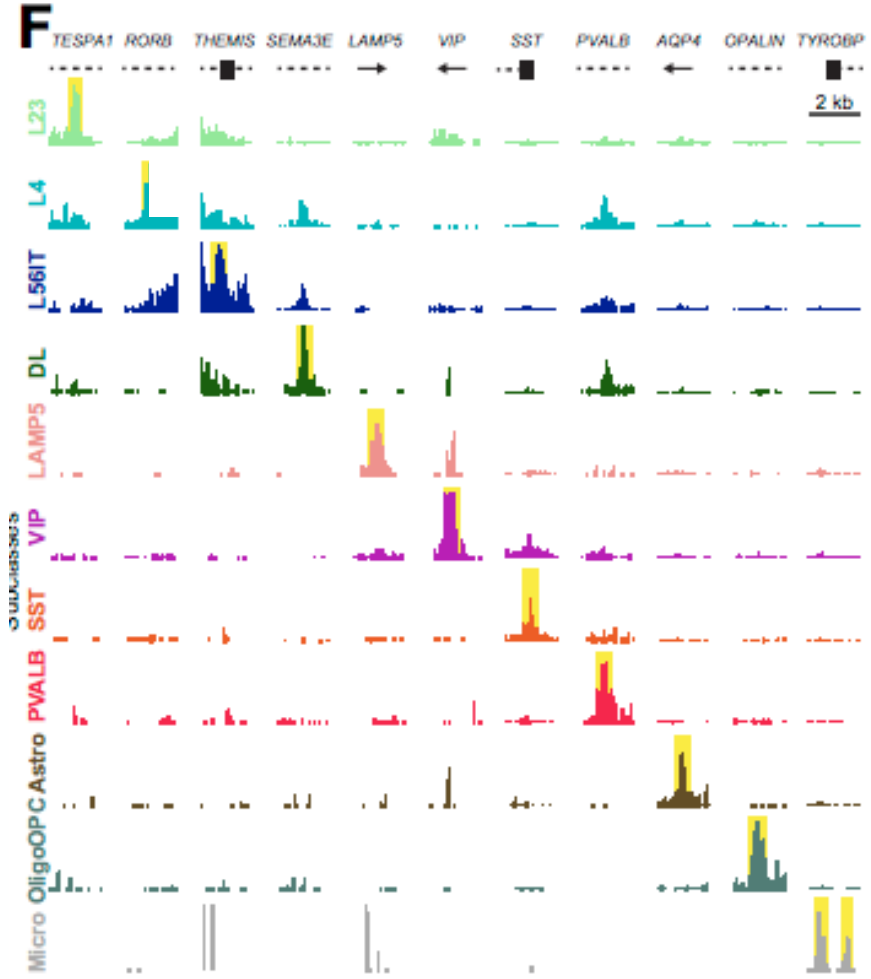
Intersectional approaches:

Transgene X +
Virus Y (titer + delivery route) →
Access to cell type Z

enhancer selection for genetic tools/

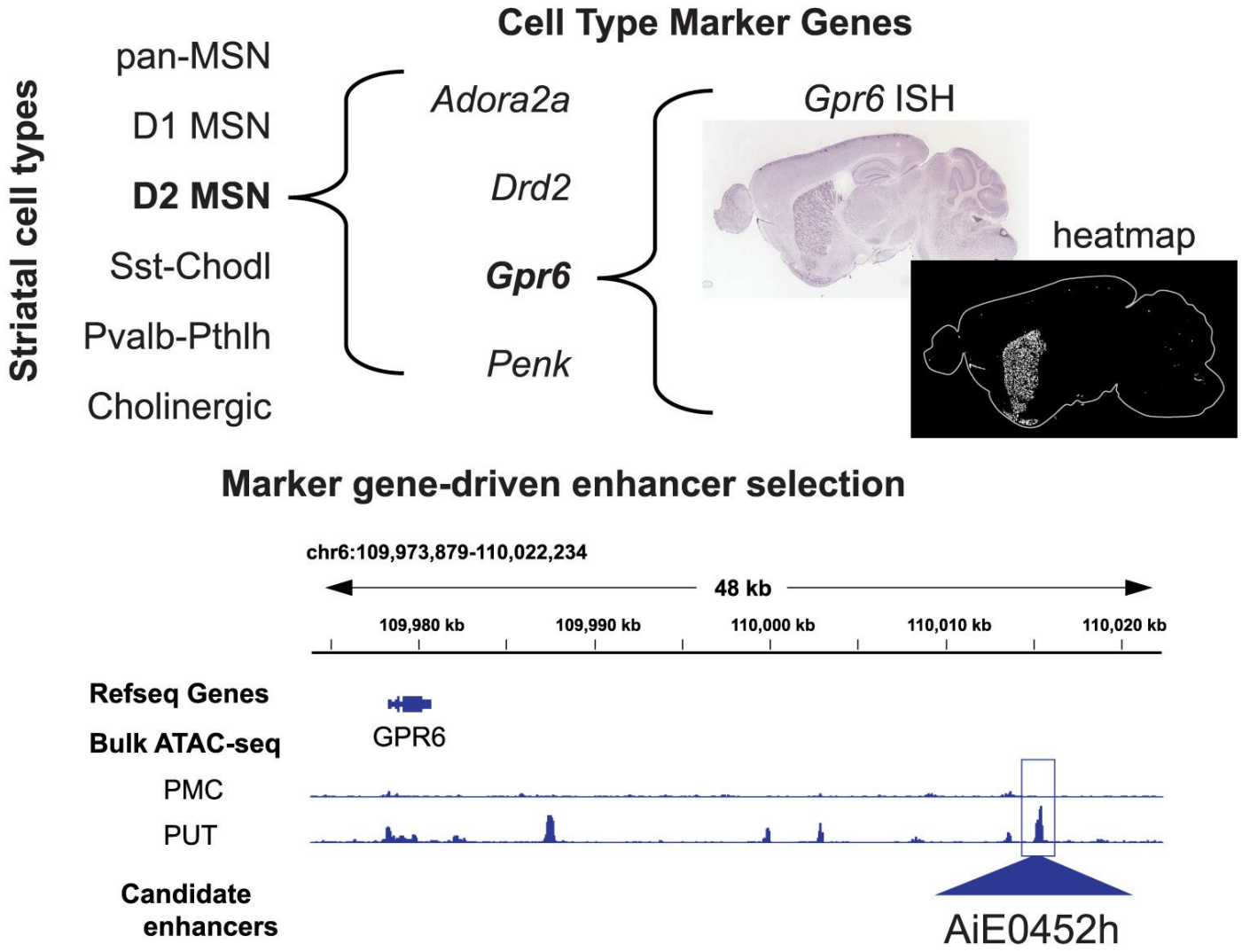
Goal: Identify cell type-specific regions of open chromatin using pseudo-bulk scATAC-seq





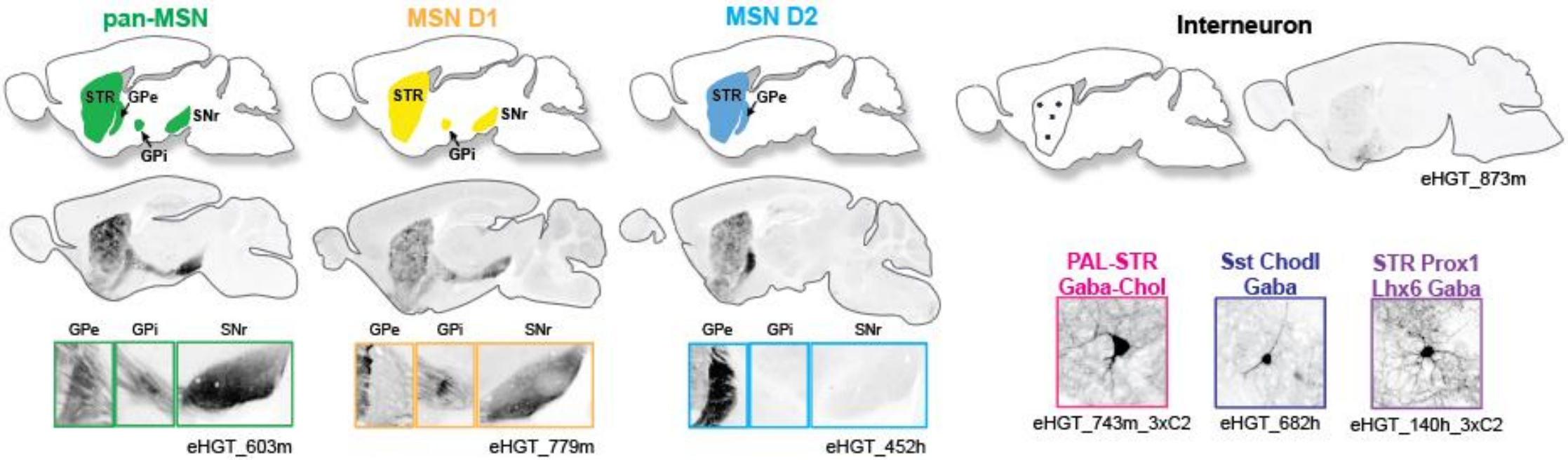
Mich, et al. (2021)

enhancer selection for genetic tools/marker-based approach



Hunker, Wirthlin, et al. (2025) Neuron

first collection of BG-relevant tools/published May 2025

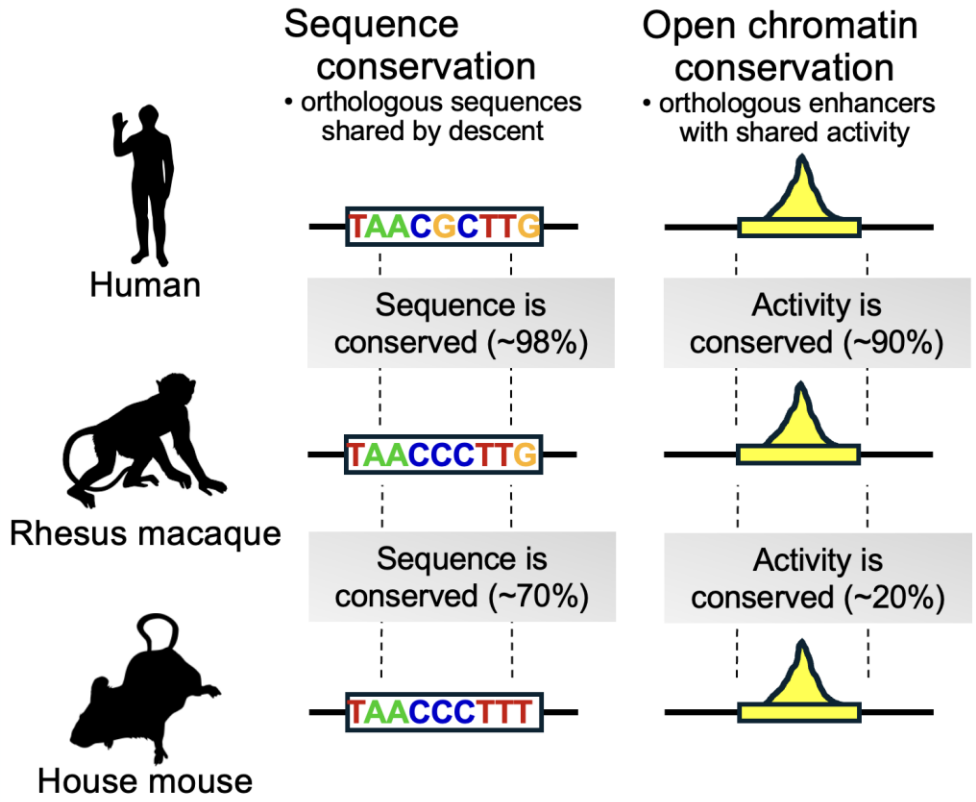


- Best-in-class tools for canonical MSNs (aka SPNs) and striatal interneurons
- Enhancer peaks selected proximal to marker genes
- First dataset in Genetic Tools Atlas

Hunker, Wirthlin, et al. (2025) Neuron

enhancer selection for genetic tools/evolutionary approach

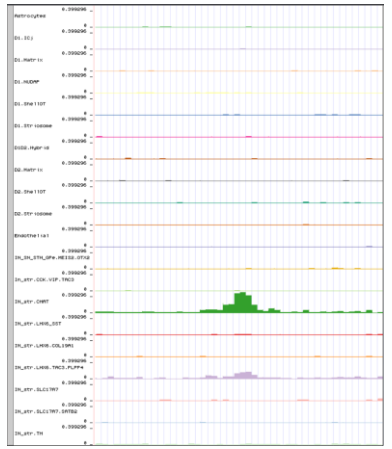
- We developed a new **C**ross-species **E**nhancer **R**anking **P**ipeline (**CERP**)
- CERP preferentially selects for strong, shared peaks between macaque and mouse where open chromatin (snATAC) activity is conserved in homologous cell types

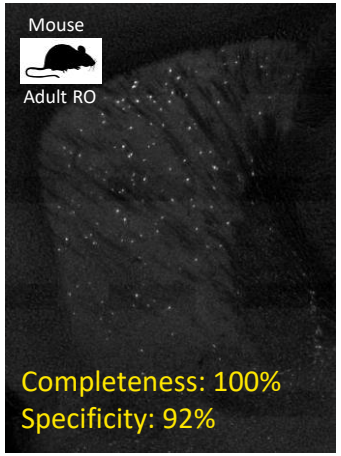


CERP IN-CHAT peak, rank #62

VGT-Validated Enhancer A

IN-CHAT snATAC

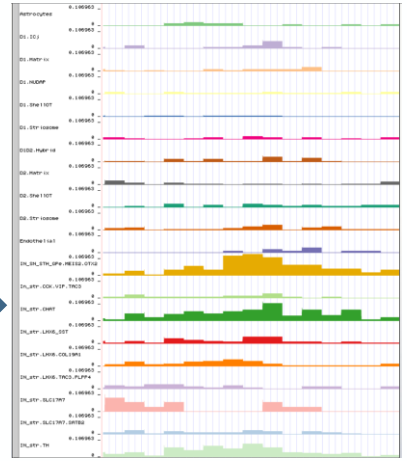


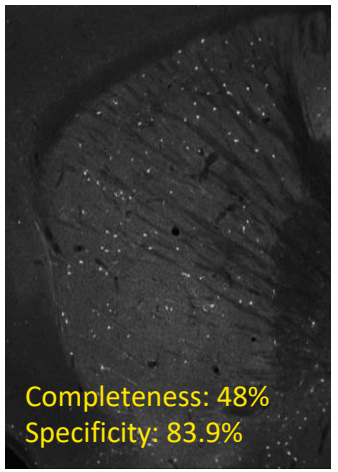


Peak did not pass CERP filtering

VGT-Validated Enhancer B

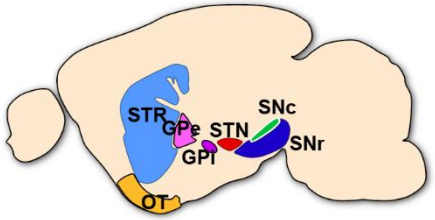
IN-CHAT snATAC





expansion of BG genetic tools/beyond classical MSNs and striatal interneurons

Common Literature Name(s)	AiT21 Taxonomy	HMBA Consensus Names	Region
Pvalb-Pthlh, fast-spiking interneurons	STR Prox1 Lhx6 Gaba	STR_FS_PTHLH-PVALB_GABA	STR
	STR Lhx8 Gaba	GPe-NDB-SI_LHX6-LHX8-GBX1_GABA	PAL
Sst-Chodl, low-threshold spiking interneurons	Sst Chodl Gaba	STR_SST-CHODL_GABA	STR
	NDB-SI-MA-STRv Lhx8 Gaba	GPe_SOX6-CTXND1_GABA	PAL
Cholinergic interneurons	PAL-STR Gaba-Chol	STR_Cholinergic_GABA	STR
Tac3 interneurons	STR Lhx8 GABA	STR_TAC3-PLPP4_GABA	STR
	GPe-SI Sox6 Cyp26b1 Gaba	GPe_MEIS2-SOX6_GABA	PAL
GPe Cholinergic	PAL-STR Gaba-Chol	GPin-BF Cholinergic GABA	PAL
D1 Islands of Calleja	OT D3 Folh1 Gaba	OT_D1_ICj	STR
D1 MSN	STR D1 Gaba	STR_D1_MSN	STR
D1 MSN NAc Shell	STR D1 Gaba	STRv_D1_MSN	STR
D2 MSN	STR D2 Gaba	STR_D2_MSN	STR
D2 MSN NAc Shell	STR D2 Gaba	STRv_D2_MSN	STR
D2 MSN Hybrid	STR D2 Gaba	STRd_D2_StrioMat_Hybrid_MSN	STR
pan-MSN	STR D1 Gaba & STR D2 Gaba	STR_MSN	STR
Striosome MSN	STR D1 Gaba & STR D2 Gaba	STR_Striosome_MSN	STR
Matrix MSN	STR D1 Gaba & STR D2 Gaba	STR_Matrix_MSN	STR
D1/D2 MSN Hybrid	STR D1 Sema5a Gaba	STR_D1D2_Hybrid_MSN	STR
D1 NUDAP	STR-PAL Chst9 Gaba	STRv_D1_NUDAP_MSN	STR
GPi Core	ZI Pax6 Gaba	GPi_Core	PAL
GPi Shell	GPi Tbr1 Cngb3 Gaba-Glut	GPi_Shell	STR
STN Glutamatergic	STN-PSTN Pitx2 Glut	STH_PVALB-PITX2_Glut	HY
Posterior SNr GABA	SNr-VTA Pax5 Npas1 Gaba	SN_GATA3-PAX8_GABA	MB
Anterior SNr GABA	SNr Six3 Gaba	SN_GATA3-PVALB_GABA	MB
Dopaminergic	SNc-VTA-Ramb Foxa1 Dopa	M_Dopa	MB



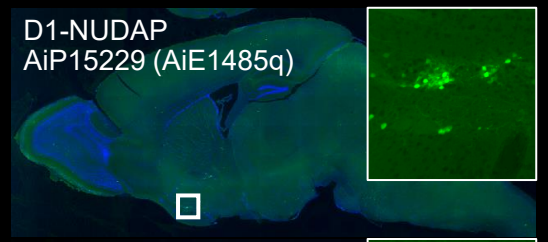
Most enhancers in this new collection were selected by CERP and from Allen macaque 10x multiome data

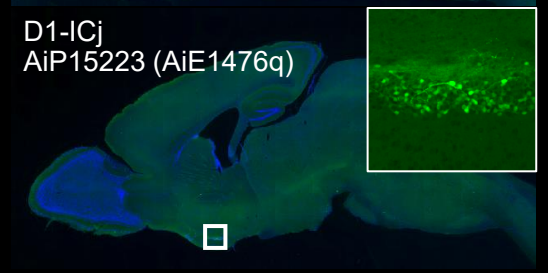
Others were selected proximal to known marker genes using various sc/snATAC-seq datasets

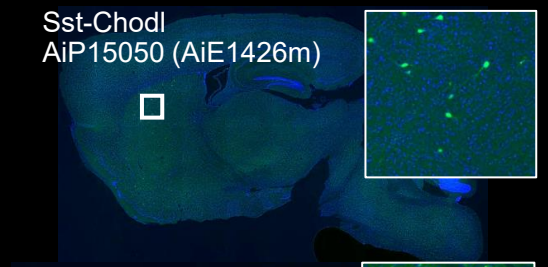
506 tested tools, 72 with exemplary specificity, across 17 cell types

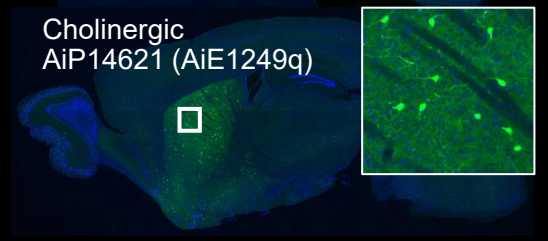
Wirthlin, Hunker, et al. (2026) bioRxiv

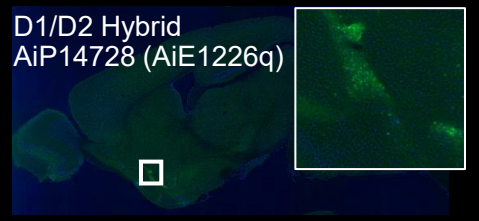
expansion of BG genetic tools/beyond classical MSNs and striatal interneurons

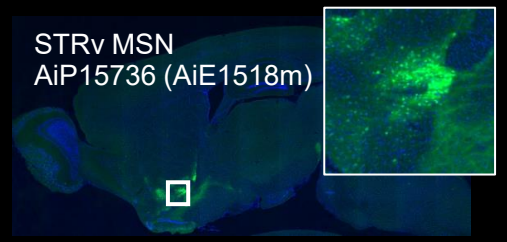


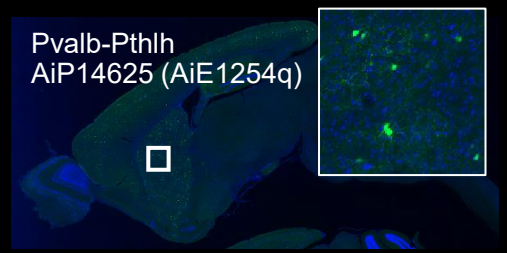


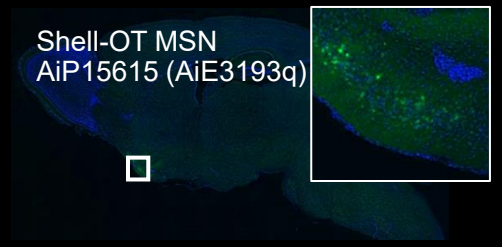


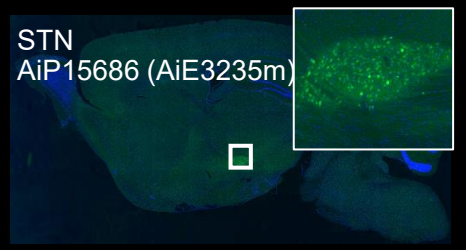


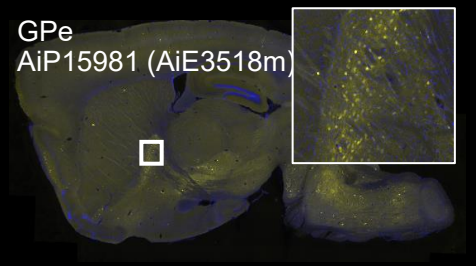


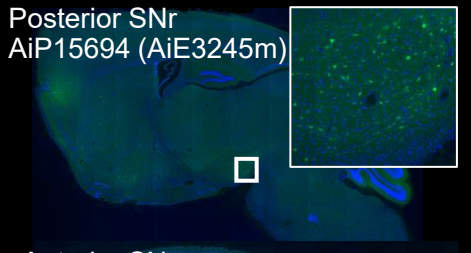


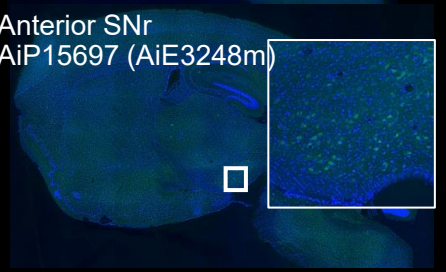


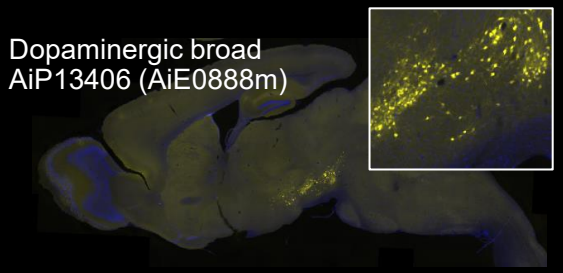


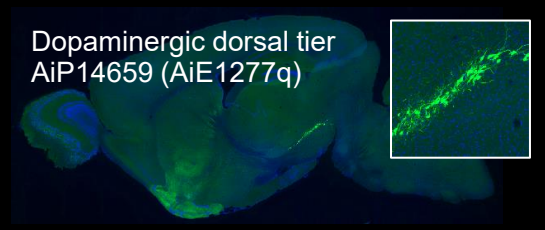


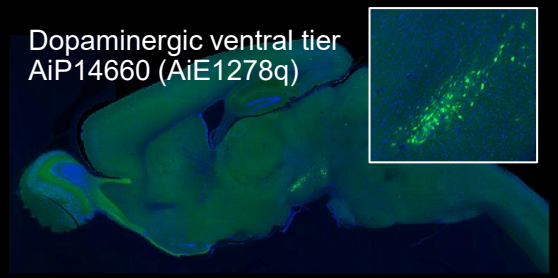




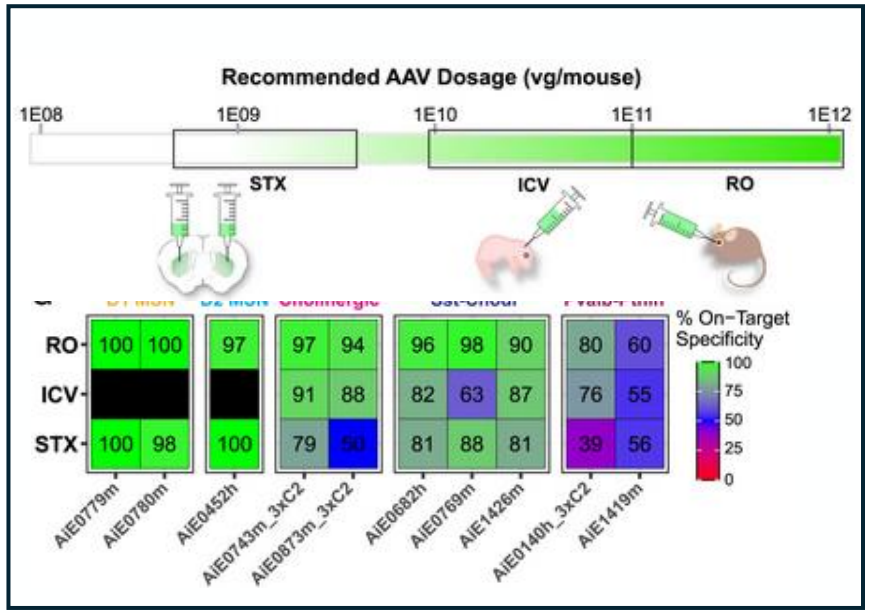


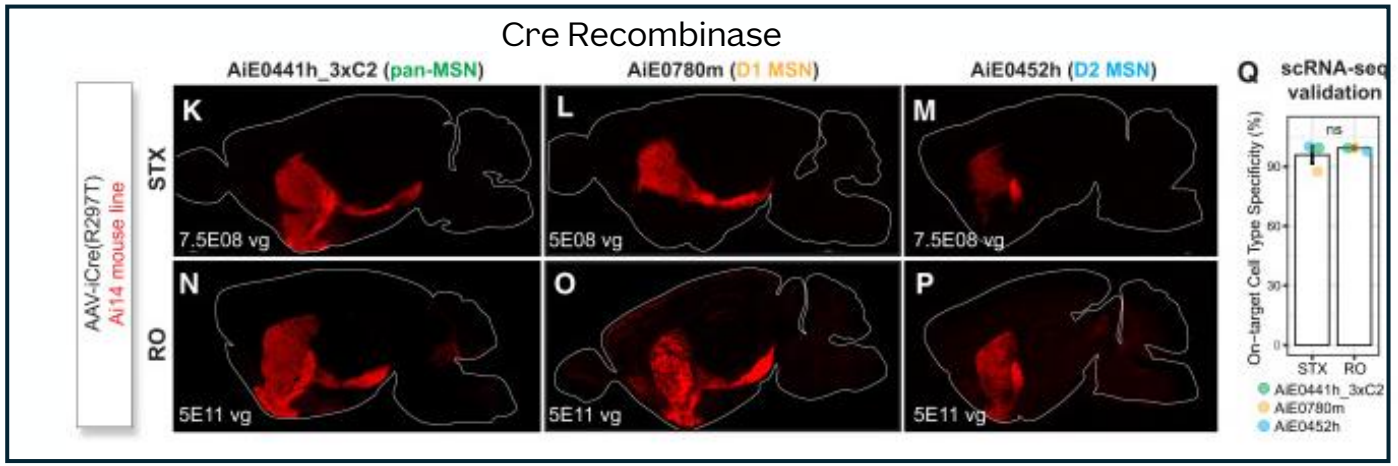


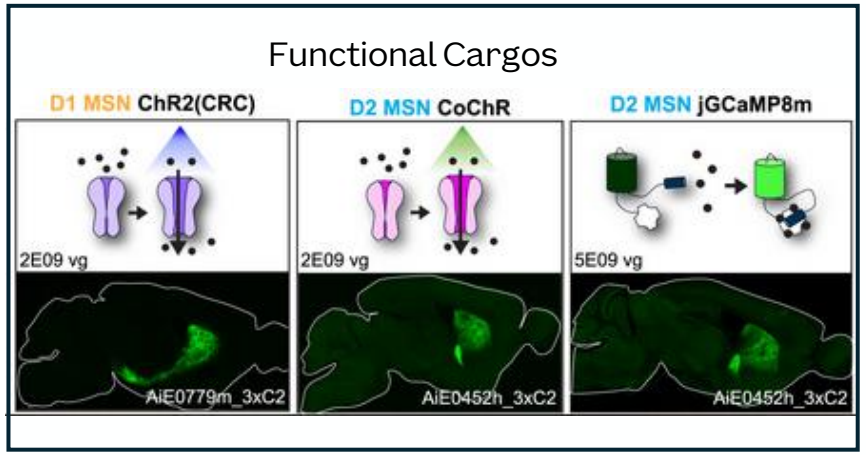


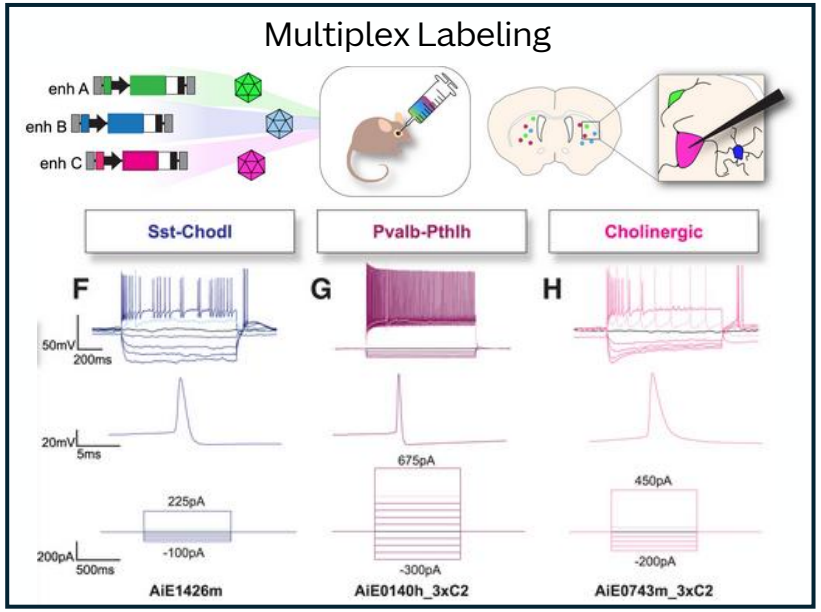


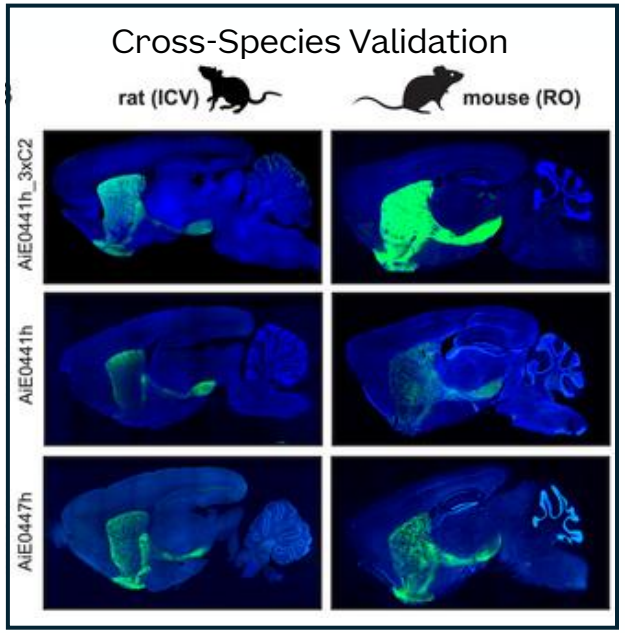
striatal tools NeuroResource/broad utility of viral genetic tools











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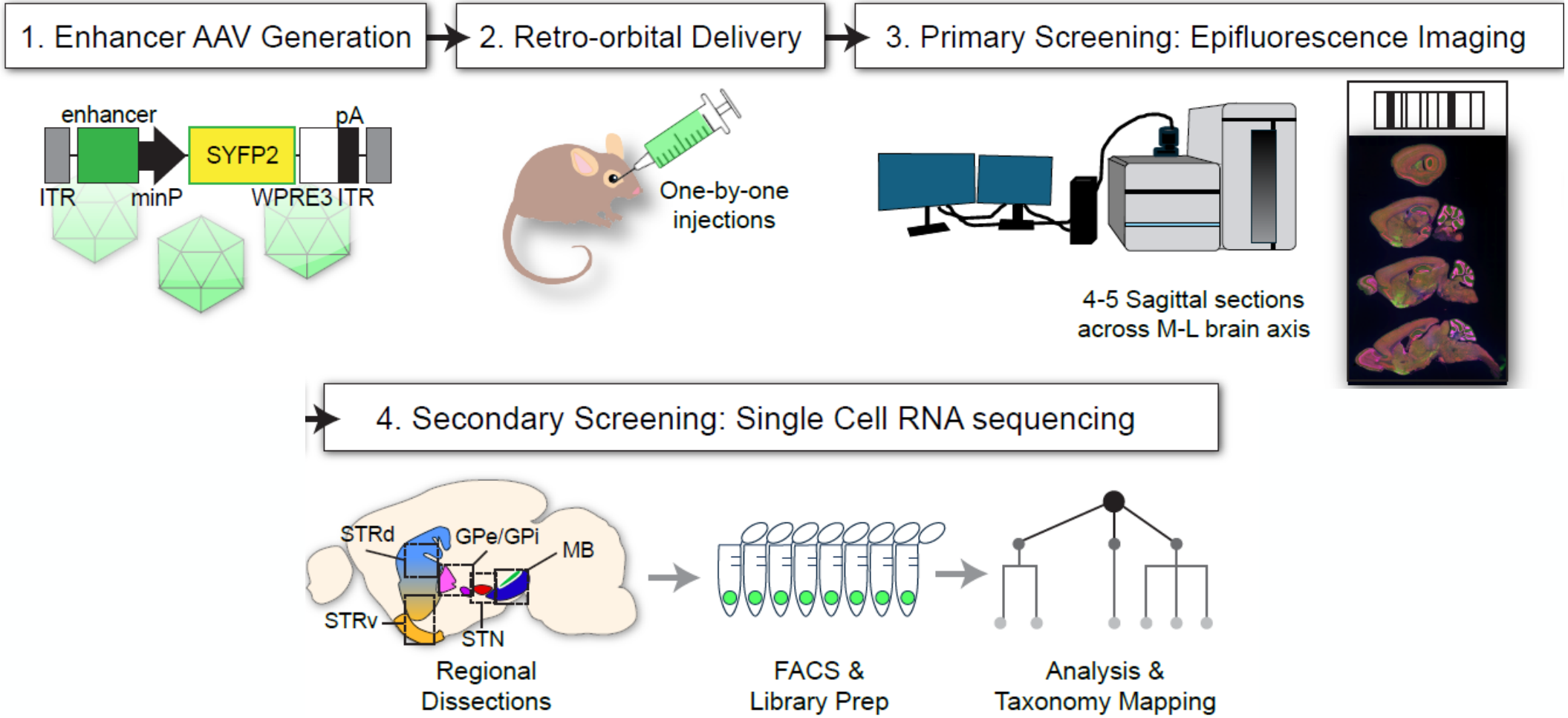
experimental methodological overview/



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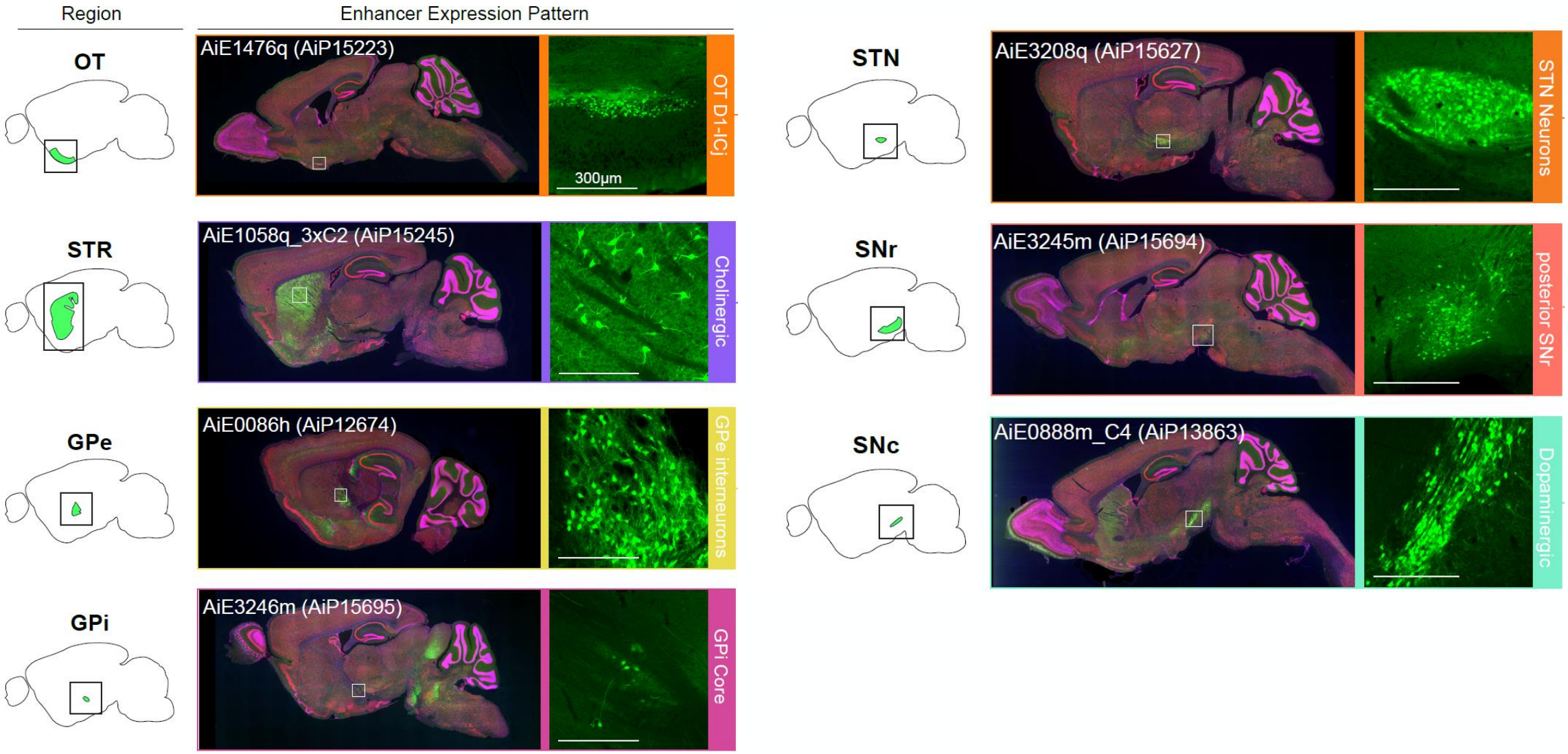
‘best-in-class’ tools highly specific to BG cell types/



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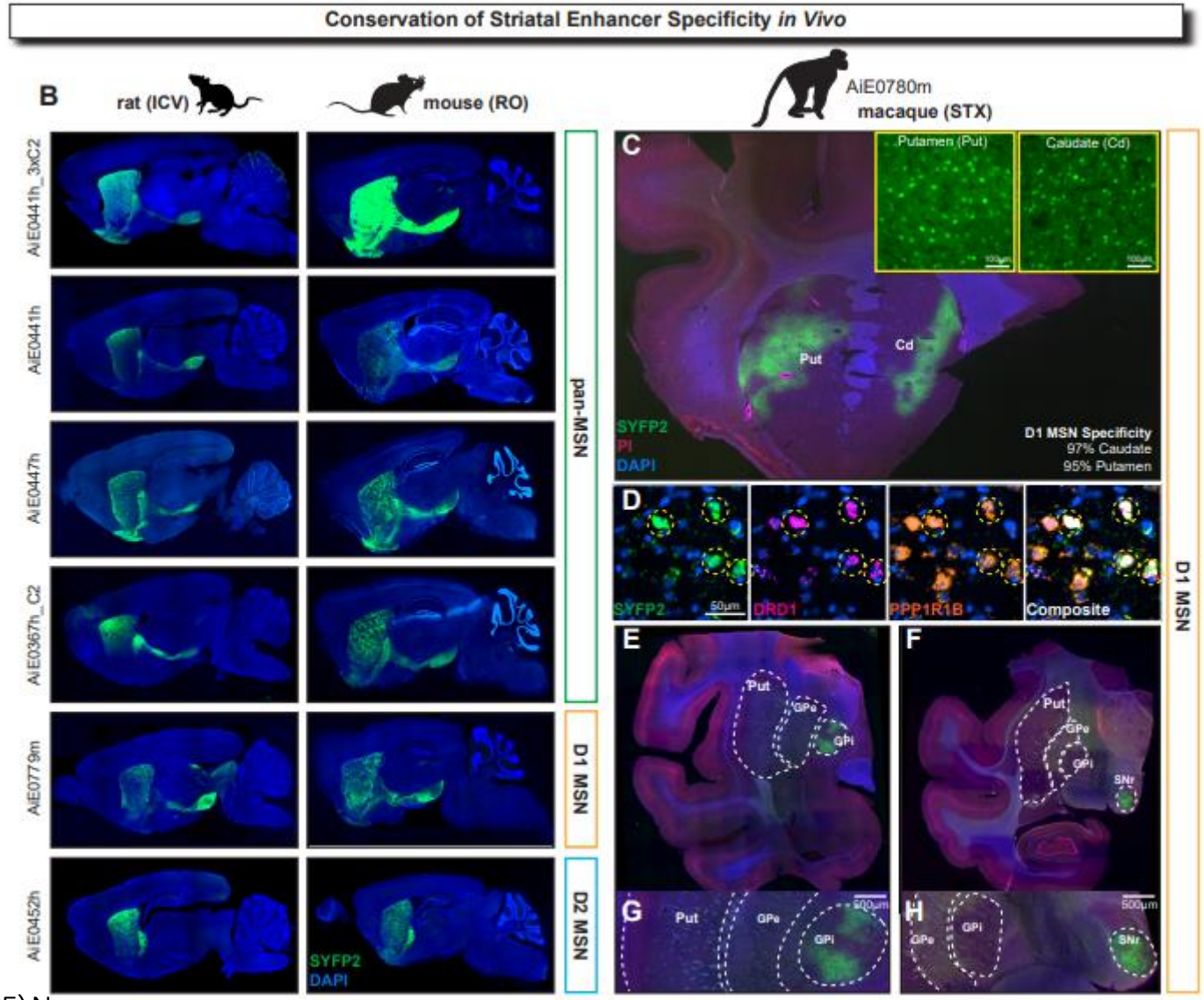
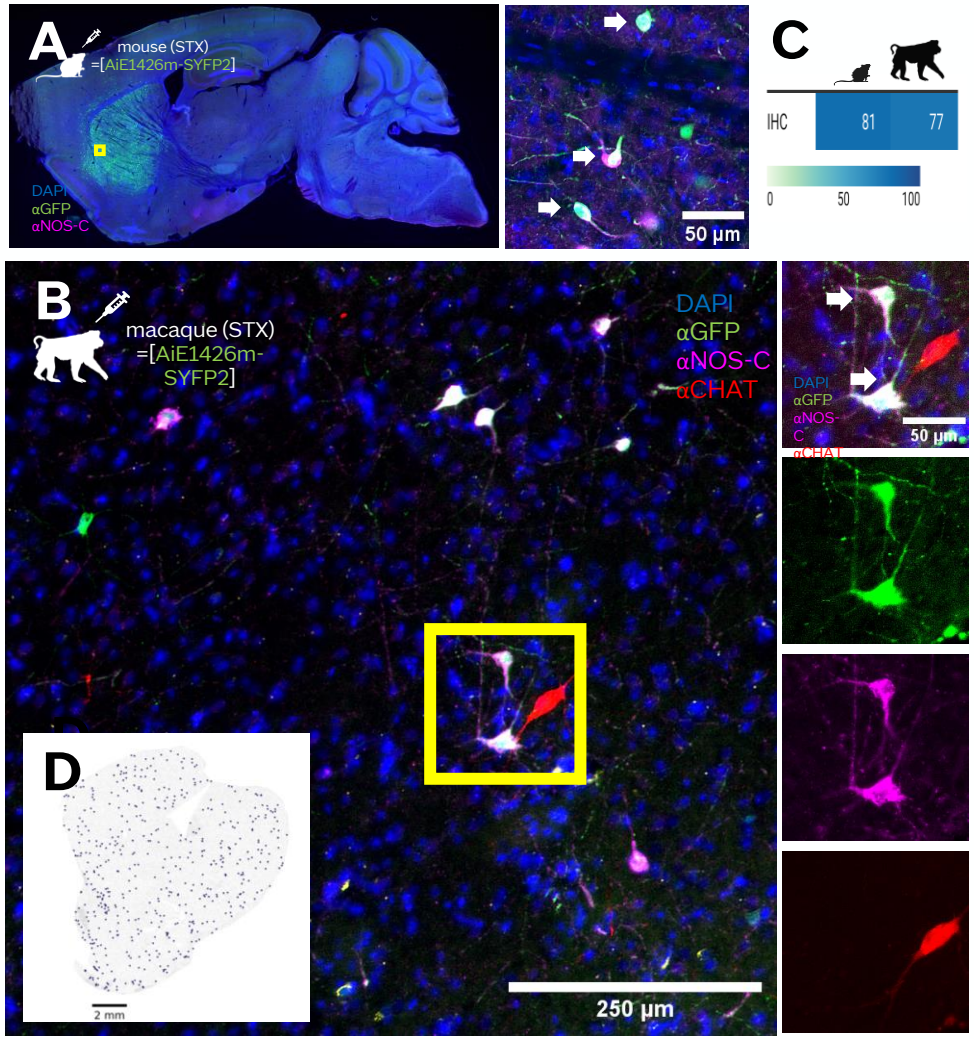
cross-species validation of striatal MSN enhancers/



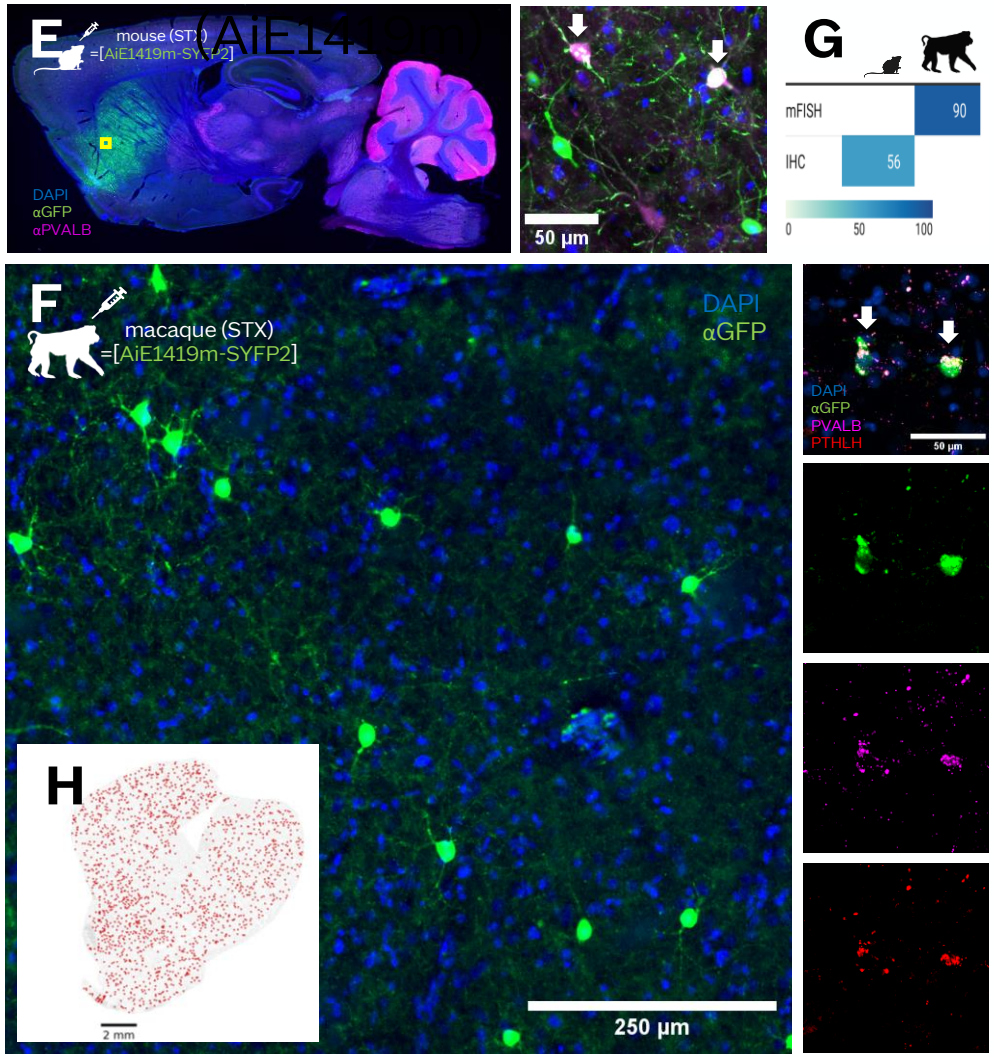
Hunker, Wirthlin, et al. (2025) Neuron

cross-species validation of striatal interneuron enhancers/

SST-CHODL (AiE1426m)



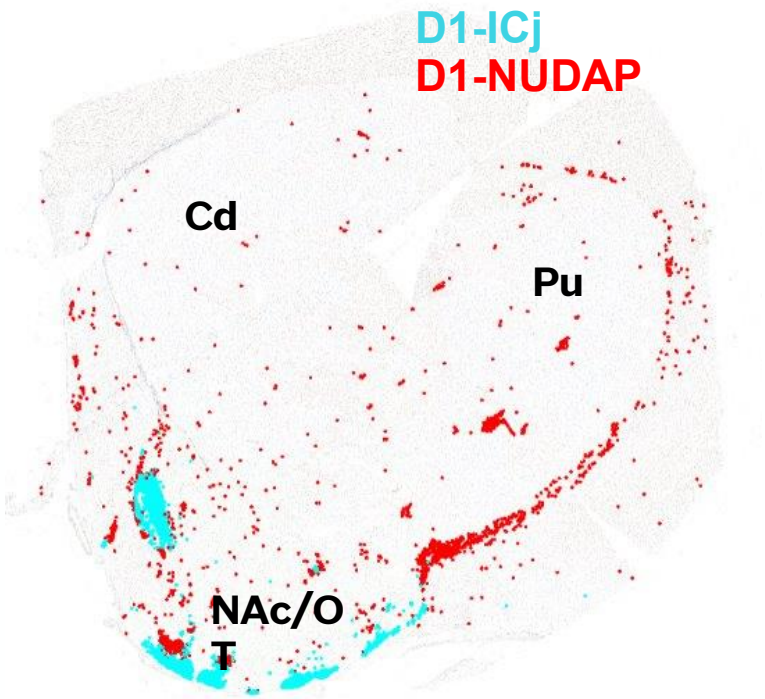
PVALB-PTHLH

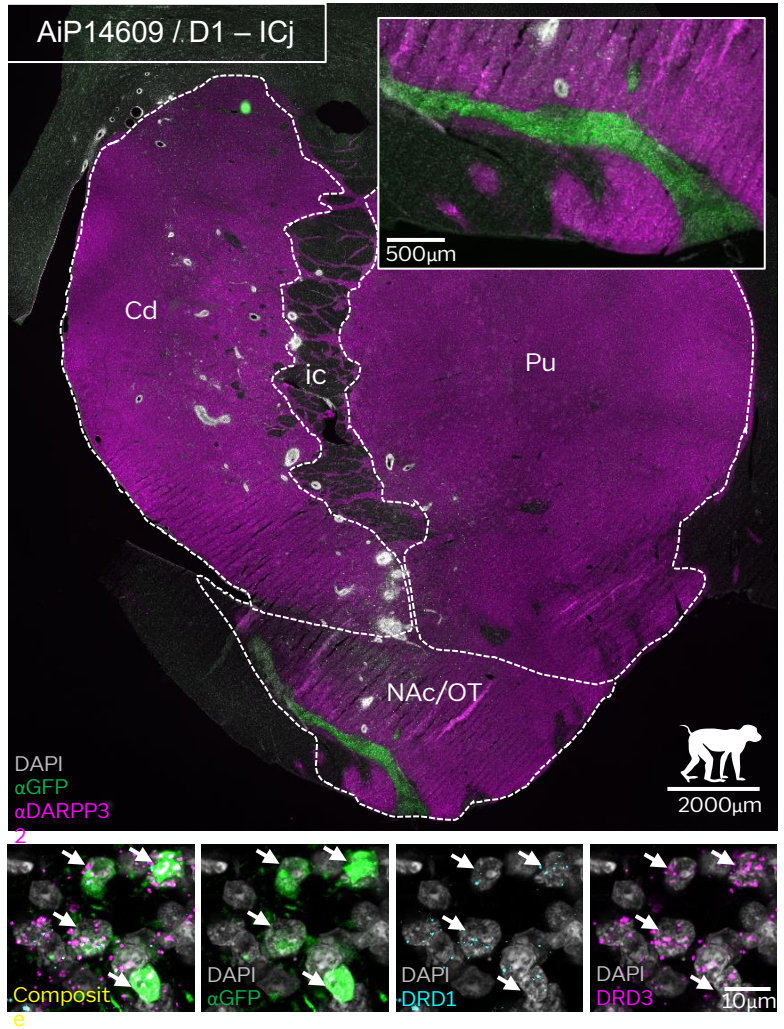


Nathaly Lerma, Will Laird, Jonathan Ting, Naz Taskin, Greg Horwitz, Yoshiko Kojima

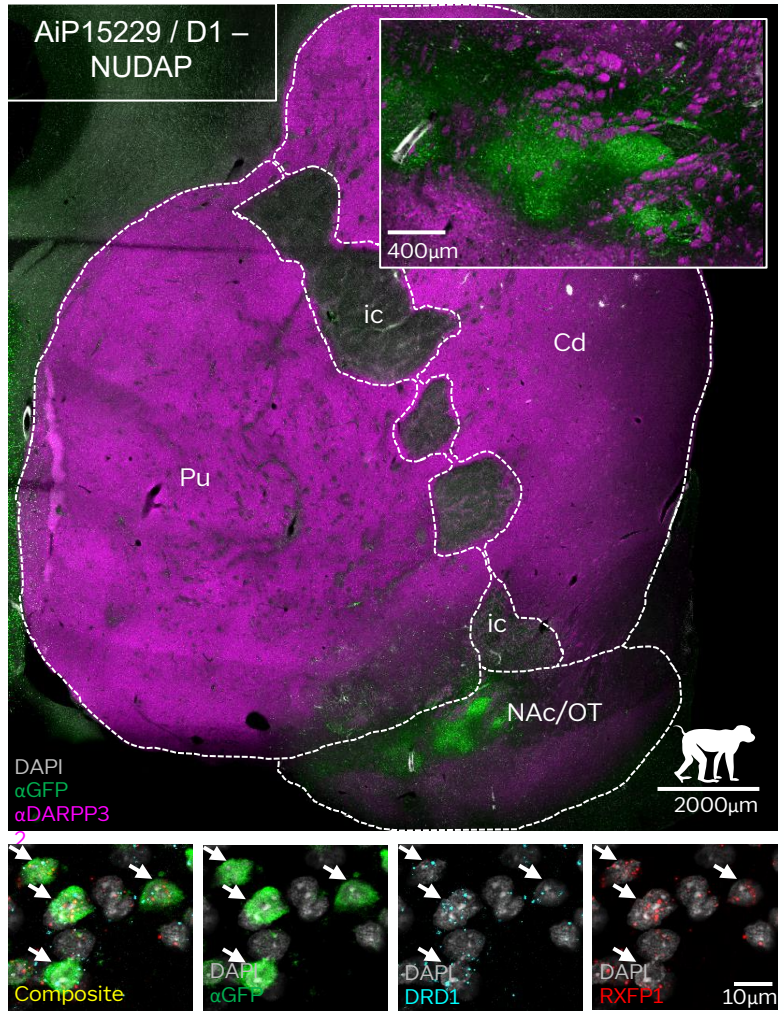
cross-species validation of D1-NUDAP and D1-ICj enhancers/

Allen Institute macaque spatial data





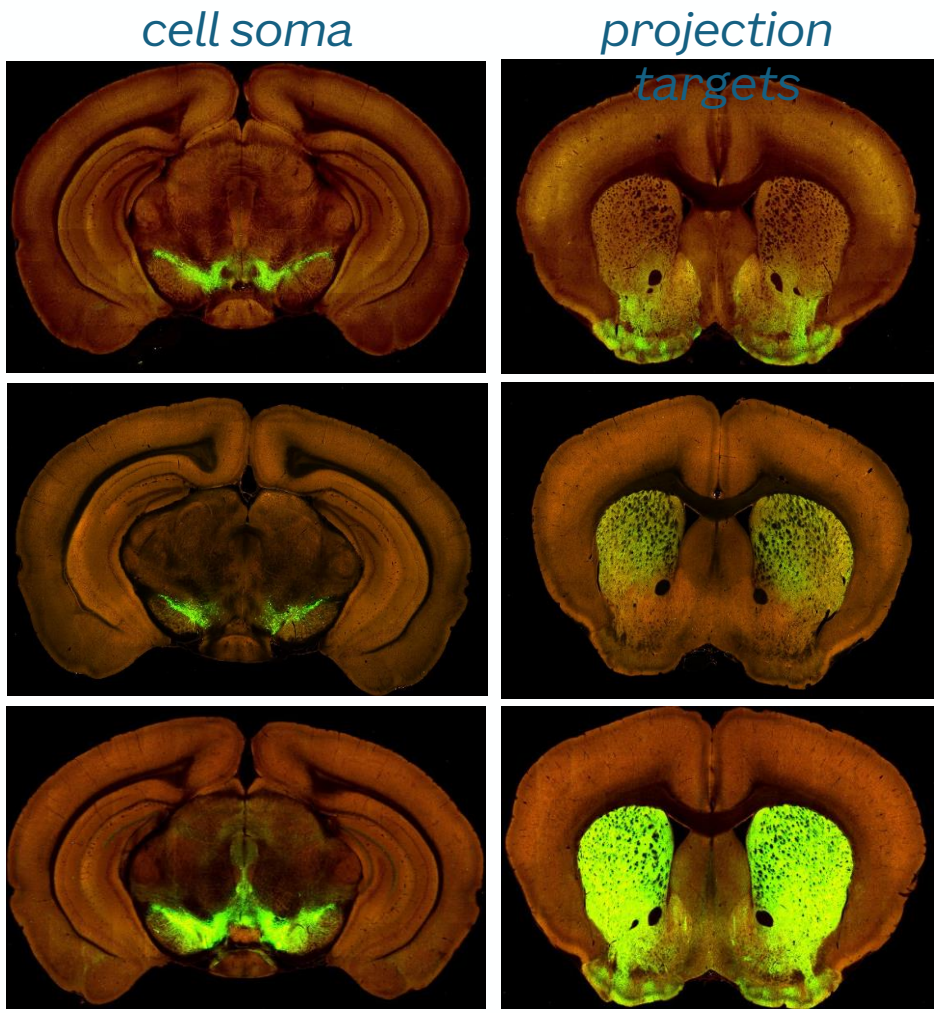
96% specific in Island of Calleja
5% MSNs labeled near ICj

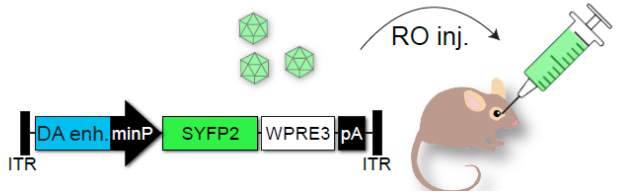


98% specific in D1-NUDAP island
5% of MSNs labeled near D1-NUDAP island

Will Laird, Nathaly Lerma, Jonathan Ting, Naz Taskin, Greg Horwitz, Yoshiko Kojima

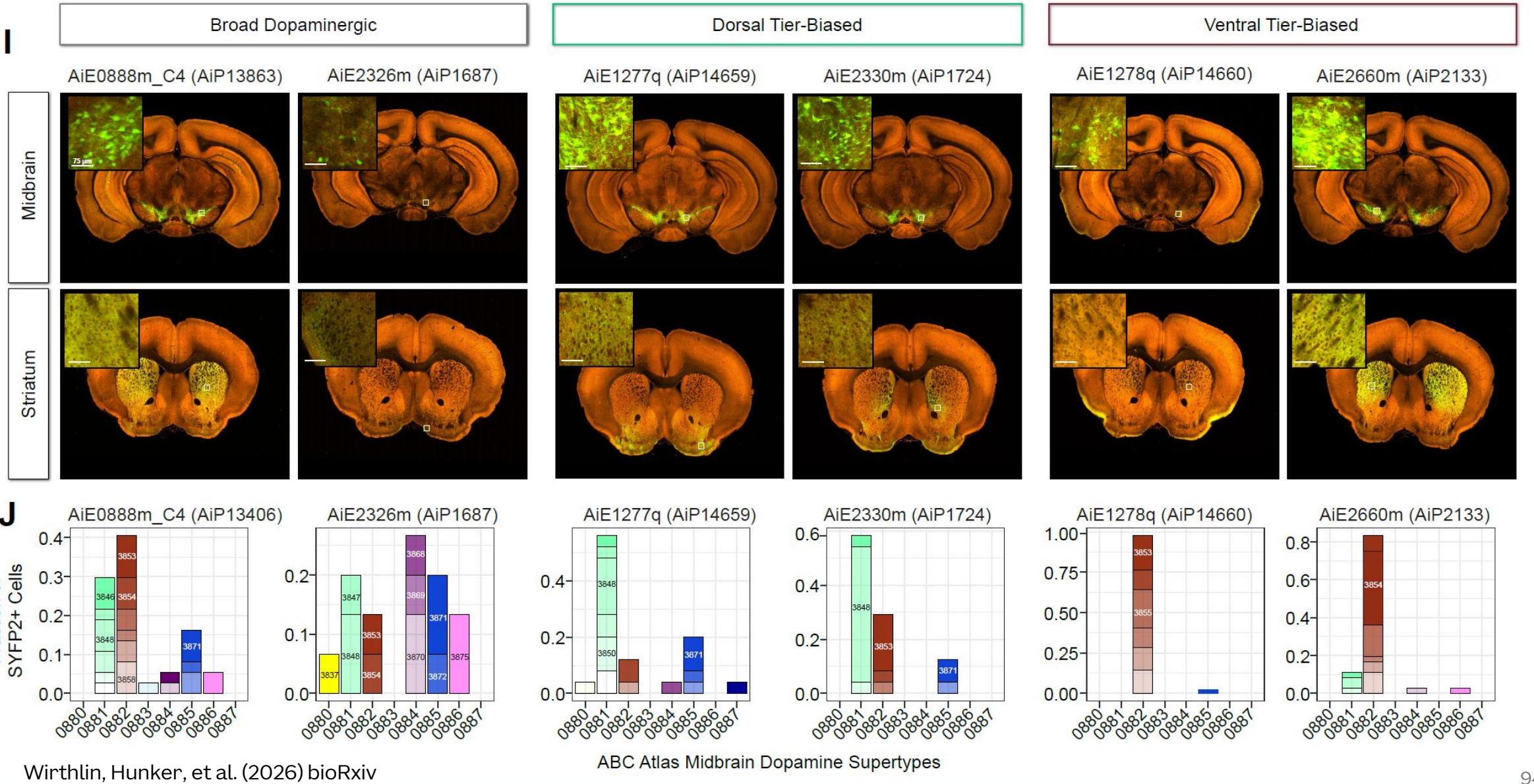
characterizing enhancers for labeling dopamine neuron supertypes/





Avery Hunker & Jonathan Ting team

characterizing enhancers for labeling dopamine neuron supertypes/



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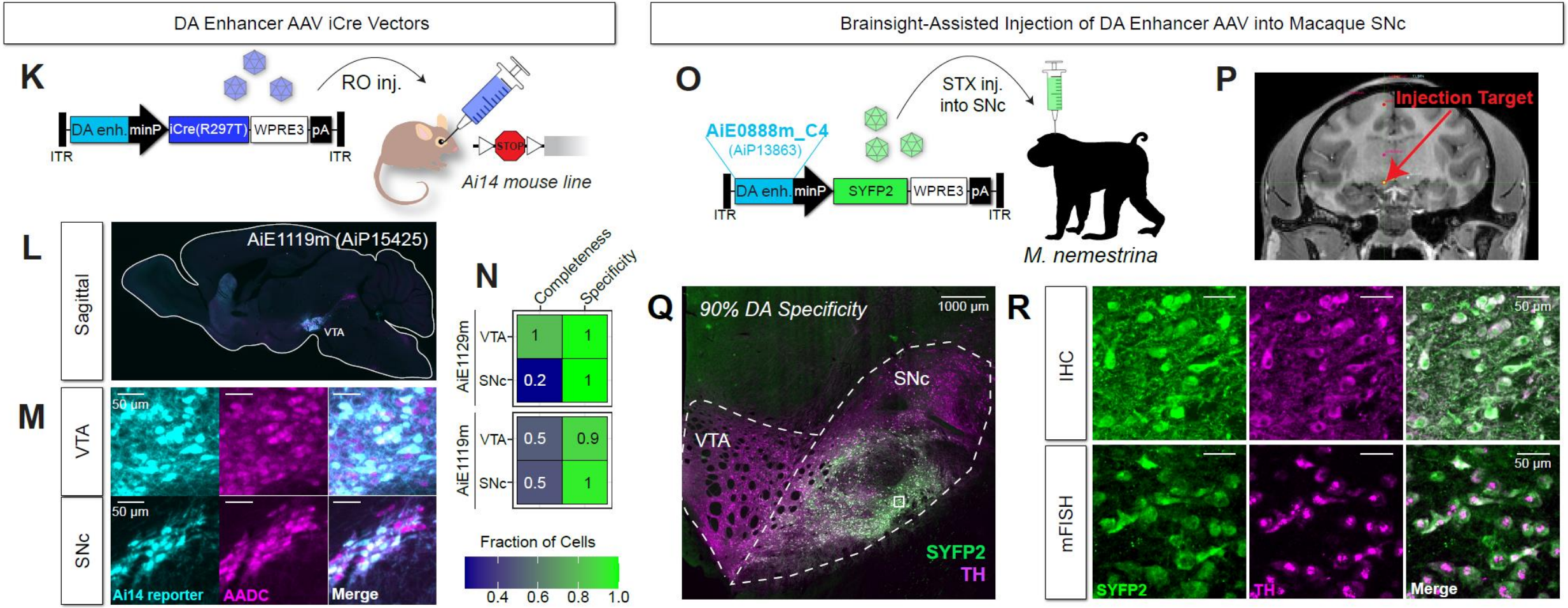
midbrain DA neuron tools exhibit high conservation of specificity in mouse and macaque brain in vivo/



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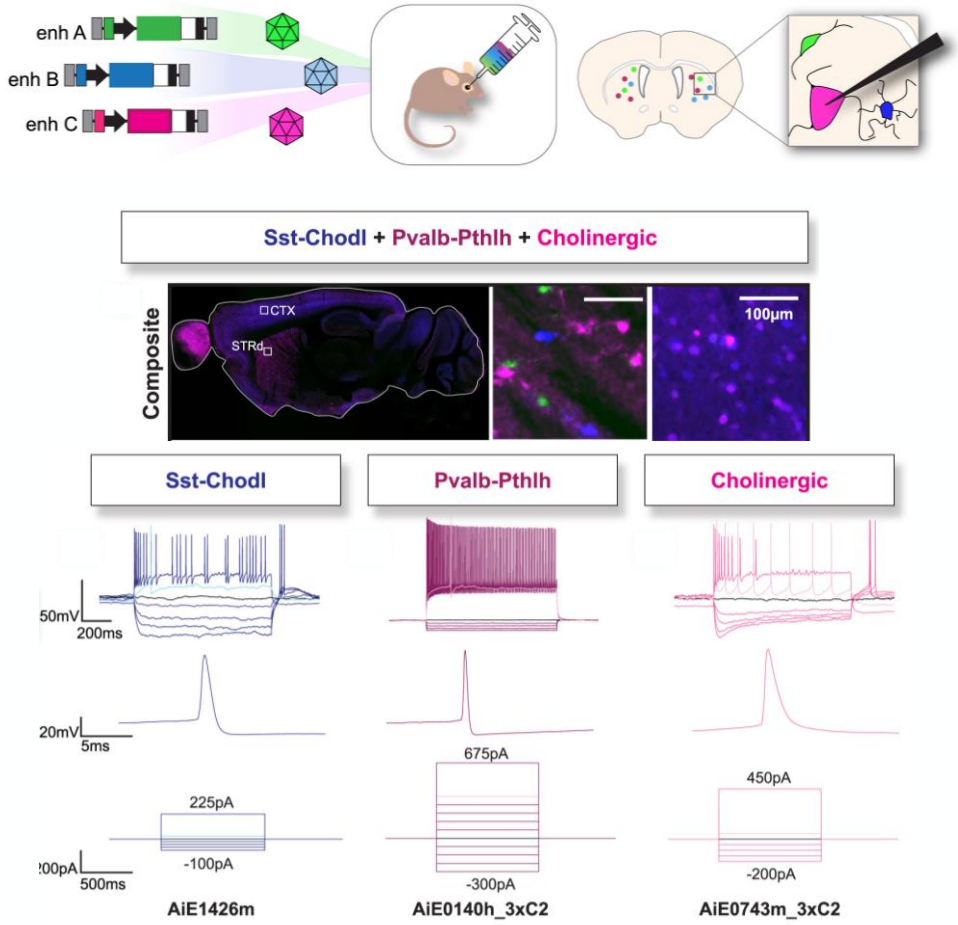
multiplexed enhancer-AAVs/co-injection of multiple enhancers show mutual exclusivity and independent experimental access to complementary cell types



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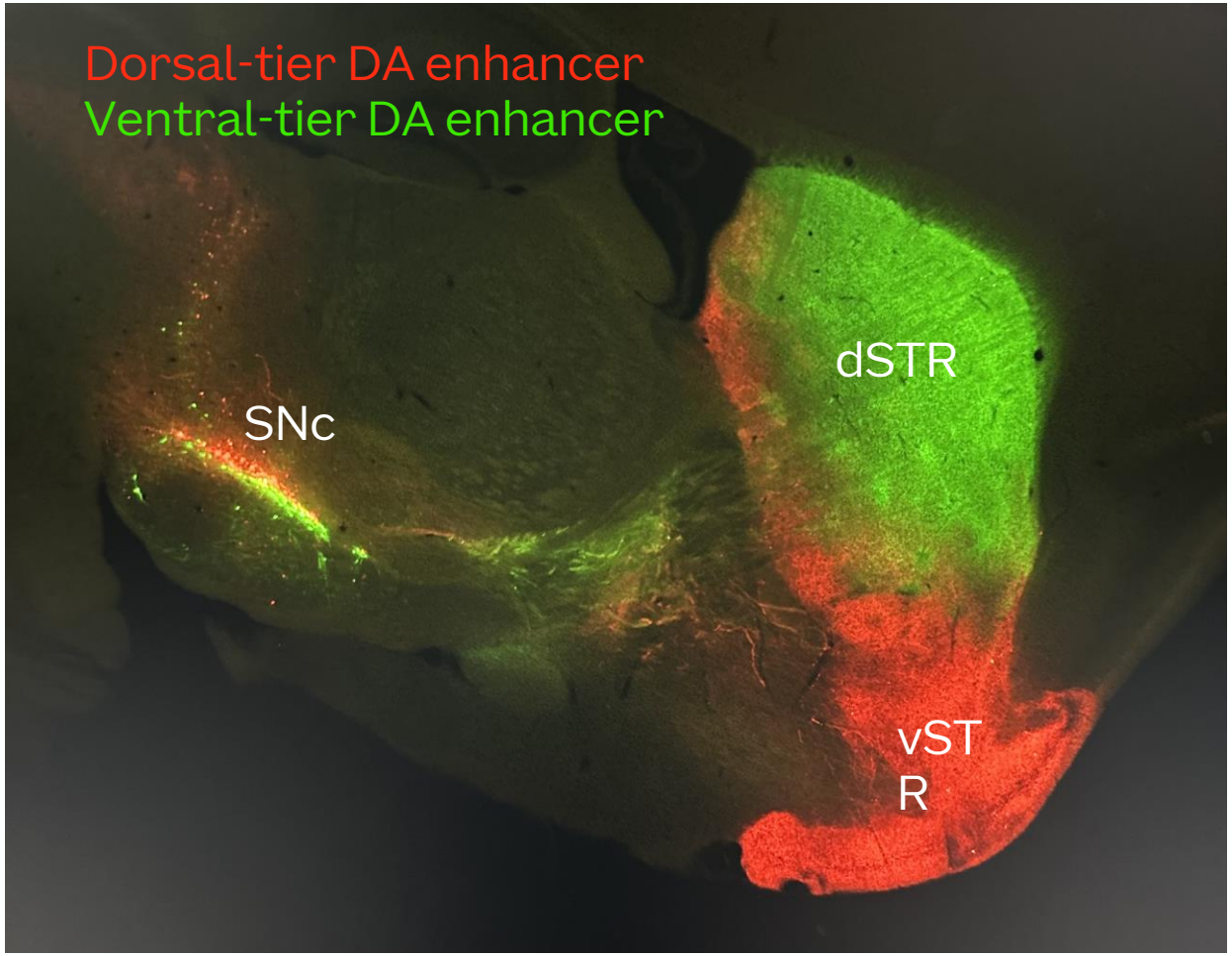
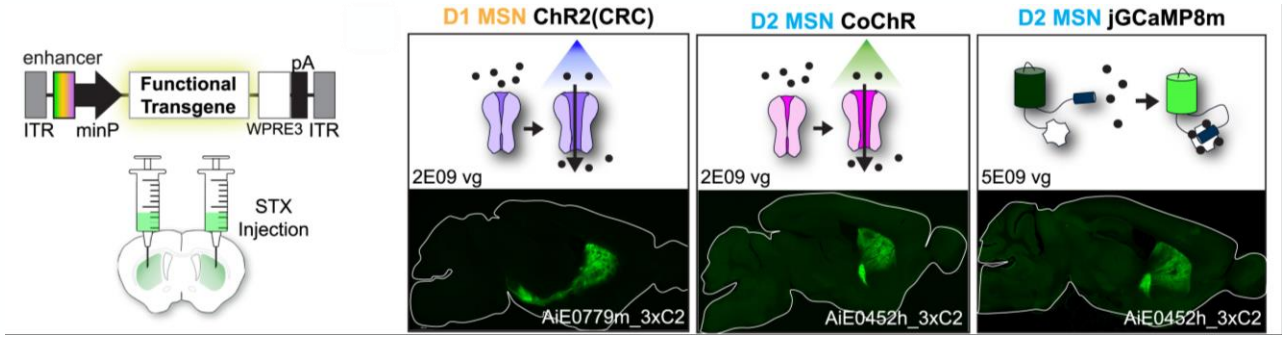
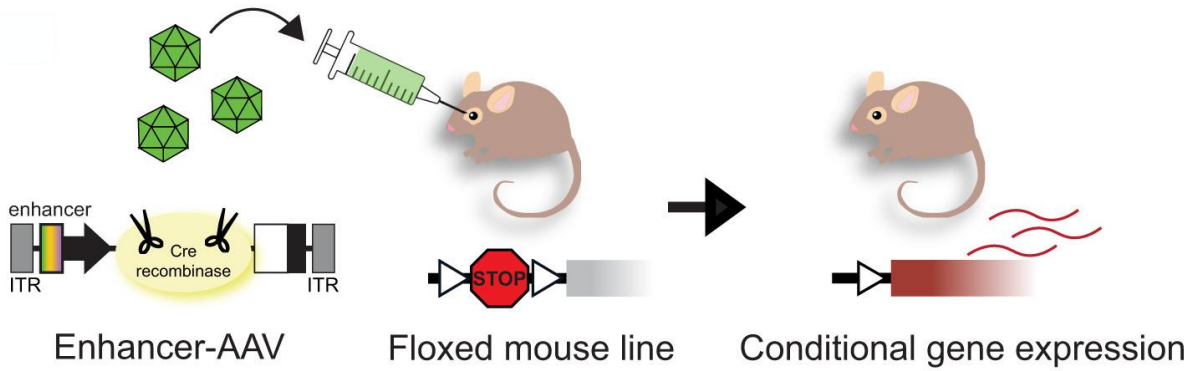
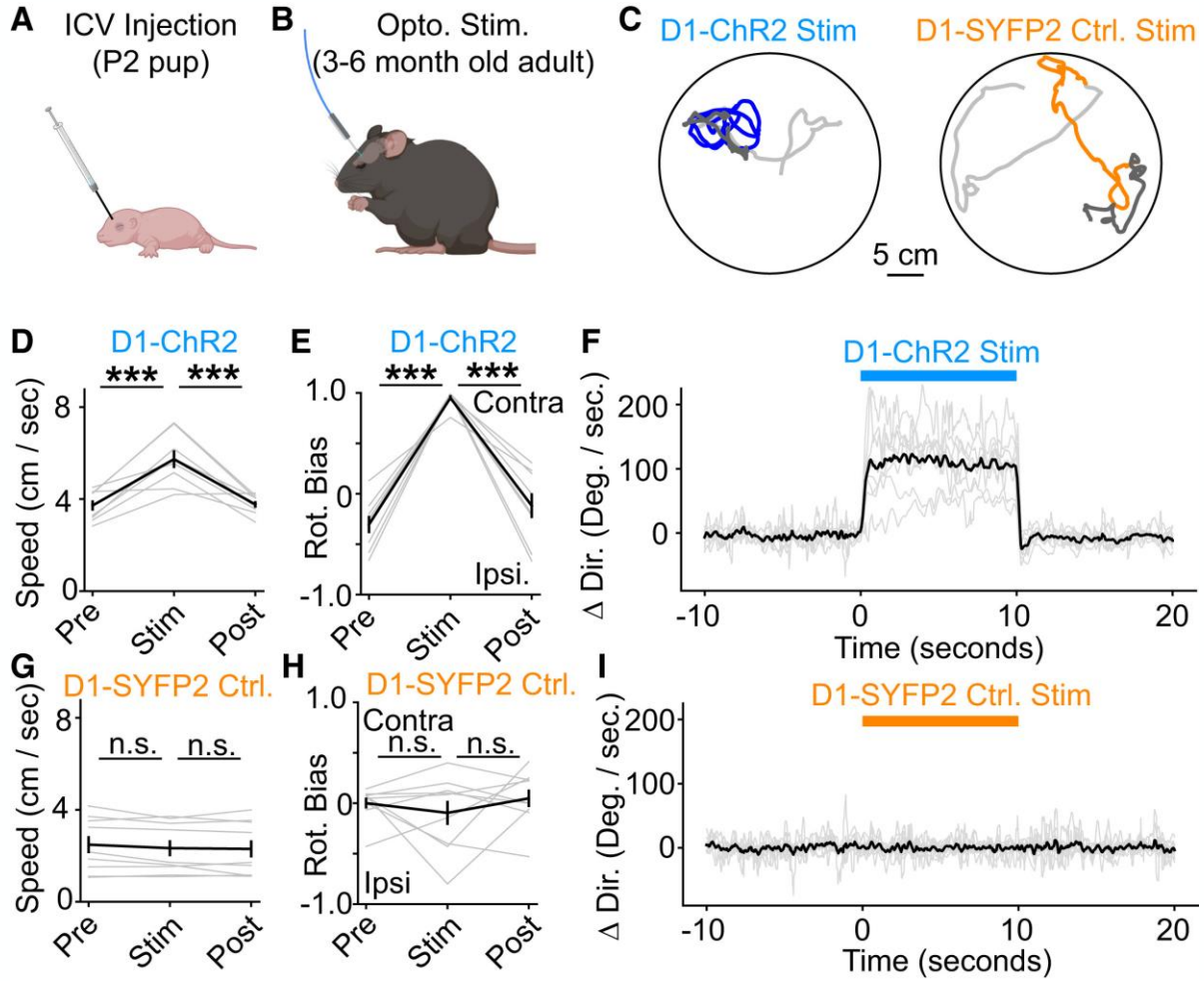


Jonathan Ting

striatal enhancer-AAVs are ready to use for experimental interventions, and can drive robust transgene expression for behavioral assays/







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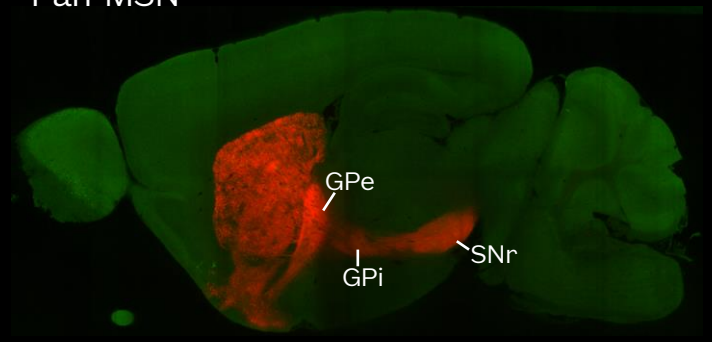
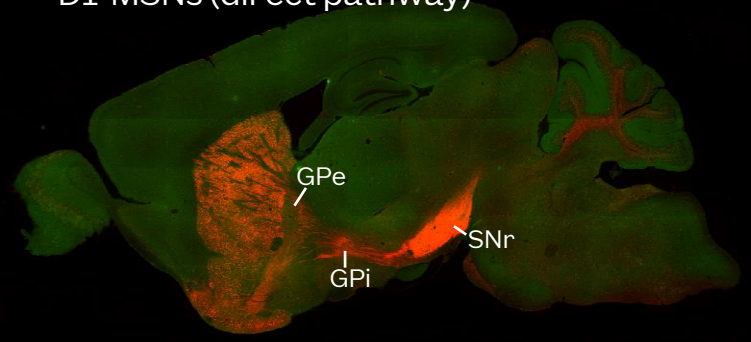
many BG cell type enhancer AAVs achieve specific Cre recombination/

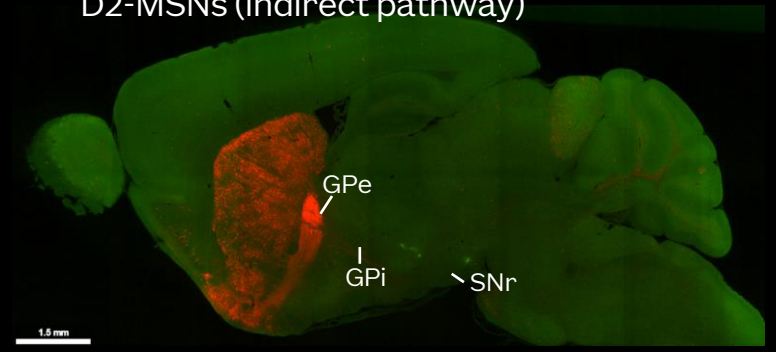
Pan-MSN



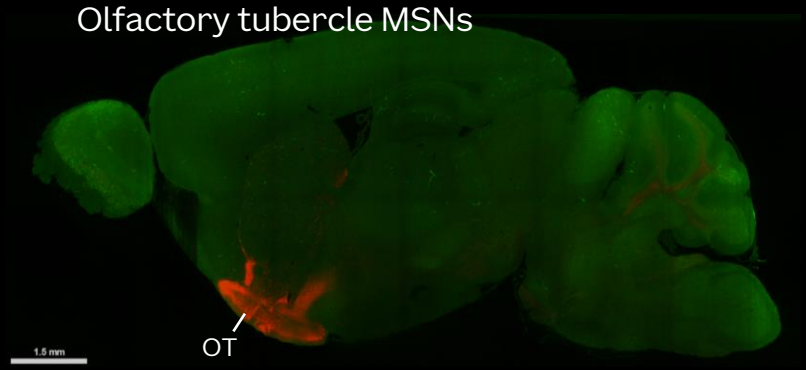
D1-MSNs (direct pathway)



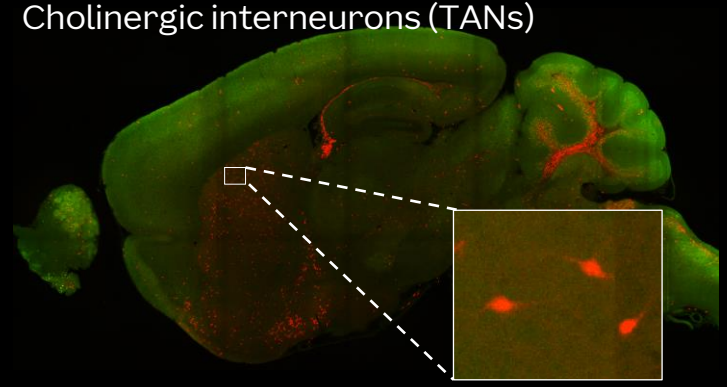
D2-MSNs (indirect pathway)



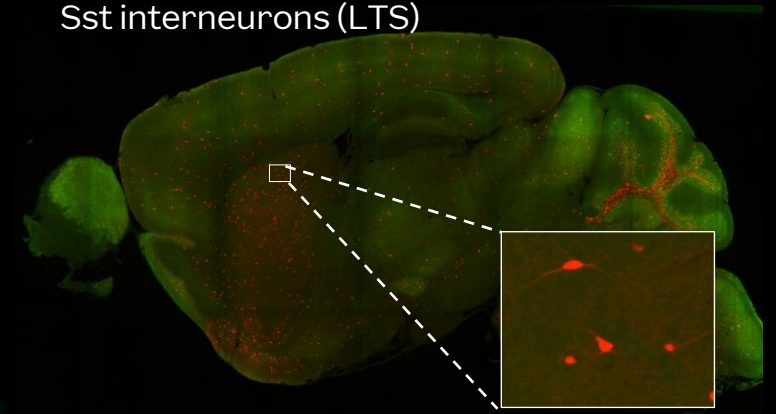
Olfactory tubercle MSNs



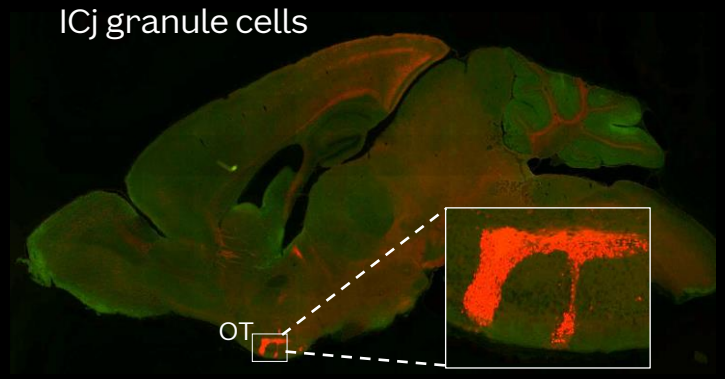
Cholinergic interneurons (TANs)



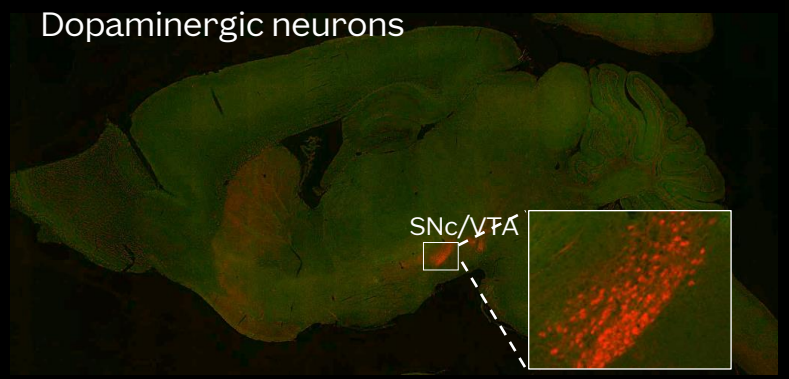
Sst interneurons (LTS)

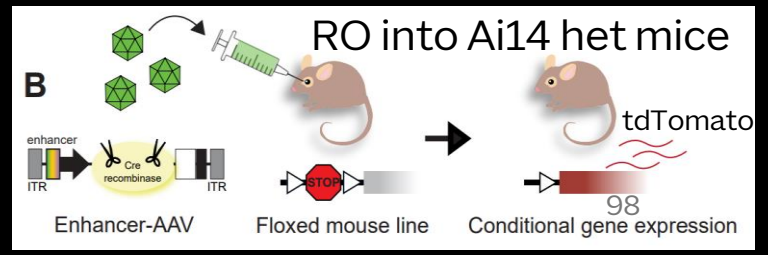


ICj granule cells

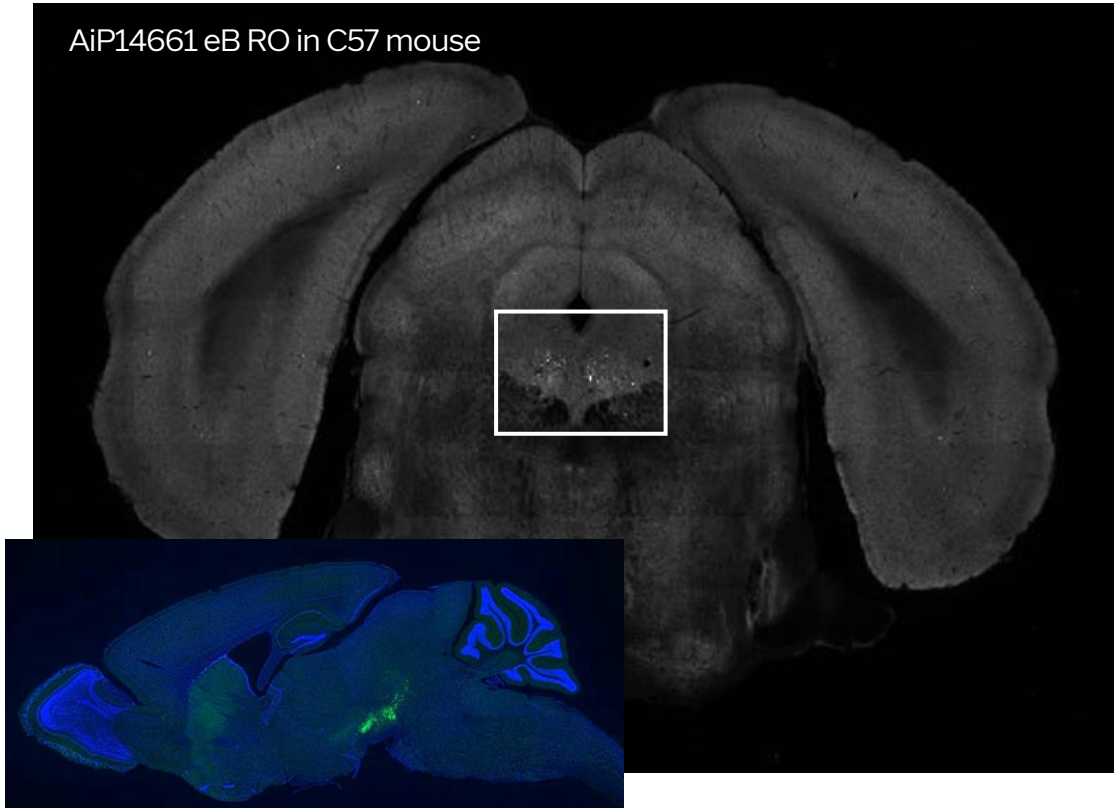


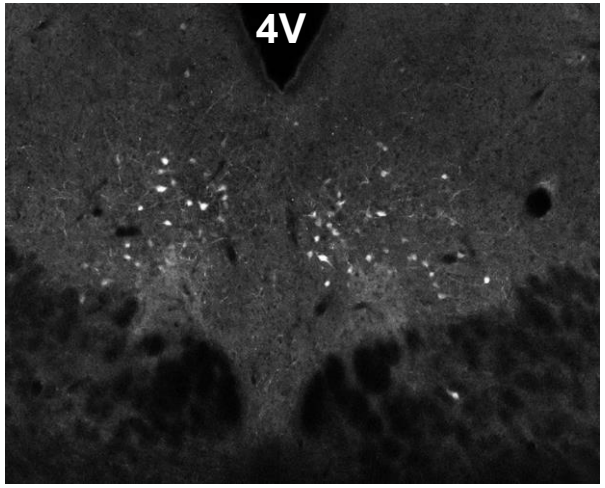
Dopaminergic neurons



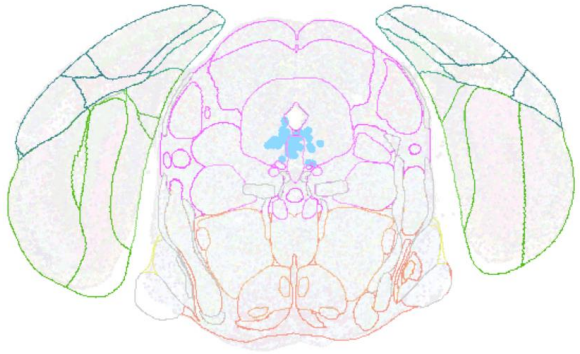


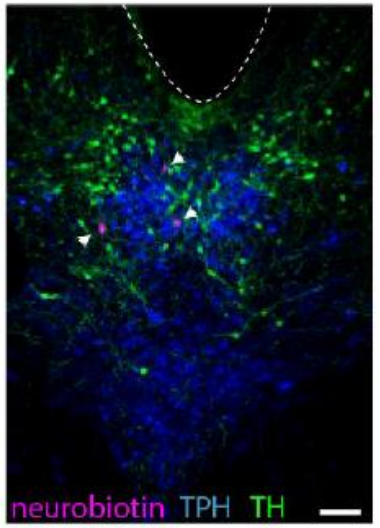
many incidental discoveries/AiE1279q enhancer labels putative dorsal raphe DA neurons (in addition to SNc and VTA DA neurons)





Spatial supertype 0883?

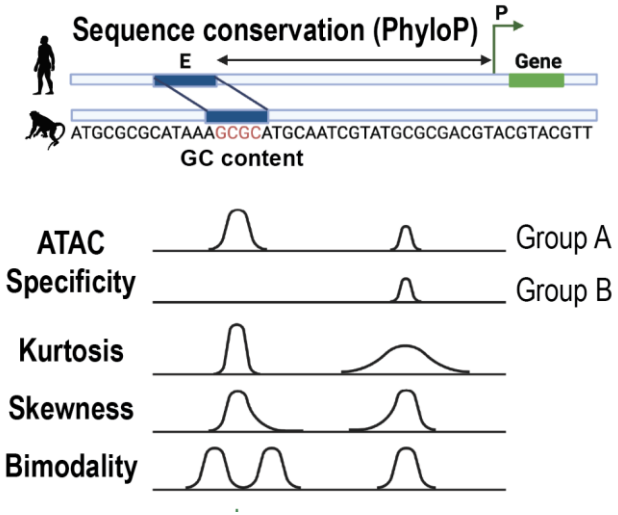




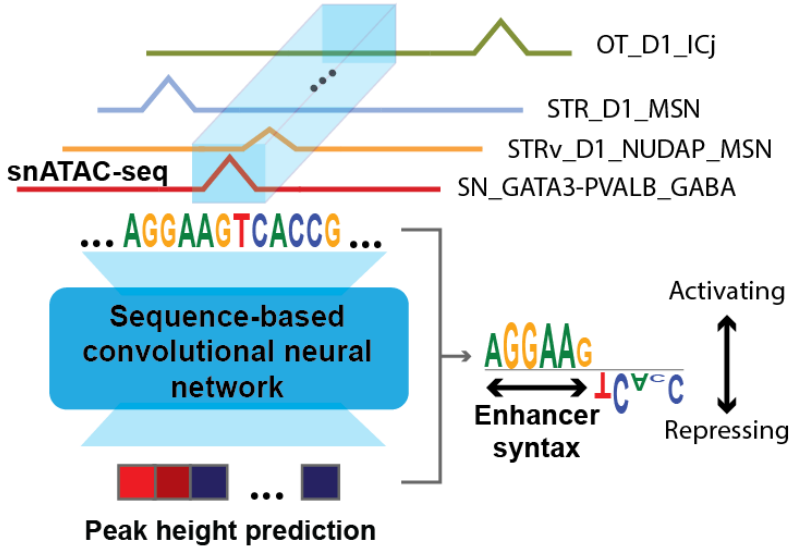
Boi et al, 2023 eLife

sequence modeling enables computational prediction of on-target activity/

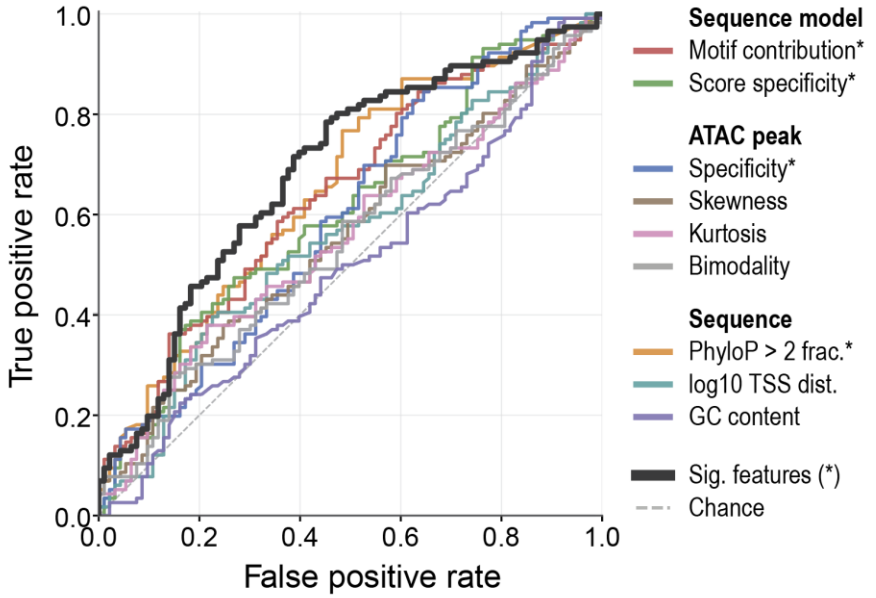
DNA sequence features



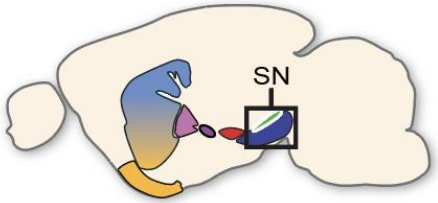
Sequence-based model



Model predicts on-target tool performance



sequence modeling enables activity prediction in cross-species enhancer orthologs/

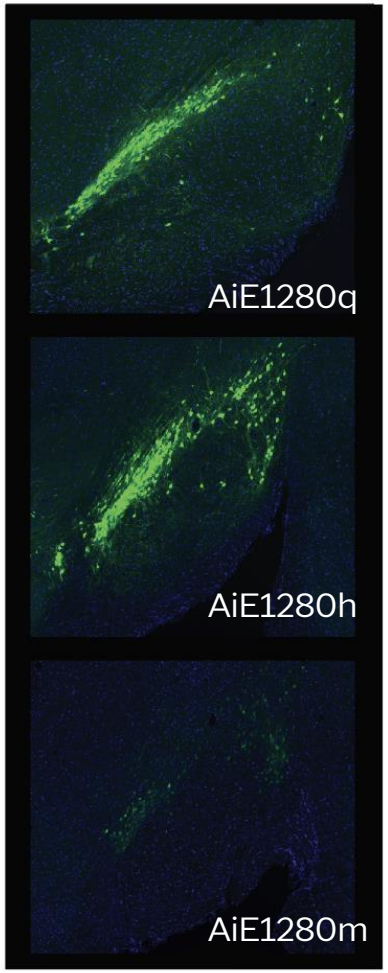


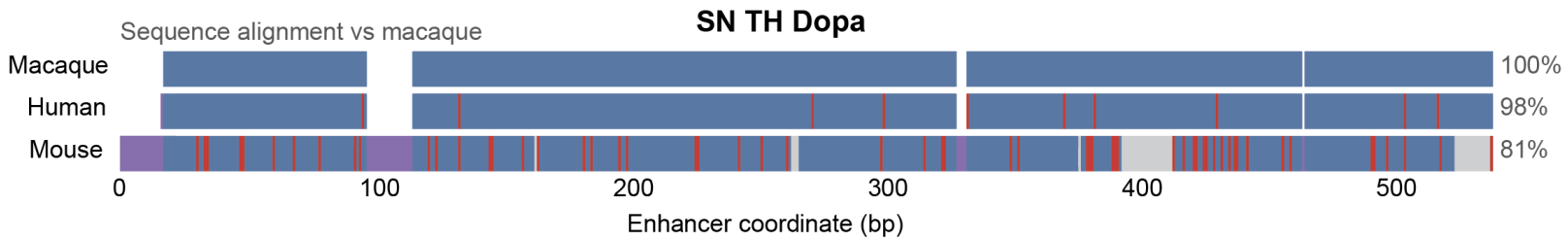
Macaque

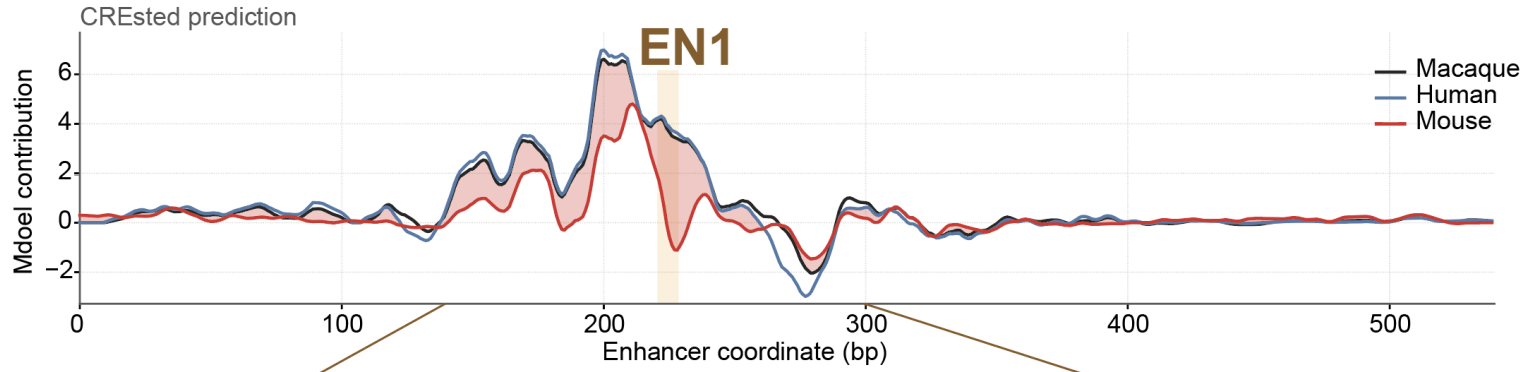
Human

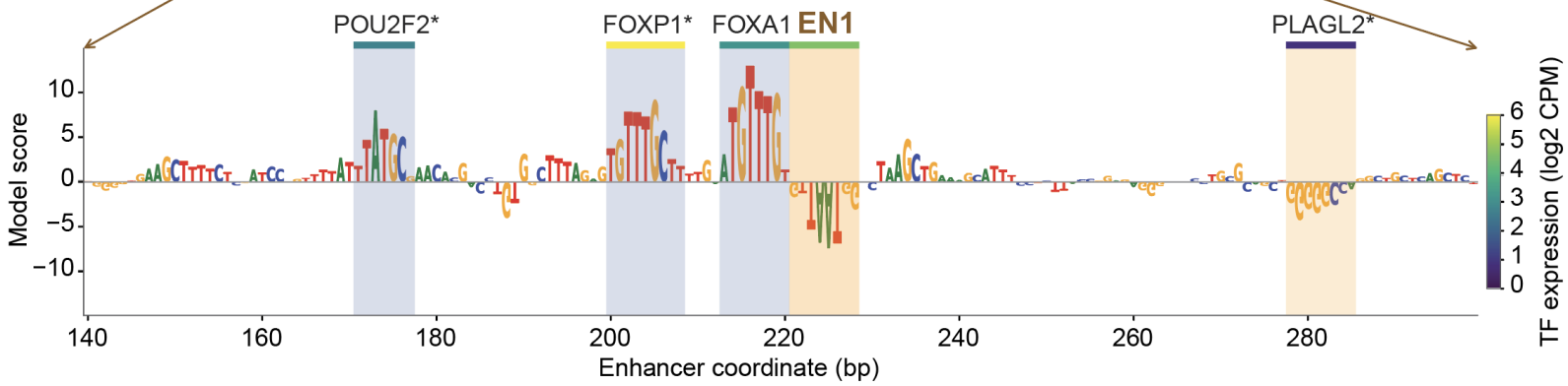
Mouse

SN TH Dopa









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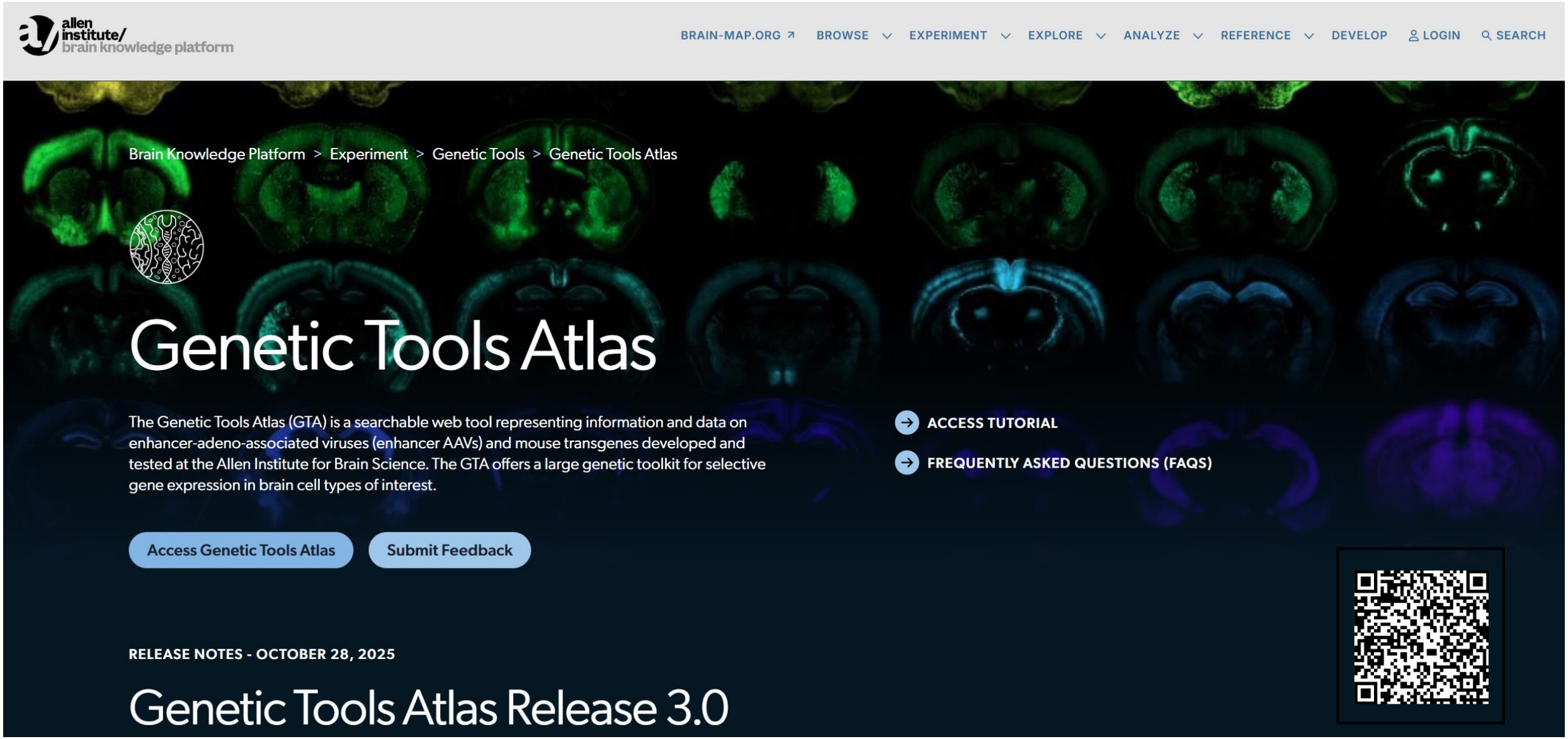
A Cross-Species Enhancer-AAV Toolkit for Cell Type-Specific Targeting Across the Basal Ganglia

Morgan E. Wirthlin, Avery C. Hunker, Saroja Somasundaram, M. Nathaly Lerma, William D. Laird, Victoria Omstead, Naz Taskin, Niklas Kempynck, Matthew T. Schmitz, Yuan Gao, Emma Thomas, Marcus Hooper, Yoav Ben-Simon, Refugio A. Martinez, Ximena Opitz-Araya John Mich, Aaron Oster, Deepanjali Dwivedi, Erin Groce, Jada Roth, Bargavi Thyagarajan, Sharon Way, Avalon Amaya, Angela Ayala, Stuard Barta, Darren Bertagnolli, Madeline Bixby, Trangthanh Cardenas, Tamara Casper, Michael Clark, Nicholas Donadio, Nadezhda I. Dotson, Tom Egdorf, Erica L. Peterson, Jessica Gloe, Jeff Goldy, Conor Grasso, Warren Han, Samantha D. Hastings, Madeleine Hewitt, Daniel Hirschstein, Windy Ho, Alvin Huang, Tye Johnson, Danielle Jones, Atlas Jordan, Zoe C. Juneau, Matthew Jungert, Madhav Kannan, Jaimie Kenney, Shannon Khem, Rana Kutsal, Samantha Lee, Henry Loeffler, Nicholas Lusk, Rachel McCue, Robyn Naidoo, Dakota Newman, Nhan-Kiet Ngo, Katrina Nguyen, Sven Otto, Ben Ouellette, Alana Oyama, Nick Pena, Elliot Phillips, Christina Alice Pom, Lydia Potekhina, Shea Ransford, Christine Rimorin, Dana Rocha, Augustin Ruiz, Raymond Sanchez, Stephanie Seeman, Lucas Suarez, Mike J. Taormina, Michael Tieu, Alex Tran, Tulip E. O'Neill, Natalie Weed, Josh Wilkes, Jeanelle Ariza, Anish Bhaswanth Chakka, Forrest Collman, Nick Dee, Tyler Mollenkopf, Nadiya V. Shapovalova, Kimberly Smith, Jack Waters, Ali Williford, Shenqin Yao, Ina Ersing, Marcella Patrick, Nelson J. Johansen, Jeremy Miller, Hongkui Zeng, Ed S. Lein, Yoshiko Kojima, Gregory D. Horwitz, Boaz P. Levi, Tanya L. Daigle, Bosiljka Tasic, Trygve E. Bakken, Jonathan T. Ting

doi: https://doi.org/10.64898/2026.02.23.706695

the Genetic Tools Atlas/ https://portal.brain-map.org/genetic-tools/genetic-tools-atlas



Viral Genetic Tools co-leads/

Avery Hunker (Novartis)

Nathaly Lerma (University of Texas, El Paso)

Jonathan Ting

Naz Taskin

Will Laird

Sami Lee (University of Washington)

Regulatory Genomics (RegX) team/

Inkar Kapen
Jessie Puckett (UChicago)
Alma Dubuc (Yale)
Matthew Schmitz
Nelson Johansen
Niklas Kempynck (KU Leuven)
Trygve Bakken
Morgan Wirthlin
Yuanyuan Fu
Saroja Somasundaram

(Left to right)

thank you to our project contributors/

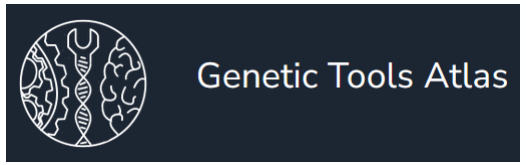
AIBS enhancer AAV pipeline: Viral Genetic Tools team

Genetic Tools Atlas: Collaborative effort between AIBS Data and Technology team and Viral Genetic Tools team

Macaque experiments: Collaborative effort between Allen Institute BRAIN AAV toolbox grant contributors and UW/WaNPRC co-Investigators Greg Horwitz and Yoshiko Kojima. Thanks to WaNPRC staff as well.

Enhancer AAV distribution: Collaborative effort between Allen Institute BRAIN AAV toolbox grant contributors and Addgene co-Investigators Marcy Patrick and Ina Ersing

Spatial transcriptomics and genomics teams: cross team collaboration on BG cell type taxonomies, spatial types, and cell type annotation



Check out enhancer AAVs on GTA!

thank you!

NIH BRAIN Armamentarium UF1MH128933

Allen Institute Teams/
AI Applications (OCTO)
Data and Technology
Flow Cytometry
Histology
Human Cell Types
Imaging
Informatics & Data Science
Molecular Biology
Molecular Genetics
Neurosurgery and Behavior
Project Management
Tissue Processing
Viral Technology

NIH National Institutes of Health
The BRAIN Initiative®

10 Years of BRAIN
A Decade of Innovation

agenda/morning

- 9:00-9:30 | check-in / breakfast
- 9:30-9:40 | welcome, logistics, and overview of allen institute
- 9:40-10:05 | cell typing at allen institute: taxonomies, data sets, and use cases
- 10:05-10:30 | human and mammalian brain atlas (HMBA)
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- 11:55-12:05 | Overview of brain knowledge platform (BKP) tools & resources
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- 12:10-1:10 | lunch

Seattle Alzheimer's Disease Brain Cell Atlas in the brain health accelerator/

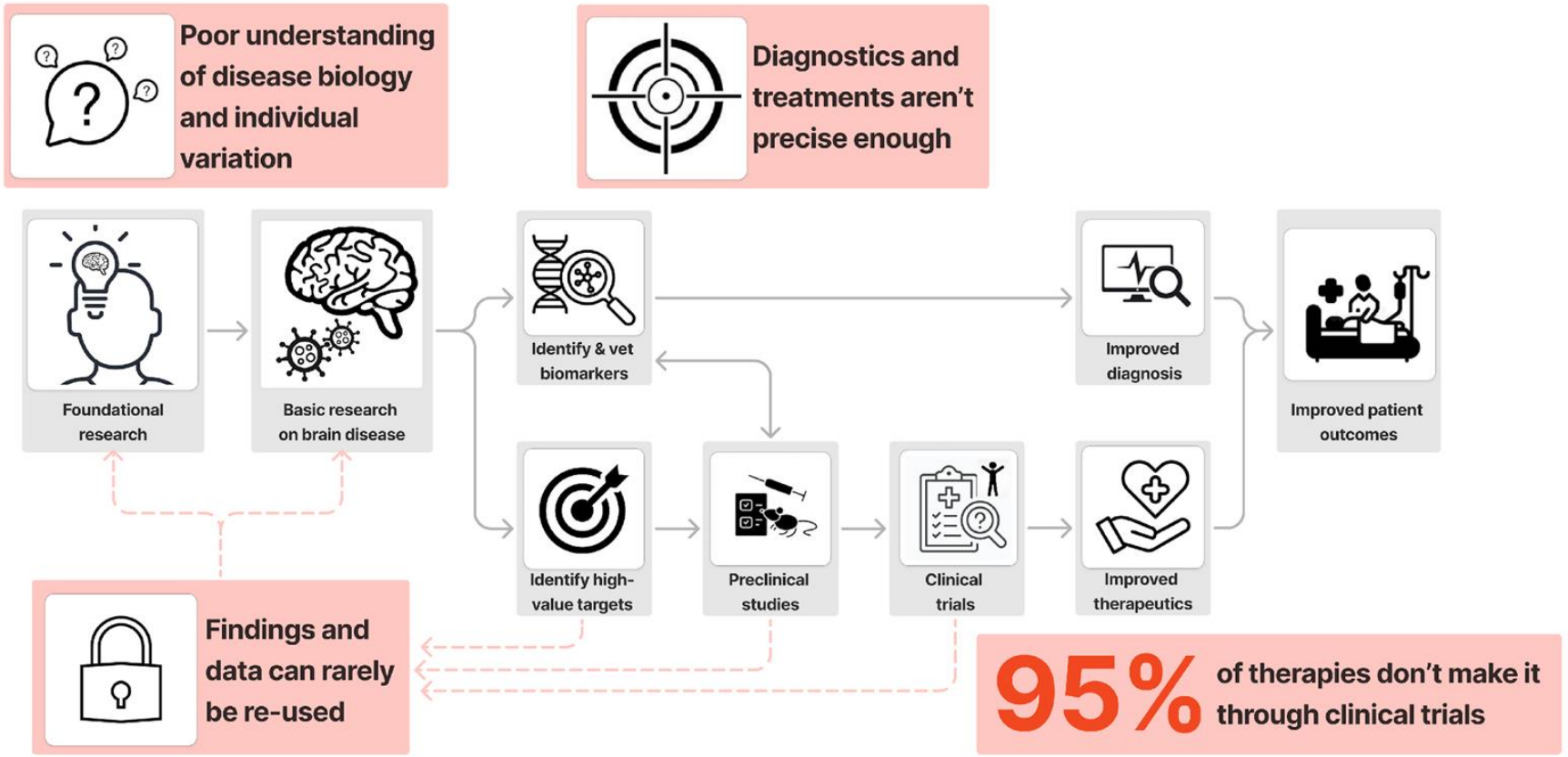
Kyle Travaglini, Assistant Investigator

allen institute/

allen institute

The time is right for a brain health accelerator/

Why? >3 billion people suffer from tragic neurological conditions



Why now? New technologies and knowledge we have developed are ready for impact

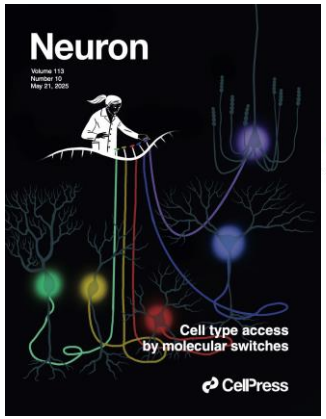
Why us? Model of Big, Open, Team science is perfect for a new global collaborative model

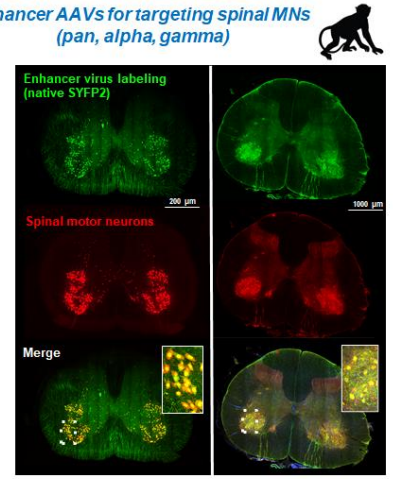
Technologies driving rapid advances in human neuroscience

Genome-style foundational human and cross-species cell and anatomical atlases (BICAN)

The BRAIN Initiative®

Cell type- and circuit-targeting tools for precision genetic therapies





Kussick et al., Cell Reports 2025

The BRAIN Initiative®

Deep characterization of disease trajectories and vulnerable cell types in Alzheimer's disease (SEA-AD)

SEA-AD

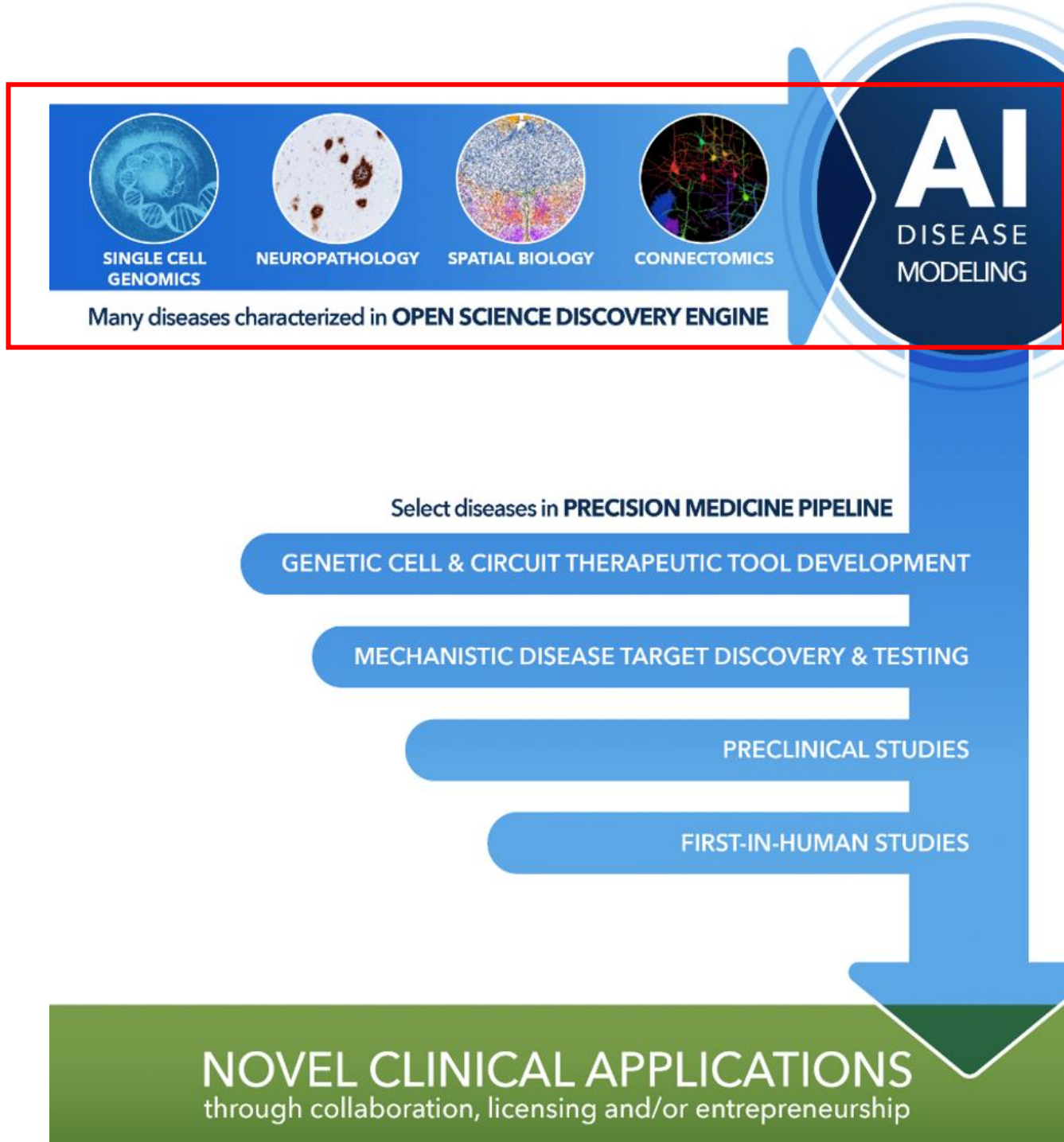
NIH National Institute on Aging

Pillars within brain health

From “Human-First” brain disease characterization to “First-in-Human” therapeutics strategies

Focus initially on:

- ALS
- Alzheimer’s disease
- Parkinson’s disease
- Lewy body dementia
- Huntington’s disease

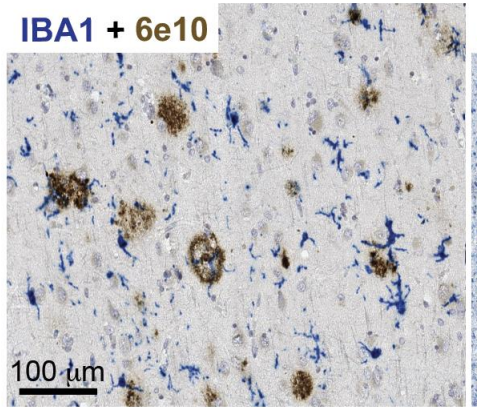


Dementia

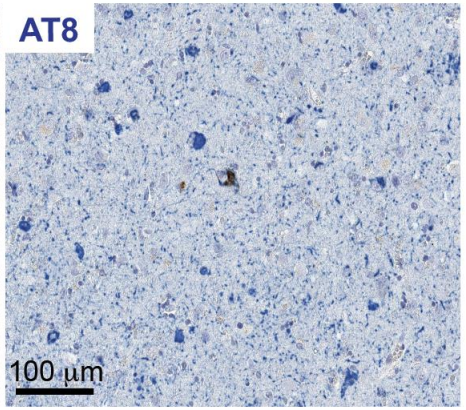
is an umbrella term for several diseases affecting memory, other cognitive abilities and behavior that interfere significantly with a person's ability to maintain their activities of daily living.

Alzheimer's disease accounts for 60-70% of dementia cases. Presents as changes in thinking and memory and is charactered by abnormal deposits of amyloid plaques and tau tangles.

Amyloid Beta Plaques (6e10)



Neurofibrulary Tangles (AT8)



Gabitto, Travaglini et al (2024)

Other Dementias

Vascular Dementia
Prevalence: 20%
Conditions, such as blood clots, disrupt blood flow in the brain

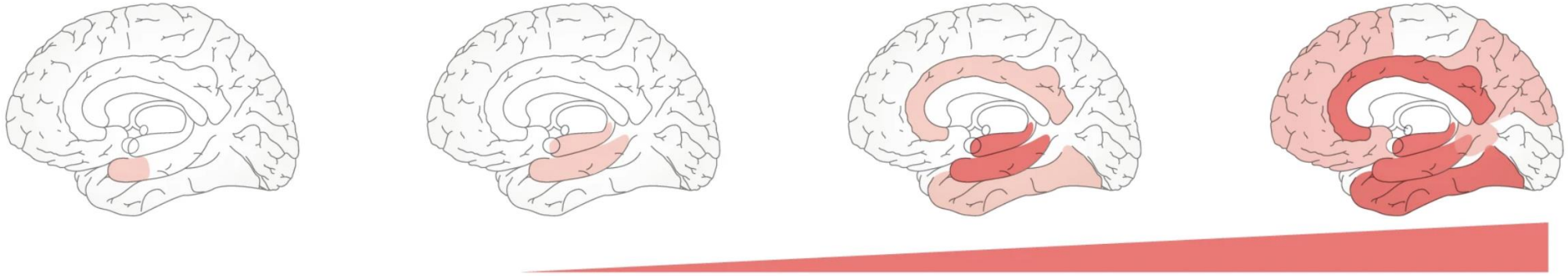
Lewy Body Dementia
Prevalence: 10-15%
Abnormal deposits of alpha-synuclein proteins, called "Lewy bodies"

Frontotemporal dementia
Prevalence: 2%
Abnormal tau and TDP-43 protein in the frontal and temporal lobes

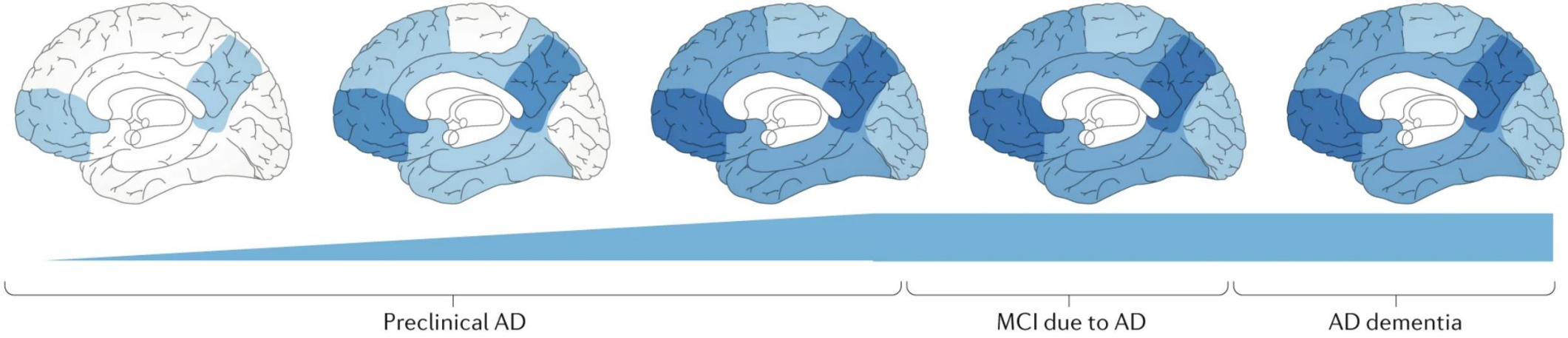
nia.nih.gov

Plaques and tangles follow stereotyped patterns of accumulation across brain regions

Tangle pathology



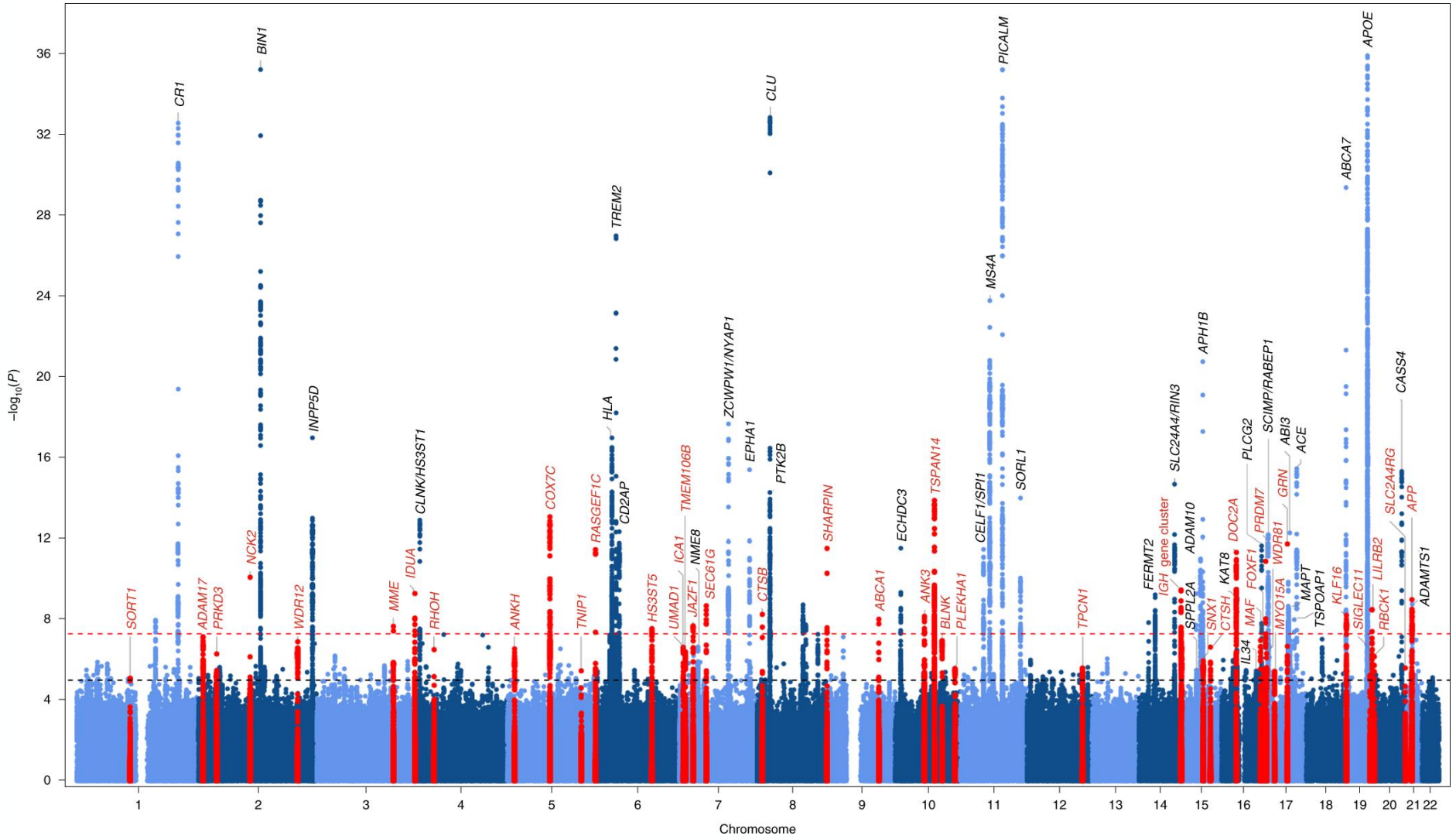
Plaque pathology



van der Kant et al (2019)

Genetics provide clues for drivers of Alzheimer's disease (AD)

AD dementia genome wide association study



Bellenguez et al (2022)

BICAN: A reference taxonomy of the whole human brain that includes locations, morphologies, and electrical properties

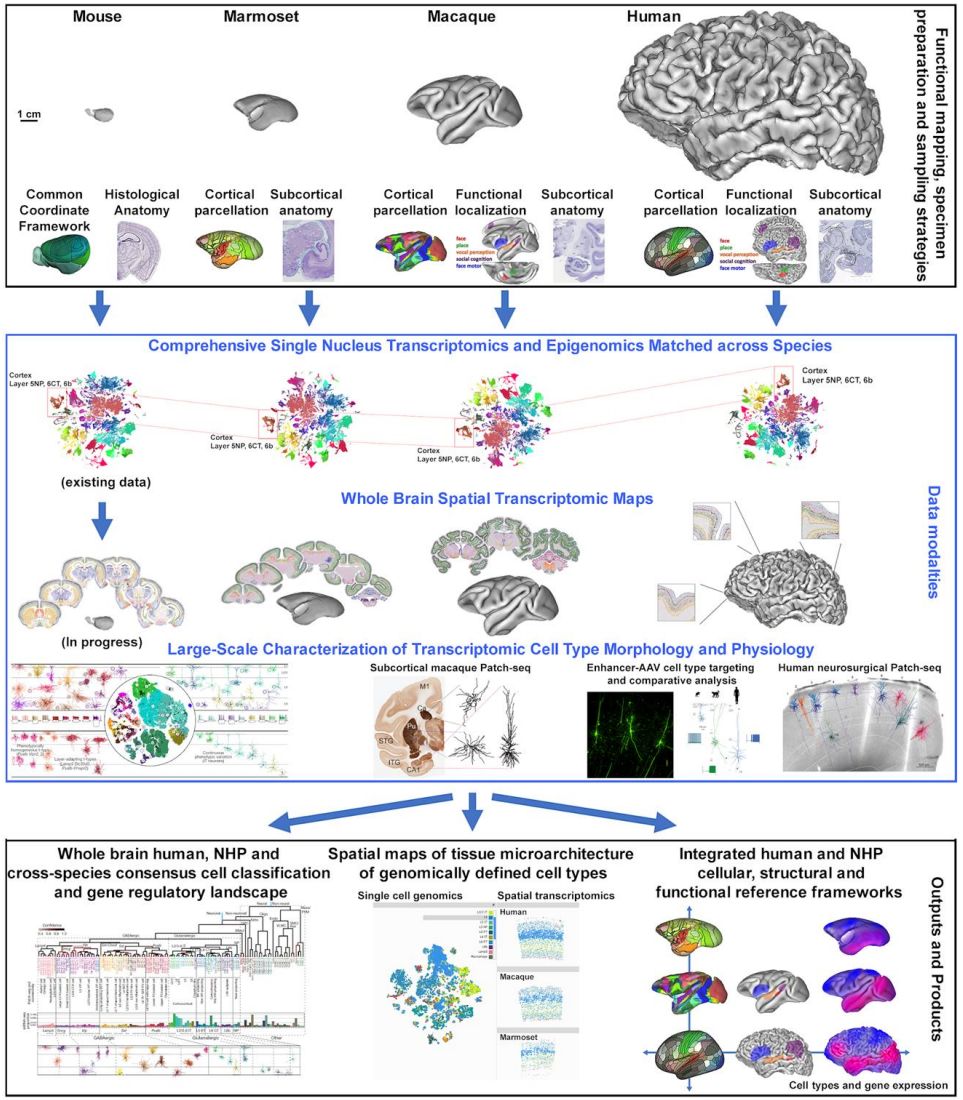
fMRI

Single cell genomics

Spatial transcriptomics

Patch-seq

Products



HMBA

Human and Mammalian Brain Cell Atlas

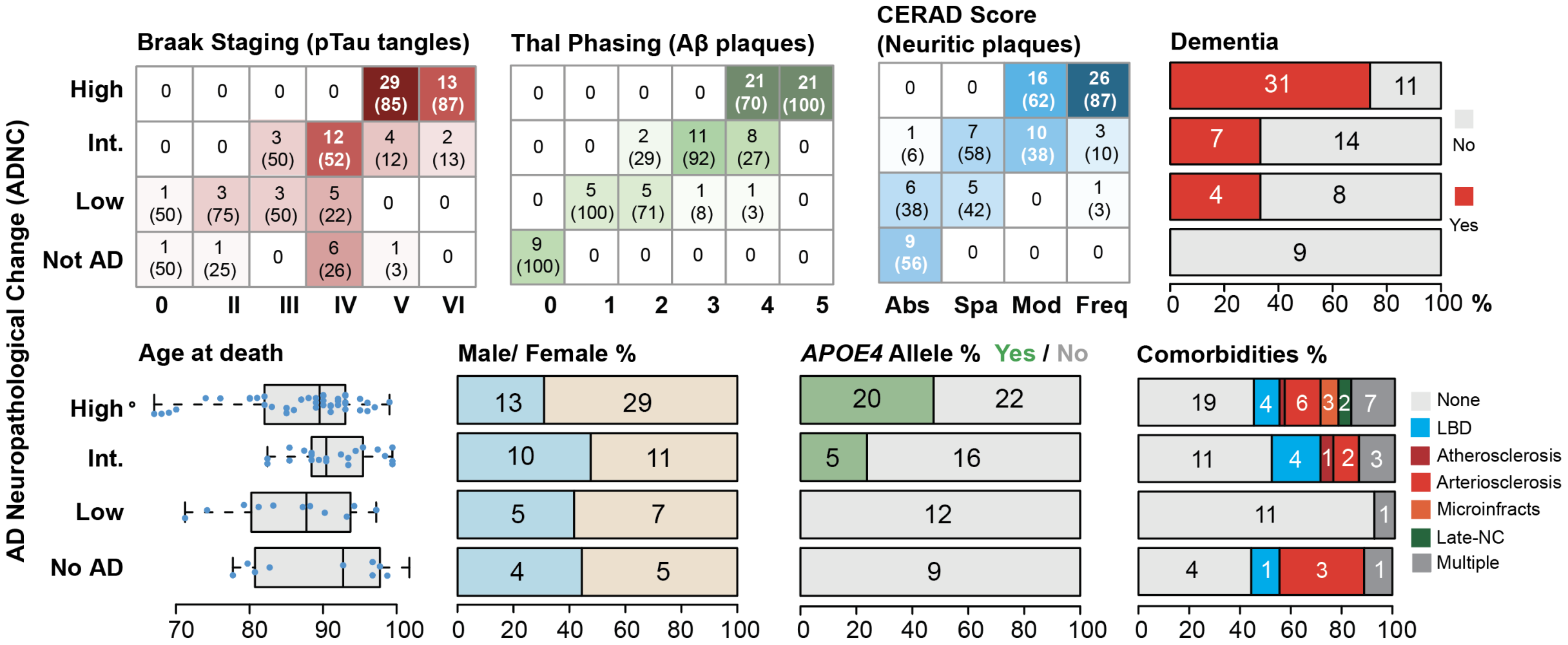
Seattle Alzheimer's Disease Brain Cell Atlas (SEA-AD)

The Seattle Alzheimer's Disease Brain Cell Atlas (SEA-AD) consortium is supported by the National Institutes on Aging (NIA) grant 2U19AG060909.

SEA-AD: Primary Goals

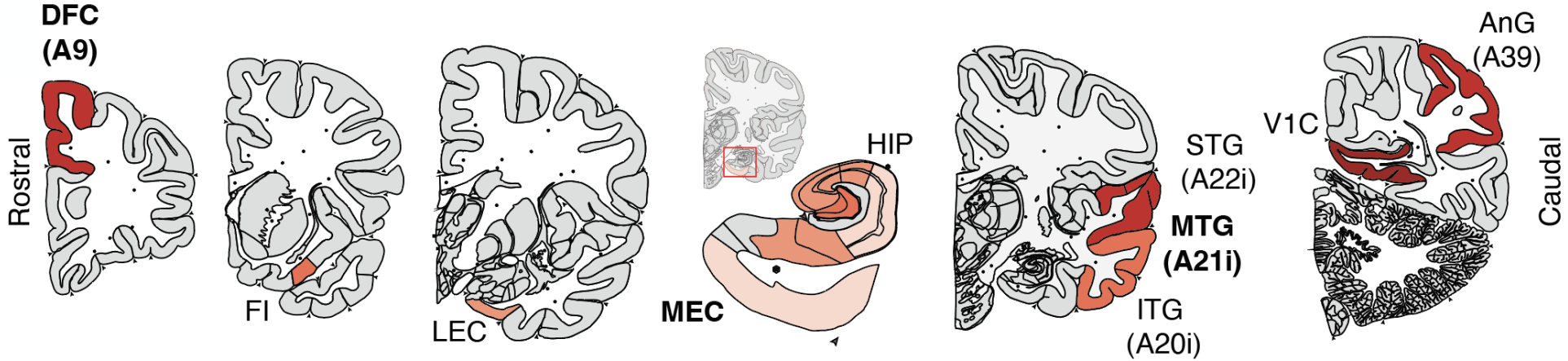
- Move beyond a case-control paradigm using data-driven representation of AD
- Determine and stage the molecular and cellular changes that occur in AD
- Prioritize cellular and molecular changes related to AD initiation and progression
- Nominate intervention strategies for notable cellular and molecular targets

Moving beyond case-control to capture the spectrum of AD pathological and cognitive phenotypes



Gabitto, Travaglini et al (2024)

Multimodal characterization of each brain region

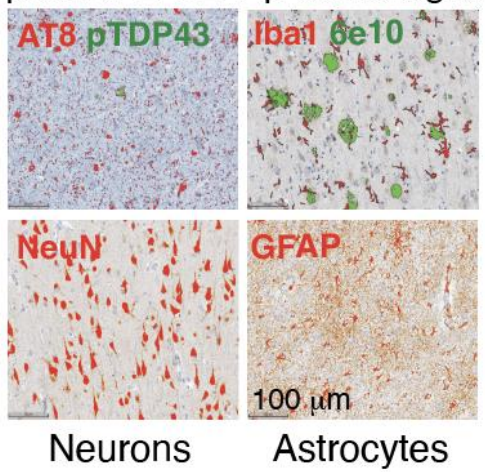


Braak I/II: MEC, CA1 (HIP) Braak III: Sub (HIP), CA2-3 (HIP), LEC Braak IV: CA4 (HIP), ITG, MTG, FI

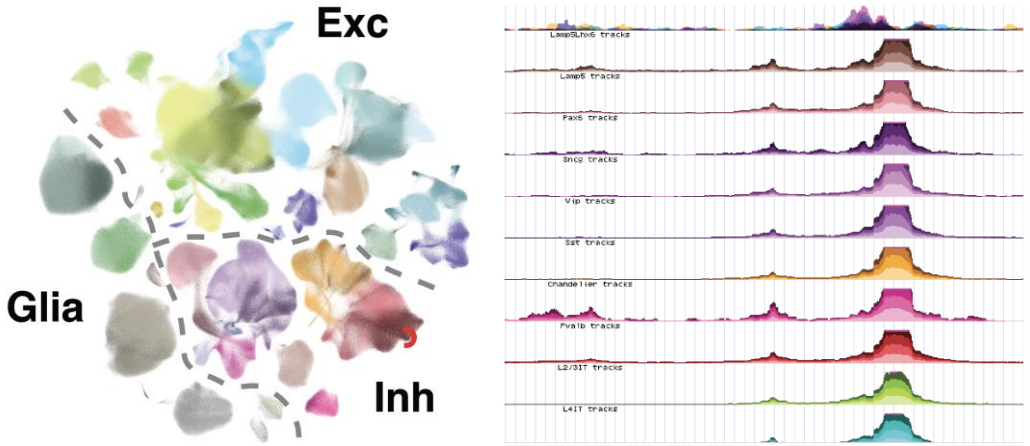
Braak V: STG, DFC, AnG Braak VI: V1C AD spectrum only (n=43) / **AD spectrum and co-morbidities (n=84)**

Quantitative Neuropathology

pTau & LATE Aβ & Microglia

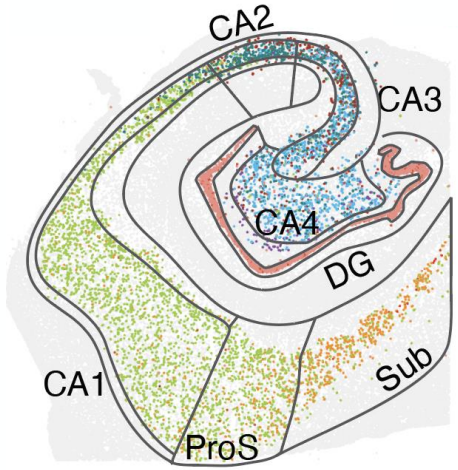


Single nucleus multiomics (RNAseq and ATACseq)



Whole genome sequencing

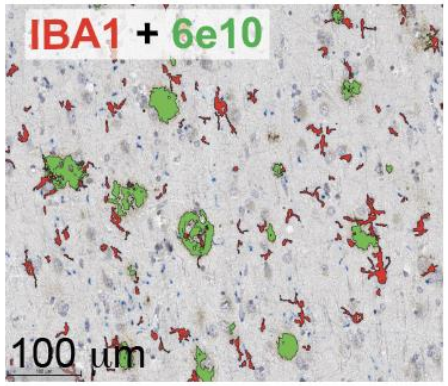
Targeted spatial genomics



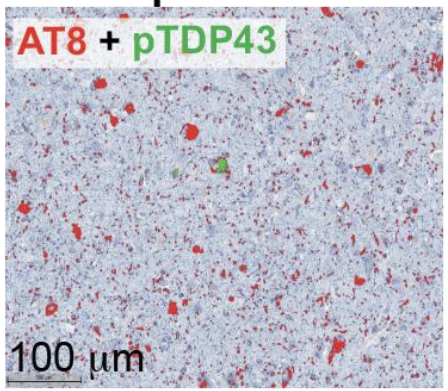
Travaglini, Gabitto, Ding et al (2026)

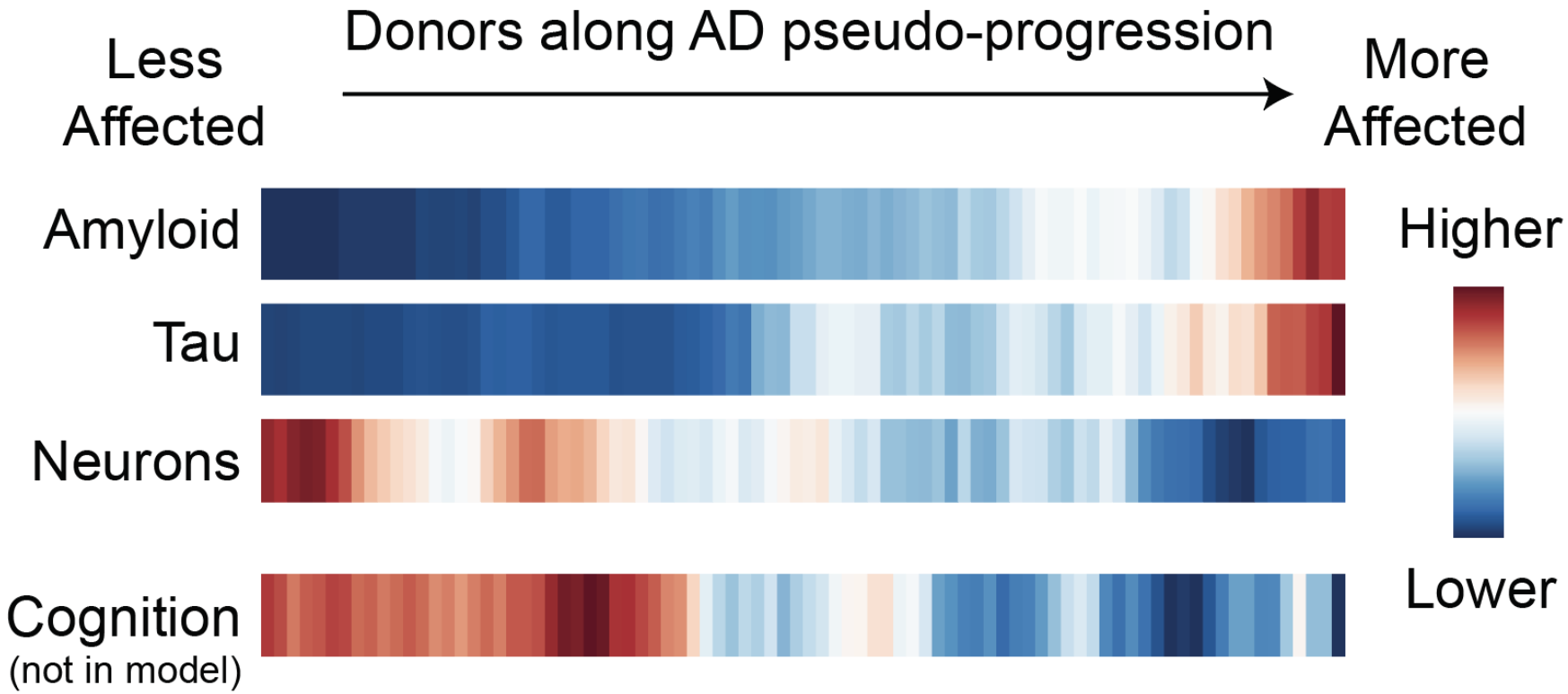
Biophysically inspired models of plaque and tangle accumulation learn an AD “pseudo-progression” across donors

Amyloid



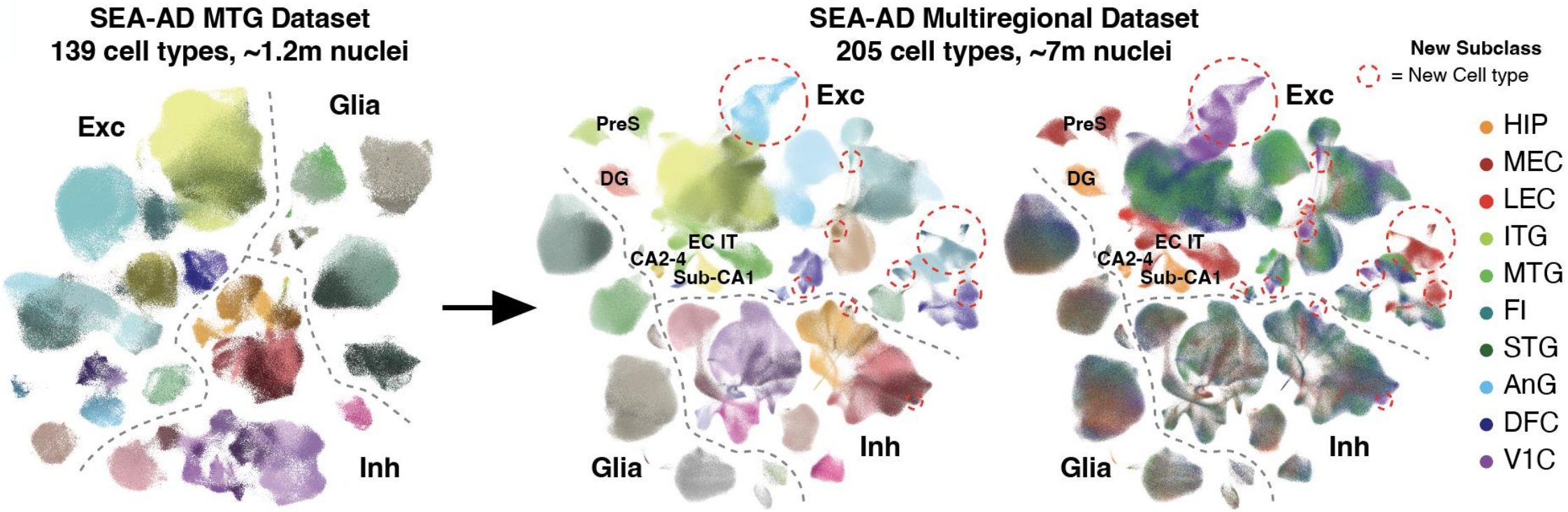
pTau





Agarwal et al (2025)

All -omics data mapped to an expanded BICAN taxonomy to account for regional cell types and AD-specific states



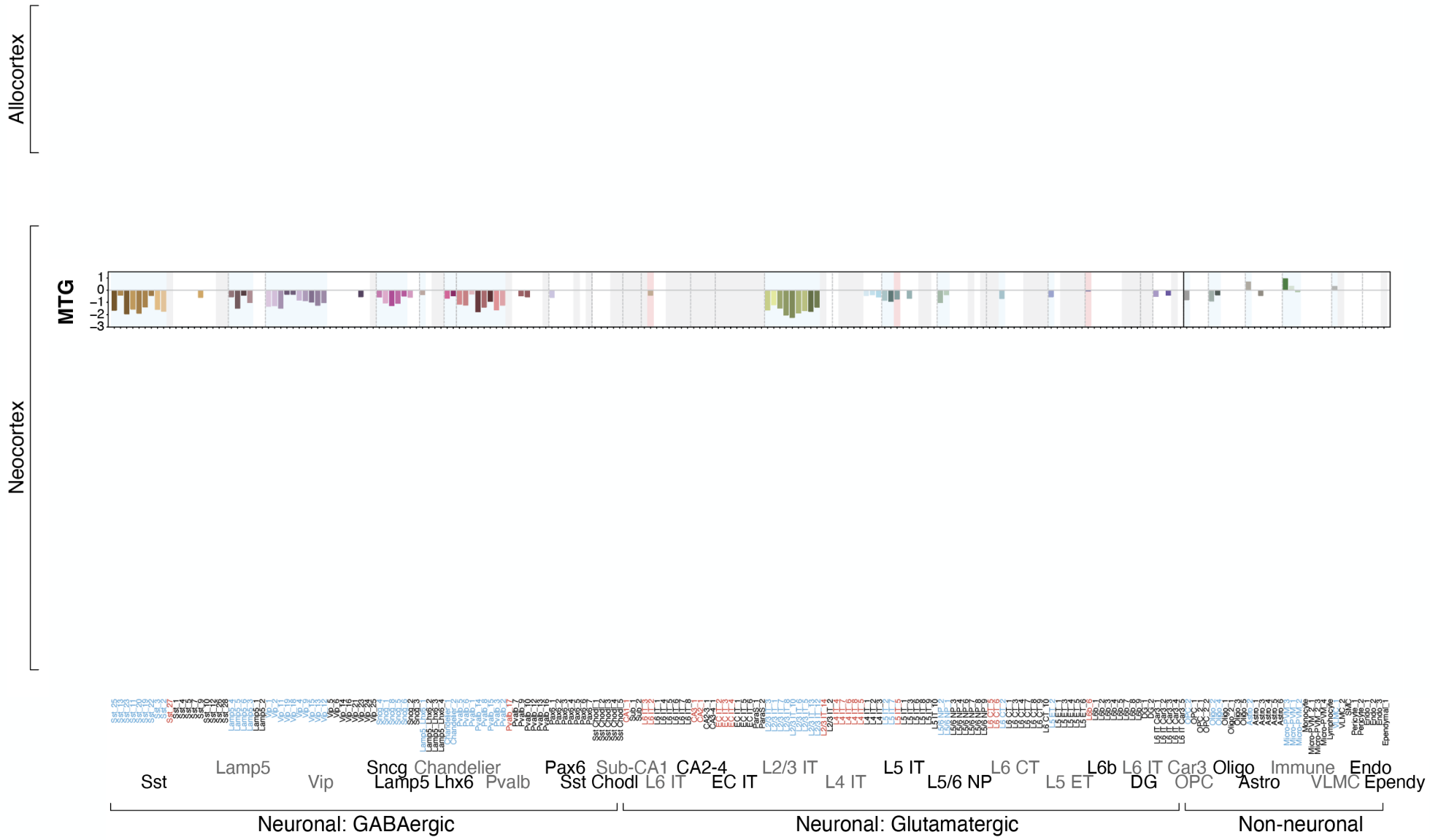
Common changes to cell types shared across brain regions

AD Spectrum+
AD Spectrum only

Cellular Abundances ~ **Local PS** + Sex + Age at Death + APOE4 Status + PMI

type not in region
regional change
shared change

Effect size Local CPS of each cell type across brain regions



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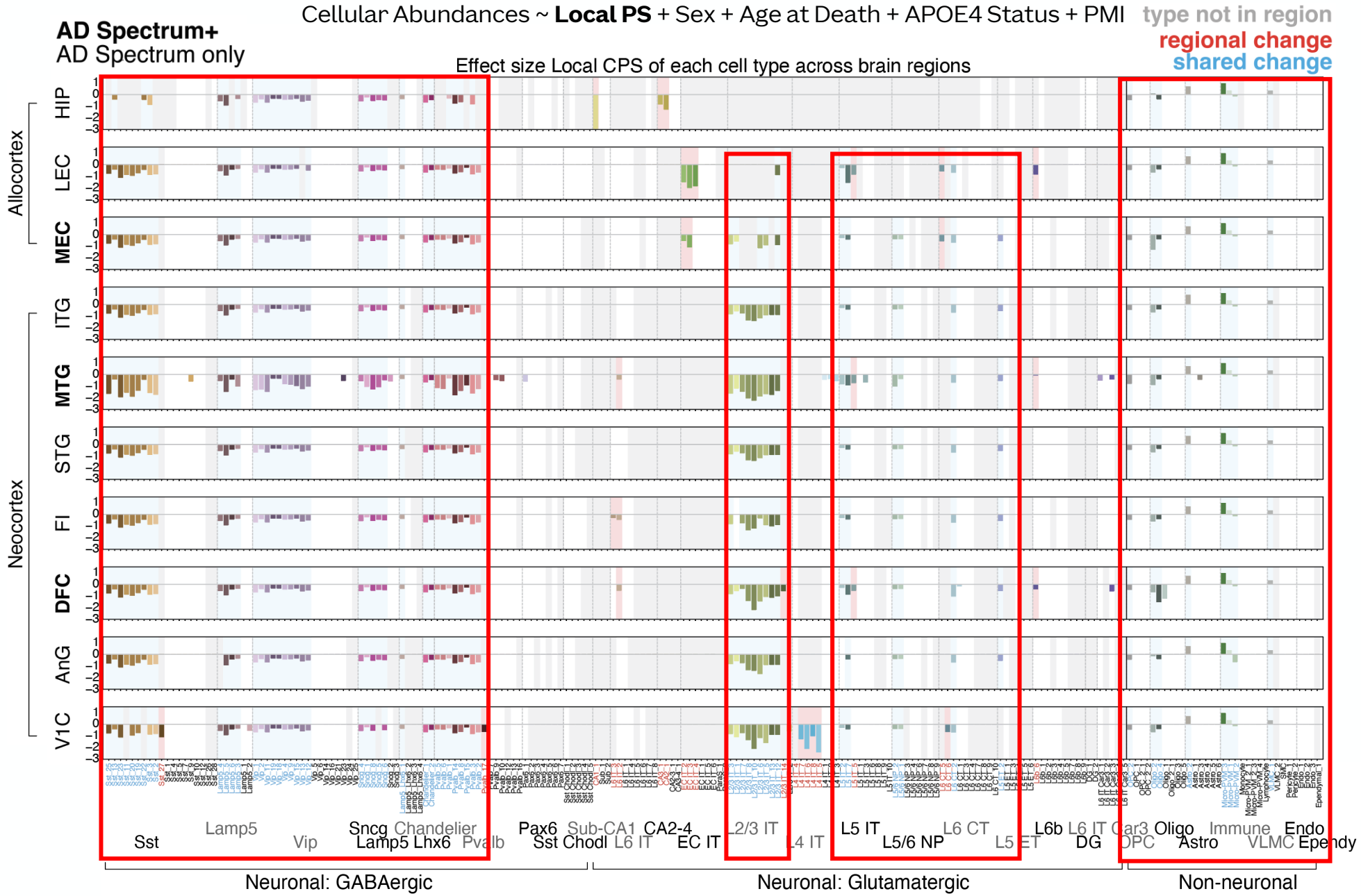
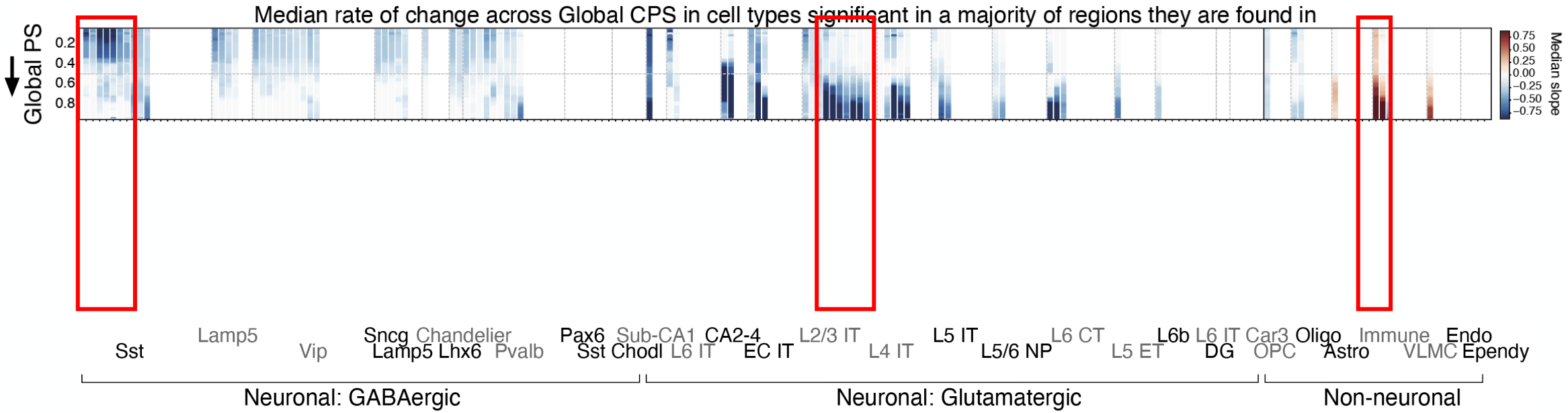
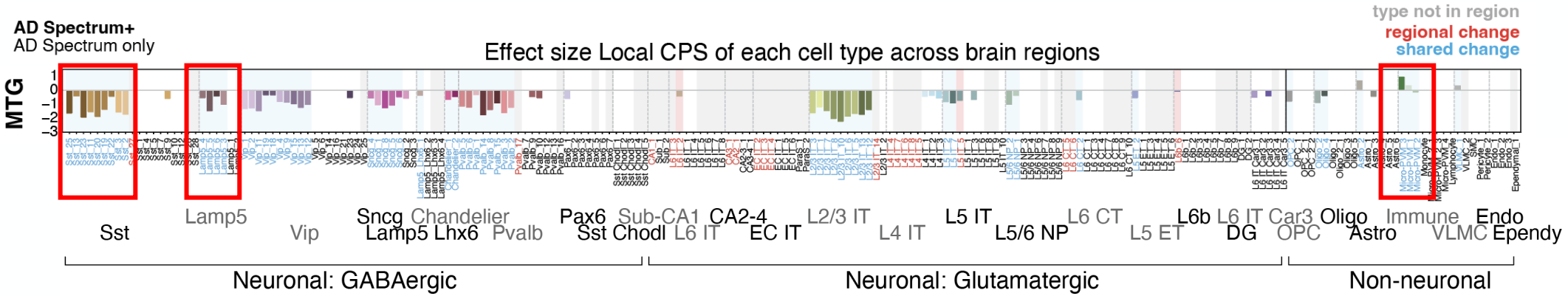
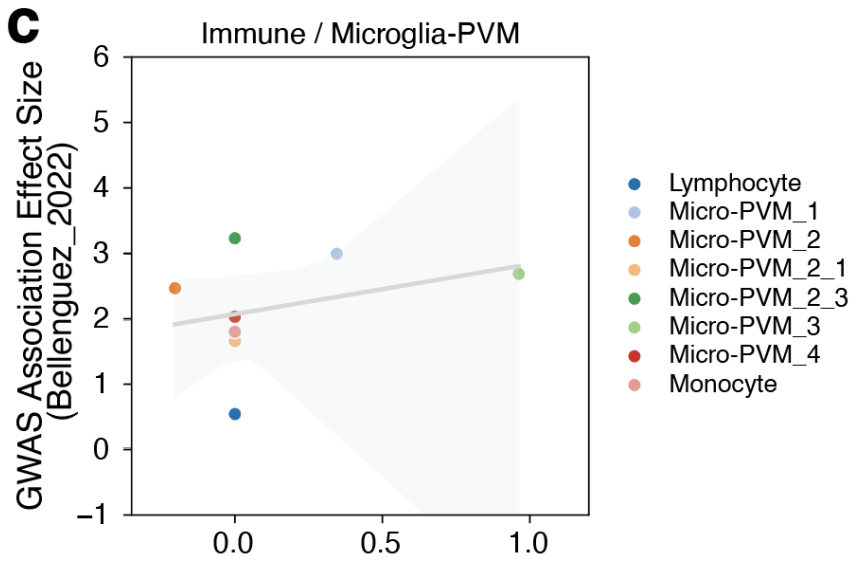


Inhibitory interneurons loss occurs early in pseudo-progression and is more associated with amyloid accumulation



Early affected Sst and Lamp5 interneurons have enriched expression of genes prioritized by AD GWAS

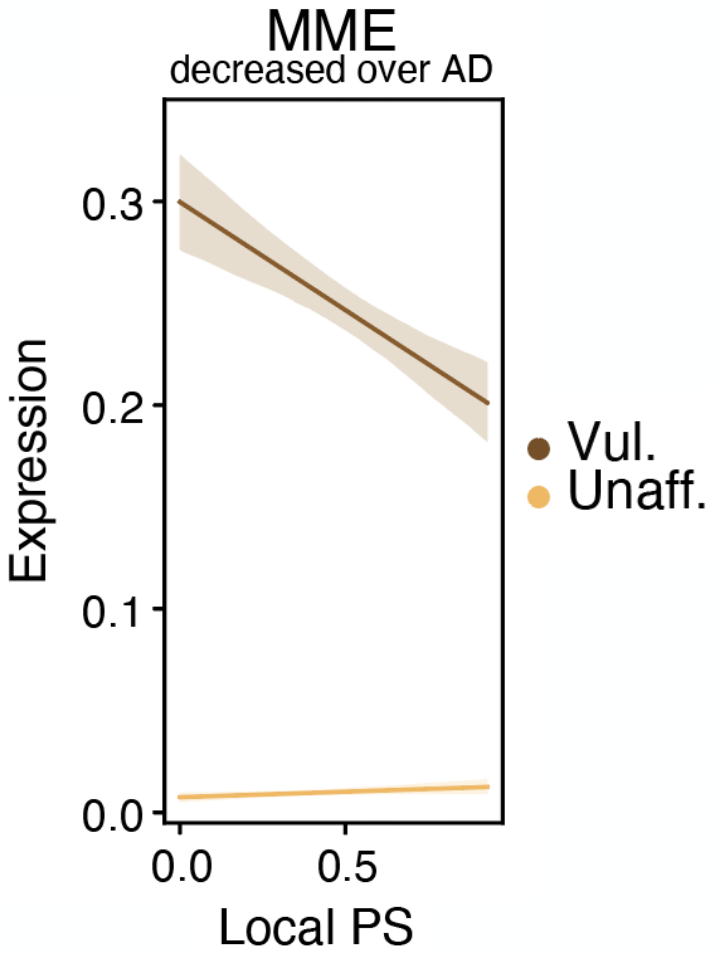




In collaboration with Laramie Duncan and Hazal Senturk

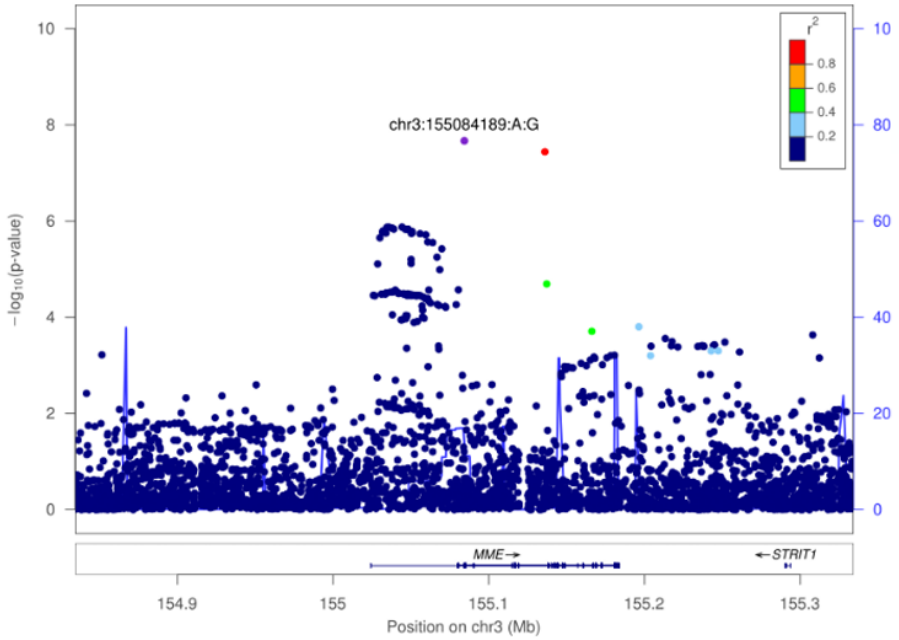
Including of MME in Sst, which encodes a protease that degrades amyloid beta plaques

Expression ~ **Local PS** + ...
+ (1|Region) + (1|Donor ID_Reigon)

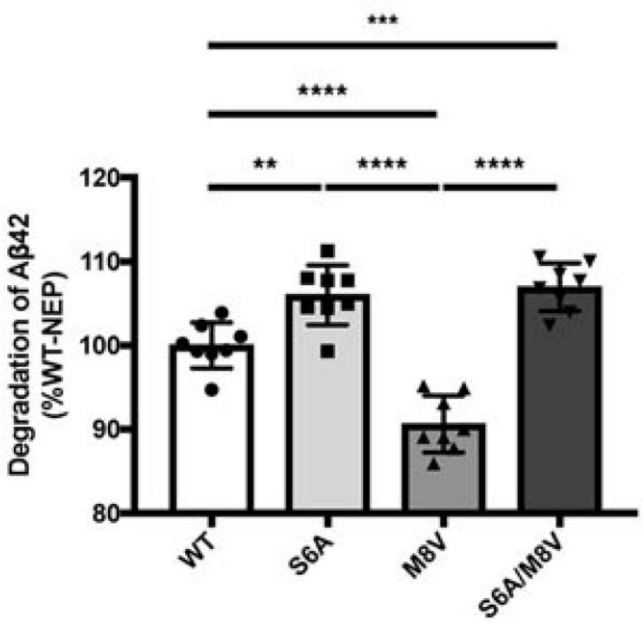


M8V coding variant in MME increases risk of AD (OR=1.16) and decreases enzyme activity towards amyloid beta (in culture)

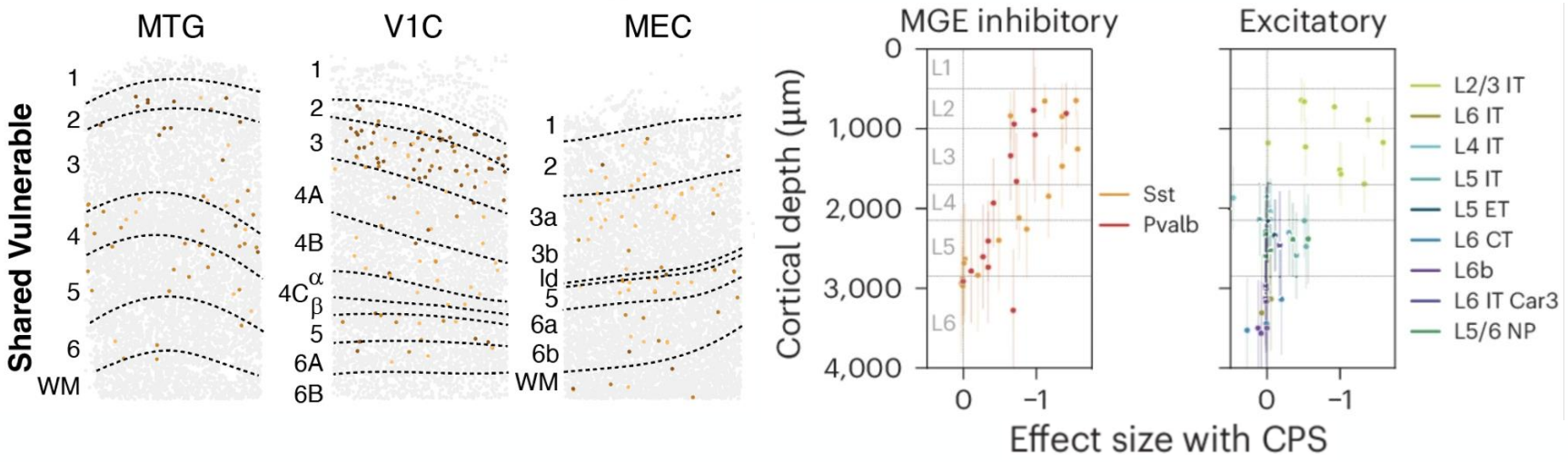
Bellenguez et al (2022) *Nature Genetics*

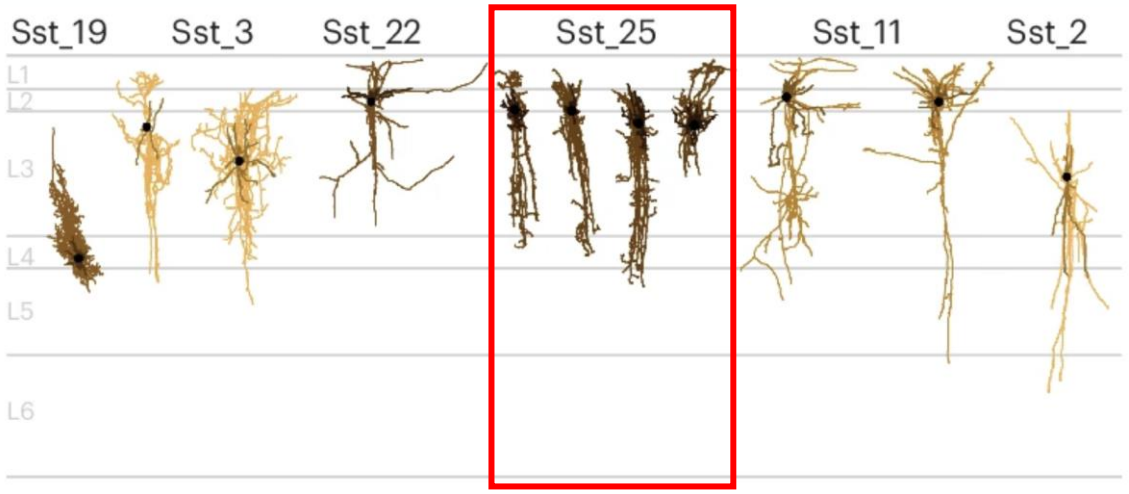


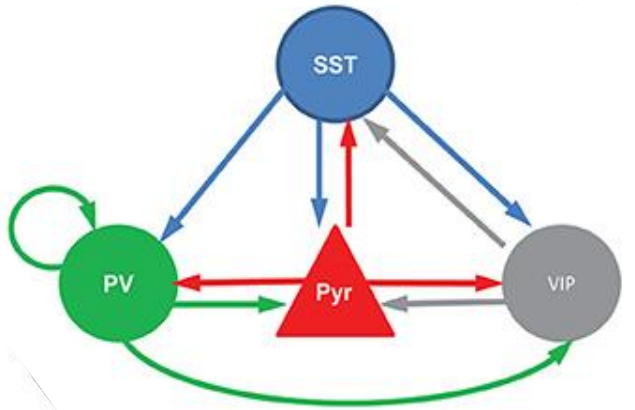
Morito T. (2025) *J Neuro*



Vulnerable Sst neurons are lost early in AD and localize to upper cortical layers, nearby excitatory neurons lost later in AD

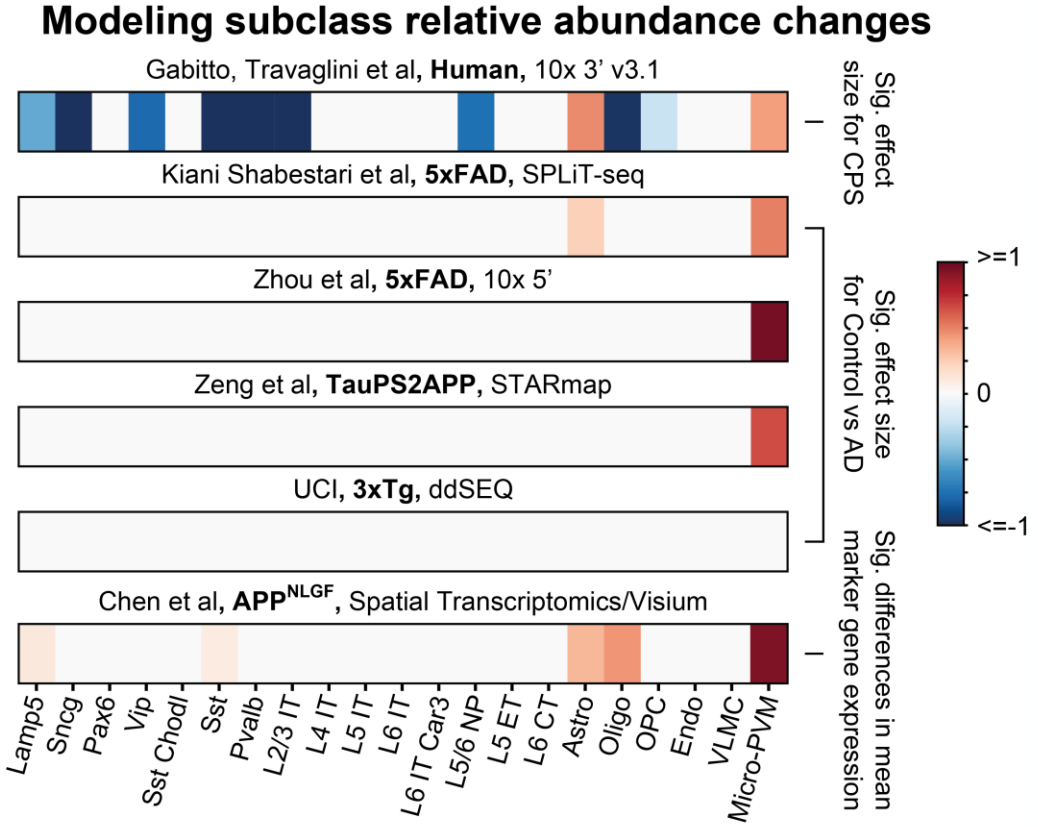


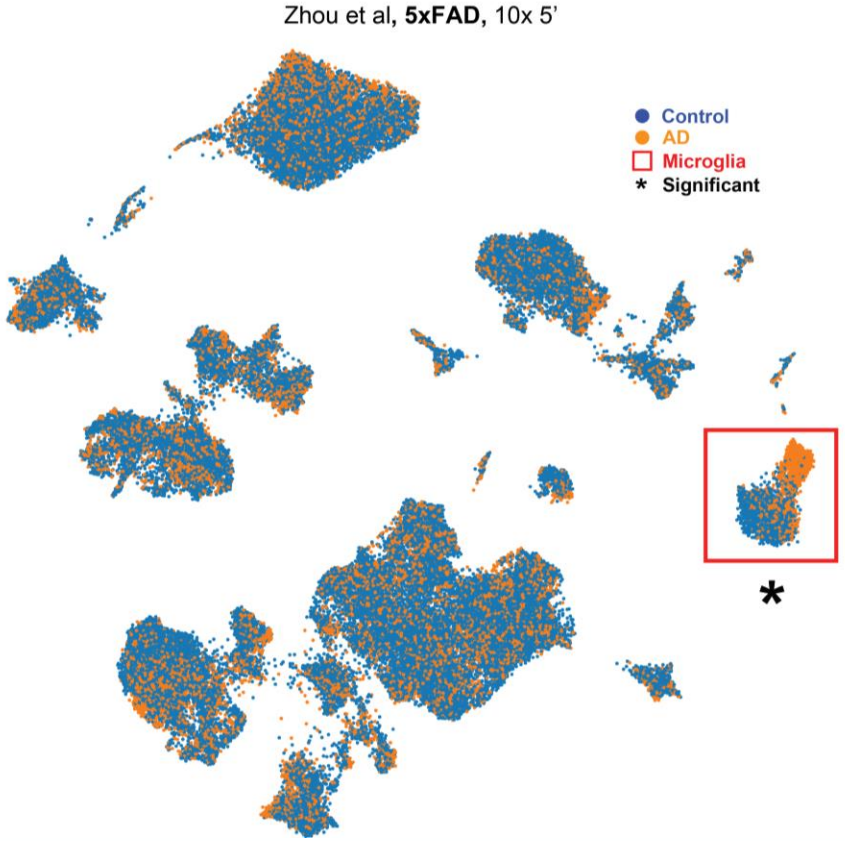




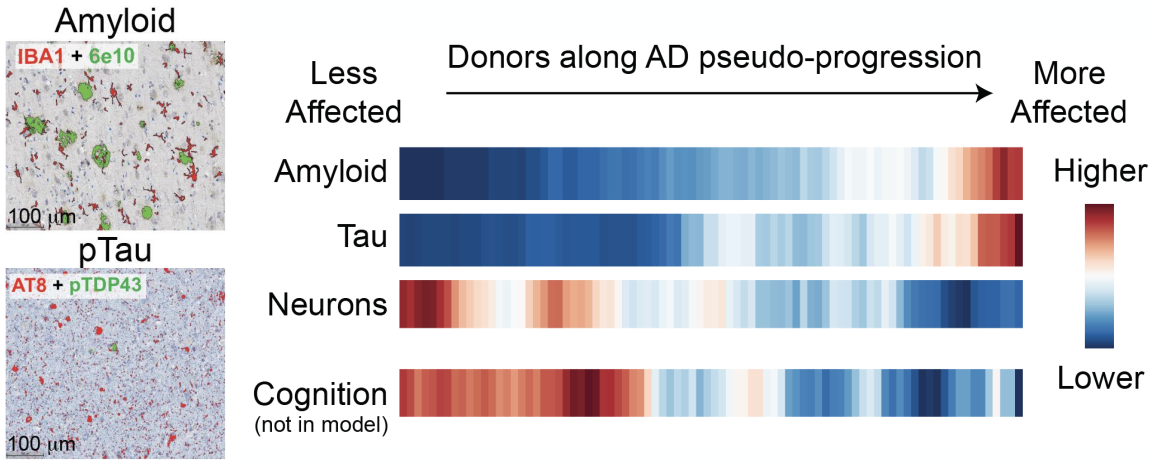
Lee et al (2017)

Similar re-processing in datasets from AD mouse models capture microglial states but not selective neuronal loss

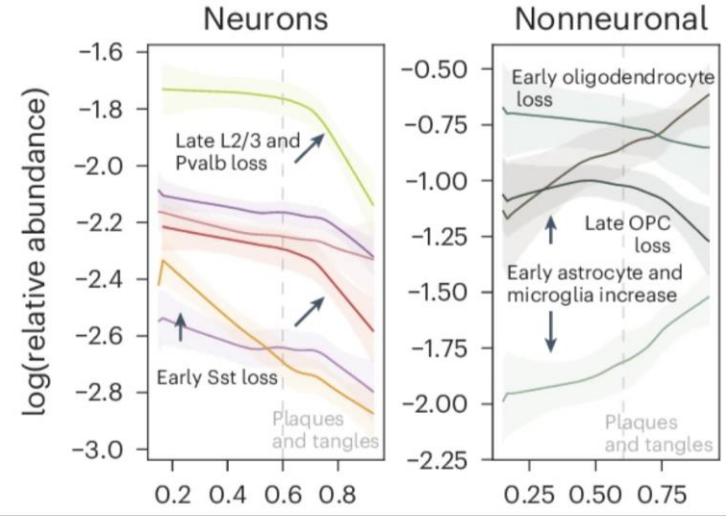




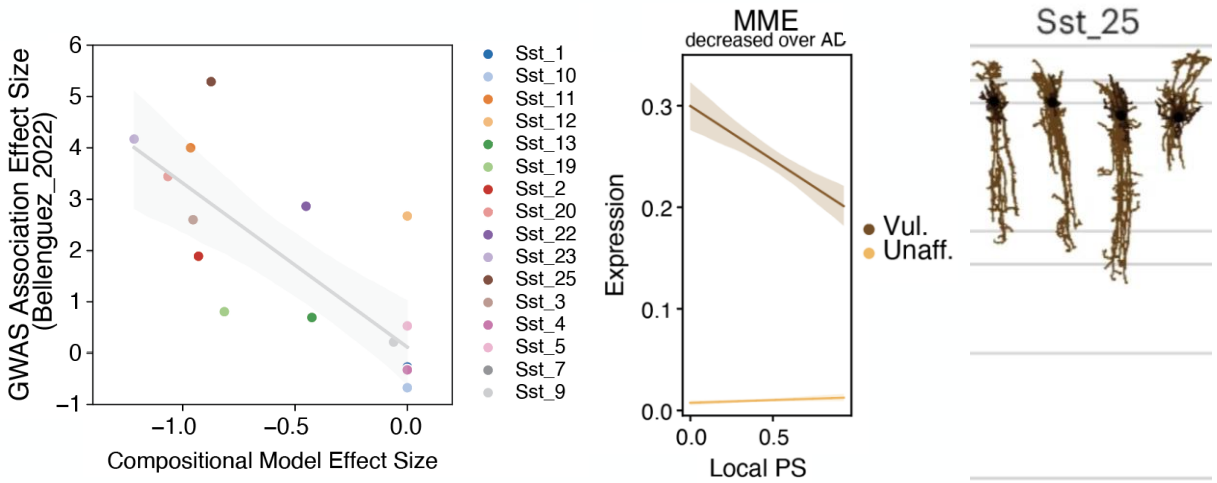
Learn a data-driven “pseudo-progression” across donors with quantitative neuropathology



Determine cellular changes associated with pseudo-progression and validate them



Prioritize cellular and molecular changes more likely to related to early disease progression



Seattle Alzheimer's Disease Brain Cell Atlas (SEA-AD)

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agenda/morning

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What's in the Brain Knowledge Platform?

Rachel Hostetler, Scientist I

allen institute/

allenins

brain knowledge platform (BKP)/

brain-map.org/bkp

browse

Find data in the Data Catalog

explore

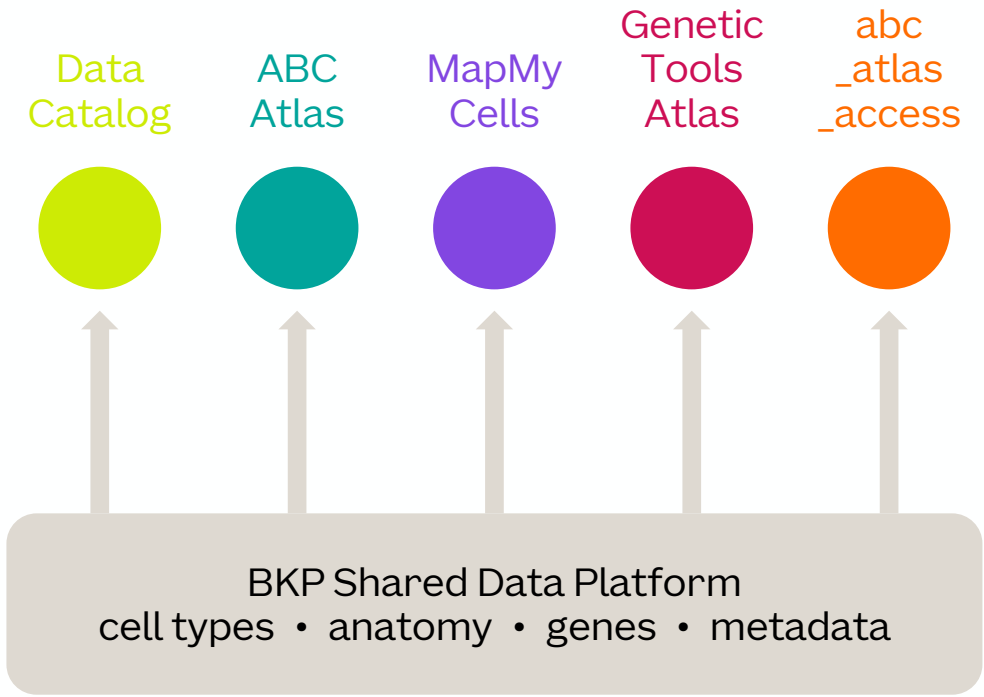
Visualize and compare data in ABC Atlas

analyze

Label and align your data with MapMyCells

experiment

Design experiments with the Genetic Tools Atlas



develop

Download and work with data programmatically with abc_atlas_access

bkp/ browse

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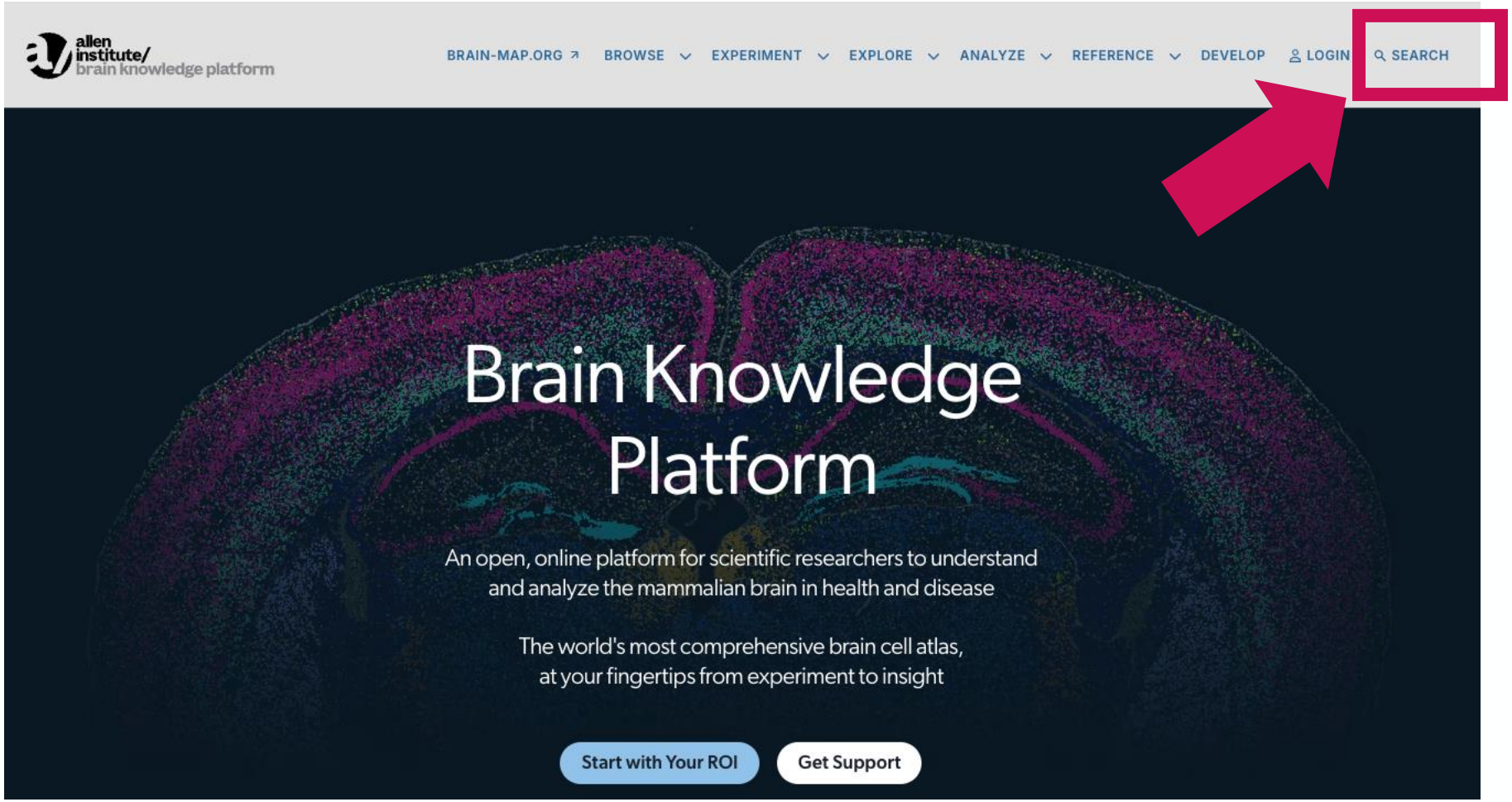
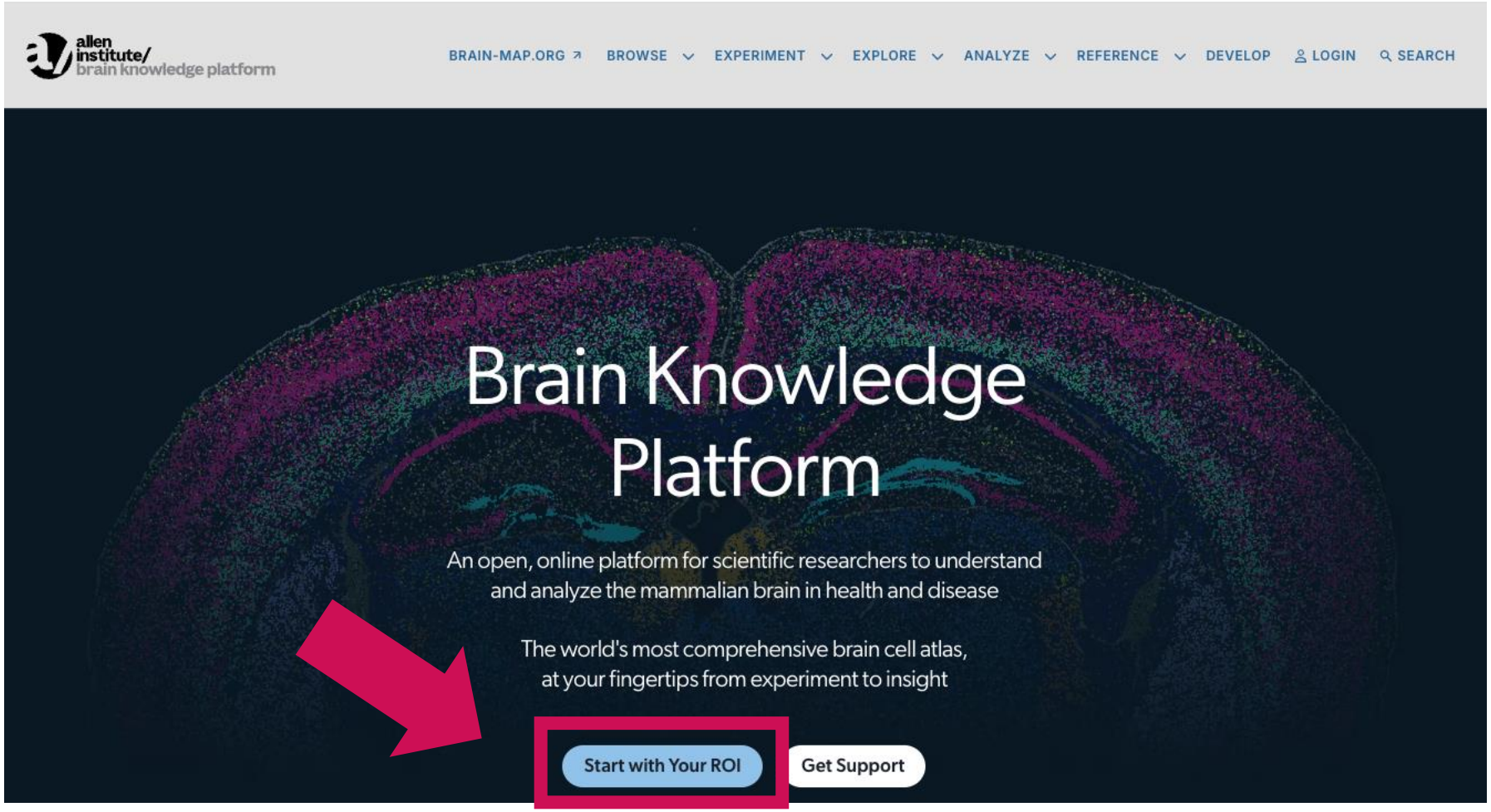


brain-map.org/bkp

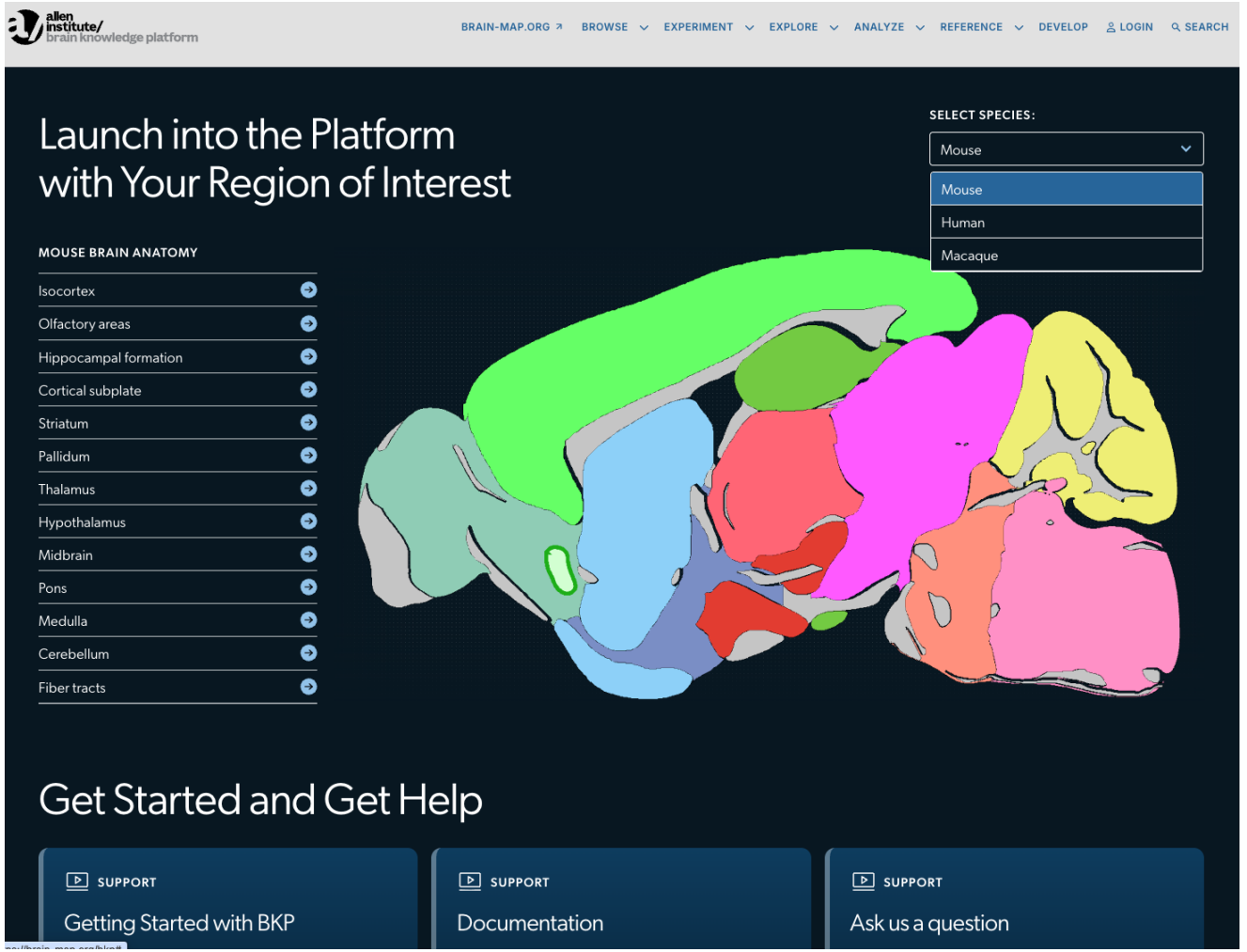
bkp/ browse



brain-map.org/bkp

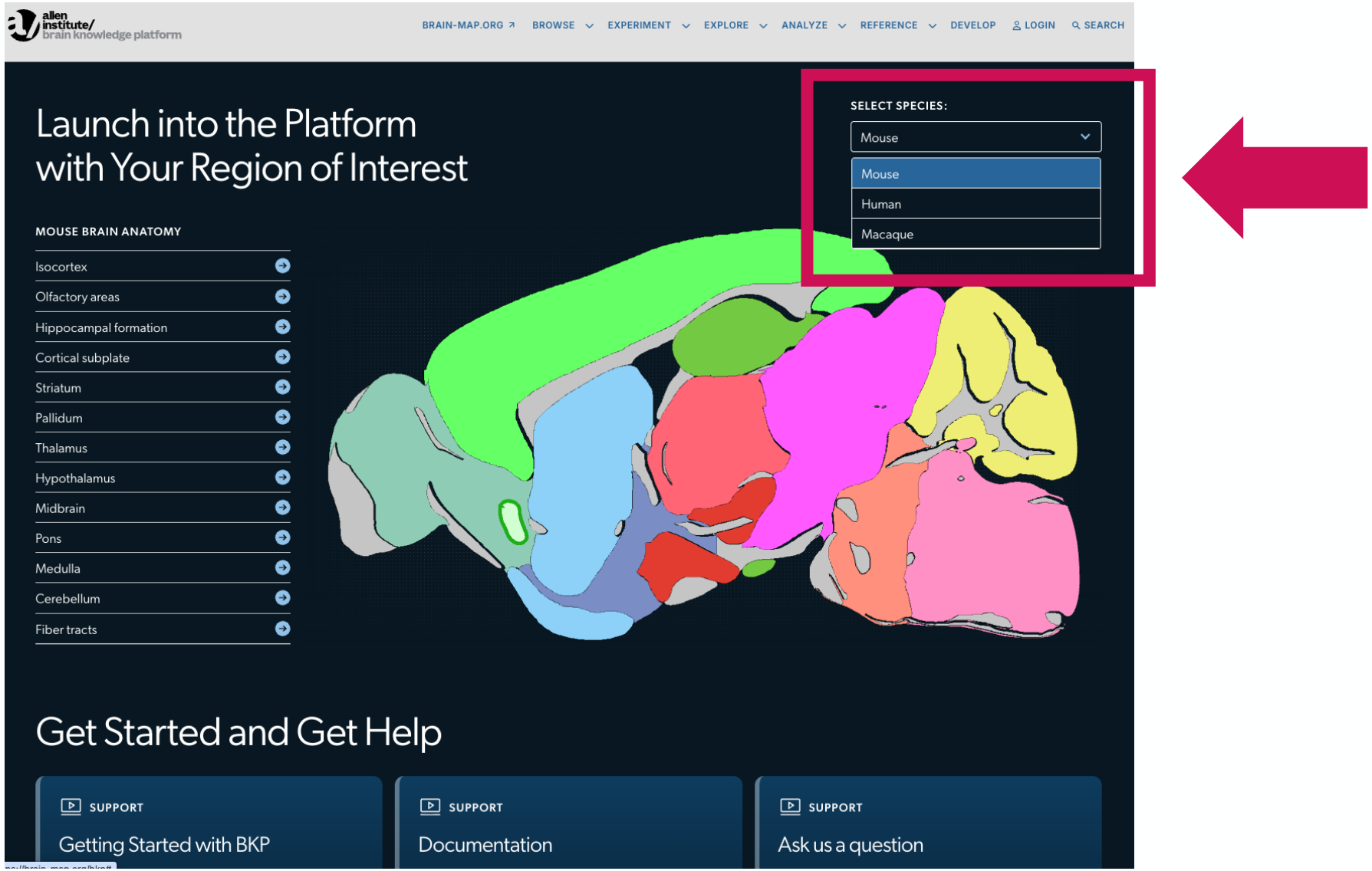
bkp/ browse

- The Brain Knowledge Platform's Data Catalog offers access to a wide range of projects & data collections
- Find relevant-to-you datasets to accelerate your research
- Filter and search by species, modality, technique, author, specimen type, and more



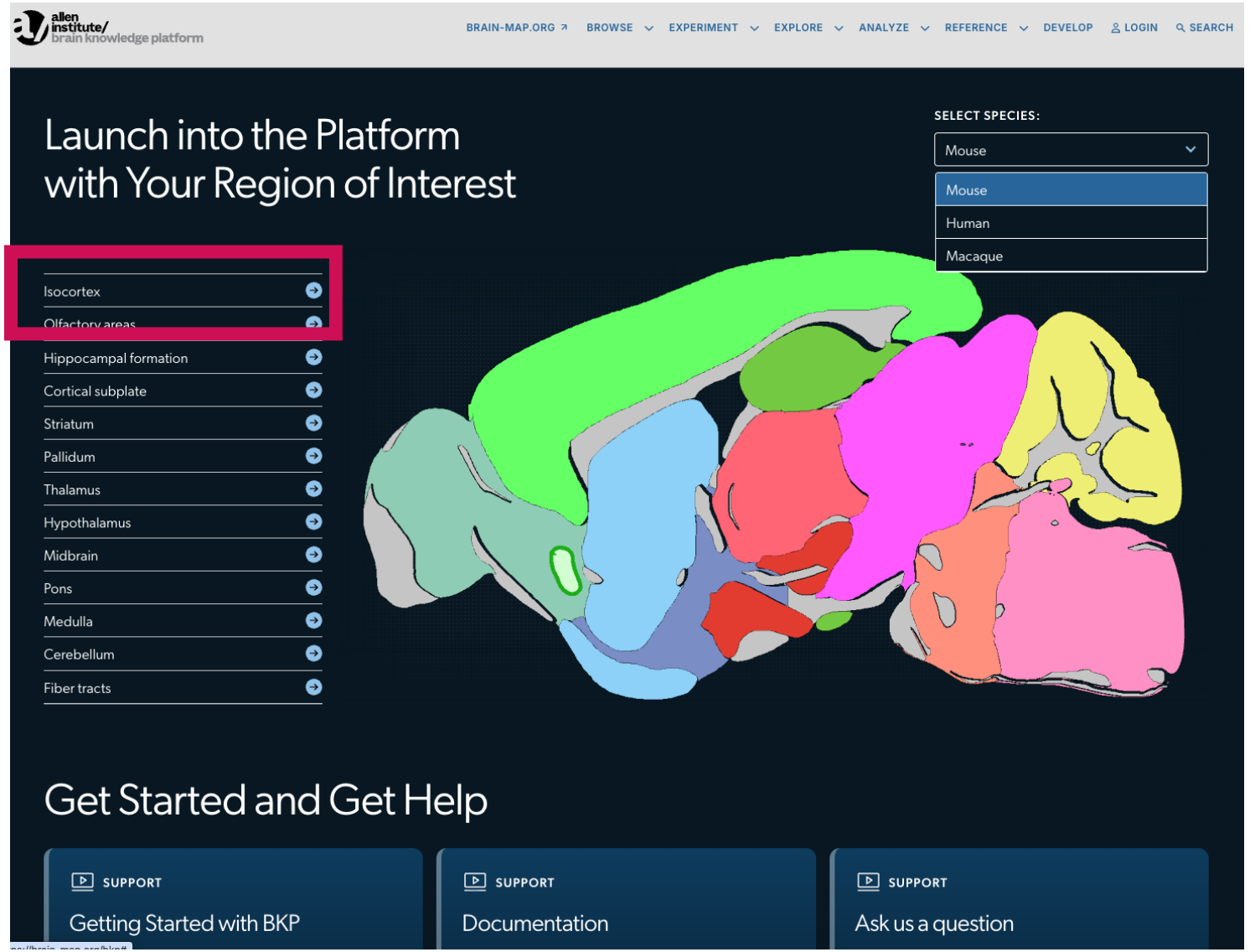
brain-map.org/bkp

bkp/ browse



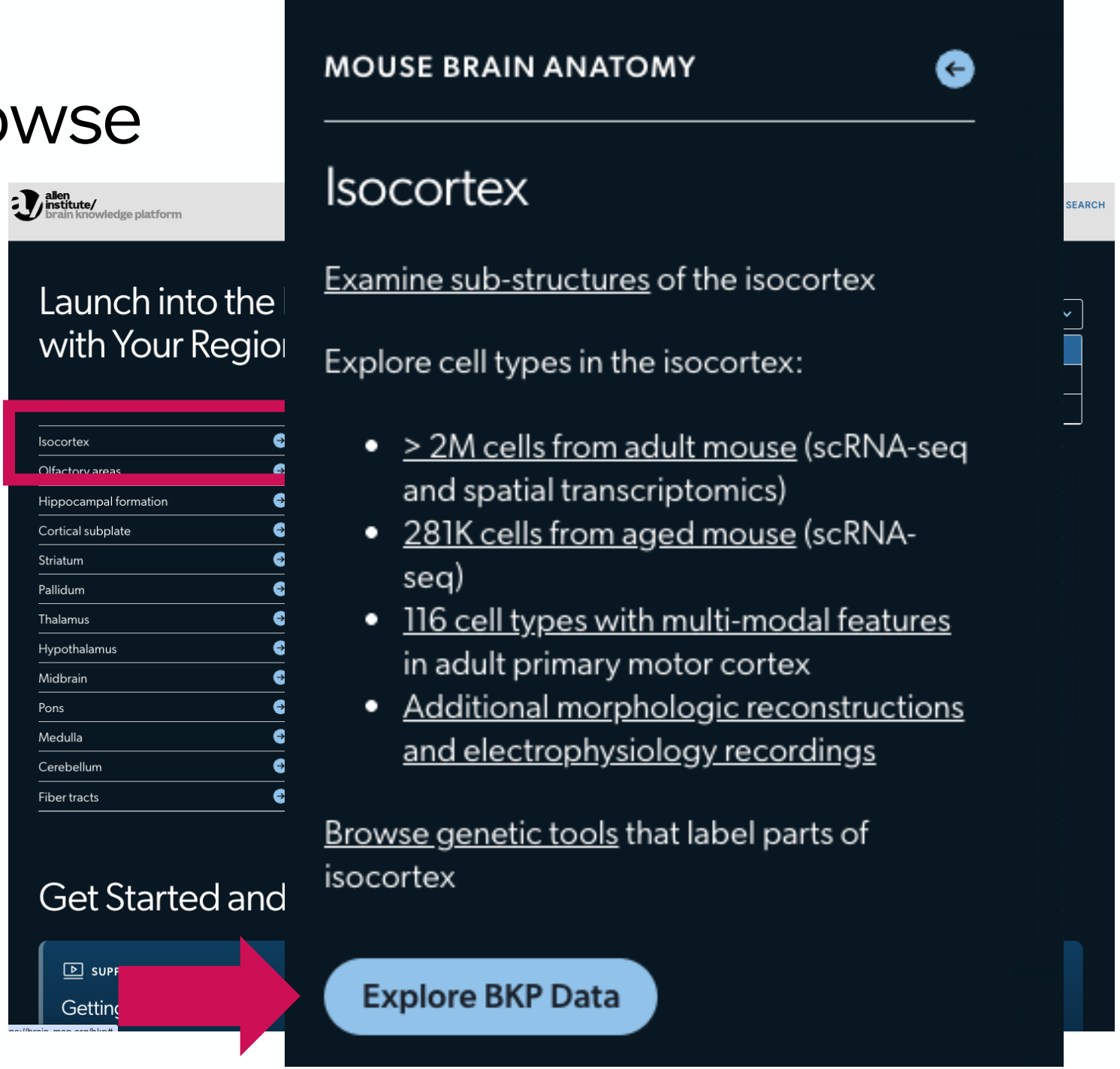
brain-map.org/bkp

bkp/ browse



brain-map.org/bkp

bkp/ browse



brain-map.org/bkp

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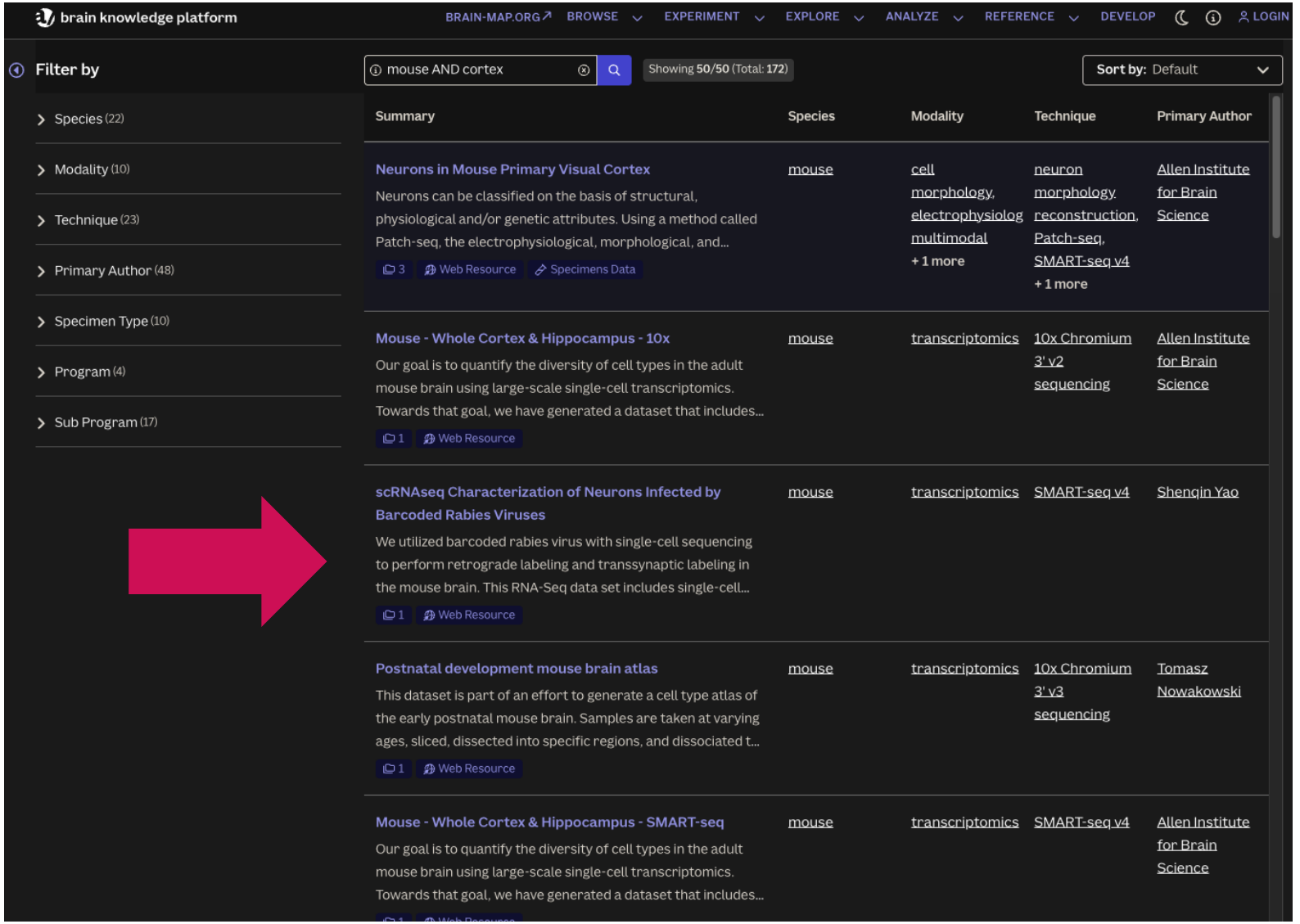
brain knowledge platform
BRAIN-MAP.ORG
BROWSE
EXPERIMENT
EXPLORE
ANALYZE
REFERENCE
DEVELOP
LOGIN
Filter by
mouse AND cortex
Showing 50/50 (Total: 172)
Sort by: Default
Species (22)
Modality (10)
Technique (23)
Primary Author (48)
Specimen Type (10)
Program (4)
Sub Program (17)
Summary
Species
Modality
Technique
Primary Author
Neurons in Mouse Primary Visual Cortex
Neurons can be classified on the basis of structural, physiological and/or genetic attributes. Using a method called Patch-seq, the electrophysiological, morphological, and...
3
Web Resource
Specimens Data
mouse
cell morphology, electrophysiolog multimodal
+ 1 more
neuron morphology reconstruction, Patch-seq, SMART-seq v4
+ 1 more
Allen Institute for Brain Science
Mouse - Whole Cortex & Hippocampus - 10x
Our goal is to quantify the diversity of cell types in the adult mouse brain using large-scale single-cell transcriptomics. Towards that goal, we have generated a dataset that includes...
1
Web Resource
mouse
transcriptomics
10x Chromium 3' v2 sequencing
Allen Institute for Brain Science
scRNAseq Characterization of Neurons Infected by Barcoded Rabies Viruses
We utilized barcoded rabies virus with single-cell sequencing to perform retrograde labeling and transsynaptic labeling in the mouse brain. This RNA-Seq data set includes single-cell...
1
Web Resource
mouse
transcriptomics
SMART-seq v4
Shenqin Yao
Postnatal development mouse brain atlas
This dataset is part of an effort to generate a cell type atlas of the early postnatal mouse brain. Samples are taken at varying ages, sliced, dissected into specific regions, and dissociated t...
1
Web Resource
mouse
transcriptomics
10x Chromium 3' v3 sequencing
Tomasz Nowakowski
Mouse - Whole Cortex & Hippocampus - SMART-seq
Our goal is to quantify the diversity of cell types in the adult mouse brain using large-scale single-cell transcriptomics. Towards that goal, we have generated a dataset that includes...
mouse
transcriptomics
SMART-seq v4
Allen Institute for Brain Science

bkp/ explore

browse

Find data in the Data Catalog

explore

Visualize and compare data in ABC Atlas

analyze

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Design experiments with the Genetic Tools Atlas

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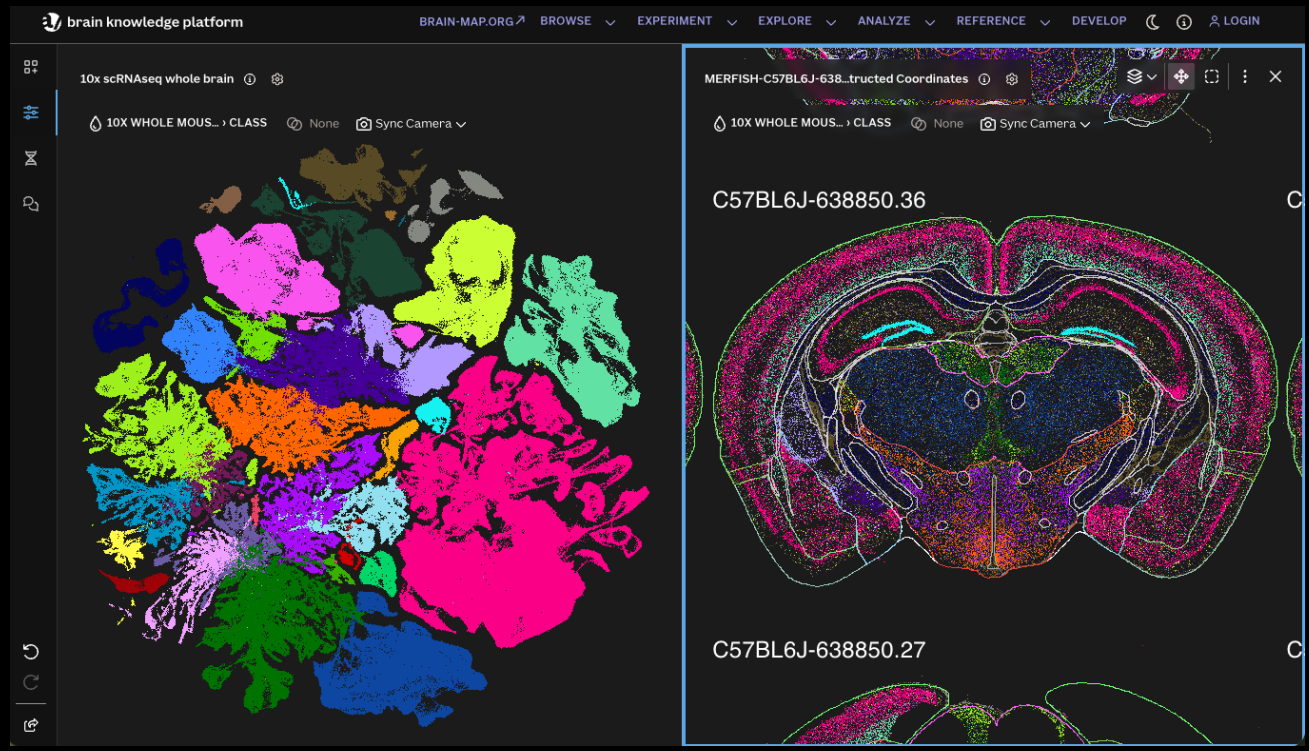
Download and work with data programmatically with abc_atlas_access

ABC atlas/ what is it?

- a comprehensive data visualization platform/

- 28 high-quality data sets
- 20+ million cells



performance at scale
supports viewing hundreds of millions of points simultaneously and many thousands of cell type clusters

integrated
visualize connected data across species, modality, development, and condition, linked by shared metadata

accessible
free, web-based access serving ~3,500 monthly users

brain-map.org/explore

ABC atlas/ what's in it?

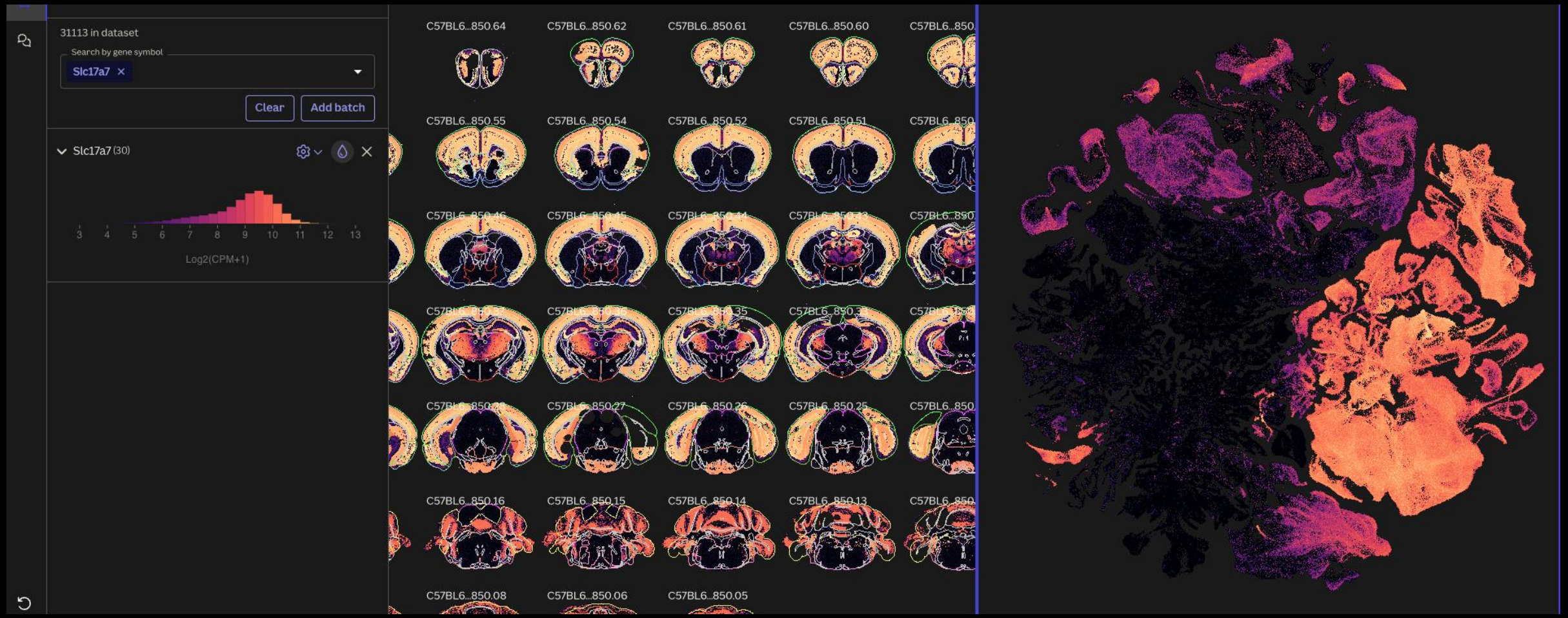
- mouse

- Whole Mouse Brain (scRNAseq and MERFISH data)
  - Yao et al., 2023, Zhang et al., 2023 – Nature
- Consensus Whole Mouse Brain (scRNAseq)
  - Yao et al., 2023, Langlieb et al., 2023 – Nature
- Aging Mouse (scRNAseq)
  - Jin et al., 2025 – Nature
- Developing Mouse Visual Cortex (scRNAseq)
  - Gao et al., 2025 – Nature

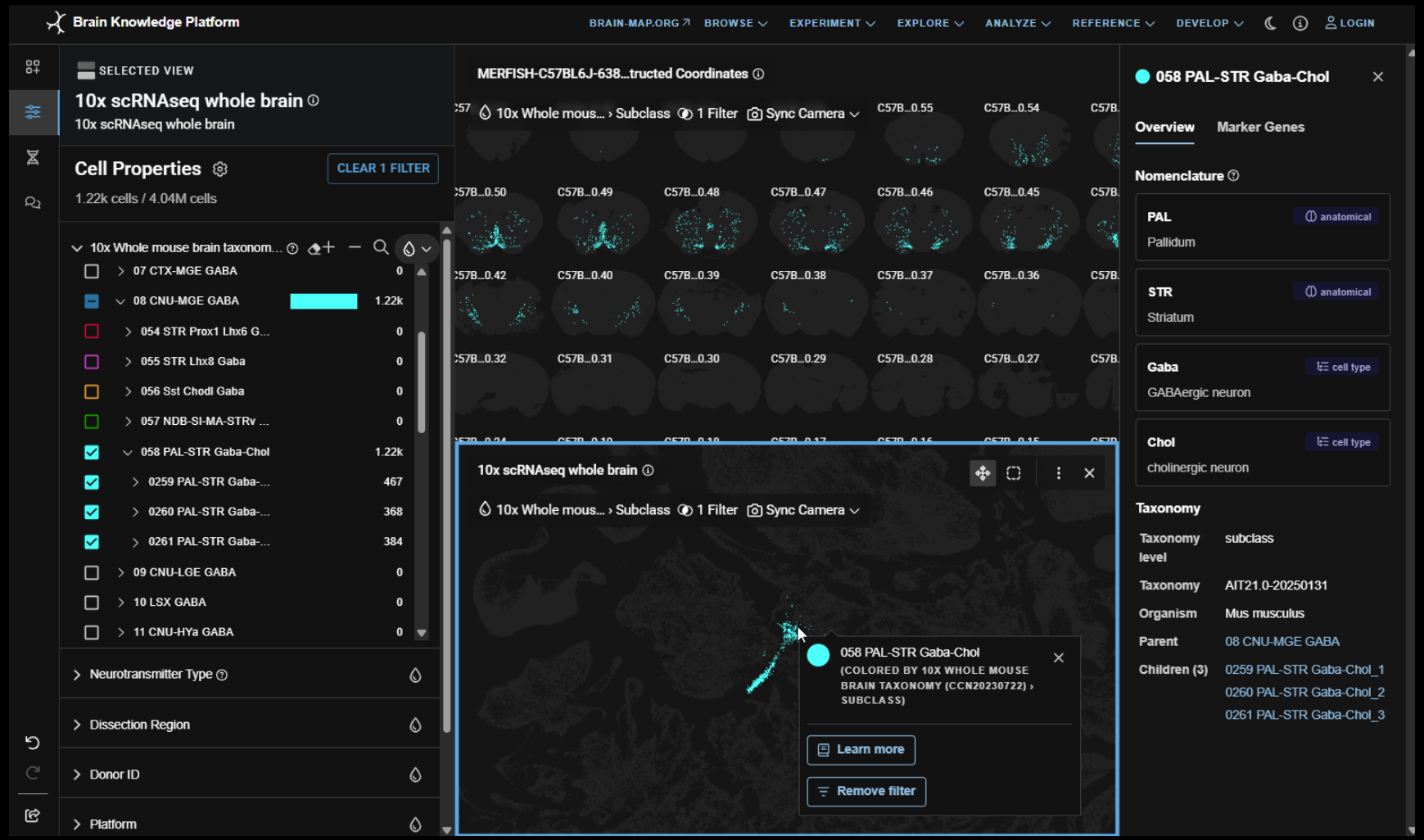
- human & NHP

- Whole Human Brain (snRNAseq)
  - Siletti et al., 2023 – Science
- Human Alzheimer's Disease – MTG and DLPFC (SEA-AD) (snRNAseq and MERFISH data)
  - Gabitto & Travaglini et al., 2024 – Nature Neuroscience
  - Travaglini, Gabitto, Ding, et al 2026 – bioRxiv
  - Kana et al 2026 - bioRxiv
- Human Parkinson's Disease (ASAP Human Postmortem BrainSeq) (snRNAseq)
  - (No publication—collaboration with MJFF/ASAP)
- Cross-species HMBA Basal Ganglia (snRNAseq and MERFISH data)
  - Johansen & Fu et al, 2026
  - Hewitt & Turner et al, 2026 - bioRxiv

ABC atlas/ what can you do/ view gene expression



ABC atlas/ what can you do/ explore cell types



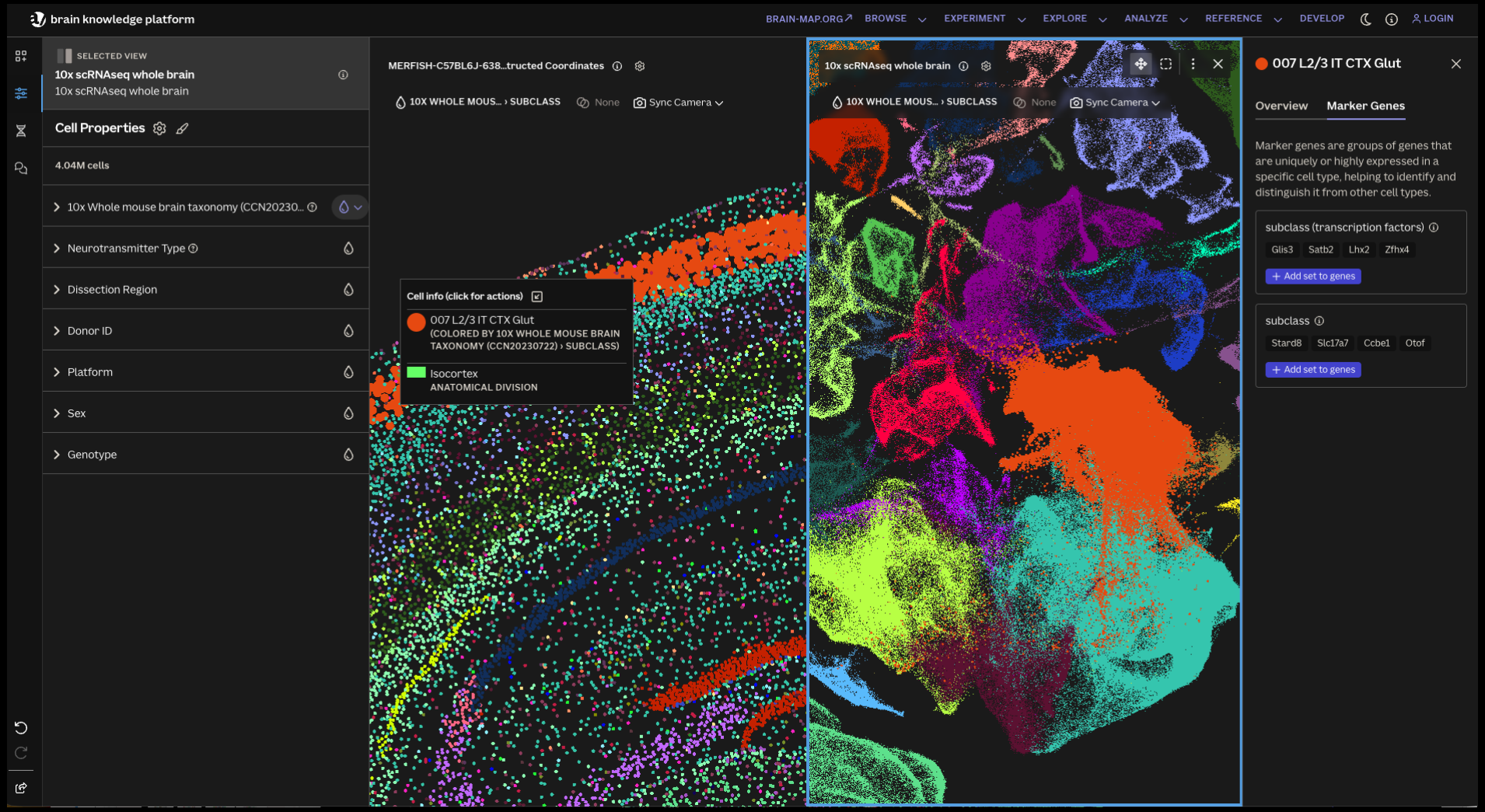
What do the acronyms mean?

How is this cell type defined?

What else is known about this cell type?

brain-map.org/explore

ABC atlas/ what can you do/ explore cell types



brain-map.org/explore

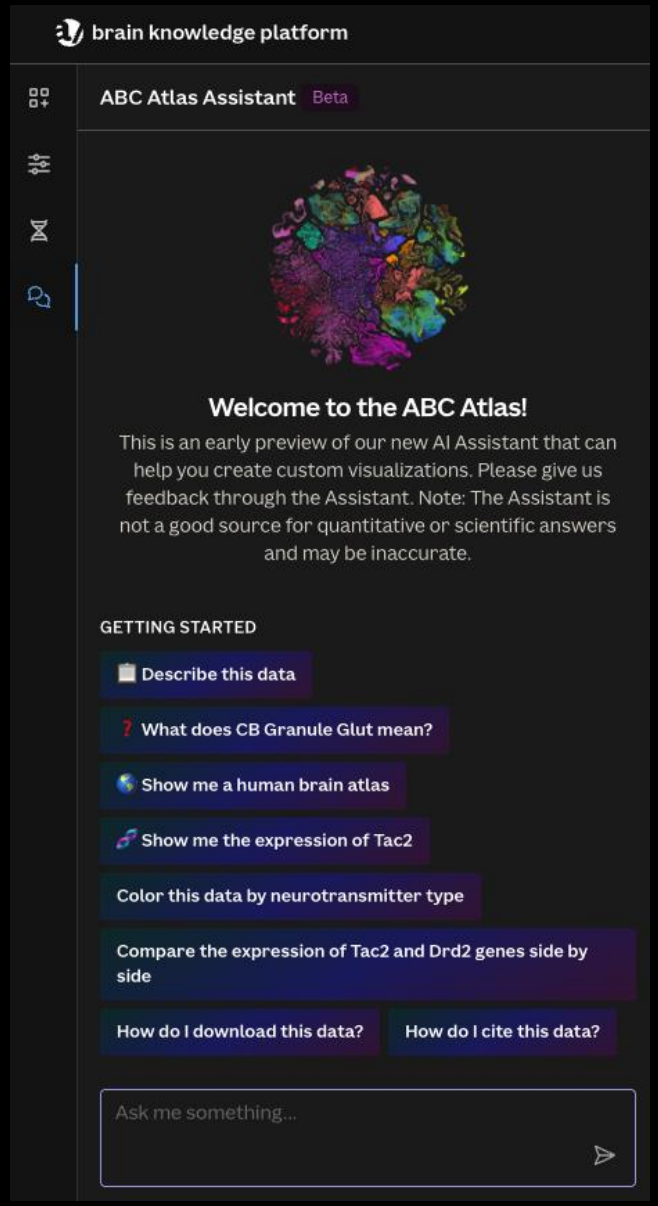
ABC atlas/ ask the ABC atlas assistant!

the AI Assistant uses natural language input to help users create custom visualizations faster

- simplifies complex tasks for new users
- accelerates advanced controls for experienced users

the Assistant can:

- describe, suggest, and open data sets
- adjust colors and filters across multiple data sets based on user instruction
- offer “how to” instructions



brain-map.org/explore

bkp/ explore

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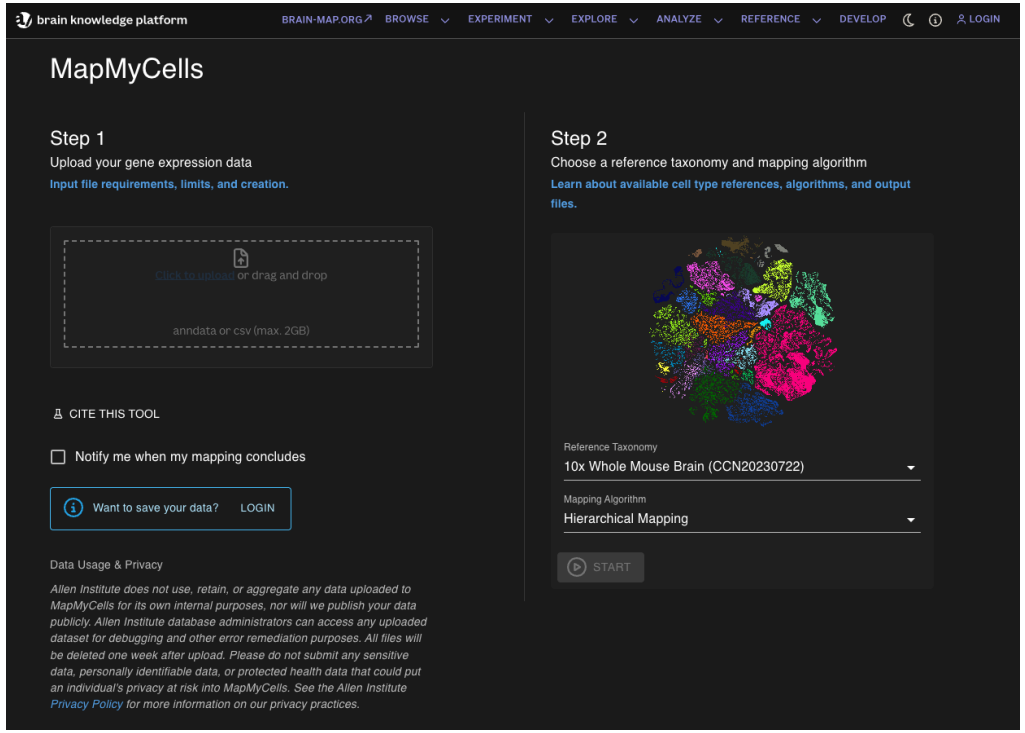
MapMyCells/ map your own RNAseq data onto an Allen reference taxonomy

- 6 cell type taxonomies
- 3 mapping algorithms



input

.h5ad file, .csv file, or cellxgene matrix

output

- .csv file
- cell IDs with allen institute taxonomy labels & probabilities
- .JSON file (metadata)

also available for offline programmatic access:
github.com/AllenInstitute/cell_type_mapper

brain-map.org/bkp/analyze

MapMyCells/ map your own RNAseq data onto an Allen reference taxonomy

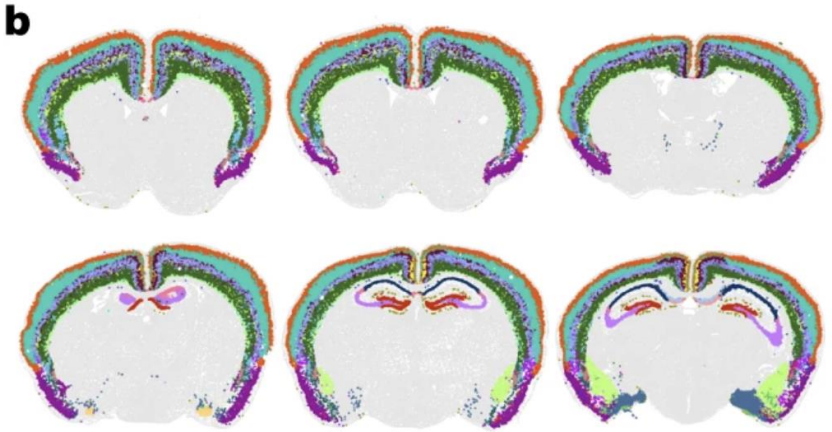
Purpose: "I have unlabeled cell-by-gene data. What cell types are in my data, according to the Allen Institute?“

Example use cases:

I have spatial transcriptomic data. What cell types are represented there? Do cell types segregate spatially? (actual use case in Yao et al.)

I have data annotated with cell types by [some other method]. What cell types does the Allen Institute think I have? How do the two annotations align?

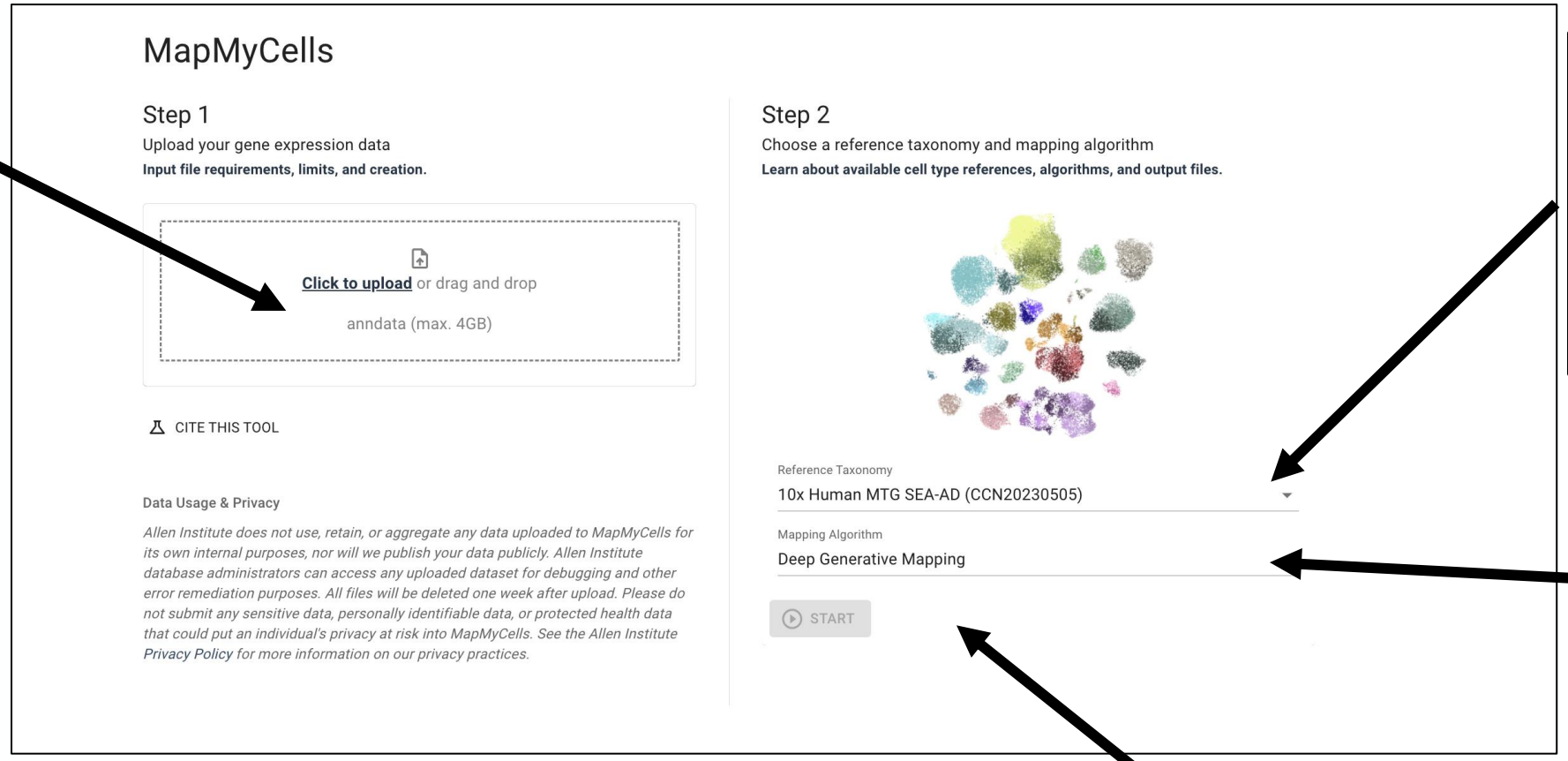
I have developmental or disease model data. How does the distribution of cell types (according to MapMyCells) change with time?



brain-map.org/bkp/analyze

MapMyCells/ how does it work?

1) Drag-and-drop input file here



2) Select taxonomy to which to map
- **Whole Mouse Brain**
- **SEA-AD (human MTG)**

3) Select mapping algorithm

4) Click "Start"

https://knowledge.brain-map.org/mapmycells/process/

MapMyCells/ what is the output?

CSV output

Rows denoted by '#' are metadata

Each row is a cell from the input file.

Confidence in mapping

Machine-readable assignments

Human-readable assignments

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	P
1	# metadata =	ensembl_dot_10xWholeMouseBrain(CCN20230722)_HierarchicalMapping_UTC_1697746016681.json													
2	# taxonomy h	"CCN202307	"CCN202307	"CCN20230722_CLUS"]											
3	# readable ta	"subclass"	"supertype"	"cluster"]											
4	# algorithm: 'hierarchical'; codebase: http://github.com/AllenInstitute/cell_type_mapper; version: 1.0.0														
5	cell_id	class_label	class_name	class_bootstr	subclass_labe	subclass_nam	subclass_boo	supertype_la	supertype_na	supertype_bo	cluster_label	cluster_name	cluster_alias	cluster_bootstrapping_probability	
6	cell_0	CS20230722	24 MY Glut	0.39	CS20230722	243 PGRN-PA	0.56	CS20230722	0979 PGRN-F	0.52	CS20230722	4398 PGRN-F	24411	0.52	
7	cell_1	CS20230722	30 Astro-Epe	0.4	CS20230722	318 Astro-NT	0.66	CS20230722	1159 Astro-N	0.56	CS20230722	5208 Astro-N	14869	0.56	
8	cell_2	CS20230722	10 LSX GABA	0.65	CS20230722	071 LSX Prdn	0.63	CS20230722	0327 LSX Prd	0.63	CS20230722	1161 LSX Prd	917	0.63	
9	cell_3	CS20230722	24 MY Glut	0.72	CS20230722	252 DMX VII	0.73	CS20230722	1010 DMX VI	0.91	CS20230722	4551 DMX VI	3893	0.91	
10	cell_4	CS20230722	20 MB GABA	0.98	CS20230722	205 SC-PAG I	0.38	CS20230722	0843 SC-PAG	0.27	CS20230722	3676 SC-PAG	3505	0.27	
11	cell_5	CS20230722	14 HY Glut	0.45	CS20230722	133 PVH-SO-	0.4	CS20230722	0586 PVH-SC	0.62	CS20230722	2371 PVH-SC	2117	0.62	
12	cell_6	CS20230722	10 LSX GABA	0.53	CS20230722	070 LSX Prdn	0.42	CS20230722	0318 LSX Prd	0.77	CS20230722	1127 LSX Prd	953	0.77	

MapMyCells/ what are the mapping algorithms?

Available algorithms:

- **Deep Generative mapping**
  - Use variational autoencoders to map labeled and unlabeled data to the same latent space. Assign cell types in that space.
  - Supported taxonomies:
    - SEA-AD (human MTG)
- **Hierarchical mapping**
  - Use Pearson's correlation coefficient between unlabeled cells and cell types to assign cell types hierarchically (e.g. class, then subclass, then cluster)
  - Supported taxonomies:
    - Whole Mouse Brain
    - SEA-AD (human MTG)
- **Flat mapping**
  - Hierarchical mapping, but maps directly to leaf node of taxonomy
  - Supported taxonomies:
    - Whole Mouse Brain
    - SEA-AD (human MTG)

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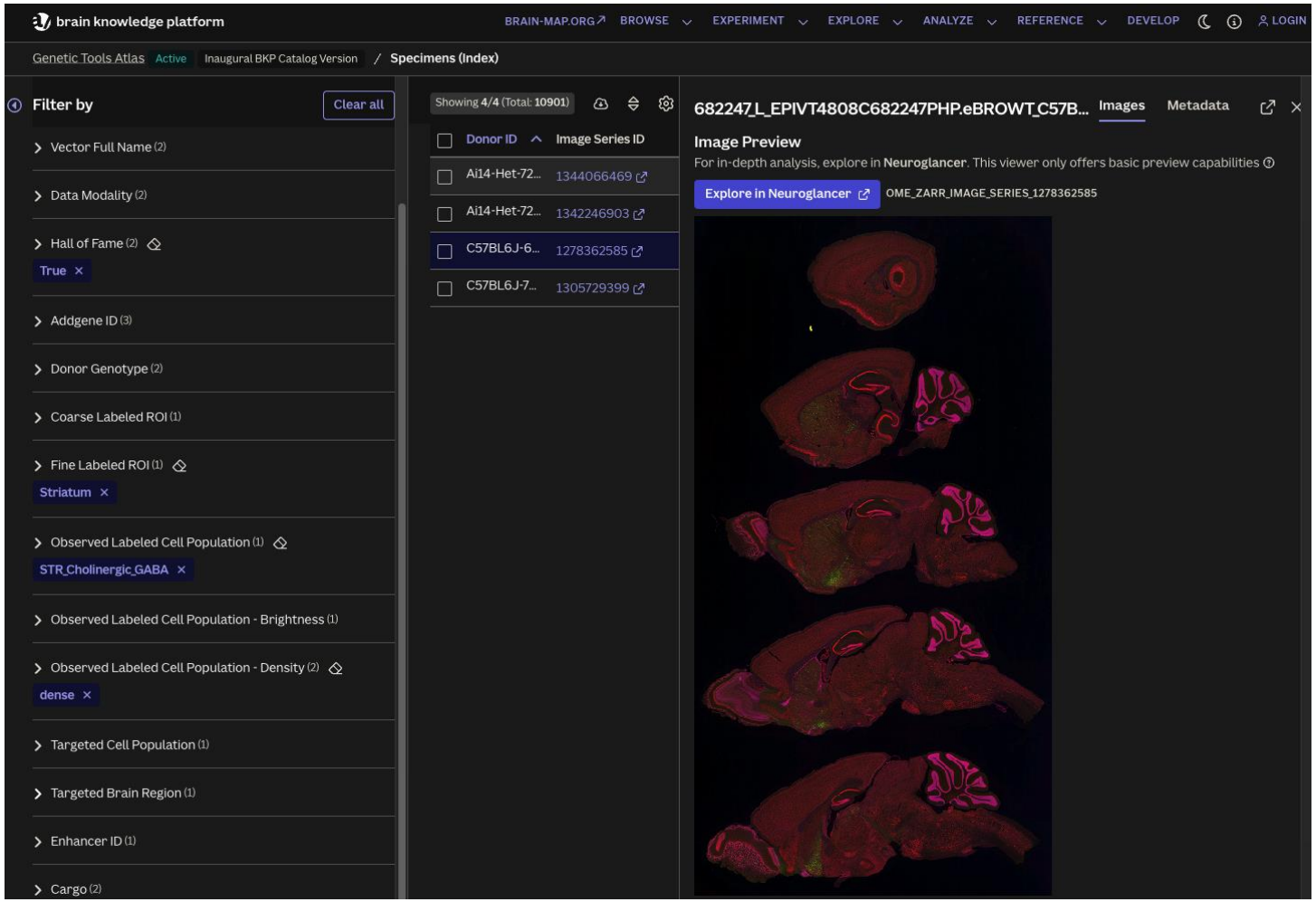
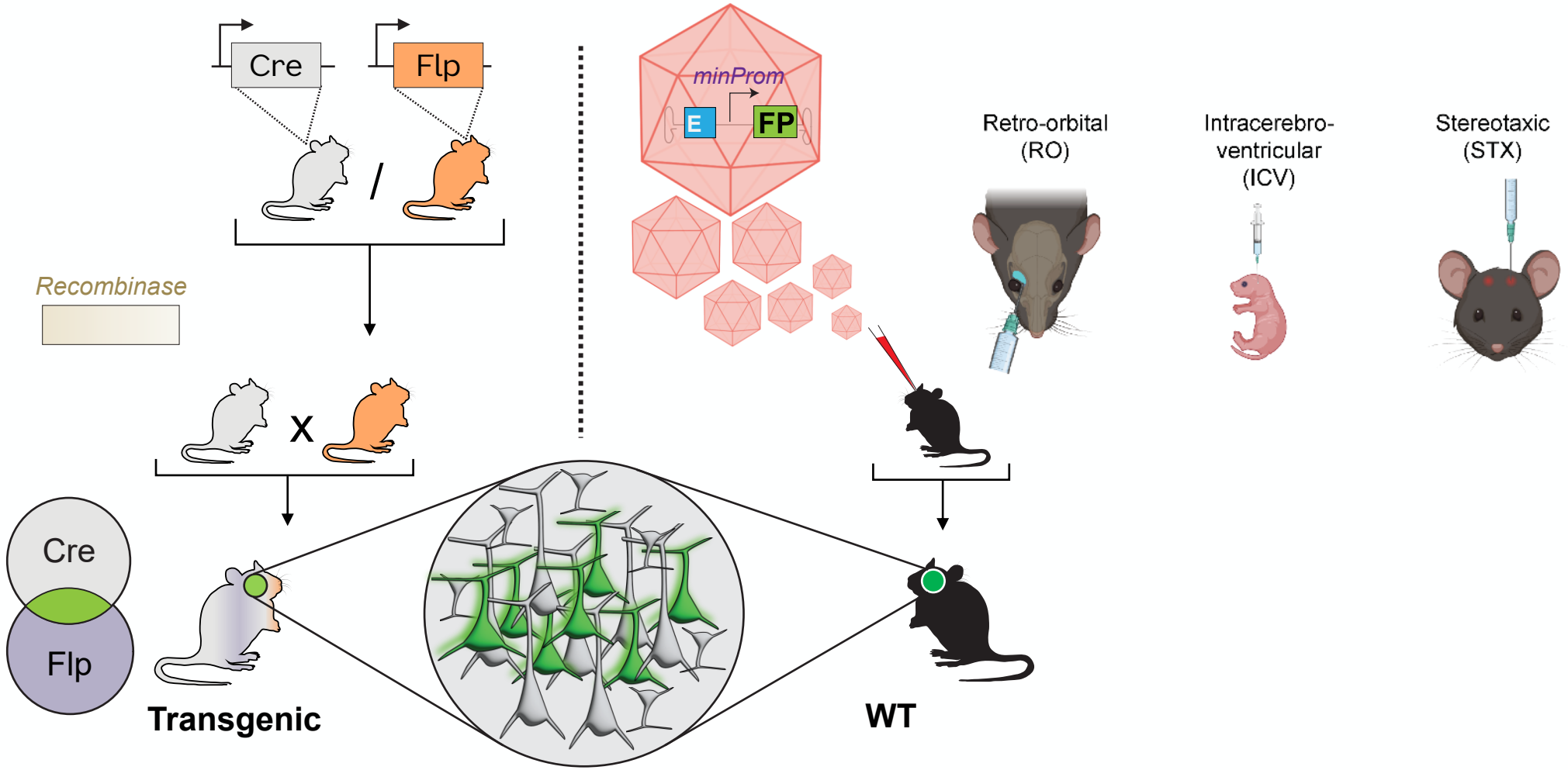
genetic tools atlas (GTA)/

- large genetic toolkit for selective gene expression in brain cell types of interest
- Searchable web tool
  - information and data on enhancer-adeno-associated viruses (enhancer AAVs)
  - information and data on mouse transgenes
- all tools developed and tested at the allen institute
  - Some are available to purchase from Addgene!



brain-map.org/bkp/experiment

genetic tools atlas (GTA)/ modalities used



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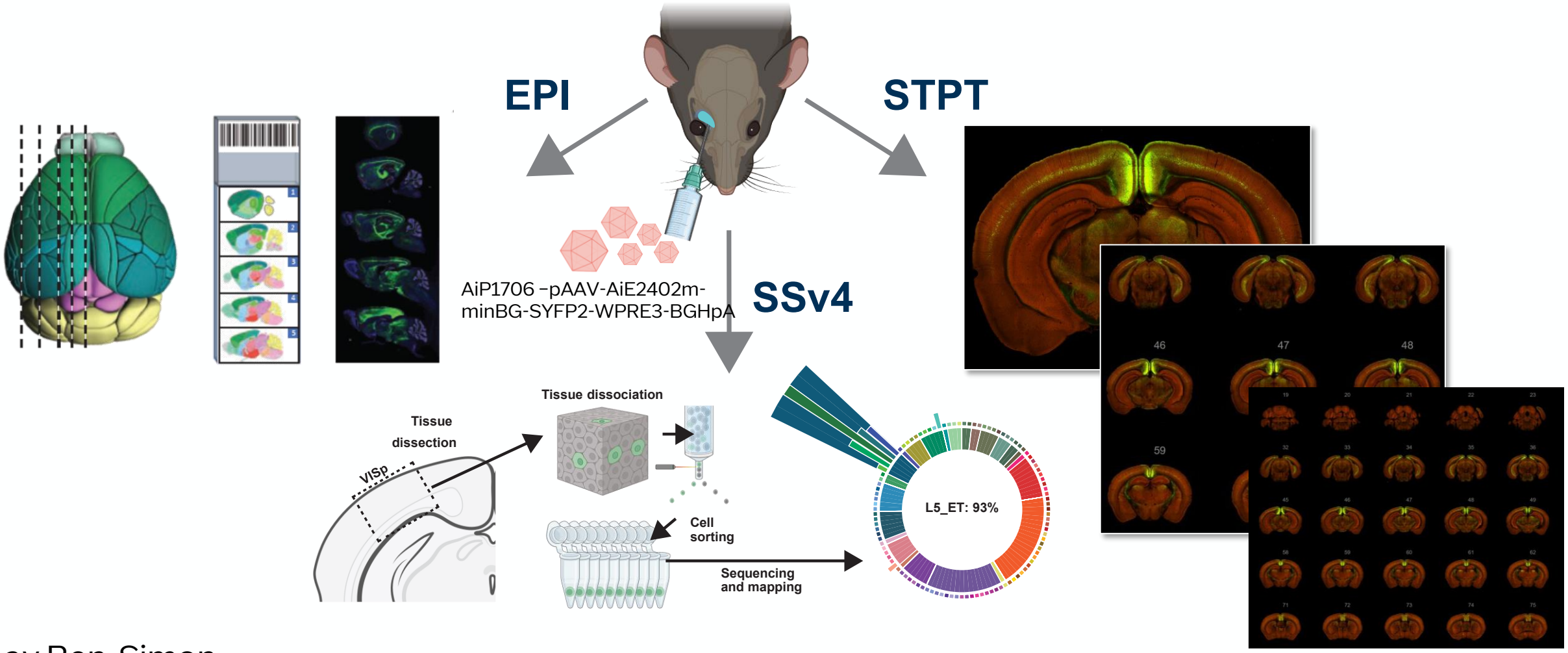
genetic tools atlas (GTA)/ enhancer screening pipeline

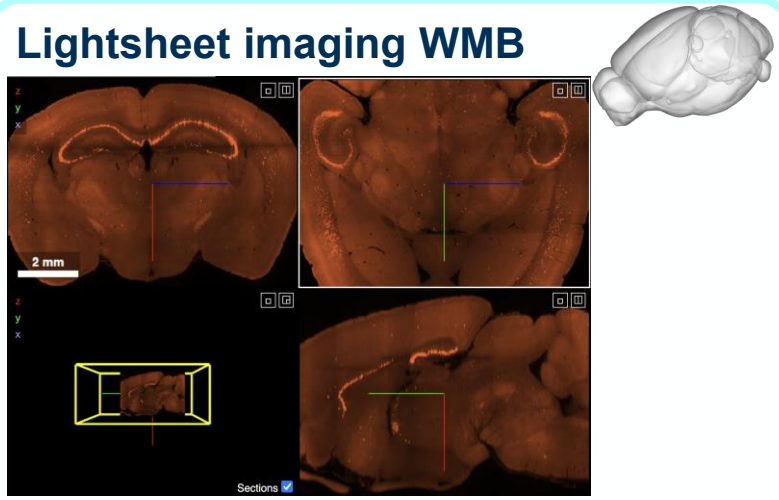


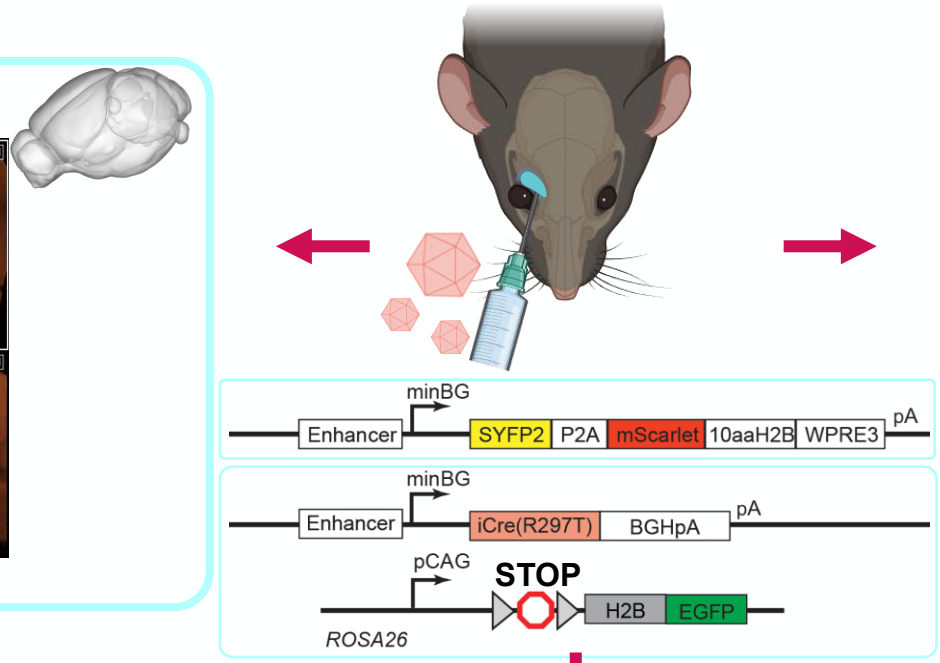
Yoav Ben-Simon

brain-map.org/bkp/experiment

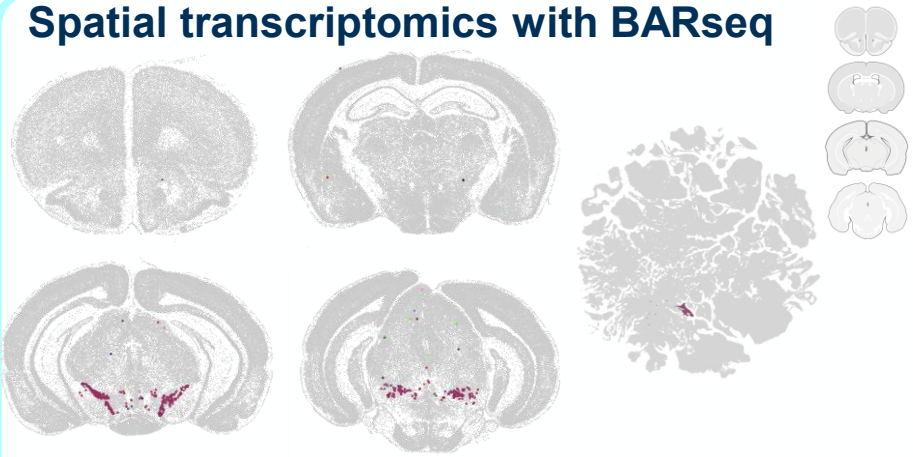
genetic tools atlas (GTA)/ new modalities in v2.0

Lightsheet imaging WMB



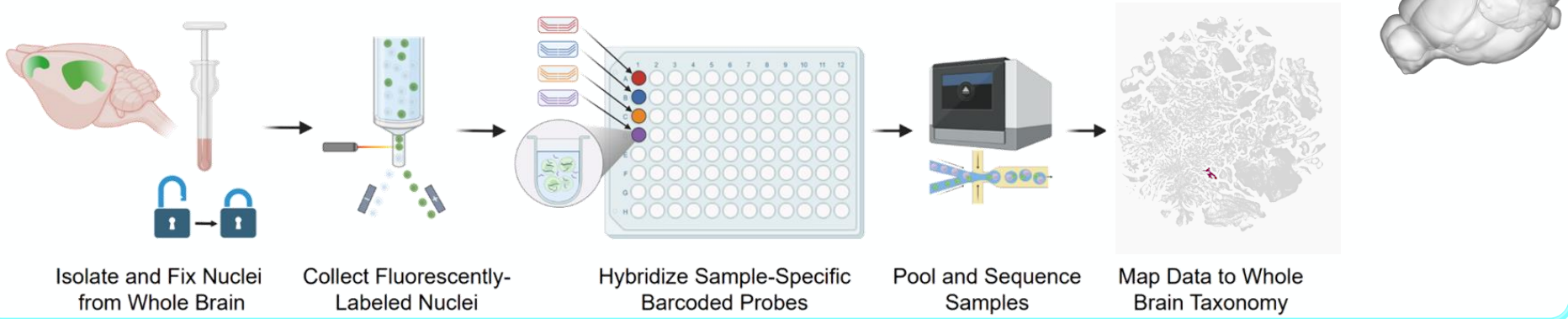


Spatial transcriptomics with BARseq



Map transcriptome to the WMB AiT21

10x Flex on WMB



brain-map.org/bkp/experiment

genetic tools atlas (GTA)/ acknowledgments

Leadership
- Bosiljka Tasic
- Jonathan Ting
- Boaz Levi
- Yoav Ben-Simon
- Tanya Daigle
- Trygve Bakken

Project management
- Sharon Way
- Bargavi Thyagarajan

Cloning
- Refugio Martinez
- Ximena Optiz-Araya
- Jada Roth
- Noah Greisman

Analysis
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- Sujatha Narayan
- Avery Hunker
- John Mich
- Deepanjali Dwivedi
- Morgan Wirthlin
- Nelson Johansen

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- LAS (Melissa Reding)
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- Imaging (Jack Waters)
- Tissue processing (Nick Dee)
- MolBio (Kim Smith)

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- Tyler Mollenkopf
- Sven Otto
- Sean Meharry
- Nicholas Lusk

Data management
- Aaron Oster
- Raymond Sanchez

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allen institute github repo/ abc_atlas_access

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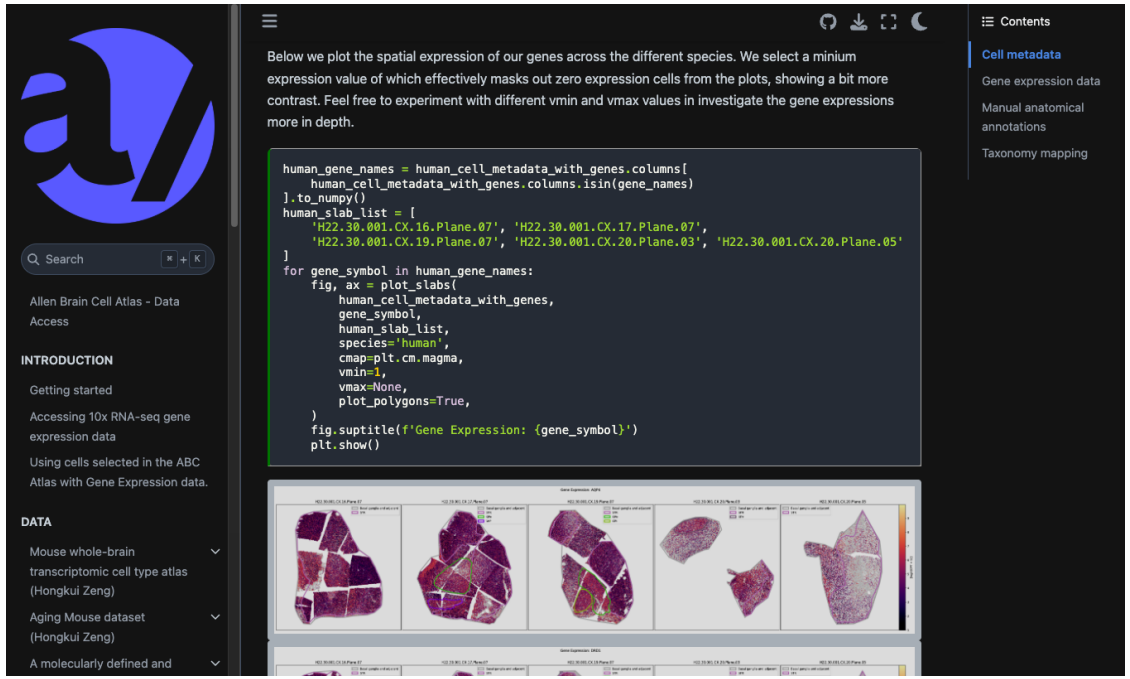
How do I create custom differential gene expression analysis?

How do I create custom visualizations using Python?

allen institute github repo provides an overview of the available data & how to download and use it

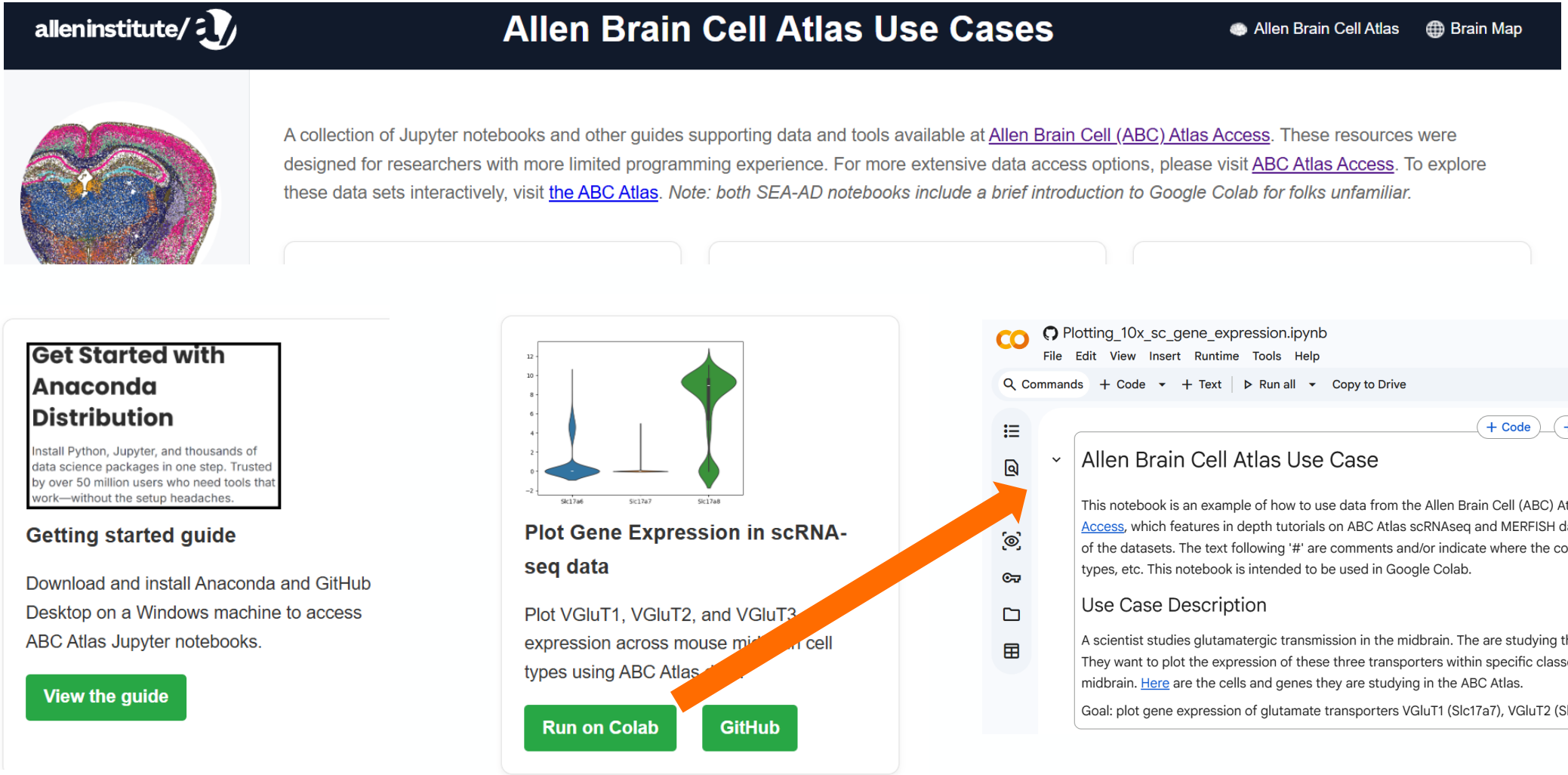
- Download full and partitioned data sets
- Access data definitions and schemas
- Tutorial Python notebooks with example queries



arn:aws:s3:::allen-brain-cell-atlas

alleninstitute.github.io/abc_atlas_access/intro.html

allen institute github repo/ SPOCTT



https://alleninstitute.github.io/SPOCTT/ABCAtlas_UseCases

agenda/morning

- 9:00-9:30 | check-in / breakfast
- 9:30-9:40 | welcome, logistics, and overview of allen institute
- 9:40-10:05 | cell typing at allen institute: taxonomies, data sets, and use cases
- 10:05-10:30 | human and mammalian brain atlas (HMBA)
- 10:30-10:40 | break
- 10:40-11:05 | localizing cell types with spatial transcriptomics
- 11:05-11:30 | cell-type targeted enhancer development
- 11:30-11:55 | Seattle Alzheimer's Disease Brain Cell Atlas in the brain health accelerator
- 11:55-12:05 | Overview of brain knowledge platform (BKP) tools & resources
- **12:05-12:10 | Preview of afternoon**
- 12:10-1:10 | lunch

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agenda/afternoon – breakout rooms

Session 1: connecting multimodal data to brain cell types

Room 1255-1265

1:10-1:50 | defining and using common coordinate frameworks (reference atlases, mesoscale)

1:50-2:00 | patch-seq overview and tools for analysis

2:00-2:20 | Troubleshooting Annotation with the BKP

2:20-2:30 | break

2:30-2:40 | ABC Atlas demo & external use case examples

2:40-3:40 | user exploration of allen institute tools

3:40-4:00 | feedback survey

4:00 – 5:00 | additional time for user exploration

Session 2: analysis of your own sequencing data using allen institute resources.

Room 1165 - 1185

1:10-1:35 | Using a reference cell type taxonomy to understand Alzheimer's disease

1:35-2:00 | cross-species alignment and comparative analyses

2:00-2:20 | other interactive tools for -omics data exploration (CellxGene, Cytosplore, etc.)

2:20-2:30 | break

2:30-2:40 | ABC Atlas demo & external use case examples

2:40-3:40 | user exploration of allen institute tools

3:40-4:00 | feedback survey

4:00 – 5:00 | additional time for user exploration